



CE364

Mechanics of Materials



CE364 MECHANICS OF MATERIALS

ABET eCOURSEBOOK

AY2025-2026

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SECTION I

COURSE ASSESSMENTS



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TAB A

AY2025 COURSE ASSESSMENT

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I Course Administration

A. Redbook Description

The screenshot below was taken from the Redbook on 8 JUL 25.

CE364 (Version: 2026 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
CE364	MECHANICS OF MATERIALS	2026	1	Civil and Environmental Engineering	3.5 (BS=0.0, ET=3.5, MA=0.0)
SCOPE					
This course studies the behavior of a variety of materials under normal, shear, torsional, bending and combined loads. The concepts of stress, strain, failure theory and failure mechanisms are explored. The loading, geometry, functional environment and material properties of machine or structural parts are used to relate the forces applied to a body to the resulting internal forces and deformations so that performance can be evaluated. Practical applications involving the design and adequacy of mechanical and structural elements under various loading and environmental conditions are emphasized. The course includes several laboratory experiences that require students to develop and conduct experiments, analyze and interpret data, and use engineering judgement to draw conclusions.					
LESSONS: 25 @ 75 min (1.560 Att/wk) LABS: 10 @ 75 min					
SPECIAL REQUIREMENTS:					
None					

Terms Offered and Enrollments:

Table I.A.1

AY	Term	Enrollment
20	1	--
	2	128
21	1	--
	2	118
22	1	--
	2	145
23	1	--
	2	143
	7	2
24	1	--
	2	144
25	1	--
	2	172
26*	1	--
	2	135

* forecast in Redbook as of 3 JUN 25.

2. Based on projected enrollments for AY26-2, the total number of sections will reduce from ten to eight. Current lab facilities will not permit more than two sections to run concurrently. Of note, 26-2 is the last time that ME students will take the course. 27-2 will be offered to Civil, Nuclear, and Chemical Engineering students.

Switching to course offerings on both Day 1 and Day 2 in AY 24-2 significantly complicated course administration. If possible, the Course Director strongly recommends that all course offerings be confined to either Day 1 or Day 2 (not both). To minimize conflict with Civil and Environmental Engineering Capstone Design offerings, Day 1 is preferred for MC364 for AY 26-2.

3. Proposed Changes to Redbook Description. None.

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B. Course Objectives

1. Current Course Objectives.
 - a. Calculate the internal forces, internal stresses, and deformations of axially loaded members, circular members in torsion, thin-walled pressure vessels, prismatic beams in bending, columns, and members subjected to combined loading and/or thermal effects.
 - b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure subjected to loading and/or thermal effects.
 - c. Apply appropriate theories of failure to analyze and design thin-walled pressure vessels and members subjected to loading and/or thermal effects.
 - d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.
 - e. Given a state of stress at a point, determine the principal stresses, the maximum in-plane shear stress, the angle to the principal plane, and the state of stress on any plane.
 - f. Apply deformation—force, strain—stress, and stress—force relationships to analyze and design structural members and machine components.
 - g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.
2. Proposed Changes to Course Objectives. No proposed changes for 26-2 since ME remains in the course. For 27-2, it is recommended to add a block of instruction on materials science, which was taken out of the course at request of the ME majors.

C. Textbook(s)

1. The following text(s) were used this year:

Hibbeler, R.C. *Statics and Mechanics of Materials*, 6th ed. With included Mastering Engineering subscription (required). \$119.99, but since this was the required text for MC300 in 25-1, this does not represent a required cost for MC364 this semester. ISBN 9780138124069.
2. Adjustments to the text requirement for next year: No changes are recommended for AY26-2. The MC300 and MC364 Course Directors and other faculty have already worked with Pearson to develop a custom reader which has prevented students from having to rent or buy a second textbook to cover only a few lessons of MC364. In preparation for the upcoming changes to CE300 and CE364 in future years, it is recommended to create a new custom reader that includes a block of instruction about failure theories and materials science. This change needs to be implemented concurrently with CE300 for Fall 27-1 textbook order.

D. Follow-up from Previous Course Assessment.

1. Change Lesson 17 on statically indeterminate beams to a dedicated lesson on deflection by superposition. **Implemented and recommend sustain (with adjustments).** Lesson 17 and 18 were removed and replaced with deflection by superposition and a deflection problem solving lesson. The recommendation is to keep statically indeterminate bending members out of MC364, but not focus so heavily on the method of superposition by relying on the beam tables. Recommend changing the deflection block of instruction to a three part block covering the integration method, discontinuity functions, and superposition.
2. Move WPR 3 earlier in the semester. **Implemented and recommend change.** WPR3 was placed on lesson 30 this semester to give more space between WPR3 and the TEE. However, the turnaround between WPR2 and WPR3 was too fast, with only five lessons in

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- between the two. Additionally, three lessons that were covered on WPR3 were taught before WPR2 was given to the cadets, causing confusion amongst studying priorities. The recommendation is to remove WPR3 and only give two WPRs. The point allocation on each WPR should increase and the time between WPRs should increase.
3. Add an additional lesson on Statically Indeterminate Axial Members. **Implemented and recommend change.** This lesson felt unnecessarily redundant and could be replaced with additional instruction on more critical topics such as deflection and combined loading.
 4. Test steel and cast iron in MC364 uniaxial tension test labs: **implemented and recommend sustain.** This choice highlights on Lesson 2, before Cadets are prepared to understand their implications, that there is a difference in the behavior of ductile and brittle materials. This experiential learning is called on again near the end of the course when separate theories of failure for ductile and brittle materials are introduced. The current Course Direction also recommends adding a brittle specimen for comparison purposes to the torsion lab.

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II Course Content

A. Outline of Lessons and Labs

Table II.A.1

Lesson	Title/Topic	Supported Course Objectives						
		a	b	c	d	e	f	g
1	Course Intro & MC300 Review	a						
2	Uniaxial Tension Lab	a					f	g
3	Axial Deformation	a					f	
4	Buckling	a					f	
5	Statically Indeterminate Axially Loaded Members I	a	b				f	
6	Statically Indeterminate Axially Loaded Members II	a	b				f	
7	Statically Indeterminate Axially Loaded Members III	a	b				f	
8	Thermal Effects on Axially Loaded Members	a					f	g
9	Torsion Lab	a					f	
10	Introduction to Torsion	a					f	
11	Solving Torsion Problems	a	b				f	
12	WPR 1 (Axial and Torsion)	a					f	g
13	Beam Lab I	a					f	
14	Flexural Normal Stress	a					f	
15	Flexural Shear Stress	a					f	g
16	Beam Lab II	a					f	
17	Deflection of Beams	a	b				f	
18	Statically Determinate Bending Members	a	b				f	
19	Beam Analysis	a					f	
20	Beam Engineering Design Project	a					f	g
21	Stress Elements	a	b				f	
22	Stress Transformation	a					f	
23	Principal Stresses and Max Shear Stress	a					f	
24	WPR 2 (Flexure)	a				e	f	
25	Mohr's Circle	a				e	f	
26	Strain Transformation	a			d	e	f	
27	Thin-Walled Pressure Vessels I	a			d	e	f	g
28	Thin-Walled Pressure Vessels II	a			d	e	f	g
29	Combined Loading (Axial and Torsion)	a				e	f	
30	WPR 3 (Stress & Strain Transformation and TWPV)	a		c	d	e	f	
31	Principal Stresses in Flexural Members	a				e	f	
32	General Combined Loading	a				e	f	
33	Theories of Failure - Ductile Material	a		c		e	f	
34	Theories of Failure - Brittle Material	a		c		e	f	
35	Course Review and Wrap-up	a	b	c	d	e	f	

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Table II.A.2

Lesson	Laboratory Exercise	Supported Course Objectives						
		a	b	c	d	e	f	g
2	Uniaxial Tension & Compression Lab	a					f	g
9	Torsion Lab	a					f	g
13	Beam Lab I	a					f	g
16	Beam Lab II	a					f	g
20	Beam Engineering Design Project	a					f	g
27	TWPV (Strain Gage Mounting)	a			d	e	f	g
28	TWPV Lab	a			d	e	f	g

Field Trips: MC364 requires enrolled Cadets to attend Projects Day. Civil and Mechanical Engineering majors are required to attend the D/CME Projects Day; Cadets with other majors are welcome at the D/CME event but are authorized to attend their own or another Department's event if they choose. A Reading Quiz (10 points) due after Projects Day requires Cadets to relate the details of the presentations they visited and to reflect on what they learned and their own plans for capstone projects or research.

In past years, Cadets were placed on trip section to facilitate their attendance. This year, a trip section was not submitted since last year's trip section was disapproved by the Registrar's Office. For future years, it is recommended to continue using the quiz as motivation for cadets to attend projects day and leave the ownership on the cadets to prioritize their scheduling.

B. Proposed Changes to Lessons and Labs.

1. **Replace** MC300 review (LSN1) with Axial Stress/Strain review to lead into the UTT lab on LSN2. Since the majority of students completed MC300 in the previous fall semester, an entire lesson devoted to review is not necessary. Instead, a Homework 0 covering MC300 review topics could be assigned.
2. **Remove** Indeterminate Axial Member III (LSN7). This lesson felt unnecessarily redundant and could be replaced with additional instruction on more critical topics such as deflection and combined loading.
3. **Remove** Statically Determinate Bending Members (LSN18). This lesson relied too heavily on the beam tables for using the method of superposition. In place of this lesson, **add** a second lesson on Deflection of Beams (LSN 28 AY26-2). This new lesson should cover continuity equations and the fourth-order integration method.
4. **Remove** WPR3 and only give two WPRs. The point allocation on each WPR should increase and the time between WPRs should increase. Three WPR's added a significant grading load on the instructor team and there was a very quick turnaround between WPRs 2 and 3.
5. **Move** buckling to the end of the course (LSNs 31 & 32 AY 26-2) and add a second lesson to cover the derivation of Euler's buckling equation. Placing the topic after flexure theory allows students to better understand the foundational principles behind Euler buckling. For future years (AY27-2) Buckling I & II could be moved to CE300 to make room for materials science.
6. **Move** Beam EDP to the end of the course (LSN35) to better balance course assignments. Positioning the EDP after the deflection block enables more thorough discussion of serviceability checks in beam design.

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Table II.B.1

Lesson	Title/Topic	Supported Course Objectives						
		a	b	c	d	e	f	g
1	Course Intro & Axial Stress-Strain	a						
2	Uniaxial Tension Lab	a					f	g
3	Axial Deformation	a					f	
4	Statically Indeterminate Axially Loaded Members I	a	b				f	
5	Statically Indeterminate Axially Loaded Members II	a	b				f	
6	Thermal Effects on Axially Loaded Members	a	b				f	
7	Torsion Lab	a					f	g
8	Introduction to Torsion	a					f	
9	Solving Torsion Problems	a					f	
10	Beam Lab I	a			d		f	g
11	Flexural Normal Stress	a					f	
12	Flexural Shear Stress	a					f	
13	Beam Lab II	a			d		f	g
14	WPR 1 (Axial, Torsion, Flexure)	a					f	
15	Combined Loading (Axial and Torsion)	a					f	
16	Combined Loading (Flexure and Axial)	a					f	
17	Stress Elements	a	b			e	f	
18	Stress Transformation	a				e	f	
19	Principal Stresses and Max Shear Stress	a				e	f	
20	Mohr's Circle	a				e	f	
21	Principal Stresses in Flexural Members	a				e	f	
22	Strain Transformation	a			d	e	f	
23	Thin-Walled Pressure Vessels I	a		c	d	e	f	g
24	Thin-Walled Pressure Vessels II	a		c	d	e	f	g
25	General Combined Loading	a				e	f	
26	WPR 2 (Combined Loading, Stress Elements, TWPV)	a		c	d	e	f	
27	Deflection of Beams I – Integration Method	a	b				f	
28	Deflection of Beams II – Discontinuity Functions	a	b				f	
29	Deflection of Beams III – Superposition	a	b				f	
30	Beam Design	a				e	f	
31	Buckling I	a					f	
32	Buckling II	a					f	
33	Theories of Failure - Ductile Material	a		c		e	f	
34	Theories of Failure - Brittle Material	a		c		e	f	
35	Course Review and EDP Testing	a	b	c	d	e	f	g

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Table II.B.2

Lesson	Laboratory Exercise	Supported Course Objectives						
		a	b	c	d	e	f	g
2	Uniaxial Tension Lab	a					f	g
7	Torsion Lab	a					f	g
10	Beam Lab I	a					f	g
13	Beam Lab II	a					f	g
23	TWPV (Strain Gage Mounting)	a					f	g
24	TWPV Lab	a			d	e	f	g
35	Beam Engineering Design Project	a			d	e	f	g

C. Graded Requirements

1. Summary of Graded Requirements.

Table II.C.1

Requirement	Qty	Points	Total	% of Total
Homework	8	25	250	12.5%
Reading Quizzes	11	10	110	5.5%*
Engineering Design Project	1	100	100	5%
Laboratory Reports	4	100	400	20%
Written Partial Reviews	3	200	600	30%
MATLAB Codes via AI	4	10	40	2%
Term End Examination	1	500	500	25%
	32	TOTAL=	2000	100%

* The lowest three reading quizzes were dropped.

7. Proposed Changes to Graded Requirements:
 - a. Remove reading quizzes and replace them with in-class quizzes (Canvas) covering FE-style questions during each major instructional topic.
 - b. Reduce lab report requirement to three (remove the UTT lab report), while maintaining the existing lab structure. Reallocate these points toward homework assignments. The data from the UTT lab report can be reported in a homework assignment.
 - c. Remove the MATLAB AI assignments. Integrate MATLAB coding into homework assignments.
 - d. Remove WPR3 and increase the point value of WPR1 and WPR2 to 250 points each.

Table II.C.2

Requirement	Qty	Points	Total	% of Total
Homework	9	50	450	22.5%
In-Class Quizzes	5	20	100	5%
Engineering Design Project	1	150	150	7.5%
Laboratory Reports	3	100	300	15%
Written Partial Reviews	2	250	500	25%
Term End Examination	1	500	500	25%
	21	TOTAL =	2000	100%

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III Course Assessment

A. Course Director Assessment of Course Objectives.

Scale:

5 = Objective Clearly Met. 100% of students achieved the objective and can be expected to demonstrate proficiency.

4 = Objective Solidly Met. Over 90% of students met the objective.

3 = Objective Satisfactorily Met. Over 70% of students met the objective.

2 = Objective Marginally Met. Roughly 50% of students met the objective or most students minimally met the objective.

1 = Objective Not Met. Over 75% of students did not meet the objective.

The definition of “meeting a course objective” is achieving at least a “C” level (i.e., $\geq 70.0\%$) on the task.

Table III.A

Course Objective	Current Assessment (AY 25)	Previous Assessment (AY 24)	How Evaluated and Remarks
a. Calculate the internal forces, internal stresses, and deformations...	4	4	WPR1: 86% Avg. & 93.6% > C WPR2: 87.4% Avg. & 91.9% > C WPR3: 84.5% Avg. & 90.1% > C TEE: 80.6% Avg. & 86.7% > C
b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure...	4	3	WPR1: 86% Avg. & 93.6% > C WPR2: 87.4% Avg. & 91.9% > C TEE Problem 1: 88% Avg. & 87% > C
c. Apply appropriate theories of failure to analyze and design...	2	2	WPR3: 84.5% Avg. & 90.1% > C TEE Problem 6a: 53% Avg. & 24% > C TEE Problem 6b: 73% Avg. & 65% > C Low rating may be caused by few assessments. Let's discuss alternatives.
d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.	3	4	WPR3: 84.5% Avg. & 90.1% > C TEE Problem 6a: 53% Avg. & 24% > C
e. Given a state of stress at a point, determine the principal stresses, the maximum in-plane shear stress, the angle to the principal plane, and the state of stress on any plane.	4	4	WPR3: 84.5% Avg. & 90.1% > C TEE Problem 5: 83.7% Avg. & 86% > C
f. Apply deformation—strain, strain—stress, and stress—force relationships to analyze and	4	4	WPR1: 86% Avg. & 93.6% > C WPR2: 87.4% Avg. & 91.9% > C

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design structural members and machine components.			TEE Problem 3: 83% Avg. & 88.3% > C
g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.	4	5	LAB1: 92% Avg. & 98% > C LAB2: 93% Avg. & 100% > C LAB3: 93% Avg. & 96% > C LAB4: 85% Avg. & 100% > C

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B. Assessment of ABET Program-Level Student Outcomes Directly Supported by this Course

Standard Grade Percentage Cutoffs:		Performance Description	Performance Rating
A+	96.67%	Outstanding	5
A	93.33%		
A-	90.00%	Very Good	4
B+	86.67%		
B	83.33%	Good	3
B-	80.00%		
C+	76.67%	Fair	2
C	73.33%		
C-	70.00%		
D	65.00%	Poor	1
F	0.00%		
--	--	Not Assessed	0

1. An ability to identify, formulate, and **SOLVE** complex engineering problems by applying principles of engineering, science, and mathematics; N/A.
2. An ability to apply engineering **DESIGN** to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors; N/A.
3. An ability to **COMMUNICATE** effectively with a range of audiences;

Table III.B.3

Embedded Performance Indicator (Event Points Possible)	# CE Cadets Assessed by Grad Year	Performance Ratings by Bin, TECOM, and West Point Writing Program						
MC364: Thin-Walled Pressure Vessel (100 Points)	55 ('27)	Performance Level Rating (# Cadets in Bin)						Average Score (%)
		Outstanding 5	Very Good 4	Good 3	Fair 2	Poor 1	Not Assessed	
		4	23	8	20	0	0	84.9%
		Category	TECOM Rating (# Cadets in Bin)					Average
			Exemplary 4	Satisfactory 3	Marginal 2	Not Proficient 1	Not Assessed	
		Function	28	27	0	0	0	3.5
		Correctness	24	31	0	0	0	3.4
		Depth	33	20	2	0	0	3.6
		Research	38	17	0	0	0	3.7
		Creativity	40	15	0	0	0	3.7
		Category	WPWR Rating (# Cadets in Bin)					Average
			Exemplary 4	Satisfactory 3	Marginal 2	Not Proficient 1	Not Assessed	
		Substance	6	49	0	0	0	3.1
		Organization	25	28	2	0	0	3.4
		Style & Presentation	6	47	2	0	0	3.1
		Mechanics & Correctness	4	42	8	1	0	2.9
		Documentation	20	26	9	0	0	3.2
		Overall Proficiency	17	38	0	0	0	3.3

4. An ability to recognize **ETHICAL** and **PROFESSIONAL** responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts; N/A.

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5. An ability to function effectively on a **TEAM** whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives; N/A.

6. An ability to develop and conduct appropriate **EXPERIMENT**ation, analyze and interpret **DATA**, and use engineering judgment to draw conclusions;

Table III.B.6

Embedded Performance Indicator	Event Points Possible	# CE Cadets Assessed in AY by Grad Year	Performance Level Rating (# Cadets in Bin)						Average Score (%)
			Outstanding 5	Very Good 4	Good 3	Fair 2	Poor 1	Not Assessed	
MC364: Thin-Walled Pressure Vessel (100 Points)	100	55 ('27)	4	23	8	20	0	0	85.2%

7. An ability to acquire and apply **NEW KNOWLEDGE** as needed, using appropriate learning strategies. N/A.

C. Assessment of Changes Made this Semester.

1. *Text-centric vs. Video-centric flipped class format.* Lesson videos developed in prior years made extensive reference to the previous textbook and were deemed no longer appropriate for use with the new textbook. They were posted to Canvas for Cadets' optional use, but the required pre-class preparation included only readings from the new textbook and a reading quiz designed to reward preparation. This year, the cadets widely criticized them as misaligned with preparation goals. Concerns included lack of clarity, inconsistent difficulty, and time constraints to complete them on time. In lieu of reading quizzes, it is recommended to **sustain** the flipped class structure, but with some **modifications**. The cadets had overwhelming positive feedback about the amount of board work offered in the course. However, they expressed a desire for more instruction at the beginning of class to cover difficult concepts prior to taking boards. Therefore, the course director recommends replacing the reading quizzes with a fewer amount of in-class quizzes conducted on Canvas at the end of class. These in-class quizzes would be FE style problems and only worth 5% of the cadets' overall grade.
2. *WPR 3 moved to Lesson 30.* The previous two years, the WPR was moved forward in the course by about two weeks, to Lesson 30. The lessons covered by the WPR were adjusted so that it did not cover Combined Loading and Theories of Failure, which were taught after the WPR. Advantages of this approach are that it gives the instructor team more time to grade the WPRs and discuss the results, and more time to remediate any shortfalls in Cadet understanding prior to the TEE. However, WPR2 was placed too late in the course (LSN24) and was administered after three lessons were taught that covered content for WPR3. The Course Director recommends **improving** this change by eliminating WPR3 and spacing out WPR1 and WPR2 more evenly across the semester, reducing the grading load on the instructor team.
3. *Adoption of the FE Reference Manual as the only authorized exam reference.* Previous iterations of this class relied on an issued formula card and one letter-sized cadet-created notes sheet. With the change to the new textbook, the reference card had to be heavily revised to accommodate changes in variables and other differences between the two textbooks. At the same time, instructors frequently noticed that Cadets were using their notes sheets not for formulas, but to copy solutions of example problems and then using these solutions to pattern-match during the major graded events. The migration to the new text provided an opportunity to solve both problems with the adoption of an established reference document, the Fundamentals of Engineering Exam Reference, as the only authorized reference for major graded events, eliminating the self-created notes sheet. The FE Reference Manual is a very long document, so

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for ease of copying and distribution it was reduced to an excerpt which contained only the formulas relevant to MC364. Advantages of this approach included giving Cadets the opportunity to become familiar with a resource they will need when they take the Fundamentals of Engineering Exam. It also afforded an opportunity to discuss in class the importance of professional licensure and the process required to attain it. It eliminated the copying of problem solutions and placed all Cadets on equal footing in the exams. Disadvantages include the loss of the educational benefit of distilling course notes to the one page of most important material. In addition, the FE Reference Manual sometimes presents concepts differently and uses different variables from the textbook, which could cause confusion if not carefully introduced in class. The Course Director recommends **sustaining** this change.

4. *Further Reduction in Scope of Lab Report supporting calculations.* This year's Cadet feedback still included many observations that the labs were too complicated for their point value. This year's lab exercises did not change from last year's and the lab reports were descoped to include reducing the number of supporting calculations. The Course Director recommends further descoping the lab report requirements by replacing the first lab report with a problem set style submission requirement. The UTT lab is already conducted in MC300 (albeit with only a single specimen), so the cadets are already getting exposure to producing a lab style report for a Uniaxial Tension Test. It is recommended to keep the point values of the other three lab reports the same and reallocate the points from the first lab report to the WPRs and the homeworks.
5. *Late changes to academic schedule.* Due to unforeseen circumstances early in the semester (President Carter's funeral) and the inability to hold class on Friday before spring break, scheduling became quite difficult across both Day 1 and Day 2 sections. At some points in the semester, the Day 2 sections were three lessons ahead of Day 1. In order to mitigate future issues with course scheduling, it is recommended to merge CE364 into one academic day. Day 1 is preferable to not conflict with capstone hour and to not overwhelm the lab support team. For future semesters (27-2), it may be worth considering moving to a 40x55 lesson schedule with lab periods. Once the course changes in 27-2, there will be fewer sections required (4 to 5 max) so it won't be necessary to hold the course on both Day 1 and Day 2.

D. Assessment of Resources and Impacts.

1. *Laboratory Technician and Resource Support.* A quarter of cadets' grade in this course is devoted to the laboratories and EDP (500/2000). It is critical to have the support of a laboratory technician for the setup and execution of all laboratory activities. Many of the labs require setup in multiple classrooms. It is important that there is enough equipment for each classroom. With the current shortage of classroom space and sharing classrooms with other academic programs, the ability to set up laboratory equipment in the classroom is significantly restricted. It is recommended to shift the course schedule to Day 1's with two sections per hour to reduce the demand on laboratory technician support.

2. *Equipment.* Much of the equipment for MC364 is aging and has not been serviced in several years. The tension machines, torsion machines, and aluminum beam specimens are all experiencing issues which have had significant impacts on the lab delivery. These machines must be checked each semester.

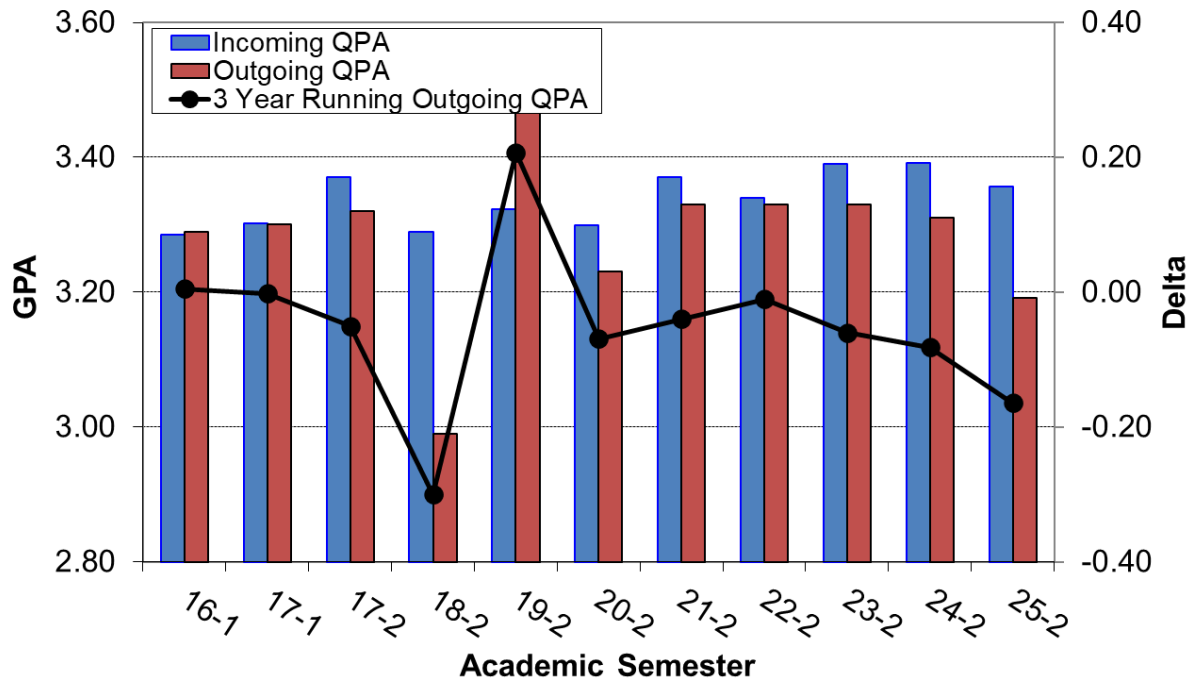
3. *Material.* Raw material is needed for many of the laboratory and EDP executions. The uniaxial tension laboratory for MC300 and MC364 needs samples of steel, aluminum and cast iron. For the torsion laboratory, we need samples of brass and cast iron. For the Beam EDP, we need balsa wood (36"x3"x1/4" & 36"x3"x1/8"). For the TWPV Lab, we need Coca Cola Cans, hot dogs, and strain gages.

4. *Lab Space.* Currently, all tension machines are in the primary cadet material testing lab in D10. All torsion machines are stored in C19 and then are moved into D10 for execution of the lab. D10 will not permit more than two sections to run the lab simultaneously. If time or classroom space constraints require us to run more sections concurrently, we will need more lab space.

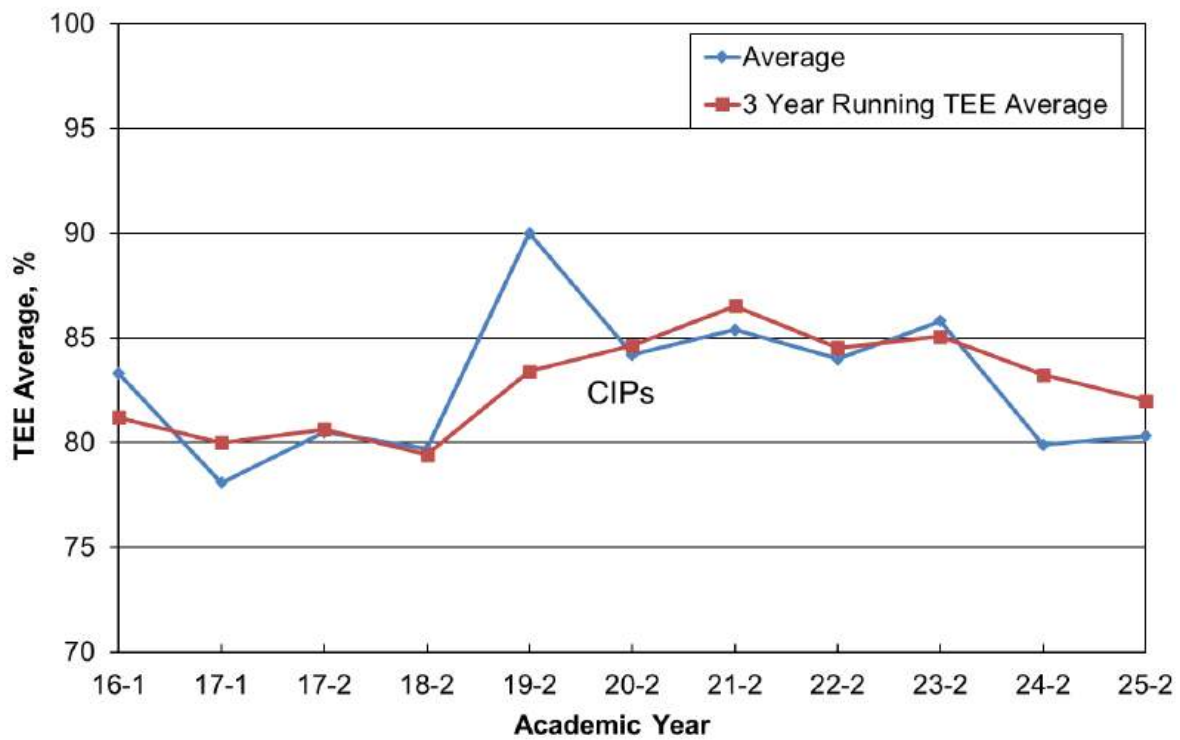
MC364 Course Assessment Package, AY25

MC364 Course Assessment Package, AY25

E. Course QPA Graph



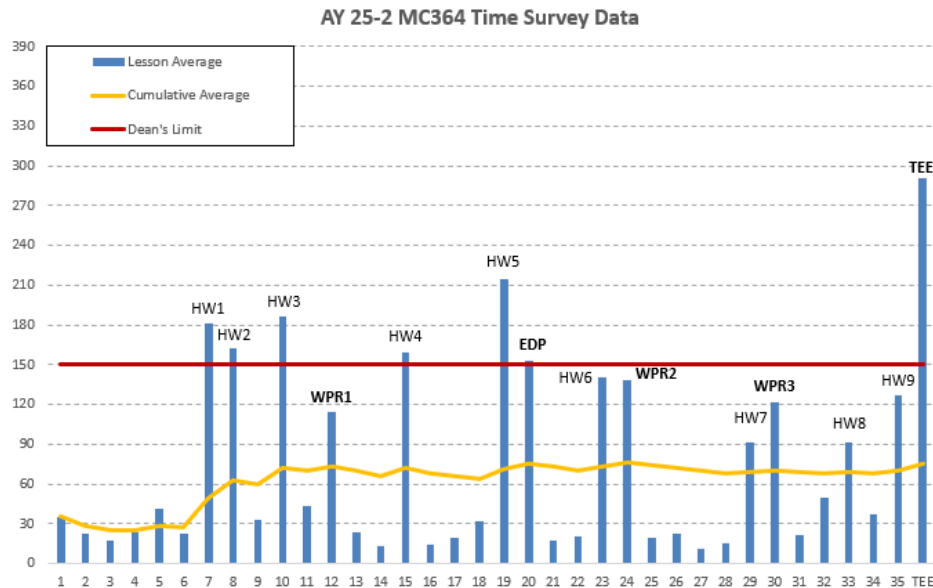
F. TEE Grades Graph



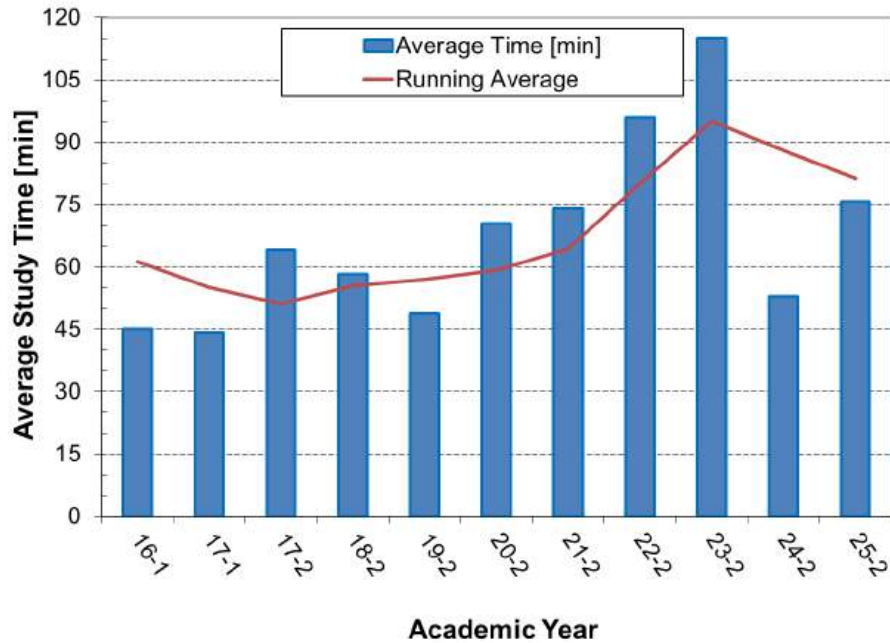
G. Course Time Study Graph

MC364 Course Assessment Package, AY25

1. Current Semester time survey data. (AVG: 68 minutes/lesson)



2. Historical Time Survey Data



3. Any comments on time survey data if needed. Per-lesson engagement is slightly higher than last year. There are significant peaks surrounding the homework due dates, which were significantly larger than the time spent preparing for WPRs. Based on end of course feedback, it is recommended to space out graded assignments more efficiently and place more weight on the homework to better balance the time to point ratio.

H. Course End Feedback. No course-specific questions were included in the course end survey.

MC364 Course Assessment Package, AY25

IV. Program Director's Current Areas of Emphasis

- A. *If applicable, describe how the course applies materials science. [Guidance provided by ASCE](#) states that: materials science is the scientific study of the properties of solid materials and how a material's composition and structure determine those properties.*

Currently, MC364 does not have a dedicated block of instruction devoted to materials science. There is room for discussion about material properties during the laboratory exercises. For example, the discussion of the difference between the molecular structure of steel and cast iron during the Uniaxial Tension Lab was included, but not formally taught. It is recommended for 27-2 to include a block of materials science.

- B. *If applicable, describe the course's requirements for written communication. Provide commentary on whether cadets' written communication skills seem adequate to complete these requirements.*

This course requires cadets to complete four written lab reports that emphasize technical communication, including the clear presentation of theory, means and methods, interpreting results, and drawing conclusions. Cadets are expected to use appropriate engineering terminology, logical organization, and concise explanations to convey their findings in two pages or less.

Overall, cadets' written communication skills appear adequate to meet these requirements. Most cadets demonstrate the ability to articulate technical content clearly and follow the prescribed lab report format. While some reports would benefit from improved clarity and more concise writing, the majority of cadets effectively communicate their understanding of the engineering concepts.

- C. *If applicable, describe prerequisite technical knowledge and skills which are important to cadets' success in the course. Provide commentary on whether cadets come to the course with an appropriate foundation in these areas.*

MC364 relies heavily on prerequisite knowledge from MC300 and MA204. From MC300, cadets are expected to have a solid understanding of how to apply equations of equilibrium to rigid bodies, construct shear and moment diagrams, and comprehend the fundamental behavior of axial and flexural members. MA204, which is currently a co-requisite, provides the multivariable differential and integral calculus necessary for analyzing beam deflection and column buckling.

Overall, most cadets enter MC364 with an appropriate foundation in these areas. However, occasional gaps in conceptual understanding, particularly with shear and moment diagrams or calculus-based derivations, can pose challenges throughout in the course. It may be worth allocating more time to shear and moment diagrams in MC300/CE300.

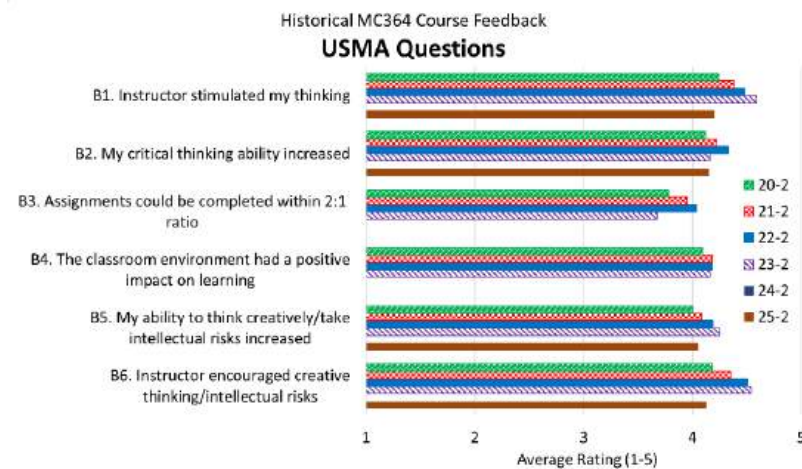
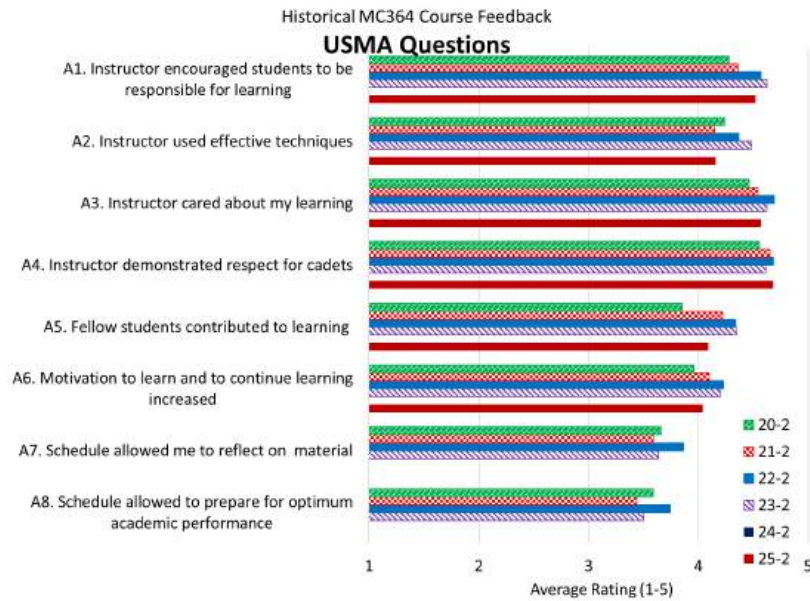
Enclosures

1. Historical Course Feedback on USMA Questions
2. Term 25-2 Course Feedback on USMA Questions
3. Historical Course Feedback on CME Questions
4. Term 25-2 Course Feedback on CME Questions
5. Course Narrative Feedback
6. Course Numeric Feedback
7. Other Feedback

MC364 Course Assessment Package, AY25

Enclosures:

1. Historical Course Feedback on USMA Questions

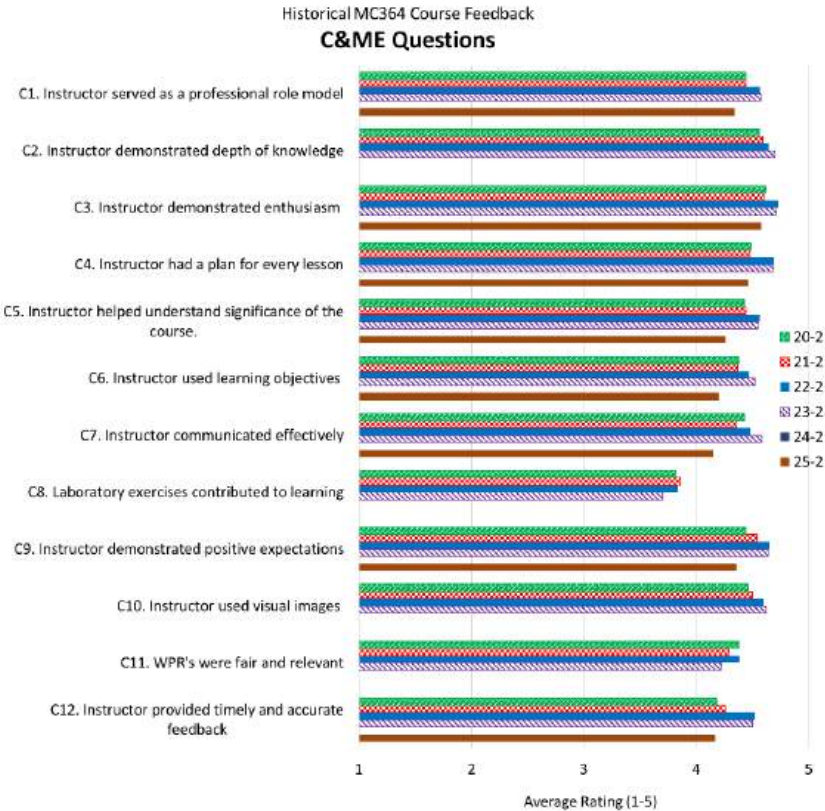


2. Term 25-2 Course Feedback on USMA Questions

Aggregated data are not yet available for cadet feedback on USMA Questions.

3. Historical Course Feedback on CME Questions

MC364 Course Assessment Package, AY25



4. Term 25-2 Course Feedback on CME Questions

Aggregated data are not yet available for cadet feedback on CME Questions.

5. All Course Narrative Feedback for the current period of assessment.

Additional comments about the course were gathered from an end of course canvas survey. Responses were consolidated using AI (ChatGPT) into strengths, improves, and overall recommendations.

MC364 Course Assessment Summary – Student Feedback Overview

Overall Sentiment:

Students found the course intellectually engaging, well-paced, and valuable for reinforcing engineering problem-solving skills. Instructional enthusiasm and the problem-solving focus were frequently praised. However, several themes emerged for potential course refinement.

Key Strengths

- Problem-Solving Focus:**
 Board work and in-class practice were consistently cited as effective learning tools. Students appreciated the logical flow and fair challenge of assignments and exams.
- Instructional Quality:**
 Instructors were recognized for their clarity, enthusiasm, and support
- Hands-On Labs:**
 Labs like the Thin-Walled Pressure Vessel (TWPV) and “hot dog” day were engaging and helped reinforce concepts.

MC364 Course Assessment Package, AY25

- **Support Resources:**

Posted solutions, pre-class videos, and review sessions were seen as highly beneficial. Mastering Engineering was also valued as a learning aid.

Areas for Improvement

- **Reading Quizzes:**

Students widely criticized them as misaligned with preparation goals. Concerns included lack of clarity, inconsistent difficulty, and time constraints.

- **Workload Management:**

Several noted uneven assignment distribution, especially around breaks and WPRs, leading to periods of high academic load.

- **Lab Alignment and Structure:**

Some labs felt disconnected from course content or overly time-consuming. Requests were made for clearer guidance and alignment with lessons.

- **Instructional Balance:**

Students requested more instructor-led problem walkthroughs and conceptual explanations before transitioning to board work.

- **Major-Specific Relevance:**

Mechanical engineering majors felt some course content leaned too heavily toward civil engineering applications.

Recommendations

- Reevaluate the structure and purpose of reading quizzes; consider alternative formats or timing.
- Improve consistency in assignment scheduling and align labs/WPRs more deliberately with lesson blocks.
- Increase conceptual instruction before board work; provide more guided examples early in lessons.
- Consider adjusting labs for relevance and efficiency; provide templates or exemplars where helpful.
- Explore minor tailoring or examples to improve perceived relevance for mechanical engineering majors.

6. All Course Numeric Feedback

N/A.



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TAB B

AY2024 COURSE ASSESSMENT

MC364 Course Assessment Package, AY24

I Course Administration

A. Redbook Description

The screenshot below was taken from the Redbook on 19 MAY 24.

MC364 (Version: 2020 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
MC364	MECHANICS OF MATERIALS	2020	1	Civil and Mechanical Engineering	3.5 (BS=0.0, ET=3.5, MA=0.0)
SCOPE					
This course studies the behavior of a variety of materials under normal, shear, torsional, bending and combined loads. The concepts of stress, strain, failure theory and failure mechanisms are explored. The loading, geometry, functional environment and material properties of machine or structural parts are used to relate the forces applied to a body to the resulting internal forces and deformations so that performance can be evaluated. Practical applications involving the design and adequacy of mechanical and structural elements under various loading and environmental conditions are emphasized. The course includes several laboratory experiences that require students to develop and conduct experiments, analyze and interpret data, and use engineering judgement to draw conclusions.					
LESSONS: 25 @ 75 min (1,560 Att/wk) LABS: 10 @ 75 min					
SPECIAL REQUIREMENTS:					
None					

Terms Offered and Enrollments:

Table I.A.1

AY	Term	Enrollment
19	1	--
	2	144
20	1	--
	2	128
21	1	--
	2	118
22	1	--
	2	145
23	1	--
	2	143
	7	2
24	1	--
	2	144
25*	1	--
	2	177

* forecast in Redbook as of 19 MAY 24.

2. Based on projected enrollments, the following issues may arise for future course offerings. Two additional sections will likely be required for AY25, for a total of ten sections. Current lab facilities will not permit more than two sections to run concurrently.

Switching to course offerings on both Day 1 and Day 2 this academic year significantly complicated course administration. If possible, the Course Director strongly recommends that all course offerings be confined to either Day 1 or Day 2 (not both). To minimize conflict with Civil and Mechanical Engineering Capstone Design offerings, Day 1 is preferred for MC364.

3. Proposed Changes to Redbook Description. None.

MC364 Course Assessment Package, AY24

B. Course Objectives

1. Current Course Objectives.
 - a. Calculate the internal forces, internal stresses, and deformations of axially loaded members, circular members in torsion, thin-walled pressure vessels, prismatic beams in bending, columns, and members subjected to combined loading and/or thermal effects.
 - b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure subjected to loading and/or thermal effects.
 - c. Apply appropriate theories of failure to analyze and design thin-walled pressure vessels and members subjected to loading and/or thermal effects.
 - d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.
 - e. Given a state of stress at a point, determine the principal stresses, the maximum in-plane shear stress, the angle to the principal plane, and the state of stress on any plane.
 - f. Apply deformation—force, strain—stress, and stress—force relationships to analyze and design structural members and machine components.
 - g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.
2. Proposed Changes to Course Objectives. N/A.

C. Textbook(s)

1. The following text(s) were used this year:

Hibbeler, R.C. *Statics and Mechanics of Materials*, 6th ed. With included Mastering Engineering subscription (required). \$119.99, but since this was the required text for MC300 in 24-1, this does not represent a required cost for MC364 this semester. ISBN 9780138124069.

Hibbeler, R.C. *Mechanics of Materials*, 11th ed. This text was required because the textbook above does not cover theories of failure. Cadets were strongly encouraged to rent access to the online textbook at a total cost of \$40 for the semester. If they purchased the text, the price was \$79.99 without access to Mastering Engineering. ISBN 9780137605385.

2. Adjustments to the text requirement for next year: The MC300 and MC364 Course Directors and other faculty have already worked with Pearson to develop a custom reader which includes material needed by both courses, which will be adopted by both courses. This will prevent next year's students from having to rent or buy a second textbook to cover only a few lessons of MC364.

D. Follow-up from Previous Course Assessment.

1. Change Lesson 17 on statically indeterminate beams to a dedicated lesson on deflection by superposition. **Partially implemented.** Lesson 17 introduces the use of the method of superposition to calculate beam deflection. Coverage emphasized that the method of superposition may be used to calculate deflection in statically determinate systems, and a homework problem was assigned to allow Cadets to practice using the method of superposition in a statically determinate system. The focus of Lesson 17 remained primarily on statically indeterminate systems to allow the separate coverage of multiple types of compatibility of deformation equations.

MC364 Course Assessment Package, AY24

2. Descope lab reports to allow cadets to improve their technical writing skills and focus on one or two targeted research questions – **implemented and recommend sustain**. In addition, the Course Director recommends that the Program consider adopting a program-wide textbook or other resource on technical writing, to be introduced as early as possible and referenced in multiple courses, so that Cadets can better understand the Department's expectations on technical writing and can observe the same standards across multiple courses.
3. Increase WPRs to 250 points each and decrease labs and EDP point value: upon careful reflection, the current Course Director **did not implement this recommendation** because the labs were deemed to be appropriate in point value for their scope. Many Cadets noted in this year's course-end feedback that the lab exercises are still time-consuming and difficult to complete. If the labs are further significantly decreased in scope, we should revisit this recommendation.
4. Adjusting the number of reported labs: the Redbook entry for the course still reports 10 labs. MC364 currently has seven labs or lab-like activities.
5. Test steel and cast iron in MC364 uniaxial tension test labs: **implemented and recommend sustain**. This choice highlights on Lesson 2, before Cadets are prepared to understand their implications, that there is a difference in the behavior of ductile and brittle materials. This experiential learning is called on again near the end of the course when separate theories of failure for ductile and brittle materials are introduced.
6. Transition MC364 from a 35x75-minute course to 40 lessons at 21x75 minutes and 19x55 minutes. **Not implemented**. The previous and current Course Directors decided not to implement this change due to the relatively large number of other changes made in the course this year. Assessing impact of this change would have been difficult.

MC364 Course Assessment Package, AY24

II Course Content

A. Outline of Lessons and Labs

Table II.A.1

Table II.

Lesson	Title/Topic	Supported Course Objectives						
		a	b	c	d	e	f	g
1	Course Intro & MC300 Review	a						
2	Uniaxial Tension & Compression Lab	a					f	g
3	Axial Deformation	a					f	
4	Buckling II	a					f	
5	Statically Indeterminate Axially Loaded Members I	a	b				f	
6	Statically Indeterminate Axially Loaded Members II	a	b				f	
7	Thermal Effects on Axially Loaded Members	a	b				f	
8	Torsion Lab	a					f	g
9	Introduction to Torsion	a					f	
10	Solving Torsion Problems	a					f	
11	WPR 1 (Axial and Torsion)	a	b				f	
12	Beam Lab I	a					f	g
13	Flexural Normal Stress	a					f	
14	Flexural Shear Stress	a					f	
15	Beam Lab II	a					f	g
16	Deflection of Beams	a					f	
17	Statically Indeterminate Bending Members I	a	b				f	
18	Statically Indeterminate Bending Members II	a	b				f	
19	Beam Analysis	a					f	
20	Beam Design	a					f	g
21	WPR2 (Flexural)	a	b				f	
22	Stress Elements	a					f	
23	Stress Transformation	a					f	
24	Principal Stresses and Max Shear Stress	a				e	f	
25	Mohr's Circle	a				e	f	
26	Strain Transformation	a			d	e	f	
27	TWPV (Strain Gage Mounting)	a			d	e	f	g
28	TWPV Lab	a			d	e	f	g
29	Combined Loading (Axial and Torsion)	a				e	f	
30	WPR 3 (Stress & Strain Transformation and TWPV)	a			d	e	f	
31	Principal Stresses in Flexural Members	a				e	f	
32	General Combined Loading	a				e	f	
33	Theories of Failure - Ductile Material	a		c		e	f	
34	Theories of Failure - Brittle Material	a		c		e	f	
34	WPR 3 (Stress & Strain Transformation and TWPV)	a		c	d	e	f	
35	Course Review and Wrap-up	a	b	c	d	e	f	

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Table II.A.2

Lesson	Laboratory Exercise	Supported Course Objectives						
		a	b	c	d	e	f	g
2	Uniaxial Tension & Compression Lab	a					f	g
8	Torsion Lab	a					f	g
12	Beam Lab I	a					f	g
15	Beam Lab II	a					f	g
20	Beam Design	a					f	g
27	TWPV (Strain Gage Mounting)	a			d	e	f	g
28	TWPV Lab	a			d	e	f	g

Field Trips: MC364 requires enrolled Cadets to attend Projects Day. Civil and Mechanical Engineering majors are required to attend the D/CME Projects Day; Cadets with other majors are welcome at the D/CME event but are authorized to attend their own or another Department's event if they choose. A Reading Quiz (10 points) due after Projects Day requires Cadets to relate the details of the presentations they visited and to reflect on what they learned and their own plans for capstone projects or research.

In past years, Cadets were placed on trip section to facilitate their attendance. This year, the Registrar's Office disapproved the trip section, stating that it was not necessary due to the academic priority already given to Projects Day. However, all Third Class Cadets – a strong majority of course enrollment – were also required to attend the "Inspiration to Serve" tour of the West Point Cemetery during Projects Day. Cadets were allowed to attend with groups other than their Company if their scheduled time conflicted with Projects Day, but this option was not well publicized. In future years, the MC364 Course Director should anticipate continued conflicts between these two important events and work with Cadets to ensure that they are able to attend both. Submitting a trip section request may still be desirable. The Course Director should create a plan for Cadets who are unable to attend Projects Day to make up the assignment.

B. Proposed Changes to Lessons and Labs.

1. N/A.

C. Graded Requirements

1. Summary of Graded Requirements.

Table II.C.1

Requirement	Qty	Points	Total	% of Total
Laboratory Assignments	4	100	400	20%
Engineering Design Project	1	100	100	5%
Reading Quizzes	14	10	110	5.5%*
Homework	8	30	240	12%
WPRs**	3	200	600	30%
TEE**	1	500	500	25%
Instructor Points	1	50	50	2.5%
		TOTAL=	2000	100%

* The lowest three reading quizzes were dropped.

** Individual Work – (WPRs + TEE): 55% of course points

2. Proposed Changes to Graded Requirements: None.

MC364 Course Assessment Package, AY24

III Course Assessment

A. Course Director Assessment of Course Objectives.

Scale:

5 = Objective Clearly Met. 100% of students achieved the objective and can be expected to demonstrate proficiency.

4 = Objective Solidly Met. Over 90% of students met the objective.

3 = Objective Satisfactorily Met. Over 70% of students met the objective.

2 = Objective Marginally Met. Roughly 50% of students met the objective or most students minimally met the objective.

1 = Objective Not Met. Over 75% of students did not meet the objective.

The definition of “meeting a course objective” is achieving at least a “C” level (i.e., $\geq 70.0\%$) on the task.
--

Table III.A

Course Objective	Current Assessment (AY 24)	Previous Assessment (AY 23)	How Evaluated and Remarks
a. Calculate the internal forces, internal stresses, and deformations...	4	3	WPR1: 89% Avg. & 98.6% > C WPR2: 90% Avg. & 96.5% > C WPR3: 85% Avg. & 95% > C TEE: 80% Avg. & 82% > C
b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure...	3	3	WPR1: 89% Avg. & 98.6% > C WPR2: 90% Avg. & 96.5% > C TEE Problem 1: 69% Avg. & 61% > C TEE Problem 4: 82% Avg. & 81% > C
c. Apply appropriate theories of failure to analyze and design...	2	4	WPR3: 85% Avg. & 95% > C TEE Problem 6a: 68% Avg. & 37.5% > C TEE Problem 6b: 84% Avg. & 65% > C Low rating may be caused by few assessments. Let's discuss alternatives.
d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.	4	4	WPR3: 85% Avg. & 95% > C
e. Given a state of stress at a point, determine the principal stresses, the maximum in-plane shear stress, the angle to the principal plane, and the state of stress on any plane.	4	4	WPR3: 85% Avg. & 95% > C TEE Problem 5: 85% Avg. & 88% > C

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f. Apply deformation—strain, strain—stress, and stress—force relationships to analyze and design structural members and machine components.	4	3	WPR1: 89% Avg. & 98.6% > C WPR2: 90% Avg. & 96.5% > C TEE Problem 3: 85% Avg. & 86% > C
g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.	5	5	LAB1: 91% Avg. & 100% > C LAB2: 92% Avg. & 100% > C LAB3: 93% Avg. & 100% > C LAB4: 94% Avg. & 100% > C

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B. Assessment of ABET Program-Level Student Outcomes Directly Supported by this Course

Standard Grade	Percentage Cutoffs:	Performance Description	Performance Rating
A+	96.67%	Outstanding	5
A	93.33%		
A-	90.00%	Very Good	4
B+	86.67%		
B	83.33%	Good	3
B-	80.00%		
C+	76.67%	Fair	2
C	73.33%		
C-	70.00%		
D	65.00%	Poor	1
F	0.00%		
--	--	Not Assessed	0

1. An ability to identify, formulate, and **SOLVE** complex engineering problems by applying principles of engineering, science, and mathematics; N/A.
2. An ability to apply engineering **DESIGN** to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors; N/A.
3. An ability to **COMMUNICATE** effectively with a range of audiences;

Table III.B.3

Embedded Performance Indicator (Event Points Possible)	# CE Cadets Assessed by Grad Year	Performance Ratings by Bin, TECOM, and West Point Writing Program						
MC364: Uniaxial Tension Lab Report (100 Points)	32 ('26)	Performance Level Rating (# Cadets in Bin)						Average Score (%)
		Outstanding 5	Very Good 4	Good 3	Fair 2	Poor 1	Not Assessed	
		12	18	2	0	0	1	92.06%
		TECOM Rating (# Cadets in Bin)						
		Category	Exemplary 4	Satisfactory 3	Marginal 2	Not Proficient 1	Not Assessed	Average
		Clear	19	11	2	0	1	3.5
		Concise	10	22	0	0	1	3.3
		Precise	11	21	0	0	1	3.3
		Purposeful	24	8	0	0	1	3.8
		Consistent	23	9	0	0	1	3.7
		WPWR Rating (# Cadets in Bin)						
		Category	Exemplary 4	Satisfactory 3	Marginal 2	Not Proficient 1	Not Assessed	Average
		Substance	16	15	1	0	1	3.5
		Organization	21	10	1	0	1	3.6
		Style & Presentation	15	13	4	0	1	3.3
		Mechanics & Correctness	4	27	1	0	1	3.1
		Documentation	25	7	0	0	1	3.8
		Overall Proficiency	11	21	0	0	1	3.3

4. An ability to recognize **ETHICAL** and **PROFESSIONAL** responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts; N/A.

MC364 Course Assessment Package, AY24

5. An ability to function effectively on a **TEAM** whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives; N/A.

6. An ability to develop and conduct appropriate **EXPERIMENT**ation, analyze and interpret **DATA**, and use engineering judgment to draw conclusions;

Table III.B.6

Embedded Performance Indicator	Event Points Possible	# CE Cadets Assessed in AY by Grad Year	Performance Level Rating (# Cadets in Bin)						Average Score (%)
			Outstanding 5	Very Good 4	Good 3	Fair 2	Poor 1	Not Assessed	
MC364: Uniaxial Tension Lab	100	32 ('26)	12	18	2	0	0	1	92.06%

7. An ability to acquire and apply **NEW KNOWLEDGE** as needed, using appropriate learning strategies. N/A.

C. Assessment of Changes Made this Semester.

1. *Change of Course Textbook.* This change was prompted by a change in the textbook adopted by MC300. To minimize the financial burden on Cadets, MC364 has often historically used the textbook adopted by MC300. It was not possible to use only this textbook, however, because Hibbeler's *Statics and Mechanics of Materials* does not include a discussion of theories of failure. Cadets were encouraged to rent access (to minimize the cost) to Hibbeler's *Mechanics of Materials* to gain access to the necessary coverage of theories of failure for ductile and brittle materials.

One advantage of the new text is the Mastering Engineering platform for student problem solving, which automated grading of 2/3 of the course's homework problems. Cadets responded positively to the support and guidance offered by this online learning system.

The chief disadvantage of the new test is that it is limited in the support materials offered by the publishers as compared to the previous textbook (Philpot and Thomas, *Mechanics of Materials*, 5th ed.). The previous text offered rich, high-production-value videos explaining course concepts and a wealth of practice problems, many with video solutions. The new textbook's online video and other supplementary offerings are not comparable.

The current MC364 Course Director recommends **sustaining** the adoption of the new textbook. Adapting the course to the new textbook required significant effort, and the effort has already been expended. MC300 and MC364 teaching staff have also spent considerable time and effort working with Pearson to develop a custom reader which is adequate to both courses' needs. Adopting the custom reader in future years represents the best use of instructor time.

2. *Text-centric vs. Video-centric flipped class format.* Lesson videos developed in prior years made extensive reference to the previous textbook and were deemed no longer appropriate for use with the new textbook. They were posted to Canvas for Cadets' optional use, but the required pre-class preparation included only readings from the new textbook and a reading quiz designed to reward preparation. Reading quizzes were revised as necessary to remove references to the course videos and ensure that they were appropriate to the new text. Students were able to prepare adequately for class using only the course text, and the Course Director recommends **sustaining** this change.

MC364 Course Assessment Package, AY24

3. *WPR 3 moved to Lesson 30.* The previous year's Cadet feedback contained several suggestions to eliminate WPR 3 or move it earlier in the semester (previously it was on Lesson 34, in the final week of classes.) This year, the WPR was moved forward in the course by about two weeks, to Lesson 30. The lessons covered by the WPR were adjusted so that it did not cover Combined Loading and Theories of Failure, which were taught after the WPR. Advantages of this approach are that it gives the instructor team more time to grade the WPRs and discuss the results, and more time to remediate any shortfalls in Cadet understanding. Students did not have to study for and take two major graded events within a week of each other. The main disadvantage was the loss of an opportunity to assess Cadets' understanding of failure theories prior to the TEE. The Course Director recommends **sustaining** this change.
4. *Adoption of the FE Reference Manual as the only authorized exam reference.* Previous iterations of this class relied on an issued formula card and one letter-sized cadet-created notes sheet. With the change to the new textbook, the reference card had to be heavily revised to accommodate changes in variables and other differences between the two textbooks. At the same time, instructors frequently noticed that Cadets were using their notes sheets not for formulas, but to copy solutions of example problems and then using these solutions to pattern-match during the major graded events. The migration to the new text provided an opportunity to solve both problems with the adoption of an established reference document, the Fundamentals of Engineering Exam Reference, as the only authorized reference for major graded events, eliminating the self-created notes sheet. The FE Reference Manual is a very long document, so for ease of copying and distribution it was reduced to an excerpt which contained only the formulas relevant to MC364. Advantages of this approach included giving Cadets the opportunity to become familiar with a resource they will need when they take the Fundamentals of Engineering Exam. It also afforded an opportunity to discuss in class the importance of professional licensure and the process required to attain it. It eliminated the copying of problem solutions and placed all Cadets on equal footing in the exams. Disadvantages include the loss of the educational benefit of distilling course notes to the one page of most important material. In addition, the FE Reference Manual sometimes presents concepts differently and uses different variables from the textbook, which could cause confusion if not carefully introduced in class. The Course Director recommends **sustaining** this change.
5. *Reduction in Scope of Lab Report supporting calculations.* The prior year's Cadet feedback included many observations that the labs were too complicated for their point value, and the reports were difficult for the instructor team to grade efficiently. This year's lab exercises were similar in scope for the actual activities, although the uniaxial compression test was removed from Lab 1 and replaced with a uniaxial tension test with a brittle material (cast iron) to demonstrate the difference in behavior of ductile and brittle materials. For all laboratory exercises, the required supporting calculations were reduced in order to focus on the ones which were absolutely fundamental to understanding the key concepts. The Course Director recommends **sustaining** this change and perhaps taking it further next year.

D. Assessment of Resources and Impacts.

1. *Laboratory Technician and Resource Support.* A quarter of cadets' grade in this course is devoted to the laboratories and EDP (500/2000). It is critical to have the support of a laboratory technician for the setup and execution of all laboratory activities. Many of the labs require setup in multiple classrooms. It is important that there is enough equipment for each classroom. With the current shortage of classroom space and sharing classrooms with other academic programs, the ability to set up laboratory equipment in the classroom is significantly restricted.

2. *Equipment.* Much of the equipment for MC364 is aging and has not been serviced in several years. The tension machines, torsion machines, and aluminum beam specimens are all experiencing issues which have had significant impacts on the lab delivery. These machines must be checked each semester.

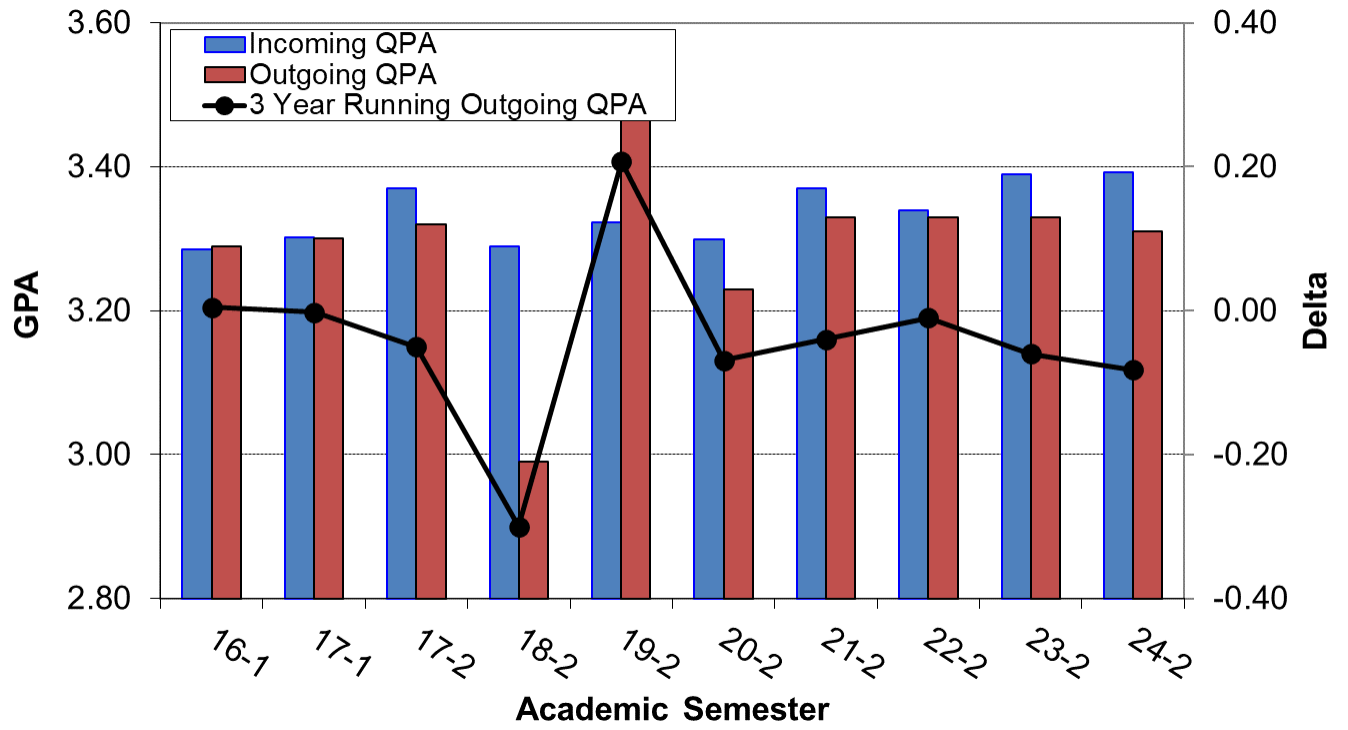
MC364 Course Assessment Package, AY24

3. *Material.* Raw material is needed for many of the laboratory and EDP executions. The uniaxial tension laboratory for MC300 and MC364 needs samples of steel, aluminum and cast iron. For the torsion laboratory, we need samples of brass. For the Beam EDP, we need balsa wood (36"x3"x1/4" & 36"x3"x1/8"). For the TWPV Lab, we need Coca Cola Cans and hot dogs.

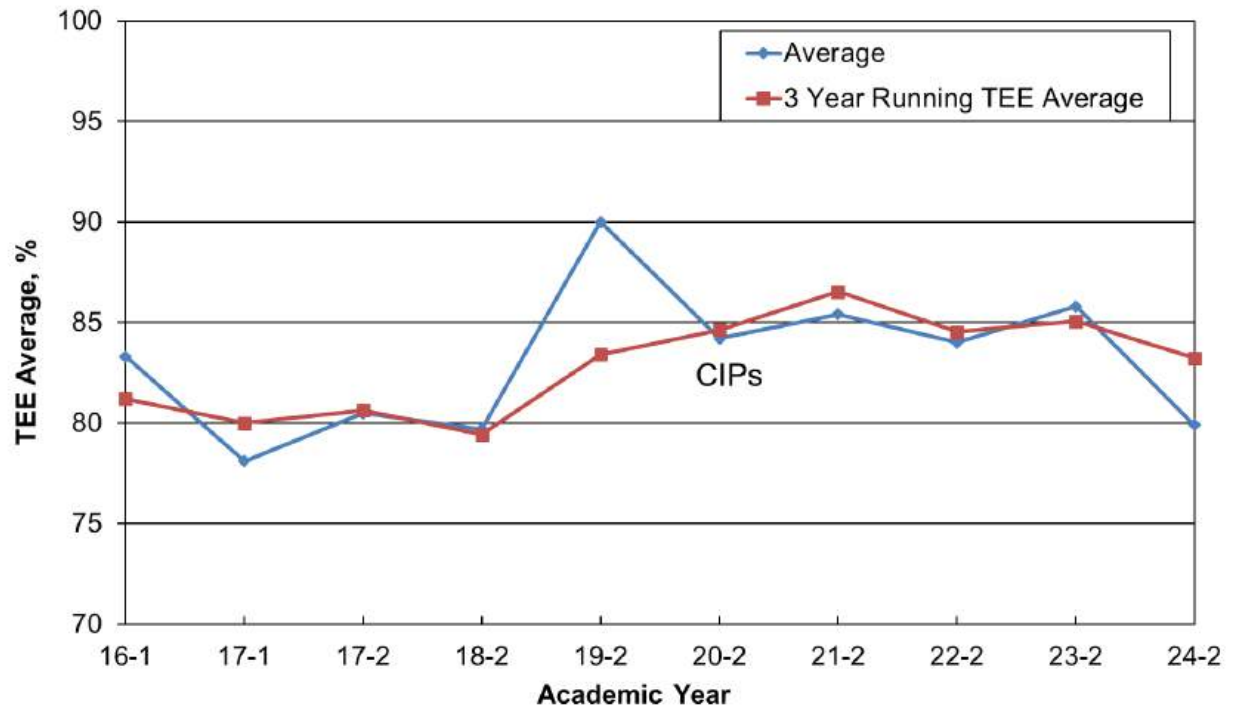
4. *Lab Space.* Currently, all tension machines are in the primary cadet material testing lab in D10. All torsion machines are stored in C19 and then are moved into D10 for execution of the lab. If the Academy sustains the large section sizes assigned in 24-2, D10 will not permit more than two sections to run the lab simultaneously. If time or classroom space constraints require us to run more sections concurrently, we will need more lab space.

MC364 Course Assessment Package, AY24

E. Course QPA Graph



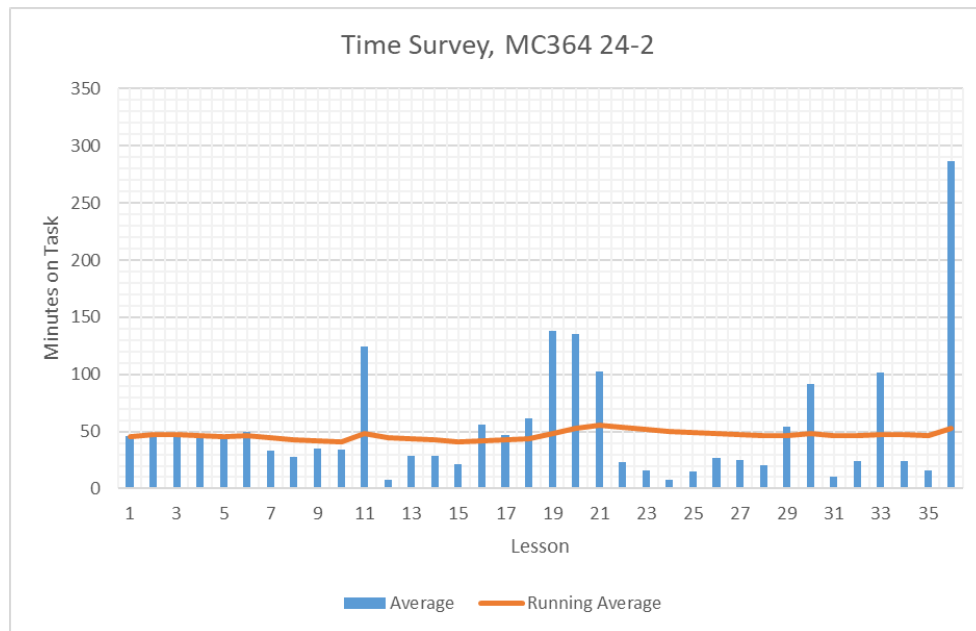
F. TEE Grades Graph



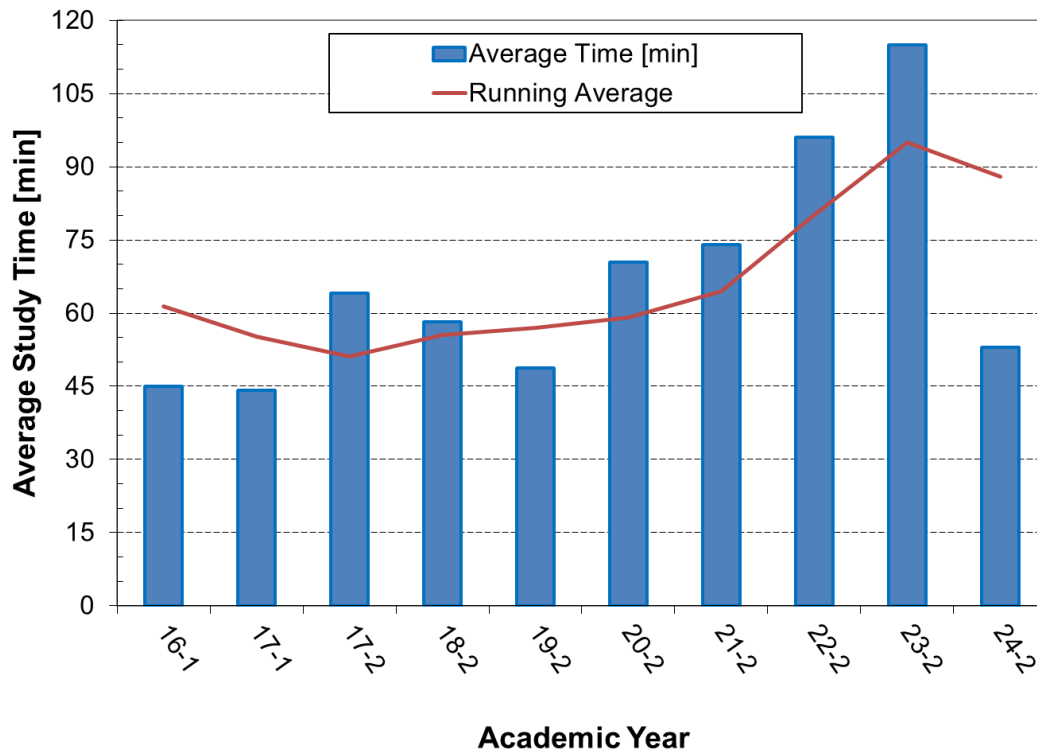
MC364 Course Assessment Package, AY24

G. Course Time Study Graph

1. Current Semester time survey data. (AVG: 53 minutes/lesson)



2. Historical Time Survey Data



3. Any comments on time survey data if needed. Per-lesson engagement is considerably lower this year than in past years. We speculate that this may be due to the shift away from focus on lesson videos for pre-class preparation. In addition, most homework assignments were due out of class on

MC364 Course Assessment Package, AY24

Thursday or Friday evenings and we speculate that Cadets may not have retained accurate memories of their time on task through the weekend to report it in class the following Monday or Tuesday.

H. Course End Feedback. No course-specific questions were included in the course end survey.

IV. Program Director's Current Areas of Emphasis

- A. *If applicable, describe how the course develops a student's ability to apply principles of the following: sustainability, risk, resilience, and diversity, equity, and inclusion (DEI).*

Resilience is defined as a material property and discussed in the course during the lessons on axial loading as the energy absorbed by the material while in the elastic region – the area under the linear portion of the stress-strain curve. This is likely not the sense of resilience intended by the Program Director's question. MC364 contains no substantive, programmed coverage of the other listed concepts.

- B. *If applicable, describe how the course develops a student's ability to apply numerical methods relevant to CE. The CE Program conducted an initial review of this topic in AY23, but additional ASCE guidance warrants further investigation. In summary, ASCE views numerical methods as solving continuous problems using numeric approximation, providing approximate but accurate numeric solutions when exact solutions are either impossible or prohibitively expensive to calculate.*

MC364 contains no formal discussion of numerical methods.

- C. *Generative Artificial Intelligence (AI). Please summarize the course guidance provided in AY24 regarding the use of AI. If AI was not allowed on particular assignments, which were they and why?*

The use of generative AI is permitted on all assignments except major graded events in MC364. Cadets are cautioned that generative AI is not yet very good at solving the course's type of problems, and that the technology has been known to generate false or misleading answers and to cite evidence that does not really exist. Cadets are required to document their use of generative AI as assistance received.

The course Admin Letter provides the following guidance on the use of software programs, including generative AI:

"Computer programs such as Mathematica or MATLAB and spreadsheets such as Microsoft Excel may be used, but a printout of the code and results, with proper documentation, must be included with your submission. In addition, please note that *Documentation and Acknowledgment of Academic Work* emphasizes that assistance provided by artificial intelligence systems (including but not limited to ChatGPT) constitutes assistance and must be documented."

Enclosures

1. Historical Course Feedback on USMA Questions
2. Term 24-2 Course Feedback on USMA Questions
3. Historical Course Feedback on CME Questions
4. Term 24-2 Course Feedback on CME Questions
5. Course Narrative Feedback
6. Course Numeric Feedback
7. Other Feedback

MC364 Course Assessment Package, AY24

Enclosures:

1. Historical Course Feedback on USMA Questions

Aggregated data are not yet available for cadet feedback on USMA Questions.

2. Term 24-2 Course Feedback on USMA Questions

Aggregated data are not yet available for cadet feedback on USMA Questions.

3. Historical Course Feedback on CME Questions

Aggregated data are not yet available for cadet feedback on CME Questions.

4. Term 24-2 Course Feedback on CME Questions

Aggregated data are not yet available for cadet feedback on CME Questions.

5. All Course Narrative Feedback for the current period of assessment.

Additional comments about the course:

- I think it does a great job of building on MC300 and does a great job of pacing
- The nature of the course, as a MechE major, the content did not stimulate my thinking, it often felt arbitrary and hard to comprehend, especially the stress and strain transformations.
- MC364 was well structured.
- MC364 has been a course where I am constantly motivated to learn more and am excited to come to class every day.
- was good
- This class can be very boring and repetitive. Sometimes it feels like we do not need to be in class for 75 minutes to learn the material.
- There should be more than a 24 hour window to complete the reading quizzes , it was too easy to miss the window and get a 0/10.
- reading quizzes were hard and tanked my grade unnecessarily.
- I like how the course was setup with canvas lessons, but I only wish there were more hands on labs or projects to apply the learning directly.
- Reading quizzes were all over the place. Sometimes the reading quizzes related directly to the readings for the next lesson, sometimes they were only on material from the past few lessons.
- Clarity
- at the beginning of the semester would help there. Quizzes were also sometimes well above the skill/comprehension level for most of the class and/or did not allow enough time to process the question and try to apply the new material to it. The information on most of the quizzes was very difficult to get out of the reading and only understandable if students watched the additional helpful videos, there might be some lack of clarity there or misunderstanding on the part of the students.
- MC364 course is heavily loaded in terms of the expected amount of time cadets are expected to spend on their assignments. A lot of times it can feel like it is out of proportion.
- I thought the reading quizzes were not an accurate reflection of the assigned readings. Most were difficult and concepts we did not focus on in class. There were also some that did not allow for enough time to do the problems, not proportional to time given on WPRs or problem sets. I think more demonstrations will be helpful to visualize all the different concepts- bending, axial, torque.
- There were also some weeks where there were multiple assignments due that was very stressful, the assignments can be better spaced out.
- I feel that this class is a great class with a great teacher but becomes unnecessarily hard when adding multiple labs and homeworks. I feel that the labs should be in class and the homeworks

MC364 Course Assessment Package, AY24

- should be strictly homework.
- I feel like the course material made even more sense with the culmination of the course.
- The course needs to be 4 credit hours if we are going to have this many labs. I think the homework and WPRs are perfect the way they are but there needs to be a change with the labs. I do not think anyone is learning anything more from the labs than they are the mandatory homework. Physics 1 and 2 can be responsible for technical writing. The labs are tedious and do not teach much to the students. The labs also result in many hours of tedious work and a lot of social loafing resulting in frustration for some. I think if we have this many labs it should be a 4 credit course. If not, there needs to be less labs as the labs are a grade killer. I feel there is less of a guarantee for points in labs than homework and even WPRs. The labs are NOT an opportunity to earn more points in the course. For me, I have to mitigate the amount of points I earn rather than looking at it as a grade boost. The labs are graded too harshly.
- Inconsistent grading.
- I would suggest making pre-class checks instead of reading quizzes to better incentivize reading for all classes while also not having as much at stake.
- Great course! Super intuitive and I enjoyed it much more fun than MC300. It challenged me and I learned. Should focus less on describing difficult problems and more on board work.
- I enjoyed this course.
- Well organized course.
- I thought the course was fine.
- At times, timelines would be all over the place, reading quizzes were not set for the right date, or they had not very useful questions.
- It is a fun course that could have been taught in a different way to make it more enjoyable.
- very well put together course.
- This course was challenging, but very engaging
- Allowed me to take my knowledge from MC300 and apply it. Interesting class, but definitely difficult to conceptualize while simultaneously taking MC300. Reading quizzes were a useful tool, but sometimes they did not cover material that would be talked about in class. The material is already difficult, so taking a quiz on something you have read and have to teach yourself is not easy. I think it might be better to have the quizzes cover concepts from the previous lesson to reinforce this topics.
- I felt like the course was challenging but interesting. The labs were very helpful in this class, this is not the case in other classes.
- Reading Quizzes are not an accurate representation of one's understanding of content. Lost heavy points even though I understood the concepts just by getting quite frankly smashed by a reading quiz.

Reading quizzes should also not be near as many points as long form answers. There needs to be a point redistribution session where points are more accurately allocated to what takes up a Cadet's time. 20 points for long form and 5 for reading quizzes, perhaps.

Labs were somewhat difficult to understand and often went deeper or more complicated than necessary compared to basic level of course content.

No extra credit opportunities that were available to all regardless of a group assignment for one of the labs. Would help to have some, especially for the high achievers that are willing to put in work for it.

The practice problems and solutions on Canvas are remarkable and help TONS for prepping for WPRs and TEE (hopefully, I guess I'll find out shortly).

- I thought that this class was a good pace but it felt a little rushed at the end. The topics were not too difficult to learn. I think the practice problems were often a lot harder than the WPR questions which made me stress when I couldn't solve them. The reading quizzes were also pretty hard and I lost a good amount of points because of them.

MC364 Course Assessment Package, AY24

- Some lab report instructions included references to outdated textbooks or instructions; for example, there were multiple instances of false or contradictory instructions in the TWPV lab. This made it difficult to complete the assignment without multiple AI sessions with teachers.

I liked that conceptual knowledge was only tested on quizzes and lab reports, while the WPRs focused solely on problem-solving! This made it much easier to know what to study.

- The course was very interesting and entertaining. I really enjoyed the group project, constructing a beam with my fellow cadets.
- I enjoyed the course, and I believe the material was taught well. I think there was a good balance of labs and lectures. I think the WPRs were fair. I think the applicability of the concepts we were learning were easy to see.

This course was excellent. I enjoyed the content.

- I think MC364 is a well paced course with the exception of Block 2, I think the Block 2 WPR was poorly timed for a lot of people and several of the key concepts had not sunk in yet which meant a lot of people I know, myself included were not ready.
- I found the course a very useful addition to the concepts in MC300. I feel like certain lessons at the end of the course, such as the combined loadings, could have had more time spent on them. In addition, I feel like more time could be spent on working through problems individually and less time on going over problems as a class.
- Love the course!
- Good course overall. Interesting material; however, it was tough to truly enjoy it since my major isn't related to this material. My least favorite part of the class were the labs since drawing out the report took a lot of time, but the subject matters of the lab were interesting. Opposed to a TEE to end the course, it may be interesting to have a final project where you have to create something sort of like the EDPs; but more intensive and a semester-long assignment.
- My favorite class this semester! I appreciate using the FEE references for WPRs and the TEE to get used to them.
- The course was well organized, and the lessons were arranged in a manner that it was easy to follow along even if you missed a class. The homework assignments were also helpful in learning the material. In addition, having the worked-out solutions to practice problems is extremely helpful when attempting to understand which skills need more practice.
- I would've preferred more visual aids. I feel like we had more visual aids in MC300. This might be hard to accomplish with the course material but it would make learning more effective and enjoyable.

6. All Course Numeric Feedback

N/A.



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TAB C

AY2023 COURSE ASSESSMENT

MC364 Course Assessment Summary, AY 23

<u>COURSE ADMINISTRATION</u> • Redbook – 6 or 7 Labs @ 75 minutes • Course Objective – “Deformation-Force” relationship instead of “Deformation-Strain”.	<u>RESOURCES</u> • Laboratory staff and equipment • Lab Space – Tension and Torsion Machines • Raw materials – cast iron & balsa wood • Updating the Videos																								
<u>COURSE OBJECTIVES ASSESSMENT</u> <table><tr><th>Course Objective</th><th>Assessment AY23</th><th>Assessment AY22</th></tr><tr><td>a. Calculate the internal forces, internal stresses, and deformations...</td><td>3↓</td><td>4</td></tr><tr><td>b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure...</td><td>3</td><td>3</td></tr><tr><td>c. Apply appropriate theories of failure to analyze and design...</td><td>4↑</td><td>2*</td></tr><tr><td>d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.</td><td>4↑</td><td>3</td></tr><tr><td>e. Given a state of stress at a point, determine the principal stresses, the maximum in-plane shear stress, the angle to the principal plane, and the state of stress on any plane.</td><td>4</td><td>4</td></tr><tr><td>f. Apply deformation—strain, strain—stress, and stress—force relationships to analyze and design structural members and machine components.</td><td>3</td><td>3</td></tr><tr><td>g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.</td><td>5*↑</td><td>4</td></tr></table>		Course Objective	Assessment AY23	Assessment AY22	a. Calculate the internal forces, internal stresses, and deformations...	3↓	4	b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure...	3	3	c. Apply appropriate theories of failure to analyze and design...	4↑	2*	d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.	4↑	3	e. Given a state of stress at a point, determine the principal stresses, the maximum in-plane shear stress, the angle to the principal plane, and the state of stress on any plane.	4	4	f. Apply deformation—strain, strain—stress, and stress—force relationships to analyze and design structural members and machine components.	3	3	g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.	5*↑	4
Course Objective	Assessment AY23	Assessment AY22																							
a. Calculate the internal forces, internal stresses, and deformations...	3↓	4																							
b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure...	3	3																							
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g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.	5*↑	4																							
• Time Survey is at 115 mins/graded event. Homeworks took on average 175 minutes. Lab Reports took 199 minutes.																									
<u>CHANGES FROM PREVIOUS YEAR</u> 1.Update the Uniaxial Tension Lab to include Compression (Sustain) 2.Changed Flipped Classroom Structure to modified flipped classroom (Sustain) 3.Changed Lab Submission (Sustain)	<u>RECOMMENDED CHANGES</u> • Add Lesson on Deflection by Superposition: remove single lesson on Statically Indeterminate Bending Members • Move Strain Transformation and TWPV to the end of the semester. • Descope Lab Report assignments • Increase weight of WPRs (250 pts) – 62.5% Individual assessments (WPRs & TEE)																								
<u>DISCUSSION POINTS:</u> <i>Flipped Classroom</i> – should we maintain this classroom format? Should we restructure the format to have a more formal instructor led portion and more targeted problems? <i>Graded Events</i> – Do we continue to administer Reading Quizzes? <i>Course Offerings</i> – Would it be possible to offer MC364 in the Fall and Spring? <i>Topics in MC364</i> – Do we need Statically Indeterminate Axially loaded members, Statically Indeterminate Beams, and Thin-walled Pressure Vessels? <i>Format</i> - Can we transition to a 40 lesson, 3.5 credit hour course? <i>WPR3</i> – Is a WPR during the last week of class effective?																									

MC364 Course Assessment Package, AY 23

I Course Administration

A. Redbook Description

The screenshot below was taken from the Redbook on 19 MAY 23.

MC364 (Version: 2020 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
MC364	MECHANICS OF MATERIALS	2020	1	Civil and Mechanical Engineering	3.5 (BS=0.0, ET=3.5, MA=0.0)

SCOPE

This course studies the behavior of a variety of materials under normal, shear, torsional, bending and combined loads. The concepts of stress, strain, failure theory and failure mechanisms are explored. The loading, geometry, functional environment and material properties of machine or structural parts are used to relate the forces applied to a body to the resulting internal forces and deformations so that performance can be evaluated. Practical applications involving the design and adequacy of mechanical and structural elements under various loading and environmental conditions are emphasized. The course includes several laboratory experiences that require students to develop and conduct experiments, analyze and interpret data, and use engineering judgement to draw conclusions.

LESSONS: 25 @ 75 min (1.560 Att/wk) **LABS:** 10 @ 75 min

SPECIAL REQUIREMENTS:

None

MC364 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
CO REQUISITE					
	MA205	2017	1	1	Y
	MA255	2016	2	2	Y
	MA366	2018	2	3	Y
DISQUALIFIER					
	CE364	2006	1	1	Y
PRE REQUISITE					
	MC300	2017	1	1	Y

1. Terms Offered and Enrollments:

Table I.A.1

AY 17	Term 17-1	169		AY21	Term 21-1	
	Term 17-2	149			Term 21-2	118
AY18	Term 18-1			AY22	Term 22-1	
	Term 18-2	112			Term 22-2	145
AY19	Term 19-1			AY23	Term 22-1	
	Term 19-2	144			Term 22-2	143
AY20	Term 20-1			AY23*	Term 22-7	2**
	Term 20-2	128			Term 23-1	
					Term 23-2	164

* forecast in Redbook as of 19 MAY 23.

** 2 ME cadets enrolled in STAP. 1 cadet is late major add and 1 cadet is a medical case.

MC364 Course Assessment Package, AY 23

2. Based on projected enrollments, the following issues may arise for future course offerings.

None

3. Proposed Changes to Redbook Description.

Reduce the number of labs to 6 or 7 @ 75 minutes

B. Course Objectives

1. Current Course Objectives.

- a. Calculate the internal forces, internal stresses, and deformations of axially loaded members, circular members in torsion, thin-walled pressure vessels, prismatic beams in bending, columns, and members subjected to combined loading and/or thermal effects.
- b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure subjected to loading and/or thermal effects.
- c. Apply appropriate theories of failure to analyze and design thin-walled pressure vessels and members subjected to loading and/or thermal effects.
- d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.
- e. Given a state of stress at a point, determine the principal stresses, the maximum in-plane shear stress, the angle to the principal plane, and the state of stress on any plane.
- f. Apply deformation—strain, strain—stress, and stress—force relationships to analyze and design structural members and machine components.
- g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.

2. Proposed Changes to Course Objectives.

- f. Apply deformation—~~strain~~force, strain—stress, and stress—force relationships to analyze and design structural members and machine components.

C. Textbook(s)

1. The following text(s) were used this year:

Textbook Changes: Mechanics of Materials, 5th Edition by Timothy A. Philpot and Jeffery S. Thomas. ISBN: 978-1-119-60301-6

2. Make following adjustments to the text requirement for next year:

No changes to the text.

D. Follow-up from Previous Course Assessment.

None

MC364 Course Assessment Package, AY 23

II Course Content

A. Outline of Lessons and Labs

Table II.A.1

Lesson	Title/Topic	Supported Course Objectives						
		a	b	c	d	e	f	g
1	Course Intro & MC300 Review	a						
2	Uniaxial Tension & Compression Lab	a					f	g
3	Axial Deformation	a					f	
4	Buckling II	a					f	
5	Statically Indeterminate Axially Loaded Members I	a	b				f	
6	Statically Indeterminate Axially Loaded Members II	a	b				f	
7	Thermal Effects on Axially Loaded Members	a	b				f	
8	Torsion Lab	a					f	g
9	Introduction to Torsion	a					f	
10	Solving Torsion Problems	a					f	
11	WPR 1 (Axial and Torsion)	a	b				f	
12	Beam Lab I	a					f	g
13	Flexural Normal Stress	a					f	
14	Flexural Shear Stress	a					f	
15	Beam Lab II	a					f	g
16	Deflections of Beams	a					f	
17	Statically Indeterminate Bending Members I	a	b				f	
18	Statically Indeterminate Bending Members II	a	b				f	
19	Beam Analysis	a					f	
20	Beam Design	a					f	g
21	WPR2 (Flexural)	a	b				f	
22	Stress Elements	a					f	
23	Stress Transformation	a					f	
24	Principal Stresses and Max Shear Stress	a				e	f	
25	Mohr's Circle	a				e	f	
26	Strain Transformation	a			d	e	f	
27	TWPV (Strain Gage Mounting)	a			d	e	f	g
28	TWPV Lab	a			d	e	f	g
29	Combined Loading (Axial and Torsion)	a			d	e	f	
30	Principal Stresses in Flexural Members	a			d	e	f	
31	General Combined Loading	a			d	e	f	
32	Theories of Failure - Ductile Material	a		c		e	f	
33	Theories of Failure - Brittle Material	a		c		e	f	
34	WPR 3 (Stress & Strain Transformation and TWPV)	a		c	d	e	f	
35	Course Review and Wrap-up	a	b	c	d	e	f	

Field Trips:

No field trips in this course. All cadets were put on trip section for Project's Day. Is it within our jurisdiction to put cadets on trip section for Project's Day?

MC364 Course Assessment Package, AY 23

B. Proposed changes to Lessons and Labs.

Table II.

Lesson	Title/Topic	Supported Course Objectives						
		a	b	c	d	e	f	g
1	Course Intro & MC300 Review	a						
2	Uniaxial Tension & Compression Lab	a					f	g
3	Axial Deformation	a					f	
4	Buckling II	a					f	
5	Statically Indeterminate Axially Loaded Members I	a	b				f	
6	Statically Indeterminate Axially Loaded Members II	a	b				f	
7	Thermal Effects on Axially Loaded Members	a	b				f	
8	Torsion Lab	a					f	g
9	Introduction to Torsion	a					f	
10	Solving Torsion Problems	a					f	
11	WPR 1 (Axial and Torsion)	a	b				f	
12	Beam Lab I	a					f	g
13	Flexural Normal Stress	a					f	
14	Flexural Shear Stress	a					f	
15	Beam Lab II	a					f	g
16	Deflections by Integration	a					f	
17	Deflection by Superposition	a					f	
18	Statically Indeterminate Bending Members	a	b				f	
19	Beam Analysis	a					f	
20	Beam Design	a					f	g
21	WPR2 (Flexural)	a	b				f	
22	Stress Elements	a					f	
23	Stress Transformation	a					f	
24	Principal Stresses and Max Shear Stress	a				e	f	
25	Mohr's Circle	a				e	f	
26	Combined Loading (Axial and Torsion)	a				e	f	
27	Principal Stresses in Flexural Members	a				e	f	
28	General Combined Loading	a				e	f	
29	Strain Transformation	a			d	e	f	
30	TWPV (Strain Gage Mounting)	a			d	e	f	g
31	TWPV Lab	a			d	e	f	g
32	Theories of Failure - Ductile Material	a		c		e	f	
33	Theories of Failure - Brittle Material	a		c		e	f	
	DROP							
34	WPR 3 (Stress & Strain Transformation and TWPV)	a		c	d	e	f	
	DROP							
35	Course Review and Wrap-up	a	b	c	d	e	f	

MC364 Course Assessment Package, AY 23

1. **Deflection by Superposition:** remove single lesson on Statically Indeterminate Bending Members and replace with a lesson on deflection by superposition for determinate beams.
2. **Move Strain Transformation and TWPV:** Move Strain Transformation and TWPV to the end of the semester as they are disconnected from the combined loading topics.

C. Graded Requirements

1. Summary of graded requirements for current semester.

Table II.C.1

Requirement	Qty	Points	Total	% of Total
Laboratory Assignments	4	100	400	20%
Engineering Design Project	1	100	100	5%
Reading Quizzes	14	10	110	5.5%*
Homeworks	8	30	240	12%
WPRs**	3	200	600	30%
TEE**	1	500	500	25%
Instructor Points	1	50	50	2.5%
		TOTAL=	2000	100%

*The lowest three reading quizzes were dropped

** Individual Work - 55% Individual Assessment (WPRs & TEE)

2. Proposed changes to graded requirements:

Table II.C.2

Requirement	Qty	Points	Total	% of Total
Laboratory Assignments	4	80	320	16%
Engineering Design Project	1	80	80	4%
Reading Quizzes	12*	10	90	4.5%
Homeworks	8	30	240	12%
WPRs**	3	250	750	37.5%
TEE**	1	500	500	25%
Instructor Points	1	20	20	1%
		TOTAL=	2000	100%

*The lowest three reading quizzes were dropped

** Individual Work – 62.5% Individual Assessment (WPRs & TEE)

1. **Descope Laboratory Assignments:** Lab assignments took a significant amount of time and effort. The lab assignments will maintain the same technical writing report format, but the amount of supporting calculations and research questions that the cadets will be required to answer will be decreased.
2. **Less points for EDP:** The amount of points allocated to the EDP will be decreased.
3. **Online Reading Quizzes:** The reading quizzes will be given to cadets via online LMS (Canvas). There will be two less reading quizzes.
4. **Increase weight of WPRs:** In an effort to ensure accurate assessment of cadets' knowledge and reflection of grades, the weight of the WPRs will increase from 200 pts to 250 pts (50%-62.5%).
5. **Instructor Points:** Decrease the amount of instructor points available in the course. The instructor points were effective in ensuring cadets completed the course end survey.

MC364 Course Assessment Package, AY 23

III Course Assessment

A. Course Director Assessment of Course Objectives.

1= Objective Not Met. Objective clearly not met, most (75%) of the students did not achieve it.

2= Objective Marginally Met. Objective met by roughly half the students or minimally by most of them.

3= Objective Satisfactorily Met. Objective clearly met by a solid majority (70%) of the students.

4= Objective Solidly Met. Objective clearly met by the vast majority (90%) of students.

5= Objective Clearly Met. All students have achieved the objective and can be expected to demonstrate it.

The definition of “meeting a course objective” is achieving a “C” level (70%) on the task.

Table III.A

Course Objective	Current Assessment (AY 23)	Current Assessment (AY 22)	How Evaluated and Remarks
a. Calculate the internal forces, internal stresses, and deformations...	3	4	WPR1: 84% Avg. & 86% > C WPR2: 86% Avg. & 68% > C WPR3: 86% Avg. & 95% > C TEE: 85% Avg. & 91% > C
b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure...	3	3	WPR1: 90% Avg. & 96.5% > C WPR2: 84% Avg. & 89% > C TEE: 82% Avg. & 75% > C
c. Apply appropriate theories of failure to analyze and design...	4	2*	WPR3: 93% Avg. & 97% > C TEE: 92.5% Avg. & 89% > C
d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.	4	3	WPR3: 86% Avg. & 94% > C TEE: 86% Avg. & 88% > C
e. Given a state of stress at a point, determine the principal stresses, the maximum in-plane shear stress, the angle to the principal plane, and the state of stress on any plane.	4	4	WPR3: 88% Avg. & 96.5% > C TEE: 90% Avg. & 94% > C
f. Apply deformation—strain, strain—stress, and stress—force relationships to analyze and design structural members and machine components.	3	3	WPR1: 83% Avg. & 84.5% > C WPR2: 86% Avg. & 73% > C TEE: 88% Avg. & 94% > C
g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.	5*	4	LAB1: 88% Avg. & 100% > C LAB2: 90% Avg. & 100% > C LAB3: 89.5% Avg. & 100% > C LAB4: 90% Avg. & 100% > C

MC364 Course Assessment Package, AY 23

B. Student (Program) Outcomes Directly Supported By This Course

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
 - **None**
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors;
 - **None**
3. an ability to communicate effectively with a range of audiences;
 - **Laboratory Report (TWPV) (bin, avg)**
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts;
 - **None**
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives;
 - **Laboratory Groups (Survey)**
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;
 - **None**
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
 - **None**

C. Assessment of Embedded Indicators.

Table III.C.1

Course Indicator	No. of CE Cadets	Indicator Points	Average (%)	High (%)	Low (%)
MC364 Lab4	36	50	88	100	66

Table III.C.2

Course Indicator	No. of CE Cadets	D or F	C	B	A	Indicator Points	Avg (%)
MC364 Lab4	36	2	5	11	17	50	88

Table III.C.3

Student self-assessment on 1-5 Likert Scale: (1) strongly disagree to (5) strongly agree.	Avg. Response
1.) My team worked in a positive collaborative manner	4.3
2.) I provided leadership in completing team assignments.	4.2
3.) My team had to address conflict to complete team assignments.	3.54
4.) Teamwork activities were effective in learning teamwork skills	4.06
5.) Teamwork activities were effective in learning course material	3.99

The only embedded indicator for the course is the MC364 TWPV Lab. As long as this lab continues to be executed in its current form, it is a good assessment tool. The TWPV Lab is the fourth lab, so the cadet teams have received a lot of feedback on their technical writing.

MC364 Course Assessment Package, AY 23

D. Assessment of Changes Made this Semester.

1. Uniaxial Tension and Compression Lab (LSN 2). This semester we modified the uniaxial tension lab to also include a compression test of the same material. This supported comparison between the same sample tested under different types of loading. The same steel torsion specimens were tested in tension, compression and then torsion (LSN6). This helped give the students a baseline comparison for the material properties. However, due to issues with the grips and the extensometers, there were several errors with the lab that led to cadets not trusting the lab. We will work with the lab technicians to modify the extensometers to avoid issues. The scope of the labs will also be decreased so that the cadets are only expected to answer one or two research questions in their report.

2. Restructuring of the Flipped Class. During the previous iteration of MC364 cadets noted that they would prefer to have more in-person instruction and that there were too many videos to watch outside of class. During this semester, for a majority of the lessons, the cadets would be required to watch a single video covering the basics of the concept for the days lesson. At the start of class, the instructors would walk the cadets through an example problem at the board. The remainder of the lesson would be spent by the cadets working at the board. The cadets appreciated the in-person instruction from the instructors.

3. Update Lab Report Format. The lab report format was updated from an ExSum with additional questions and calculations that were required to be submitted to a single coherent technical report submission with an Introduction, Theory, Results, Conclusion, and Appendix section. This format produced more effective technical writing submissions but required a significant amount of additional work. The time survey for the Labs increased from approximately 160 minutes to 200 minutes per lab report. In the next iteration of MC364, the scope of the lab reports will be reduced.

E. Assessment of Resources and Impacts.

1. Laboratory Technician and Resource Support. A quarter of cadets grade in this course is the laboratory and EDP(500/2000). It is critical that the laboratory technician support the setup and execution of all laboratory activities. Many of the labs require set up in multiple classrooms. It is important that there is enough equipment for each classroom. It is also important that no other courses are using the classrooms in between MC364 sections. Otherwise, the labs must be torn down and reset up.

2. Equipment. Much of the equipment for MC364 is aging and has not been serviced in several years. The tension machines, torsion machines, and aluminum beam specimens are all experiencing issues which have had significant impacts on the lab delivery. These machines must be checked each semester.

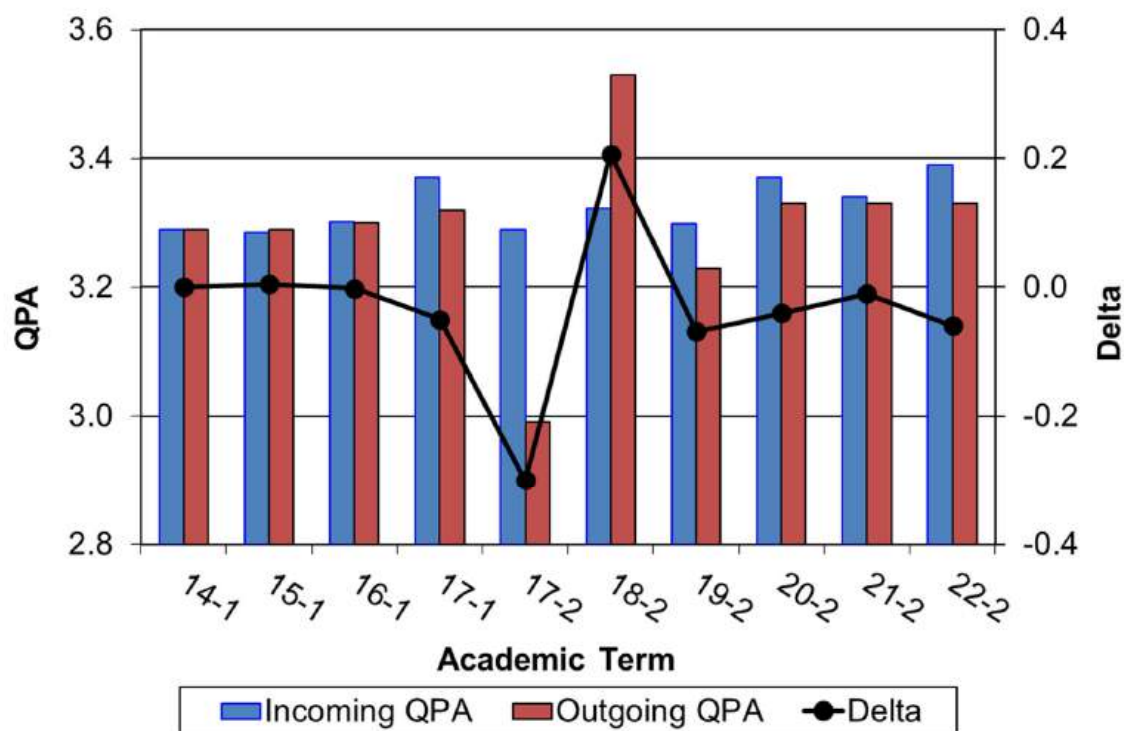
3. Material. Raw material is needed for many of the laboratory and EDP executions. For the uniaxial tension laboratory for MC300 and MC364, needs samples of steel, aluminum and cast iron. For the torsion laboratory, we need samples of steel, aluminum, and cast iron. For the Beam EDP, we need balsa wood (36"x3"x1/4" & 36"x3"x1/8"). For the TWPV Lab, we need Coca Cola Cans and hot dogs.

4. Lab Space. Currently, all tension machines are in the primary cadet material testing lab in D10. All torsion machines are stored in C19 and then are moved into D10 for execution of the lab. This space is tight for execution of three sections simultaneously. I would like to prepare this space this summer to be more flexible where machines, cabinets and equipment can be rolled out when not in use. As space becomes difficult to find, D10 may be used as a flexible space for other courses, independent studies, or capstones to also utilize.

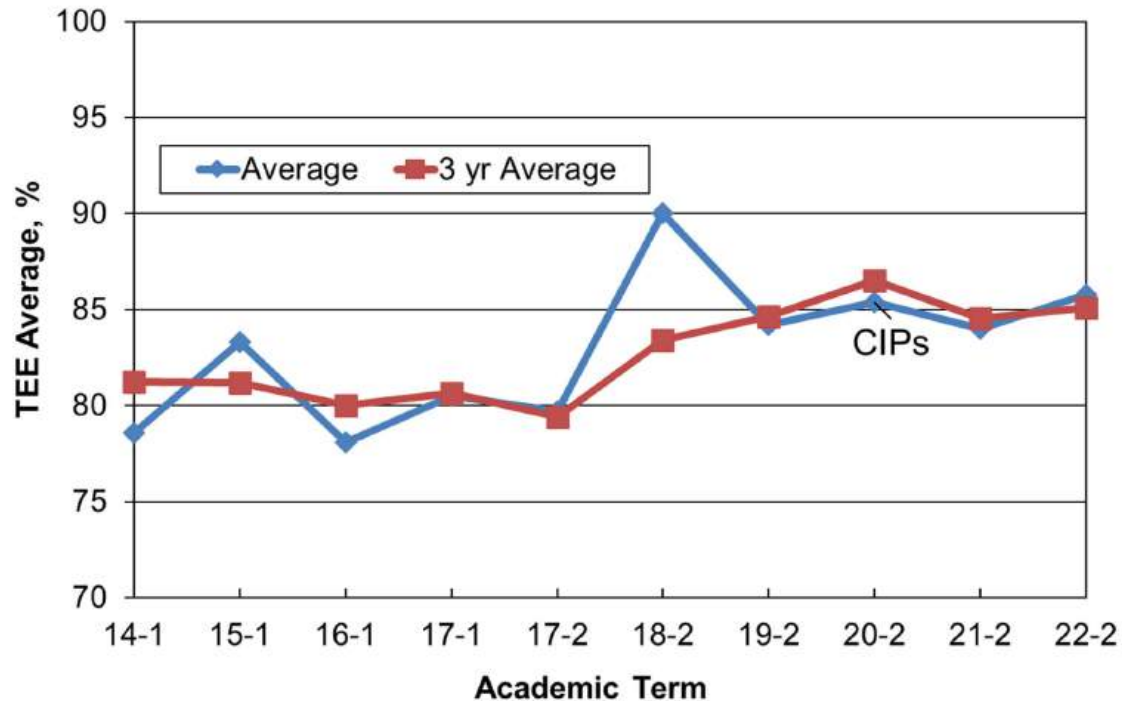
5. Videos. If MC364 remains a flipped class, there may come a time when the videos need to be updated to reflect new material, content, or objectives. This will require a significant amount of effort to develop and edit these videos.

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F. Course QPA Graph



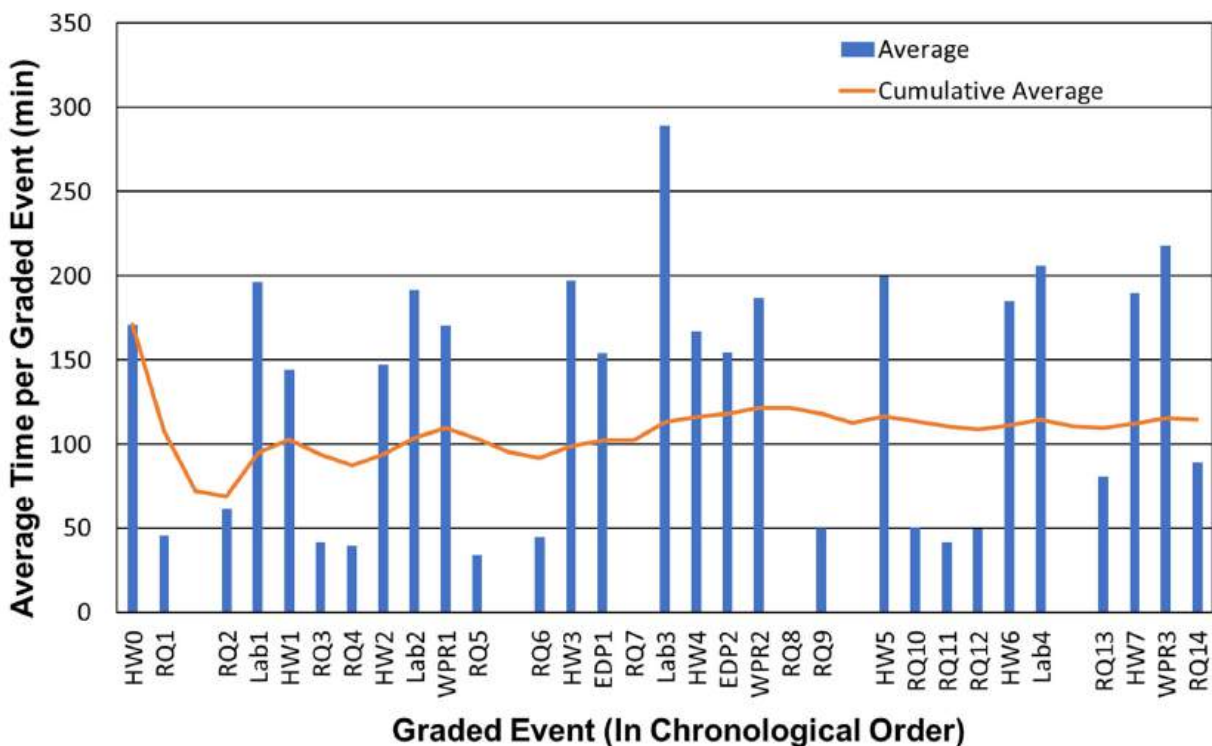
G. TEE Grades Graph



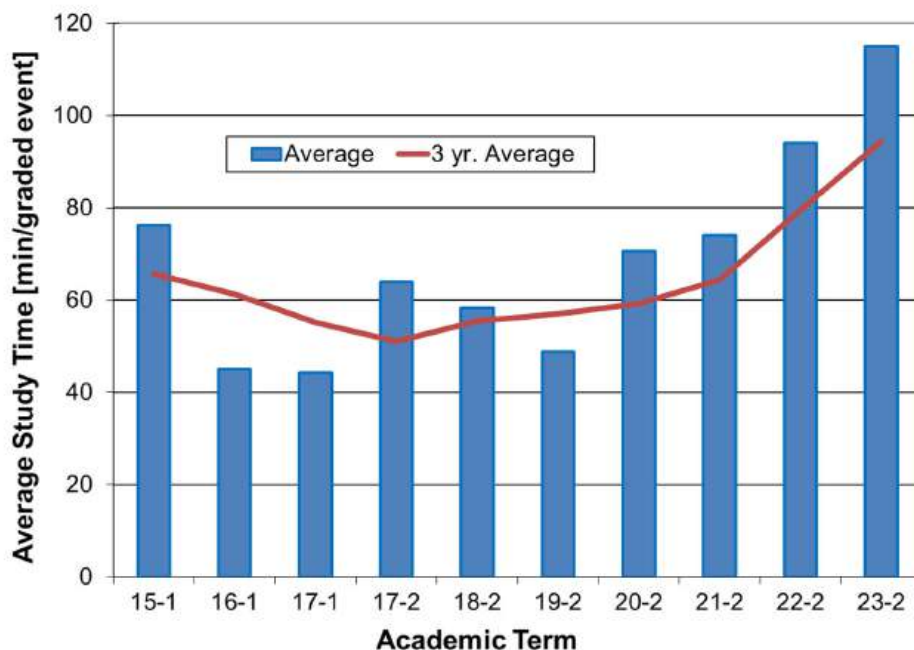
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H. Course Time Study Graph

1. Current Semester time survey data. (AVG: 115 minutes/graded event)



2. Historical Time Survey Data



3. Any comments on time survey data if needed.

The time survey in MC364 significantly increased this semester to an average of 115 minutes per graded event compared to 94 minutes per graded event in AY22. The increase in time spent may be attributed to two factors. First, the homework assignments increased by

MC364 Course Assessment Package, AY 23

approximately 60 minutes. For this academic year, the numerical solution to the homework problems was provided to the cadets, so they could work until they arrived at the correct answer. The second reason was the change to the lab format. The labs contributed to the high time survey having averaged approximately 200 minutes per lab.

I. Course End Feedback.

Question	Question Type
I provided leadership in completing team assignments	Likert
My team had to address conflict to complete team assignments	Likert
My team worked in a positive collaborative manner	Likert
Teamwork activities were effective in learning course material	Likert
Teamwork activities were effective in learning teamwork skills	Likert
Do you believe the flipped class format was effective in helping you learn the course content?	Likert
Do you believe the reading quizzes effectively assessed your learning of the course content? Why or why not?	Open
Do you believe the homework assignments were effective in helping you learn the course content? Why or why not?	Open
Do you believe working problems at the boards was effective in helping you learn how to solve problems in the course? Why or why not?	Open

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IV. Program Director's Current Areas of Emphasis

A. Communication.

No formal instruction is provided on communication, but there are several aspects of MC364 that allow cadets to work on their communication skills. First, MC364 continues to require cadets to build off of MC300 in their technical communication through the submission of homework assignments. These homework assignments are required to be submitted according to CME Standards. It is critical that cadets demonstrate strong written communication and organization skills for their problem sets. Second, there are four laboratory report submissions. These technical lab report submissions include an Introduction, Theory, Results, Conclusion, and Appendix section. It pushes cadets to the TECOM when developing their technical reports. The lab reports were significant and will be reduced in scope in the future. The third opportunity is a final oral presentation of their final EDP design. This is a short 1-Quad chart presentation, by teams of 3-4 cadets prior to testing their beam.

B. Numerical Methods.

No numerical methods are used in MC364. However, FEM is briefly mentioned when referencing capstones and the relationship between mechanics of materials.

C. Fundamentals of Engineering Exam.

Mechanics of Materials is one of the topics for the Fundamentals of Engineering Exam for Civil, Mechanical, and Other (NE). MC364 covers all topics on the FE except for Bulk Modulus, Thick-walled Pressure Vessels, Shear Flow, Composite Sections, and Elastic Strain Energy. The FE manual is not explicitly introduced in the course. However, we are looking at including FE questions on the reading quizzes and maybe on the Homework assignments.

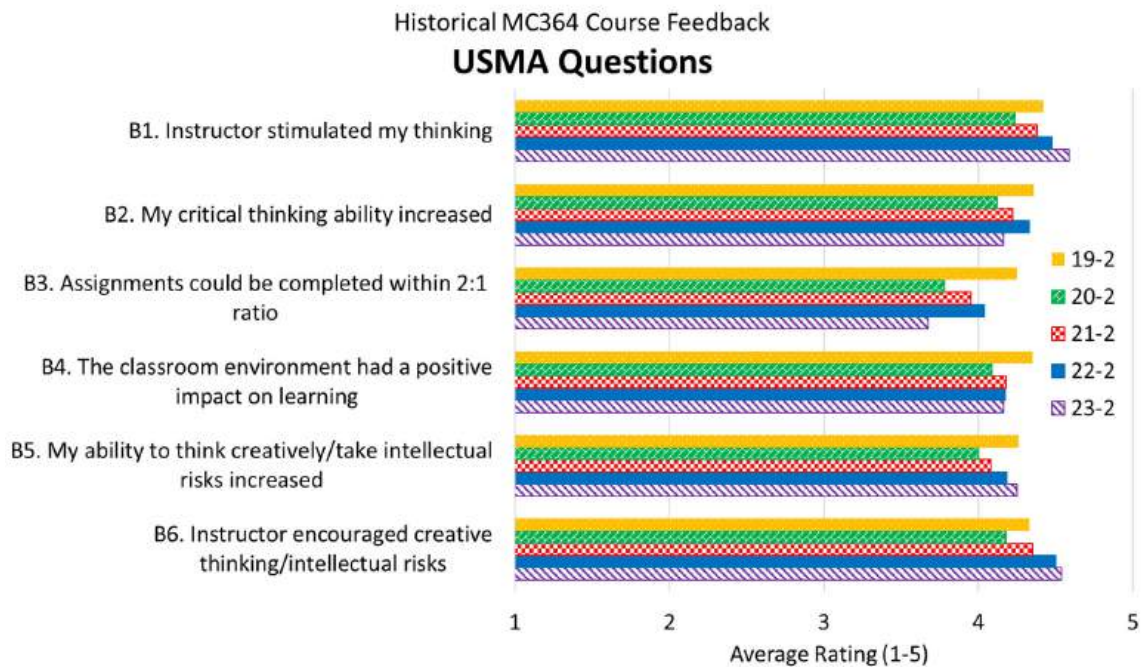
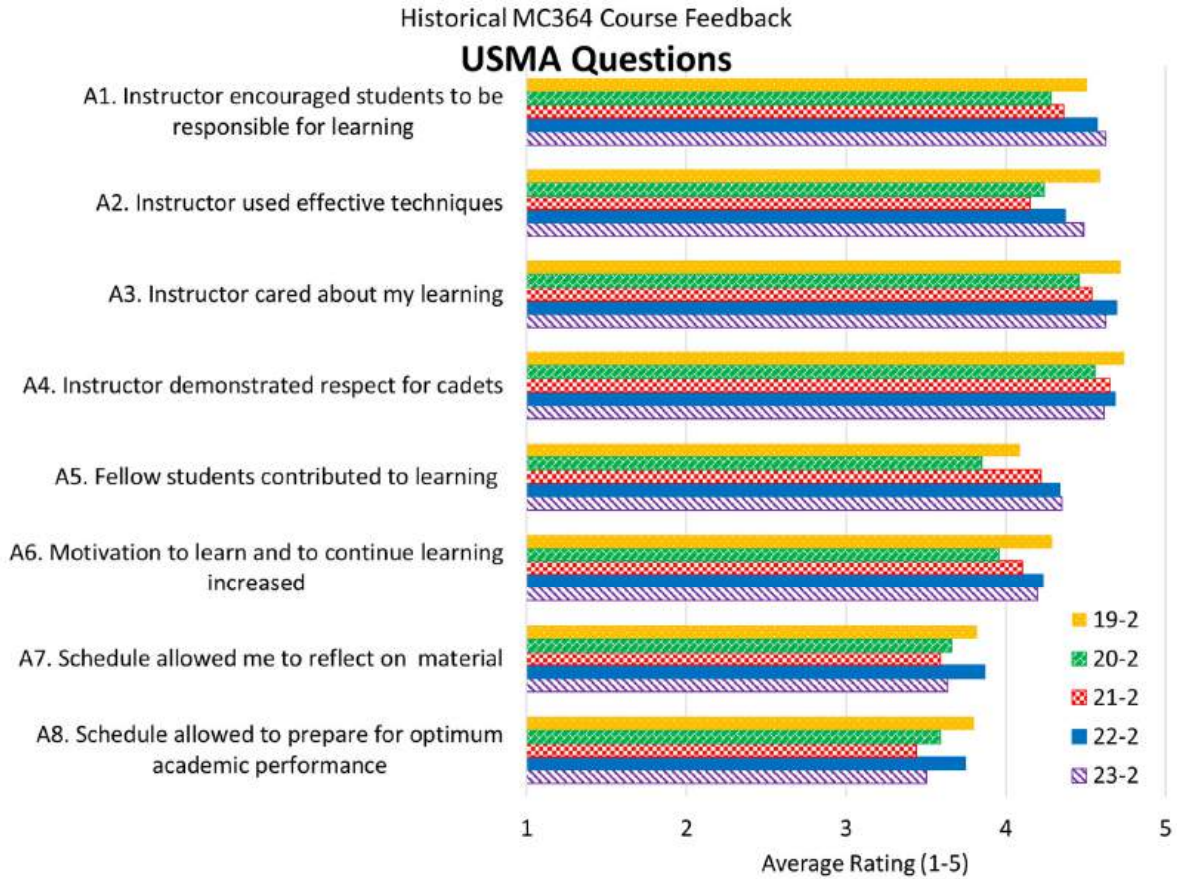
Enclosures

- 1. Historical Course Feedback on USMA Questions**
- 2. Term 23-2 Course Feedback on USMA Questions**
- 3. Historical Course Feedback on C&ME Questions**
- 4. Term 23-2 Course Feedback on C&ME Questions**
- 5. Course Narrative Feedback**
- 6. Course Numeric Feedback**
- 7. Other Feedback**

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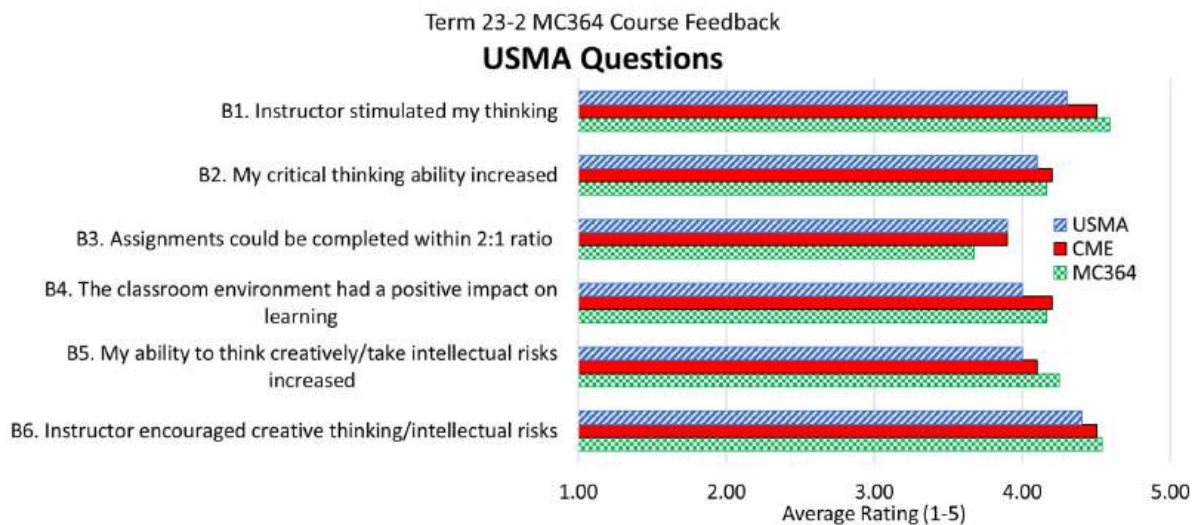
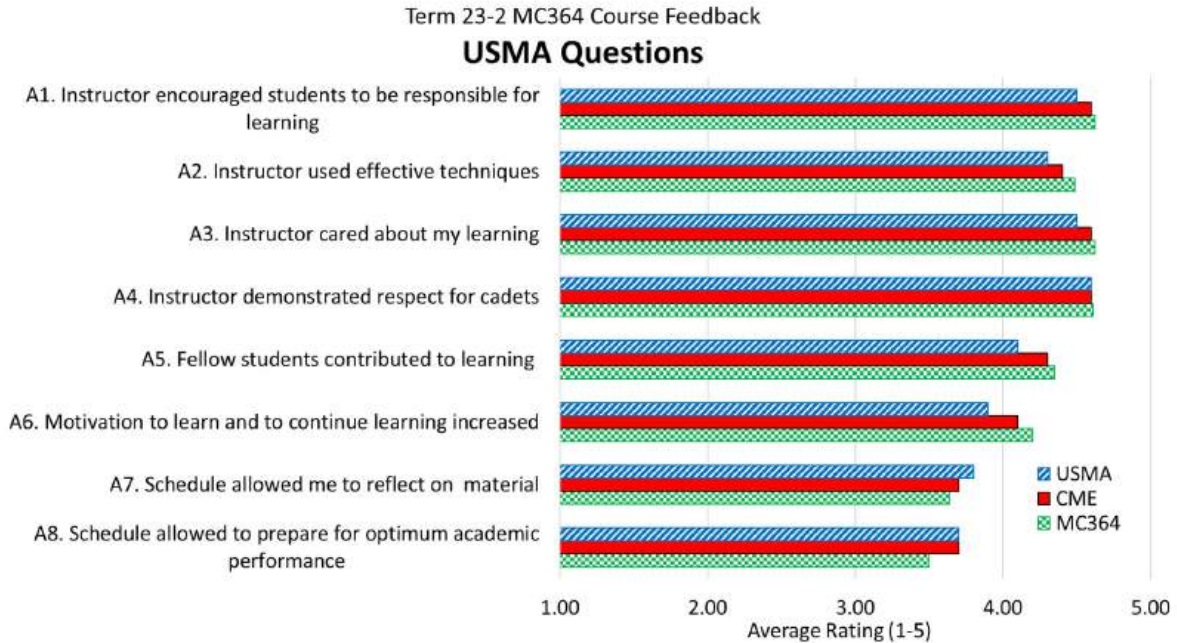
ENCLOSURES

1. Historical Course Feedback on USMA Questions



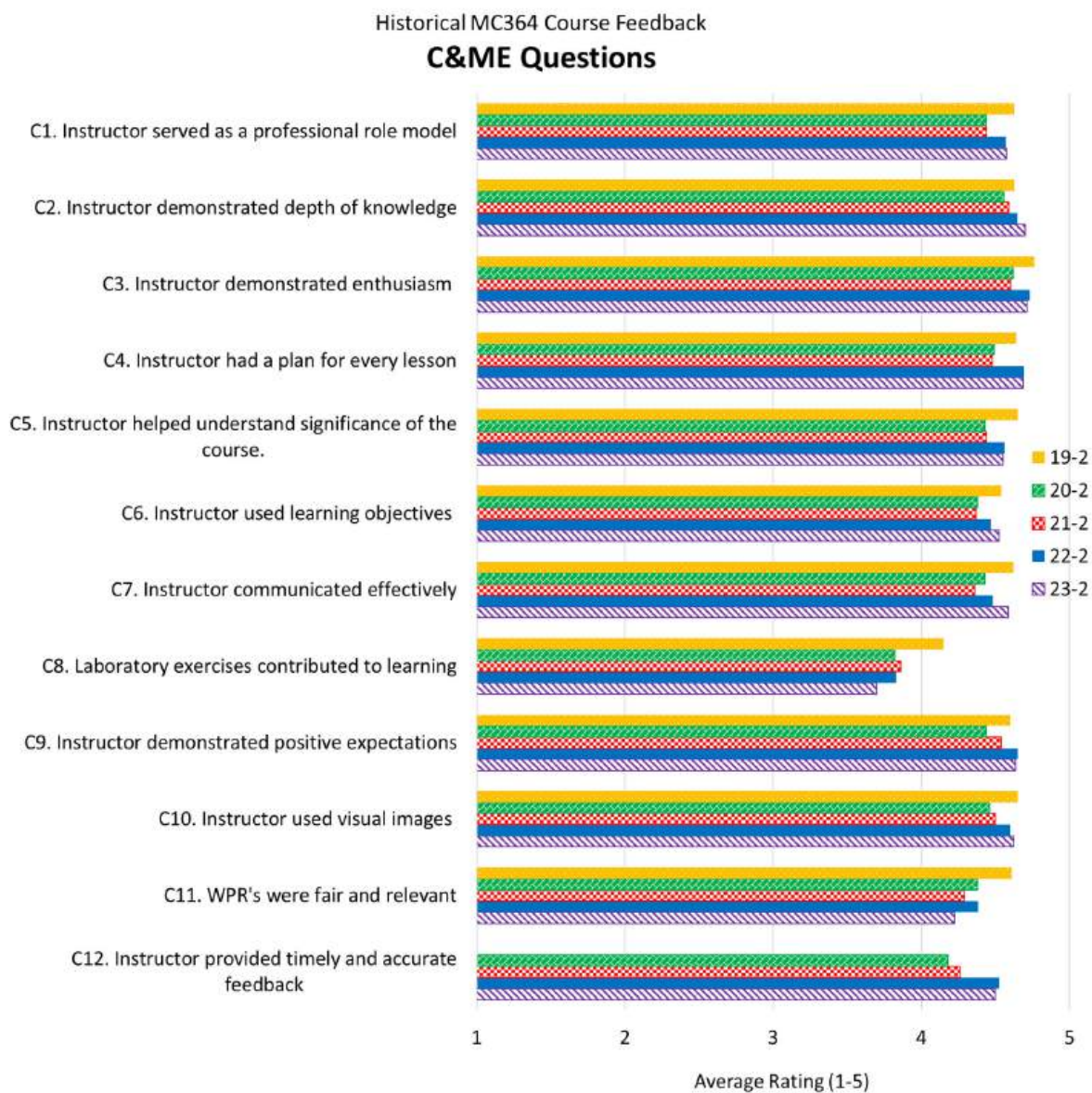
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2. Term 23-2 Course Feedback on USMA Questions



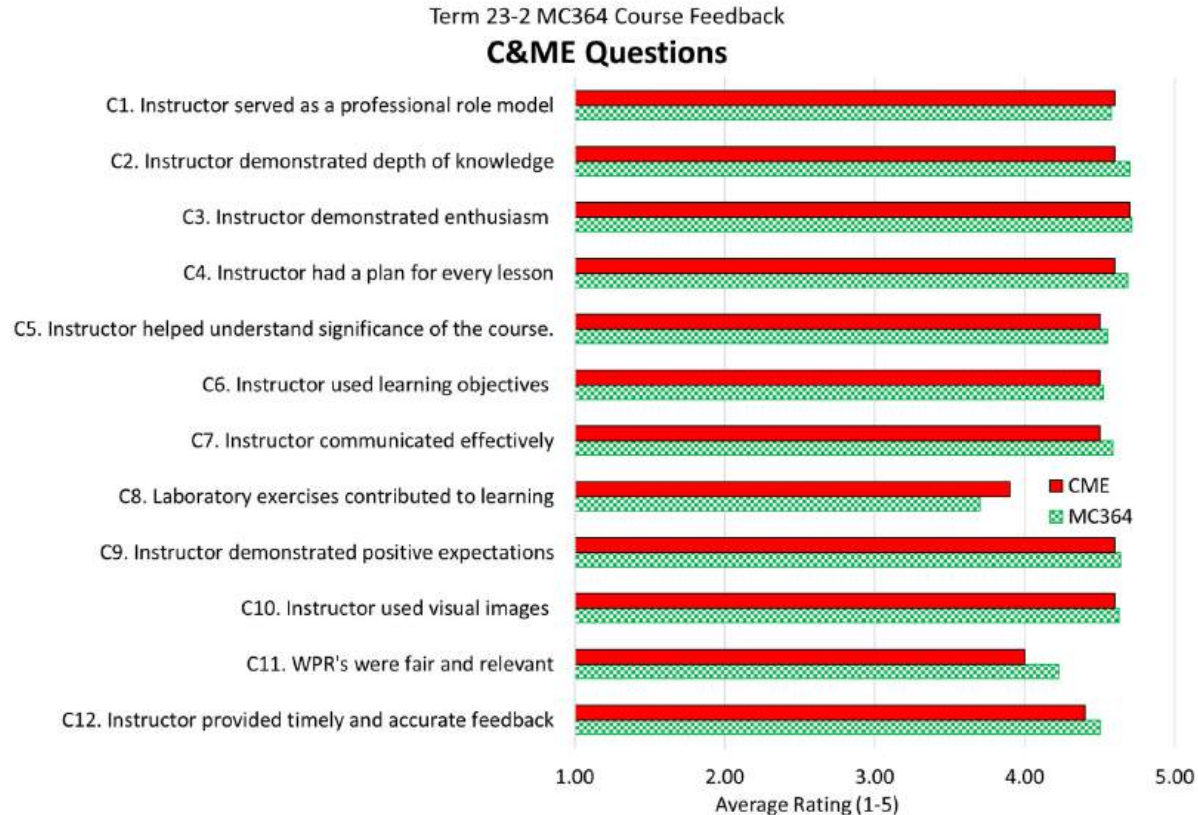
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3. Historical Course Feedback on C&ME Questions



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4. Term 23-2 Course Feedback on C&ME Questions



5. All Course Narrative Feedback for the current period of assessment.

A. Additional Comments about the Course

- Really enjoyed the hands on lessons and labs
- The high volume of assignments outside of WPRs made it extremely difficult to reflect on material conceptually, which lead to conceptual errors on the WPRs.
- My biggest issue with the course was the labs. A lot of them were very unclear as to what we should actually be trying to solve, and I don't think they helped improve my competency in the course objectives.
- I think that additional time could have been spent reviewing content from MC300- maybe 1 additional lesson at the start of the year. This would help prepare students more effectively for MC364 concepts
- WPR III right before the TEE is a little extreme.
- This course was challenging but well taught and well structured.
- There was a very high demand for our time outside of class. It is expected for this course to have a higher workload and difficulty, but another issue was that all major graded assignments and projects perfectly aligned with ME202 so that the two classes were constantly clashing for time.
- The Lab reports were a little bit too much work for what they are worth.
- Overall, they were pretty good. However, some of the tests had what I felt were trick questions on it. I think these types of questions are unfair when testing material.
- In general, I felt like errors with small conventions were docked very heavily on the WPR's.
- I think this course really increased my enjoyment for the major

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- Grading far to harsh, Often times CDT's get correct answers and lost up to a third of the points due to MUD-P, or had the right process and used wrong numbers and got 50 % on question.
- Way too many WPRs and HWs, by far the most in all my classes. Stressful more than helpful, and points do not reflect effort.
- Some classes had a set of practice problems sent to them before WPR's, that should be course wide since there is so much material in the blocks.
- Less Labs the length of the labs is fine with one to two pages.
- I think that it was a very fair course, it was just difficult to stay updated given my schedule.
- I preferred the in person RQs.
- I felt burnt out with CME in general this semester and I am not sure why. Something about this course feeling strictly methodological made me lose interest with the major I chose. Even though this class was another easy A, it was just offensively bland.
- Grades tended to be very nitpicky, and did not reward me when I wanted to be engaged in content
- Good course structure
- On teams is not a location. Homeworks often reference something that is "on teams", but teams has hundreds of files. This is not something appropriate that I could say to my commander. If the homework said "teams-MC364-DR Salem-general-files-homework" then THAT is a location. The class requires too much time for the points. In one case it took over a hour to solve a 13 point homework question DURING AI. Either make the homeworks worth more or make them easier. Maybe have fewer of them since almost every class has some form of graded homework due (no other class I have does this). Remove the time survey: my hour is much more productive than a typical college student's hour so a credit means a different thing to me. My time is also much less (taking 20.5 credits as a D1 year round athlete with a cadet military job). Too many graded events (one almost every lesson) compared to all other 3.5 credit classes. We don't need a reading quiz or homework due every day to pay attention. If I don't read and then end up failing the test, it's on me. Remove WPR3: it is a few days before the TEE. Just take it off and include the WPR3 concepts in the TEE. Then add the points to the homeworks. Because you have it though, I expect at least an answer key to be released by Friday before the TEE so we can study the questions (since we have gotten back every other WPR). Projects day: there exists no assignment on teams, the schedule, or the syllabus that allocates points to projects day. Never during all my time here could a single class mandate attendance to their department's projects day, especially when the class was another major. There should not be anything forced on us then. Textbook is expensive and too hard to access. Even with it marked as a favorite, it opens up 3 tabs before I can see it. The WPRs are also not open book, so looking up stuff in the appendix is just a waste of time only used on practice problems. I would remove the book. Teams is always a pain to use and so is canvas. The chemistry department has their own website that is much easier. Pre-lesson videos: these are posted far too late which is odd because I believe they are from past years. I wanted to knock out the daily homework over the weekend, but this is the ONLY class I have that does not have all daily homework posted at the beginning of the semester. Labs: I have never enjoyed a lab because all I can think about is the large report I have to write afterwards, and this class has some of the most and some of the biggest. Formatting labs takes more time than the points are worth. Some accepted values are provided but not all, forcing guesswork sometimes. Too many places I need to look for the lab data: just make everything I need accessible on one document (this is what you expect when I turn in a lab after all: simple to use). Equation sheet: has lots of unnecessary info and is lacking some needed info. Look at the PANE department for a good example of a reference card. Key changes: don't use teams or canvas, no WPR3, cut down on daily things due, more points toward homework, change up lab reports (make them individual, no formatting, more points), post lesson videos earlier (at least the weekend prior), get rid of the textbook or at least make it easier to open, better reference card. Keep: open note WPRs, lots of AI, DR Salem
- Very well organized and executed

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- The course material and instructor were phenomenal. However, the course structure was not as ideal. Having something due just about every class started to take a toll.
- Reading quizzes were frustrating and served to hurt my grade more than help it. Homeworks are easily enough to suffice reading comprehension. Moreover, some choices, like to put a WPR on the second to last day of class (3 days before most have the TEE) are in poor taste. Like any other course outside of CME, it is best to simply include the material covered at the end of the class in the TEE. The additional stress right before the TEE is not productive nor beneficial. It also forces cadets to study purely for the last block of course material, leaving less time to study for blocks 1 and 2. There are plenty of other ways to give out points in the course aside from assessments and quizzes. This is not on my instructor--he truly cared and did his best to motivate his students when they were frustrated with the course structure. This is a grievance regarding the course structure.
- I think that lab requirements might be helped by providing an "exemplar" (of a different subject) to base things on. While we are provided with the TECOM and CME resources, sometimes it is difficult to understand how to apply those comments in the report.
- Despite the difficulty of the course I found it really beneficial for my learning as an engineering
- do not change anything major about the course
- The videos were a great asset to my learning. They helped me learn the key concepts before class so I could solve the in-class problems with greater confidence.
- I think that having the smaller HW assignments instead of the large problem sets were a huge benefit to having knowledge checks on the material more often.
- I really liked the transition of reading quizzes into Teams to allow for more class time.
- As much as I like getting solutions up-front on the homework, there have been consistent typos and errors that become extremely frustrating in these solutions. I think this stems from changing the problems slightly from previous semesters, but they need to be better proofread to stop confusion. Both HW6 and HW7 had solution errors that were personally confusing and time-consuming.
- there should not a wpr days before the tee
- The homework's potentially took a while depending on which one. So maybe making more homework's with less problems throughout the course would be beneficial, as it would allow for continuous repetition with the content and problems throughout the semester.
- The course was wonderful. It took a little bit and made me lose points in the beginning to catch up not having enough MC300 background, but overall I enjoyed the course very much. LTC Bigerstaff is my favorite instructor at West Point.
- Taught very well by all the instructors I have been in class with. Doing notes and problems on the board with step-by-step instruction enhances the learning experience. The different colored chalk makes it very easy to follow, even after being in class for 5 hours earlier in the day.
- Make reading quizzes in class. The online ones are harder to remember to do.
- I enjoyed my instructors teaching style very much
- I appreciated all of the course resources.
- Having a WPR right before the Tee is illogical as there is not enough time to reflect on the course subje~~tc~~ before the tee.
- This course was excellent in giving all the students the resources to be successful. The only thing is that the work load doesn't allow students to get a full grasp of every subject.
- This course is well thought out, scheduled, and instructed. The material that was covered is standardized accross multiple colleges and universities which made study materials readily available.
- This class was remarkably structured and very well developed, the only thing I would critique was the lack of AC in Mahan on some days, but overall perfect otherwise.
- The flipped classroom made this class far more difficult than it should have been. Had the material just been taught instead of vaguely taught as the teacher went through a problem it would have been far easier.

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- Reading Quizzes seemed unnecessary and did not contribute to learning or amount of material retained
- Labs were a bit tedious, any type of review before the lab reports are due would be helpful for more in depth understanding

B. Additional comments about the instructor (Removed specific comments)

- One thing I didn't like was that they would often write on the boards in the back of the room and the whole class had to turn their head around and take notes which was uncomfortable
- My only improve would be that they writes a bit too low on the boards sometimes, which makes it hard to see.
- My instructor must've went to art school because they were really good at drawing.

C. Do you believe the reading quizzes effectively assessed your learning of the course content? Why or why not?

61 Yes/37 No / 9 Maybe – (19 In-Class, 10 Online)

- Difficulty - Sometimes, but often there were trick questions. Those are terrible! Trick questions can cost us multiple points when a key word is missing and we assume it is implied.
- In - class - Yes, the quizzes were directly tied to the lesson material. While the course changed to online quizzes, I prefer that quizzes be in class and physical.
- In-class - I believe in class quizzes were more effective and fair than online ones
- In-Class - I believe that the online quizzes were less effective, and overall the quizzes assessed my learning fairly well
- In-Class - I believe the reading quizzes were fair assessments of the course content. I preferred the reading Quizes to be in class because it encouraged me to study more on my own.
- In-Class - I believe they were effective at assessing my learning of the course, however, the teams format was hard to manage and keep up with.
- In-Class - I don't think the online reading quizzes were the most helpful since I cannot remember a single thing about them. The in-class quizzes at the beginning of the semester were more effective since I had to actually prepare for them.
- In-Class - I liked the reading quizzes on paper in class.
- In-Class - I preferred the in class reading quizzes because they made it so that you should watch the videos and take notes on them
- In-Class - I preferred the in-class quizzes as the online quizzes left less room for partial credit.
- In-Class - I think that the in-class reading quizzes were a better assessment of knowledge than the online ones because they only allowed one note sheet rather than being fully open note and open book.
- In-Class - I think the written quizzes did since they were actual "checks on learning." The online ones were just doing our best to flip through the textbook or notes in a hurry while not really assessing as much as we learned in class. The reading quizzes sort of became slightly to close to a hw for the difference in point values.
- In-Class - The in class quizzes, yes but the ones online did not because they were confusing
- In-Class - The in class reading quizzes did because they forced me to prepare and know the content prior to class
- In-Class - the paper ones did the online ones did not
- In-Class - Yes but I preferred the paper in person ones.
- In-Class - Yes but i preferred them in class
- In-Class - Yes, I personally didnt like the online versions even though I received higher grades in the online versions.

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- Interesting - They did not assess my learning as much as they assessed my ability to take notes from the Pre-Lesson videos. I did not really understand the content until after the inclass lesson.
- Kind of. Sometimes they weren't what i was expecting.
- N/A - I don't believe I can really say. I don't have an opinion either way.
- No - I do not think that reading quizzes effectively assessed my learning of the course content because sometimes a reading would make sense before class, and other times they wouldn't make sense until after the class. I think that changing reading quizzes to being due by the block and at your own pace would help to ensure cadets actually learn the material instead of just taking a loss on the daily quizzes.
- No - The reading quizzes did not help because you cannot review them after and the teams version was worse.
- No - The reading quizzes only covered small concepts, and although they were helpful, I don't believe they stimulated my learning.
- No because often I did not have time to ask questions about the content because I had just read it and so took a quiz on a subject I was still confused about.
- No because they varied in level of depth needed to complete the material. The pre-lesson videos were sufficient in prepping for class
- No cause the online ones you can't write anything out.
- No I think that the reading quizzes were very different from any exams and did not assist with that.
- No, a lot of them did not correlate with wpr questions since we are not tested conceptually.
- No, A lot of what I read wasn't on the quiz.
- No, because some of the questions were problem solving before we learned how to problem solve and ask questions about the topic.
- No, because the reading quiz did not give me a chance to reinforce the reading material.
- No, because they are due before the material is taught by the teacher, making it difficult to accurately gauge the amount of material understood.
- No, because we did not have the chance to retain the information learned before being tested on it. Since it is a small grade though it serves as a good way to fail and learn from that.
- No, I did not learn the material until about 24 hours after I took the quiz.
- No, I didn't enjoy the quizzes. I felt like I was taking them to taking them. More points at stake.
- No, it just wasn't beneficial
- No, it was base knowledge that by scanning the book you could find the answers without understanding why. Adding tougher questions and allowing for longer time periods to solve said questions would be a better assessment on learning.
- No, quizzes were stupid and irrelevant
- No, taking them after the lesson would have been better
- No, the online quizzes were easy to miss and were often confusing even after watching lesson videos.
- No, the reading quizzes were a good test on the reading of the course material but for me it was hard to comprehend until we got to class and talked about it.
- No, they did not. Many of the reading quizzes were problem solving and based on previous lessons. I think this is contradictory to what a reading quiz is. A reading quiz should reflect whether or not you did the reading for that lesson. We have homework and we solve problems in class, so I think that problem solving questions are pointless on the reading quizzes. Ideally, we would have it online and it would only address that lesson and the reading. That way, we would be forced to read and know what was going on before we got to class.
- No, they looked forward but seemed ineffective in gauging our learning compared to wprs.
- No, they were really bad format online, the paper ones did to make sure you read, but when you start having to solve problems that you saw for the first time on a 5 minute reading quiz, that assesses your ability to guess correctly, not your effort in preparing for class.

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- No, too many of them. See above comment.
- No. The amount of time spent to complete and prepare for the reading classes forced me to sort out specific details from videos without understanding the concepts of the videos to succeed on the reading quizzes. I did not feel as though the homework quizzes were an effective method for learning or demonstrating understanding.
- No. The online format is particularly unfair, even with open resources. I would rather take the in-class quizzes because they adequately covered the material and I did not get tripped up on sometimes trivial questions.
- No. The original style of reading quiz, in person, was helpful because it was directly correlated to the video from the night before and was conceptual based instead of solving. When the quizzes were made virtual, the questions were significantly more difficult, and the correct answers were never made available. This meant that they detracted from my grade and that I could never learn from my mistakes. Additionally, the syllabus said that 3 reading quiz grades were going to be dropped, and currently they are all in CIS instead.
- No. The reading quizzes often served to hurt my grade more than help it. Generally, I will not prepare for class ahead of time. I learn the bulk of material during class and working problems. This has been effective and successful for me in all of my classes. Even when forced to read for reading quizzes, I would skim the text and there would be little effort to truly understand the concept until it was presented in class where I could truly understand and learn it. It was especially odd to have reading quizzes while the homework assignments also asked for reading comprehension questions. Overall, reading quizzes were a waste of time and are not useful in knowing, understanding, or comprehending course material. This happens in class. I do well on homeworks, WPRs, and labs, but not on reading quizzes.
- In - Class - Not online because they were easy to forget or just search the textbook when online.
- Online - At the start no because the time was too limited for me to even provide the answers that I knew and wanted to provide, but they got better progressively
- Online - I believe the online quizzes did not but the original on paper reading quizzes did an adequate job of reviewing old course material and ideas from that days lesson video.
- Online - I felt like the online ones with the ability to have the textbook made me feel best assessed and prepared for the following class.
- Online - I felt that the computer reading quizzes did assess learning well because it helped you realize and have the chance to lookup content in your notes that you did not remember.
- Online - I think that reading quizzes were an effective assessment of what I learned in the class especially when they were taken before the class because they ensured that I understood the course material and where low stress.
- Online - I think the online quizzes, due the night before class, allowed the students to engage with the material prior to the lesson. Then, they can come to class with questions on the lesson day.
- Online - The later ones did, because they were more open-note and doing them in our rooms felt much less stressful.
- Online - The reading quizzes improved through out the semester. The time limit should be closer to 20 minutes on them and allow us to actually find answers.
- Online - Yes, although I appreciated the online quiz more to not take away from class time, the quiz always seemed to make me feel somewhat rushed for time (although this is good practice for a wpr).
- Online - I think the reading quizzes virtually were a good move and did help prepare for classes, requiring less in class time for review.
- Problems (No) - Sometimes, I thought they were better before class as general vocab and theory questions not problems. Those were better tested after the class.
- Problems(No) - I think that the reading quizzes should have been all conceptual and multiple choice.
- Problems(Yes) - Some of them did. The quizzes where we were asked to solve problems from the previous lesson were the most effective.
- Time - I just don't have time to read in depth.

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- Time - In some ways yes, but sometimes I think unfair questions were asked. The time limits could be very constraining.
- Time - Sometimes they did other times, I felt like I prepared for the lesson and I still struggled to solve the problems in time.
- Time - sometimes they were good asking for a good baseline of the material, other times they were way to hard
- Time - The reading quizzes did not effectively assess my learning of the course content because the time constraints of the quiz did not allow me to think.
- Time - The reading quizzes were helpful to make sure that I gathered the proper information from the videos. I like the online ones better (so in-class time wasn't wasted), but the 15-minute timer was sometimes too short and other times too long.
- Yes - I found that the quizzes effectively assessed my learning because it helped me understand the material for the up coming class.
- Yes - I like reading quizzes. They are really helpful and efficient.
- Yes - I think the reading quizzes did a good job of balancing the textbook readings with the pre-lesson videos and ensured that I was prepared for what we went over in class.
- Yes - I thought the course quizzes were beneficial; however, there were a few course quizzes that tested us on material which we had not learned yet, and the material in the video was not clear enough to set us up for success. In particular the video lesson on drawing the stress elements was a little unclear as to which direction the arrows went, and the quiz tested us on that material before we learned it in class.
- Yes - They helped me prepare for future lessons/review my knowledge on the lessons.
- Yes - They're as helpful as a WPR which is to say very helpful.
- Yes because they were an accurate test of the pre lesson videos. The didn't relate much to the actual reading much though.
- Yes because they were very heavily based on the readings and pre lesson videos
- Yes it highlighted the important topics for the lesson
- Yes they were fair and did not require much calculations given the time crunch.
- Yes, although sometimes they included questions that seemed kind of obscure and were not really relevant later.
- Yes, because they helped demonstrate what knowledge I had learned or knowledge I still needed to practice.
- Yes, but I would not be opposed to a small homework assignment that consists of a couple small problems similar to the Mech movies.
- Yes, I believe they were a fair assessment and that they adequately assessed how well students prepared given that students were allowed to use a note sheet they created. As such, students were to decide how prepared they wanted to be to achieve success.
- In-Class - Yes, I did prefer the in class ones though.
- Yes, I do believe they were effective because they required us to pay close attention to the pre-lesson videos and start solving future problems while ensuring we were becoming proficient in problems from the previous class.
- Yes, I think they were relevant to what was being taught at the moment.
- Yes, If you consider that you drop the worst three.
- Yes, less problems - I do, in the sense that they helped to push learning of relevant content and ensure that it was completed, but the format of having several problems that could only be solved in a short amount of time was not an effective measure of academic ability.
- Yes, sometimes they seemed different from what was going to be on the test.
- Yes, the quizzes had material relevant to what was on the pre lesson videos.
- Yes, they make you review material
- Yes, they provided a good overview of theories we covered in detail in class.
- Yes, they were always pretty close to the lesson objectives and were a good reinforcement of what should have been taken out of the pre-lesson videos.
- Yes, they were fair and could be studied for.
- Yes, they were good introduction questions to stimulate thinking about the lesson.

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- Yes, they were mostly conceptual which i thought was helpful
- Yes, they were only difficult if you did no prep
- Yes, they were relatively low point ways for me to gauge my understanding. I did prefer the in class ones to the take home ones.
- Yes. Some people would be upset when the quizzes would include some information not from the pre-lesson videos, but frankly, they were in the textbook which was an open source to use as well. Also, the homework did say to read the textbook unit, so they are responsible for the material in the long run. It was a good checkpoint to make sure you weren't slipping in the course as the semester went on.
- Yes. They helped me to identify weaknesses and know what to study.
- YN - Yes and no. The idea of a routine assignment that incentivizes lesson preparation is great. Practically, I found the reading quizzes to be frustrating and definition-based questions always felt confusing unless reading straight out of the book (meaning I was not understanding the definition or absorbing the content). Overall, though, I would mostly agree that the reading quizzes should be sustained.

D. Do you believe the homework assignments were effective in helping you learn the course content? Why or why not?

125 Yes / 6 No / 5 Maybe

- No - I don't think the homework assignment's were effective. Problem sets worth more points as opposed to homeworks which take up almost as much time as a problem set but for less points was hard on students.
- No - I think that the homework was often more difficult than anything else in the course.
- No - I would prefer fewer longer problem sets than many smaller homeworks.
- No. I feel as though I catered my understanding of the material strictly to succeed on the homework quizzes versus understanding the concepts that were more important. The timing constraints on the homework quizzes seemed to hurt more so.
- WPR - For the most part yes. I would have preferred if they were more similar to the WPRs/
- WPR - I believe they did but some of them were a bit overkill. They would be better if they were more like WPR questions.
- WPR - they were helpful but the WPR questions were always so different
- Yes - Absolutely so. I learned the most from the homeworks.
- Yes - Homework really set up us for success. They are a great help for major graded assignments.
- Yes - Homework were good testers for upcoming wpr and to wrap up all concepts needed to succeed in evaluations
- Yes - homework's were great.
- Yes - I believe homework assignments were helpful in providing practical exercises. I appreciated the answers being provided, but sometimes some conventions in the answers to the question (such as variable names) were confusing.
- Yes - I believe that the homework assignments were much more effective than the problem sets from MC300, as it allowed cadets to actually understand the material before having to test it.
- Yes - I believe the homework reinforces the work lessons taught in class and challenges the students to think through more difficult problems.
- Yes - I do because they were a good consolidation of everything that I needed to know.
- Yes - I do think the HW assignments were effective because they were often and they had the answers provided so I was able to ensure that I understood the material.
- Yes - I think the homework assignments were the most effective tool in learning the course content as they forced me to solidify, and make sure I actually did have, a good understanding of the course material. Having the answers provided also helped not just my stress around the assignments, but also enforced that I understood the class materials and was getting the right answers.

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- Yes - I think the homework assignments were very effective in helping me learn course content especially when part of the answer was given to us which helped me work toward the correct answer and understand what I was doing wrong rather than submitting a wrong answer without knowing what to correct.
- Yes - I think the HW assignments helped us learn what to know for the tests
- Yes - I thought the homework assignments were very helpful, and having the final answer really saved me time and helped me find sign/minor errors more quickly.
- Yes - lots of good difficult practice
- Yes - Once the homework assignments were posted with the answer as a goal, I was able to continue solving until I did it right.
- Yes - The homework (Problem Sets) were helpful because they were usually comprehensive and spread out well.
- Yes - The homework assignments definitely helped me with the problem solving processes that were essential on tests.
- Yes - The homework assignments were very applicable to the lesson content.
- Yes - The homework felt like review from previous lessons and prep for the next lesson or two. I liked the system.
- Yes - The homework helped me because the questions asked required me to know the concepts well.
- Yes - The homework was very applicable for the follow-on assessments.
- Yes - The homeworks were good because they provided a low stress environment for getting good reps on questions seen on the WPRs.
- Yes - They are the correct difficulty but worth FAR too few points. See above comment.
- Yes - Very much so. I hope that there was more detailed answers presented on the homeworks themselves so we can correct our work, and I wish that there were more detailed solutions to the online problems.
- Yes after it was all complete, but no I was not able to complete it on my own. I had to go to AI to get an idea of where to start. Problems seen on the homework should have similar example problems provided with better work done.
- Yes as they provided an opportunity to do questions with a minimal point load attached.
- Yes as they were helpful for reviewing for the WPRs.
- Yes because it allowed me to actually do the process for each learning objective but it also had the answers which allowed me to check my work and to see if my process was correct.
- yes because it gave me a place to practice the material we learned in class and prepare for wpr's
- Yes because they allowed you to apply the problems and practice with solving them.
- Yes because they provided a baseline understanding of course material that I needed to achieve to ensure I was on track.
- Yes because they provided unique difficult problems that taught me how to think critically on more complex problems.
- Yes because they told me what I did right and what I can fix
- yes because they were kinda like the wpr in covering the right concepts
- Yes but sometimes they were super tedious.
- yes it helped working through the problem solving steps
- Yes it was good repetitions
- Yes it was useful in getting practice with the content
- Yes sometimes the answers were kind of messed up w units and signs.
- yes the answers were very helpful.
- Yes they helped a lot
- Yes they helped a lot
- Yes they helped me review material
- Yes they matched the required problems format and in class problems.
- Yes they were good practice
- Yes they were helpful. Especially being able to check my own answers.

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- Yes! They very much helped.
- YES!!! These things saved me honestly, and were a good but not impossible challenge.
- Yes, absolutely. The homework problems are some of the most effective learning materials of any of my courses. Covering all major objectives with these assignments and publishing solutions has been extremely helpful for studying and learning the material. The size of these problem sets is perfect as well, as detailing a solution for about four or five problems is typically about a two-and-a-half hour endeavor and doesn't feel like too much of a slog. I would definitely recommend keeping the homework assignments in their current form.
- Yes, all were relevant and provided a good range of questions.
- Yes, although they should be graded a little easier. They helped me review and remember the lessons that the homework assignments covered.
- Yes, because I used them to study for my WPRs.
- Yes, because it allowed to work with peers and problem solve and develop the process to solving problems.
- Yes, because they were more bite-sized than the problem sets of MC300 and allowed me to retain the concepts better.
- Yes, because they were very similar to the content that would be covered on WPRs, and allowed time to think through the processes required to solve the problems.
- Yes, but not always as applicable to tests as they could have been. Give me the hardest problem on HW, so I can execute the easier problems later
- Yes, but the feedback can take a while
- Yes, BUT they were graded much too harshly, often times getting B's and C's on homework's where the correct answer was determined, but units or process was off. This is extremely unreasonable and unfair when the most important thing behind the homeworks should be understanding the general process, and not a negative sign or a sign for compression.
- Yes, definitely, because we got to work through full problems.
- Yes, especially giving us the answers
- Yes, good practice problems leading into the WPR in which I could see where I struggled prior to the WPR
- Yes, having a solution I was positive about also gave me a better idea that my work was correct.
- Yes, having the answers provided (but not the solution) meant that I could attempt the question myself and then check to see if the solution matched.
- Yes, homeworks were extremely helpful. They allowed me to practice what I learned in class as well as collaborate with peers. It was beneficial to have multiple homework sets consisting of around 3 questions as opposed to 3 major homeworks which would take 5-6 hours to complete. However, by 7, it felt like too much. 6 homework sets feels like a sweet spot.
- Yes, I appreciated the answers being provided so that I knew I was doing the right process.
- Yes, I believe that the problem sets were a good reflection of the material we learned in class.
- Yes, I believe they were effective because they made you practice course content and think through problems applicable to the WPRs.
- Yes, I do believe so. However, I wish they were worth more points, as they would take hours to complete only for 30ish points. I would spend more time on the homeworks than studying for the 250 point WPRs for the most part, yet they were worth a small fraction of the grade.
- Yes, I have referenced back to the homework's several times. I believe homework assignments were effective in helping me learn the course content. They provided an opportunity to practice and apply what I learned in class, which helped me to solidify my understanding of the material.
- Yes, I like how the answer was given so I could really ensure I was properly doing the questions right. I feel like it helped me get an idea of the types of problems for the WPR.
- Yes, I liked being able to go back and reflect on the content before the test.
- Yes, I looked at them as study materials too.

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- Yes, I think the homework was effective because many of the questions were multi step complex questions that did a good job of assessing our knowledge of the topics in class. However, there were some questions that I felt were too hard and it seemed like it was overboard.
- Yes, I think they were accurate representations of what was going to be on the WPR's
- Yes, I thought the homework assignments were the perfect combination of difficulty and time consumption and I felt that it reinforced my learning.
- Yes, it seems like they allowed for practice of the material we were going over at the time, but they were also challenging and seemed a bit more in-depth than the teacherled problems.
- Yes, it was a nice review of the material.
- Yes, more practice
- Yes, more practice the better understanding I can get
- Yes, practice problems are kind of how STEM is engineered.
- Yes, questions were challenging.
- Yes, the homework assignments had similar questions to the WPRs and those that we did in class so making sure one was able to do those helped solidify concepts and highlight anything that needed more focus while preparing for the WPRs.
- Yes, the homework assignments were effective. Knowing the final answers gave me more assurance that I was using the correct problem-solving methods.
- Yes, the homework assignments were very helpful.
- Yes, the homework problems were a good size and frequency to learn the problems for the course.
- Yes, the homework questions were moderately challenging and extremely useful to study off before the WPR.
- Yes, the homework was my primary method of learning.
- Yes, the homeworks are where I learned to apply the course concepts.
- Yes, the homeworks were the most useful.
- Yes, the problems covered many topics that were later tested on.
- Yes, they asked relevant questions and combined different learning objectives altogether.
- Yes, they compiled a lot of the things we learned in class and served as check on learning
- Yes, they effectively went over relevant topics. They helped because it was an opportunity to apply what we learned and see and learn how to use it.
- Yes, they problems were good practice and indicators of what would be on the WPRs.
- Yes, They provided a good knowledge check and something to look back on when studying for tests.
- Yes, they provided an opportunity to practice problems
- Yes, they provided practical problems to understand how to solve problems relevant to the class.
- Yes, they really showed what we needed to take away from the lessons. They were also spaced apart nicely to encourage getting ahead and working on them as we learn new material.
- Yes, they were comprehensive and helped to solidify lesson content
- Yes, they were effective because they were relevant and were not difficult to grasp an understanding of the material.
- Yes, they were fully worked out problems that helped me understand where I was lacking in understanding and they helped me review for WPRs.
- Yes, they were good problems to utilize the skills we learned in class.
- Yes, they were not too long and seemed accurate to the test for the most part.
- Yes, they were relevant and a good overview
- Yes, they were relevant to the material and tested the skills that would reoccur throughout the course.
- Yes, they were relevant to the topics and required critical thinking and a general solving process to learn.
- Yes, they were similar to the WPRs.

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- Yes. As long as the problems are similar to WPR problems.
- Yes. Covered all topics required to know and gave me ample time to think through the requisite process for solving each problem.
- Yes. Especially having the answers to double check your work.
- Yes. The problems were relevant and directly related to what was taught in class.
- Yes. They corresponded well to the WPR's.
- Yes. They created opportunities out of class to receive feedback on my problem-solving ability, and they forced me to seek out help from classmates.
- Yes. They did a good job of forcing us to learn all of the material.
- Yes. They were a good reinforcement of material learned in class.
- Yes. They were challenging and prepared me well for the TEE.
- Yes. They were complete assignments that really made sure I learn/review the material. They were a more complete version of the reading quizzes.
- Yes. This was the most helpful tool for studying and internalizing course material for me.
- Yes. Whenever I studied for WPR's and the TEE, I checked the homeworks, as they were an accurate representation of the learning objectives.
- Yes--they provided questions that closely mimic the WPRs and problem-solving methods used to solve course questions and those that will be asked on the TEE
- YN - They were, the assignments were not too excessive and time consuming, but were challenging enough to force you to learn the material.

E. Do you believe working problems at the boards in class was effectively in helping you learn how to solve problems in the course? Why or why not?

83 Yes / 12 No / 6 Maybe

- No - I felt like it did not help because when I would get stuck on a problem there were too many questions being asked at the same time so it was difficult to get help in a timely manner.
- No - I felt the most effective learning was when the instructor was leading problems. Doing board work diluted assistance progress on learning was slowed.
- No - I love when my instructor solves the problems on board in class. It is very useful and help us understand the concepts in depth.
- No - I think "watching" an example was much more helpful.
- No - I think the teacher-led problems were the most helpful, and working problems in your notes is good to because you can have your work saved for later.
- No - I would rather watch the instructor work problems on the board and not do any myself because I can practice problems on my own outside of class.
- No - Personally, I prefer to do instructor-led problems during class to hear a deeper explanation of the problems. I prefer to do extra problems while I'm studying after class.
- No because then I would need to take a picture of the board to have in my notes rather than just writing on my own notes and not having to transfer the information.
- No, instructor lecture was more effective
- No, it was kind of a waste of time.
- Not at all. The flipped classroom was awful. Please go back to a normal classroom model.
- Yes - Absolutely! The board problems were the best practice we had in class.
- Yes - For the most part, I think the problems done in class were good. It was really nice to see the problems done first hand and then be able to put them into my notes. However, I sometimes felt like they took too long to go over.
- Yes - I actually prefer doing the problems in class as a class led by an instructor, then going back to my room to practice some on my own/do them during the homework assignments. I believe this, because there aren't many opportunities when we have the expertise of an instructor in the same room as us, who knows the topic much better than any student. However, there are endless opportunities for me to go sit in my room and complete the

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problem on my own/with a classmate. I take a lot out of learning tips/tricks/why things are in class, and this is best accomplished when instructors are completing the problem with the cadets assisting.

- Yes - I always learn best in collaborative environments, so working at the boards with my peers and my teacher being right there to help me was very beneficial.
- Yes - I believe that the board time was what you put into it. Some days I would sandbag it and then fail to grasp the concepts needed, but then the days I used the time effectively the topic seemed easy.
- Yes - I definitely think they were effective for solving problems in the course. I like how the in-class problems are usually a bit more challenging as they ensure that I am fully prepared for WPRs and homework assignments which are generally less rigorous.
- Yes - I did because I could reference them later and understand the parts of the problem solving process.
- Yes - I do because it gave me a chance to see where my gaps in learning were while the teacher could assist. However the only issue is that all work I do on the boards does not
- Yes - I do, instructor's help on board problems helped me understand the steps to the process
- Yes - I enjoyed class problems, because it allowed me to see how to effectively solve a problem before doing it on my own.
- Yes - I found working on my iPad better because I can save me work.
- Yes - I liked doing problems on my own paper/tablet since I could keep my work for future reviewing. Stuff done on the boards was usually left behind in the classroom.
- Yes - I prefer solving problems on my tablet because then I have the work saved. I really like the form of, "When you complete this problem you can leave class." It forces participation and incentivizes good work.
- Yes - I think that working through the problems on my own was good. I never used the boards but rather did the problems at my desk to have them in my notes.
- Yes - I think the board problems helped us expose what exactly we did not know
- Yes - I typically did them in my notebook only because I like having them to refer to all in one place. I used to do them on the board and take pictures, but I never look back at pictures. I always look back in my notebook and finish unfinished problems from class time.
- Yes - Not any more effective than solving problems at my desk
- Yes - Somewhat although I like working at my desk to have notes.
- Yes - They helped with the walk through.
- Yes - Very, allowed me to work through the process.
- Yes - Working on boards helped because we could work in groups and get criticism by others or the instructor.
- Yes - Working problems in class allowed me to assess my current knowledge base and ask my peers for help, so it was very beneficial.
- Yes although I wish that the instructor would work through a problem only explaining the new concepts and solution methods instead of doing all the tedious work like every calculation for simple things that we know how to do already, like area or axial stress or converting dimensions. It would save time so we can spend more time working problems on our own.
- Yes at times, but the questions became too long and disengaging
- Yes because it helped just see how to solve a practice problem and the correct process. I could also ask questions if needed and allowed me to fully understand what to do.
- Yes because it was a good time to get practice while the instructor was present to help with any questions.
- Yes because the open work allows you to have your work checked easily and the retention of material is greater when writing on the board.
- Yes because working through practice problems is the most effective way to learn the course material
- Yes but we did not get much time at all in class to work problems on the board.
- yes great and clear instruction every day was much appreciated

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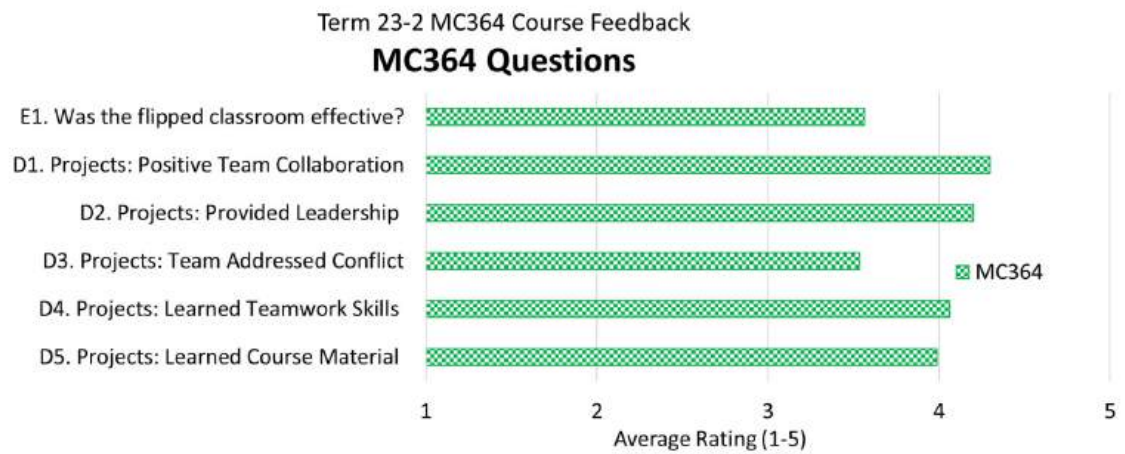
- Yes I like take boards
- Yes i liked that we could ask questions when we got stuck
- Yes it was effective because we showed up with a base of knowledge for the lesson and were able to struggle through problems with an instructor there to help us. This is better than trying problems on your own in your room.
- yes practice helped
- Yes they were helpful.
- Yes this helped me understand it.
- Yes! Practice problems are the best way for me to ingrain knowledge.
- Yes, absolutely. The style of going through problems in class resonates perfectly with my style of learning. Other math and physics-heavy classes tend to just present equations and their derivations, and this method is terrible for learning material effectively. MC364 is one of the few courses in my experience that gets learning in the classroom right, and including students throughout the solution process makes this one of the only classes that has my attention from beginning to end.
- Yes, allowed me to get direct feedback from instructor.
- Yes, as they were helpful for later reference.
- Yes, because it allowed me to address certain gaps in logic when working out a recently learned lesson and allowed the teacher to fill in those gaps with the required knowledge.
- Yes, because it allowed my instructor to comment on my work or to easily point to an area which I may need help in.
- Yes, because it forced me to get engaged.
- Yes, being able to attempt to complete problems in class while the teacher was present was helpful because any confusion or misunderstanding could be immediately clarified on the board from the teacher.
- Yes, but after we did a problem together as a class.
- Yes, but I don't like how the practice problems aren't necessary the same problems we'll see on the WPRs.
- Yes, but we often did not have enough time to do more than one problem on our own, sometimes not enough for one full problem.
- Yes, gave more ways to solve problems.
- Yes, gives us reps at the problems we wouldn't do on our own
- Yes, giving WPR and TEE tier questions in class is undoubtedly the most effective method of preparation
- Yes, having instructors there to provide immediate feedback was helpful
- Yes, I could work on my own and the teacher could provide support when I started spinning my wheels.
- Yes, I just wish we could work through one faster.
- Yes, I think these were extremely effective at figuring out where gaps in understanding were.
- Yes, it allowed us to work through what we had learned with facilitated help from our teacher.
- Yes, it gave me an opportunity to work through a problem on my own after completing an example with the class and instructor.
- Yes, it gave us time to get feedback from our instructor and practice problem solving.
- Yes, it helped to be able to see a problem successfully worked out in steps.
- Yes, it prepared us for the WPRs very well.
- Yes, most helpful for practicing concepts.
- Yes, on top of the homework, they reinforced my learning of how to use concepts and equations taught to us.
- Yes, once again, more practice problems of varying difficulty allowed better understanding of material.
- Yes, similar to the homework it provided a good environment for cadets to practice applying course concepts with other cadets and the instructor available for assistance.
- Yes, they gave us an allotted amount of time to practice the class topic while receiving instructor feedback

MC364 Course Assessment Package, AY 23

- yes, they helped me practice
- Yes, they were explained well and made sense after learning.
- Yes, they were explaining the steps necessary to address problems
- Yes, this was probably the thing that contributed most to my learning of the material. Because it forced me to work through a problem on my own and find the friction points while I was around people that I could ask about them.
- Yes, this was the best method of learning class material for me
- Yes, use those notes to assist in HW and studying
- Yes, we need that more often. It was good when it happened but it wasn't often enough
- yes, work with teacher.
- Yes, you could determine if you were capable of solving problems on your own.
- Yes. Could get immediate feedback from instructors.
- Yes. My thought process was visible and could be corrected by the instructor.
- Yes. Personally, this is my favorite thing to do in class because it allows for immediate feedback which means immediate recognition of problems and fixing of my methods.
- Yes. This helped me to get some extra time working on problems. Because of this, I was able to devote more time outside of class to other classes when I needed to.
- Yes. This was the best part of the course. Walking through board problems as a class, then moving to individual work helps tremendously. This allows cadets to go through each step of a problem then immediately attempt a similar problem.
- YN - Depended on the problems.
- YN - I think board work can help observation of the process, but having the work in your notes is better for the students to look back on later.
- YN - NA, typically did problems with instructor.
- YN - These were fine. No complaints or praises.
- YN - This was less effective for me because I prefer to have the problems written down. However, this was helpful in that it allowed my instructor to help me through problems more effectively
- YN - Yes and no, it was good practicing problems in class but it was tough to make the time crunch in some of the longer and more complex problems.

MC364 Course Assessment Package, AY 23

6. All Course Numeric Feedback





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TAB D

AY2022 COURSE ASSESSMENT

MC364 Course Assessment Summary, AY 22

<u>COURSE ADMINISTRATION</u> No changes recommended for course description, objectives, or textbook.	<u>RESOURCES</u> - Laboratory staff and equipment - Lab Space – (Tension and Torsion Machines)																								
<u>COURSE OBJECTIVES ASSESSMENT</u> <table><tr><th>Course Objective</th><th>Assessment AY22</th><th>Assessment AY21</th></tr><tr><td>a. Calculate the internal forces, internal stresses, and deformations...</td><td>4</td><td>4</td></tr><tr><td>b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure...</td><td>3</td><td>3</td></tr><tr><td>c. Apply appropriate theories of failure to analyze and design...</td><td>2*</td><td>3</td></tr><tr><td>d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.</td><td>3</td><td>1</td></tr><tr><td>e. Given a state of stress at a point, determine the principal stresses, the maximum in-plane shear stress, the angle to the principal plane, and the state of stress on any plane.</td><td>4</td><td>4</td></tr><tr><td>f. Apply deformation—strain, strain—stress, and stress—force relationships to analyze and design structural members and machine components.</td><td>3</td><td>3</td></tr><tr><td>g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.</td><td>4</td><td>4</td></tr></table> c) Theories of Failure – Only assessed on HW7 and the TEE. Improved wording of TEE question. d) Strain Transformation – will improve prelab questions for TWPV Lab to help understand the concept of measuring strain - The goal for next year is improve the structure of the flipped classroom and the balance between reading quizzes and homeworks/instructor led board work and student led boardwork . - Time Survey is at 94 mins/graded event. Homeworks took on average 118 minutes.		Course Objective	Assessment AY22	Assessment AY21	a. Calculate the internal forces, internal stresses, and deformations...	4	4	b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure...	3	3	c. Apply appropriate theories of failure to analyze and design...	2*	3	d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.	3	1	e. Given a state of stress at a point, determine the principal stresses, the maximum in-plane shear stress, the angle to the principal plane, and the state of stress on any plane.	4	4	f. Apply deformation—strain, strain—stress, and stress—force relationships to analyze and design structural members and machine components.	3	3	g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.	4	4
Course Objective	Assessment AY22	Assessment AY21																							
a. Calculate the internal forces, internal stresses, and deformations...	4	4																							
b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure...	3	3																							
c. Apply appropriate theories of failure to analyze and design...	2*	3																							
d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.	3	1																							
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g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.	4	4																							
<u>CHANGES FROM PREVIOUS YEAR</u> - Several changes made this year: - Added Lessons from MC300 (Sustain) - Added Beam EDP (Sustain) - Transition RQ to HWs (Sustain) - Changed Textbook (5ed) (Sustain) - Added Strain Rosette to TWPV Lab (Sustain)	<u>RECOMMENDED CHANGES</u> - Update the Uniaxial Tension Lab to include Compression - Change Lesson Content and Flipped Classroom structure – release basic concept video, scope down reading quizzes, develop instructor content prior to sending cadets to the board, and develop focused cadet board problems. - Change lab submission structure to improve technical communication skills. Each group writes an abstract and a results and discussion section to answer the provided research questions.																								
<u>DISCUSSION POINTS:</u> <i>Flipped Classroom</i> – should we maintain this classroom format? Should we restructure the format to have a more formal instructor led portion and more targeted problems? <i>STAP</i> – Need to make sure that all CE and ME Yearlings are enrolled in MC364 during the Spring. Late major adds should take MC300 and MC364 currently and MC300 failures may progress into MC364 if they would like to remain a CE or ME major. <i>Topics in MC364</i> – Do we need Statically Indeterminate Axially loaded members? Do we need to cover rotating rigid statically indeterminate axially loaded members? We would like to add a deflection lesson covering superposition.																									

MC364 Course Assessment Package, AY 22

I Course Administration

A. Redbook Description

The screenshot below was taken from the Redbook on 19 MAY 22.

MC364 (Version: 2020 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
MC364	MECHANICS OF MATERIALS	2020	1	Civil and Mechanical Engineering	3.5 (BS=0.0, ET=3.5, MA=0.0)

SCOPE

This course studies the behavior of a variety of materials under normal, shear, torsional, bending and combined loads. The concepts of stress, strain, failure theory and failure mechanisms are explored. The loading, geometry, functional environment and material properties of machine or structural parts are used to relate the forces applied to a body to the resulting internal forces and deformations so that performance can be evaluated. Practical applications involving the design and adequacy of mechanical and structural elements under various loading and environmental conditions are emphasized. The course includes several laboratory experiences that require students to develop and conduct experiments, analyze and interpret data, and use engineering judgement to draw conclusions.

LESSONS: 25 @ 75 min (1.560 Att/wk) **LABS:** 10 @ 75 min

SPECIAL REQUIREMENTS:

None

MC364 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
CO REQUISITE					
	MA205	2017	1	1	Y
	MA255	2016	2	2	Y
	MA366	2018	2	3	Y
DISQUALIFIER					
	CE364	2006	1	1	Y
PRE REQUISITE					
	MC300	2017	1	1	Y

1. Terms Offered and Enrollments:

Table I.A.1

AY 16	Term 16-1	109		AY20	Term 20-1	
	Term 16-2				Term 20-2	128
AY 17	Term 17-1	169		AY21	Term 21-1	
	Term 17-2	149			Term 21-2	118
AY18	Term 18-1			AY22	Term 22-1	
	Term 18-2	112			Term 22-2	145
AY19	Term 19-1				Term 22-7	4**
	Term 19-2	144		AY23*	Term 23-1	
					Term 23-2	152

* forecast in Redbook as of 19 MAY 22.

** 1 CE and 3 ME cadets enrolled in STAP. 2 cadets failed MC300 and 2 were not enrolled in MC364.

2. Based on projected enrollments, the following issues may arise for future course offerings.

The ME Division has discussed making MC364 a prerequisite for ME403. This will help present cadets who need STAP. It is also recommended that MC300 failures be allowed to continue with MC364 if their schedule allows and they would like to continue with engineering as their major.

MC364 Course Assessment Package, AY 22

3. Proposed Changes to Redbook Description.

None

B. Course Objectives

1. Current Course Objectives.

- a. Calculate the internal forces, internal stresses, and deformations of axially loaded members, circular members in torsion, thin-walled pressure vessels, prismatic beams in bending, columns, and members subjected to combined loading and/or thermal effects.
- b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure subjected to loading and/or thermal effects.
- c. Apply appropriate theories of failure to analyze and design thin-walled pressure vessels and members subjected to loading and/or thermal effects.
- d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.
- e. Given a state of stress at a point, determine the principal stresses, the maximum in-plane shear stress, the angle to the principal plane, and the state of stress on any plane.
- f. Apply deformation—strain, strain—stress, and stress—force relationships to analyze and design structural members and machine components.
- g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.

2. Proposed Changes to Course Objectives. None

C. Textbook(s)

1. The following text(s) were used this year:

Textbook Changes: Mechanics of Materials, 5th Edition by Timothy A. Philpot and Jeffery S. Thomas. ISBN: 978-1-119-60301-6

2. Make following adjustments to the text requirement for next year:

No changes to the text.

D. Follow-up from Previous Course Assessment.

The course objective (D) for strain data was changed during the Course Assessment for MC364 during AY21. The course objectives are well suited for the current form of the course. They accurately describe the course and are achievable for the cadets.

MC364 Course Assessment Package, AY 22

II Course Content

A. Outline of Lessons and Labs

Table II.A.1

Lesson	Title/Topic	Supported Course Objectives						
		a	b	c	d	e	f	g
1	Course Intro & MC300 Review	a						
2	Uniaxial Tension Lab	a					f	g
3	Axial Deformation	a					f	
4	Buckling II	a					f	
5	Statically Indeterminate Axially Loaded Members I	a	b				f	
6	Statically Indeterminate Axially Loaded Members II	a	b				f	
7	Thermal Effects on Axially Loaded Members	a	b				f	
8	Torsion Lab	a					f	g
9	Introduction to Torsion	a					f	
10	Solving Torsion Problems	a					f	
11	WPR 1 (Axial and Torsion)	a	b				f	
12	Beam Lab I	a					f	g
13	Flexural Normal Stress	a					f	
14	Flexural Shear Stress	a					f	
15	Beam Lab I	a					f	g
16	Deflections of Beams	a					f	
17	Statically Indeterminate Bending Members I	a	b				f	
18	Statically Indeterminate Bending Members II	a	b				f	
19	Beam Analysis	a					f	
20	Beam Design	a					f	g
21	WPR2 (Flexural)	a	b				f	
22	Stress Elements	a					f	
23	Stress Transformation	a					f	
24	Principal Stresses and Max Shear Stress	a				e	f	
25	Mohr's Circle	a				e	f	
26	Strain Transformation	a			d	e	f	
27	TWPV (Strain Gage Mounting)	a			d	e	f	g
28	TWPV Lab	a			d	e	f	g
29	Combined Loading (Axial and Torsion)	a			d	e	f	
30	Principal Stresses in Flexural Members	a			d	e	f	
31	WPR 3 (Stress & Strain Transformation and TWPV)	a			d	e	f	
32	General Combined Loading	a			d	e	f	
33	Theories of Failure - Ductile Material	a		c		e	f	
34	Theories of Failure - Brittle Material	a		c		e	f	
35	Course Review and Wrap-up	a	b	c	d	e	f	

MC364 Course Assessment Package, AY 22

Table II.A.2

Lesson	Title/Topic	Supported Course Objectives						
		a	b	c	d	e	f	g
1	Course Intro & MC300 Review	a						
2	Uniaxial Tension & Compression Lab	a					f	g
3	Axial Deformation	a					f	
4	Buckling II	a					f	
5	Statically Indeterminate Axially Loaded Members I	a	b				f	
6	Statically Indeterminate Axially Loaded Members II	a	b				f	
7	Thermal Effects on Axially Loaded Members	a	b				f	
8	Torsion Lab	a					f	g
9	Introduction to Torsion	a					f	
10	Solving Torsion Problems	a					f	
11	WPR 1 (Axial and Torsion)	a	b				f	
12	Beam Lab I	a					f	g
13	Flexural Normal Stress	a					f	
14	Flexural Shear Stress	a					f	
15	Beam Lab I	a					f	g
16	Deflections of Beams	a					f	
17	Statically Indeterminate Bending Members I	a	b				f	
18	Statically Indeterminate Bending Members II	a	b				f	
19	Beam Analysis	a					f	
20	Beam Design	a					f	g
21	WPR2 (Flexural)	a	b				f	
22	Stress Elements	a					f	
23	Stress Transformation	a					f	
24	Principal Stresses and Max Shear Stress	a				e	f	
25	Mohr's Circle	a				e	f	
26	Strain Transformation	a			d	e	f	
27	TWPV (Strain Gage Mounting)	a			d	e	f	g
28	TWPV Lab	a			d	e	f	g
29	Combined Loading (Axial and Torsion)	a			d	e	f	
30	Principal Stresses in Flexural Members	a			d	e	f	
31	WPR 3 (Stress & Strain Transformation and TWPV)	a			d	e	f	
32	General Combined Loading	a			d	e	f	
33	Theories of Failure - Ductile Material	a		c		e	f	
34	Theories of Failure - Brittle Material	a		c		e	f	
35	Course Review and Wrap-up	a	b	c	d	e	f	

MC364 Course Assessment Package, AY 22

Field Trips:

No field trips in this course. All cadets were put on trip section for Project's Day.

B. Proposed changes to Lessons and Labs.

1. Uniaxial Tension and Compression Lab (2 LSNs): revamp the current laboratory to test axial tension and compression samples and compare their performance.
2. Restructure flipped classroom where the instructor leads the first 20 minutes of class to ensure all cadets are prepared to actively participate. Develop targeted problems for each lesson to test critical skills. Many of the textbook problems are all encompassing. Cadets need practice with the fundamentals.
3. Update the current lab submission structure to improve technical communication skills. The current structure is an ExSum with enclosures for data and responses to questions. Instructors have found the ExSums are cookie cutter, and they are not providing conclusions or adequate answers to the questions. In the new format, each group will write an abstract and a brief results and discussion section where they would be required to provide answer and justifications for the provided research questions.

C. Graded Requirements

1. Summary of graded requirements for current semester.

Table II.C.1

Requirement	Qty	Points	Total	% of Total
Laboratory Assignments	4	100	400	20%
Engineering Design Project	1	100	100	10%
Reading Quizzes	16	20	320	16%*
Homeworks	7	20	140	7%*
WPRs**	3	200	600	30%
TEE**	1	500	500	25%
		TOTAL=	2000	100%

*The lowest three homeworks and reading quizzes were dropped

** Individual Work - 55% Individual Assessment (WPRs & TEE)

2. Proposed changes to graded requirements:

Table II.C.2

Requirement	Qty	Points	Total	% of Total
Laboratory Assignments	4	100	400	20%
Engineering Design Project	1	100	100	10%
Reading Quizzes*	18	15	225	11.3%
Homeworks	7	25	175	8.8%
WPRs**	3	200	600	30%
TEE**	1	500	500	25%
		TOTAL=	2000	100%

*The lowest three reading quizzes would be dropped

** Individual Work - 55% Individual Assessment (WPRs & TEE)

- a. Homeworks will increase in point value to 25 points each and will be separate from the reading quizzes.
- b. Reading Quizzes will decrease in point value to 15 points each. Only the lowest reading quiz grades will be dropped.

MC364 Course Assessment Package, AY 22

III Course Assessment

A. Course Director Assessment of Course Objectives.

1= Objective Not Met. Objective clearly not met, most (75%) of the students did not achieve it.

2= Objective Marginally Met. Objective met by roughly half the students or minimally by most of them.

3= Objective Satisfactorily Met. Objective clearly met by a solid majority (70%) of the students.

4= Objective Solidly Met. Objective clearly met by the vast majority (90%) of students.

5= Objective Clearly Met. All students have achieved the objective and can be expected to demonstrate it.

The definition of “meeting a course objective” is achieving a “C” level (70%) on the task.

Table III.A

Course Objective	Current Assessment (AY 22)	Previous Assessment (AY 21)	How Evaluated and Remarks
a. Calculate the internal forces, internal stresses, and deformations...	4	4	WPR1: 84% Avg. & 86% > C WPR2: 95% Avg. & 100% > C WPR3: 84% Avg. & 80% > C TEE: 86% Avg. & 90% > C
b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure...	3	3	WPR1: 85% Avg. & 94% > C WPR2: 88% Avg. & 83% > C TEE: 77% Avg. & 74% > C
c. Apply appropriate theories of failure to analyze and design...	2*	3	HW7: 91% Avg. & 97% > C TEE: 79% Avg. & 61% > C
d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.	3	1	Lab4: 93% Avg. & 100% > C HW6: 88% Avg. & 98% > C WPR3: 91% Avg. & 86% > C TEE: 78% Avg. & 61% > C
e. Given a state of stress at a point, determine the principal stresses, the maximum in-plane shear stress, the angle to the principal plane, and the state of stress on any plane.	4	4	WPR3: 90% Avg. & 95% > C TEE: 90% Avg. & 99% > C
f. Apply deformation—strain, strain—stress, and stress—force relationships to analyze and design structural members and machine components.	3	3	WPR1: 81% Avg. & 80% > C WPR2: 84% Avg. & 86% > C TEE: 89% Avg. & 95% > C
g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.	4	4	LAB1: 90% Avg. & 97% > C LAB2: 93% Avg. & 100% > C LAB3: 90% Avg. & 99% > C LAB4: 93% Avg. & 100% > C

**Failure Theories were only assessed on one HW and the TEE. The TEE question covers both TWPV and Strain Transformation. Separate performance will be tracked next AY.*

MC364 Course Assessment Package, AY 22

B. Student (Program) Outcomes Directly Supported By This Course

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
 - **None**
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors;
 - **None**
3. an ability to communicate effectively with a range of audiences;
 - **Laboratory Report (TWPV) (bin, avg)**
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts;
 - **None**
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives;
 - **Laboratory Groups (Survey)**
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;
 - **None**
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
 - **None**

C. Assessment of Embedded Indicators.

Table III.C.1

Course Indicator	No. of CE Cadets	Indicator Points	Average (%)	High (%)	Low (%)
MC364 Lab4	36	50	93	100	80

Table III.C.2

Course Indicator	No. of CE Cadets	D or F	C	B	A	Indicator Points	Avg (%)
MC364 Lab4	36	0	0	10	26	50	93

Table III.C.3

Student self-assessment on 1-5 Likert Scale: (1) strongly disagree to (5) strongly agree.	Avg. Response
1.) My team worked in a positive collaborative manner	4.29
2.) I provided leadership in completing team assignments.	4.12
3.) My team had to address conflict to complete team assignments.	2.95
4.) Teamwork activities were effective in learning teamwork skills	4.15
5.) Teamwork activities were effective in learning course material	4.08

MC364 Course Assessment Package, AY 22

D. Assessment of Changes Made this Semester.

1. Change Textbook. The textbook was changed for the second year in a row to the 5th edition of Philpot. After the death of Prof. Timothy Philpot, Wiley wanted to add an additional author to the text. Minimal changes were made. Cadets can still access the MecMovies via the Ebook. All cadets should be encouraged to purchase and register their Ebook.

2. Transitioning Reading Quizzes to Homework Assignments. The previous iteration of MC364 included only reading quizzes as assessments outside of major graded events. The only assignments that cadets were required to complete outside of class were the group laboratory reports. Feedback from cadets and instructors was that include practice problems in the flipped classroom format were not sufficient for cadets to grasp the material before major graded events. It was recommended to add short homeworks in place of reading quizzes. Instead of changing all reading quizzes 7 of the 23 reading quizzes were changed to homeworks for 16 RQs and 7 HWs. The homeworks were a great opportunity to apply their knowledge to solve complex problems, however the took significant additional effort compared to the reading quizzes as noted by the increase in the time survey from 74 minutes in AY21 to 96 minutes in AY22. The homeworks all consisted of three complete problems which on average took cadets 118 minutes to complete; between 100 minutes – 150 minutes. Anecdotally, the cadets seemed more comfortable solving problems on the WPRs as a direct result to the homework assessment. In the course end feedback survey 119 out of 127 cadets stated that the homework assignments were effective in helping them learn the course content. The homework assignments will be a sustain, however the point value will increase to place more weight on them.

3. Using a Strain Rosette for the TWPV Lab. Strain gage rosettes were purchased from the strain gage provider, Vishay with a unit cost of \$33.11 per rosette. These strain gages were used for the thin-walled pressure vessel lab. The cadets were able to glue these strain gages to a soda can to determine the entire state of stress on the can. Previously, uniaxial strain gages were used, but there was no way to confirm if the strain gage was oriented in the hoop direction. With this new set up. It reinforces the concepts of strain transformation and the importance of properly installing instrumentation. Although, this did not have a visible impact on the cadets' ability to achieve the course objective, it is a sustain. In the next iteration of the course, the lab requirements handout will be clarified to unsure better application of the theory to solve the research questions.

4. Add a Fifth Laboratory Exercise. It was decided not to add a fifth laboratory exercise, but instead incorporate the MC300 Beam EDP. This beam EDP is a great opportunity for cadets to develop a design, predict how it will perform, and then evaluate its performance based on an experimental test. It reinforces the flexural analysis concepts that are introduced in MC364 and ties in very well with the conclusions drawn from the Beam Lab. It was discussed to return the Tower of Torque lab back into MC364, however this lab is not a very well packaged and needs additional tweaks before it is ready for classroom implementation.

5. Incorporate Changes from MC300. MC300 recently underwent a significant change by introducing a more statics-based approach. Vectors were introduced to the course and most discussion on mechanics of materials, specifically flexural beam analysis was introduced solely in MC364. Shear and Moment diagrams however is still extensively covered in MC300. Three additional beam lessons were added to MC364. The first was Beam Lab 1 which is the inquiry-based learning activity where students explore the distribution of strains and stresses along the height of a beam. This is a great activity to introduce beam analysis. Two review lessons were added to the end of the beam block, covering Beam Analysis and Beam Design. These lessons helped tie together the concepts covered in the beam block and provided cadets with additional opportunities to practice solving problems. For the most part, student performance on the beam concepts was similar to previous semester. There is still a lack of understanding on Beam Deflection (previously covered in MC300) and first moment of outward area (Q), which is needed for solving flexural shear stress. In the next delivery of MC364, additional opportunities to cover beam deflection and Q will be explored.

MC364 Course Assessment Package, AY 22

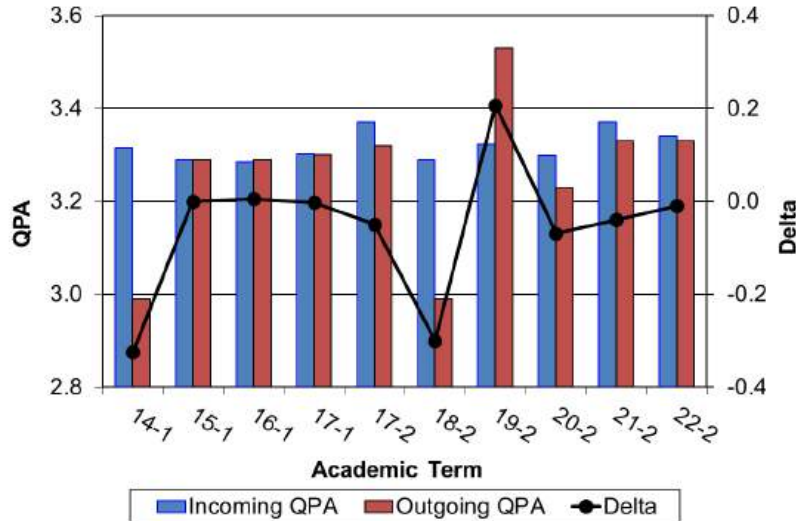
E. Assessment of Resources and Impacts.

1. *Laboratory Technician and Resource Support.* A quarter of cadets grade in this course is the laboratory and EDP(500/2000). It is critical that the laboratory technician support the setup and execution of all laboratory activities. Many of the labs require set up in multiple classrooms. It is important that there is enough equipment for each classroom. It is also important that no other courses are using the classrooms in between MC364 sections. Otherwise, the labs must be torn down and reset up.

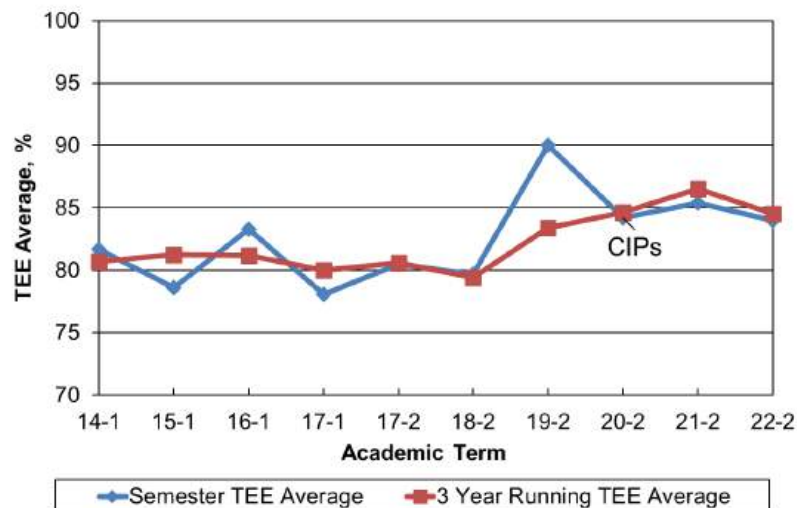
2. *Equipment.* Much of the equipment for MC364 is aging and has not been serviced in several years. The tension machines, torsion machines, aluminum beam specimens, and strain gage boxes are all experiencing issues which have had significant impacts on the lab delivery. These machines must be checked each semester.

3. *Material.* Raw material is needed for many of the laboratory and EDP executions. For the uniaxial tension laboratory for MC300 and MC364, needs samples of steel, aluminum and cast iron. For the torsion laboratory, we need samples of aluminum. For the Beam EDP, we need balsa wood (36"x3"x1/4" & 36"x3"x1/8"). For the TWPV Lab, we need Coca Cola Cans and hot dogs.

F. Course QPA Graph



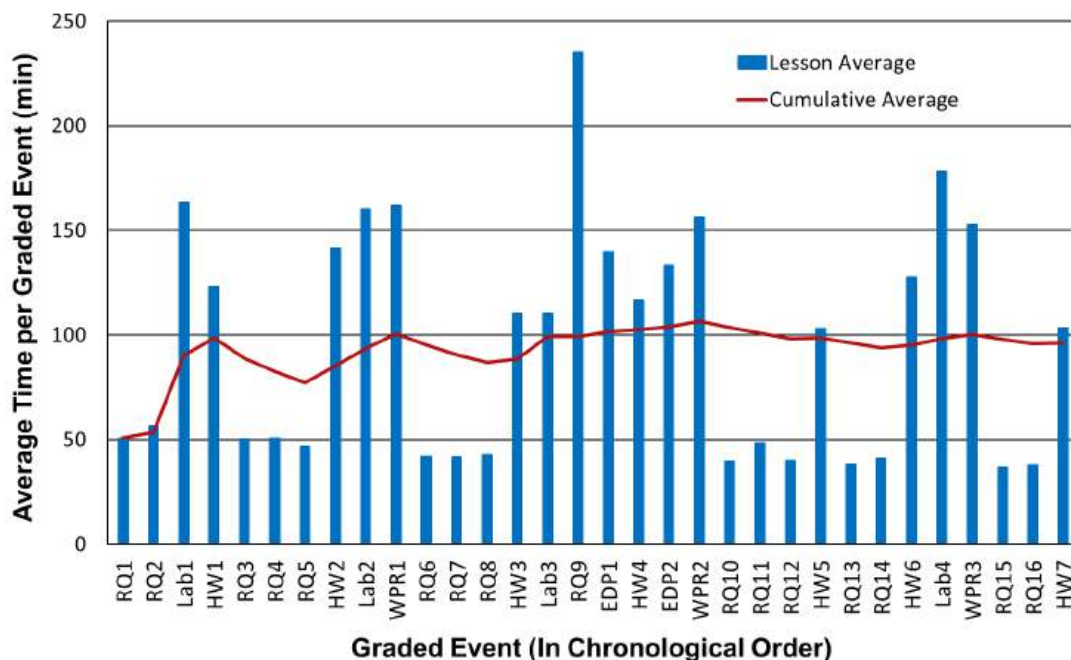
G. TEE Grades Graph



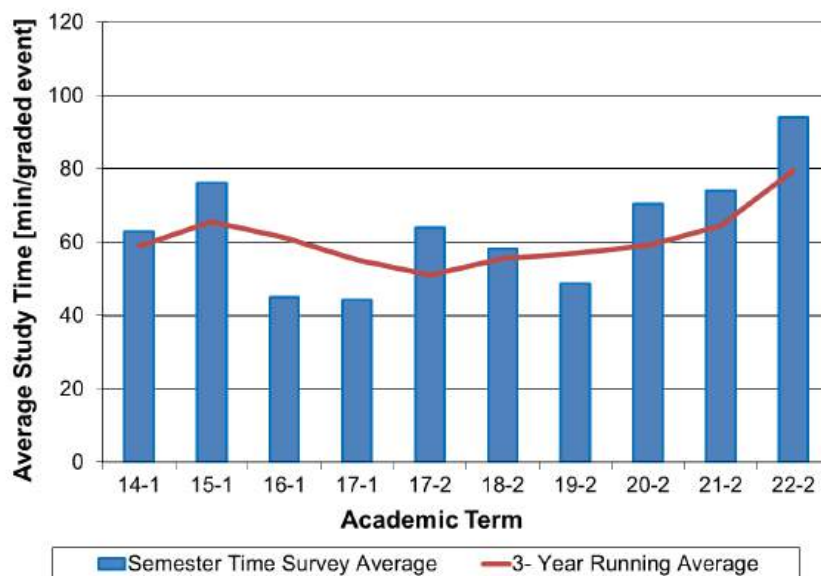
MC364 Course Assessment Package, AY 22

H. Course Time Study Graph

1. Current Semester time survey data. (AVG: 94 minutes/graded event)



2. Historical Time Survey Data



3. Any comments on time survey data if needed.

The time survey in MC364 significantly increased this semester to an average of 94 minutes per graded event compared to 74 minutes per graded event in AY21. The increase in time spent may be attributed to two factors. First, seven homework assignments were added to the flipped classroom format, so in addition to watching lesson videos the cadets were expected to complete additional out of class assignments. The cadets spent on average 44 minutes preparing for lessons with only reading quizzes. The cadets spent 118 minutes on homework assignments. Second, the beam lab was expanded. This assignment had the greatest time survey average of 235 minutes.

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I. Course End Feedback.

Question	Question Type
I provided leadership in completing team assignments	Likert
My team had to address conflict to complete team assignments	Likert
My team worked in a positive collaborative manner	Likert
Teamwork activities were effective in learning course material	Likert
Teamwork activities were effective in learning teamwork skills	Likert
Do you believe the flipped class format was effective in helping you learn the course content?	Likert
Do you believe the reading quizzes effectively assessed your learning of the course content? Why or why not?	Open
Do you believe the homework assignments were effective in helping you learn the course content? Why or why not?	Open
Do you believe working problems at the boards was effective in helping you learn how to solve problems in the course? Why or why not?	Open

IV. Program Director's Current Areas of Emphasis

A. Course Connections.

MC364 has several direct connections to MC300, Math courses, and Physics courses. We also heavily discuss the importance of mechanics of materials for follow on courses such as CE403, ME403, CE404, CE483, and MC380 Connections are made primarily in reading quizzes, out-of-class assignments and in-class instruction including board notes and PowerPoint slides. The primary topics that are connected to previous lessons information is detailed below.

1. Lesson 1, Reading Quiz 1 & 2 – Lesson 1 focuses on MC300 review and the elements that are critical to success in MC364 including solving for internal forces in structures especially trusses and frames, solving for stress and strain in axial members and review of the stress-strain curves, and shear and moment diagrams. The review hand out below is provided to the cadets to help focus their review. They are also given six practice problems which they are expected to be able to complete (these problems have no credit associated with them. The reading quiz on Lesson 2 assesses their basic knowledge on truss analysis (drawing FBDs and solving for internal forces) and the stress-strain diagram in preparation for the Axial Stress Lab on Lesson 2.

AY 22-2, MC364, MC300 Review

Trusses: structure to support loads composed entirely of two-force members

Truss Road Map

1. Identify members of interest. Zero-force members?
2. Solve for external reactions if necessary, from FBD of entire truss
3. Make appropriate cuts
 - a. Isolate a JOINT, max 2 members cut
 - b. Isolate a SECTION, max 3 members cut
4. Solve for internal forces with 2D equilibrium equations
5. Ans – MUD+P, member forces are (T) or (C)

Frames: structure to support or transmit loads composed of at least one multi-force member

Frame Road Map

1. Identify members and pins of interest. Two-force members?
2. Solve for external reactions if necessary, from FBD of entire frame
3. "Pull the pins" to blow up the structure
 - a. Draw FBDs of individual members
 - b. Apply Newton's 3rd Law to pin reactions
4. Solve for pin reactions with 2D equilibrium equations
5. Ans – MUD+P, ID member which forces act upon

Stress, Strain and Reformation (Chapter 1-3)

Normal stress – axial force divided by perpendicular cross-sectional area

$$\sigma = \frac{F}{A}$$

Shear stress – shear force divided by parallel cross-sectional area

$$\tau = \frac{F}{A}$$

Deformation – change in dimensions of a member due to applied load or temperature change

$$\delta = \epsilon \cdot L$$

Normal strain – deformation divided by original length

$$\epsilon = \frac{\delta}{L}$$

Hooke's Law – Elastic modulus equals stress divided by strain when a material is elastic

$$E = \frac{\sigma}{\epsilon}$$

Shear and Moment Diagrams (Section 7.3)

Rules for V&M Diagrams

1. Shear diagram is discontinuous @ concentrated loads
2. Change in Shear = area under distributed load curve
3. Slope of Shear = magnitude of distributed load
4. Moment diagram is discontinuous @ concentrated moment
5. Change in Moment = area under shear diagram
6. Slope of Moment = magnitude of shear diagram

General Procedure for Construction of V&M

1. Complete the load diagram
2. Construct Shear diagram – must start and end at zero
3. Locate key pts on Shear diagram – where shear is zero, max and min points
4. Construct Moment diagram – must start and end at zero

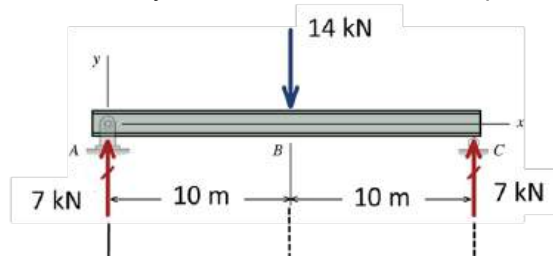
2. Lesson 2: Axial Stress Lab – Lesson 2 is the axial stress lab. This lab is set up almost identical to the axial stress lab in MC300, except the cadets are testing aluminum. This lab provides the cadets with the opportunity to reinforce their knowledge of how materials behave under tensile loading. It is not quite an exact repeat, as it is a good introduction on how to write an Executive Summary instead of an extensive laboratory report.

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3. Lesson 3: Axial Deformation – The axial deformation lesson directly builds on the axial stress and strain lessons from MC300. The axial deformation equation is derived based on the stress strain relationship: Hooke's Law.

4. Lesson 4: Buckling II – The buckling lesson in MC364 directly builds on the buckling lesson in MC300. The lesson investigates the effects of different boundary conditions and bracing on the buckling capacity of members.

5. Lesson 12: Beam Lab 1 and Reading Quiz 6 – The first lesson in the beam block is critical because there is no longer a shear and moment diagram review lesson in MC364, so cadets must know how to create shear and moment diagrams to complete Beam Lab 1. Before the start of the lab, the cadets complete Reading Quiz 6 where they need to draw a basic, complete shear and moment diagram.



6. Lesson 16: Beam Deflection by Integration – The beam deflection by integration class ties directly into calculus. It is discussed how the relationship between shear, moment, curvature, slope, and deflection can be related by integration. We also extensively discuss the importance of boundary conditions and how to solve for constants of integration based on the problem statement.

7. Homeworks and WPRs – The homeworks and WPRs included review problems to reinforce basic concepts from previous lessons and courses such as calculus, truss and frame analysis, axial stress and strain, and shear and moment diagrams.

8. Connections to Advanced Courses – Every lesson has an application which applies to advanced Mechanical or Civil engineering courses such as CE403, ME403, CE404, CE483, or MC380 or other courses. It is critical to introduce the application of each of the concepts being introduced.

B. Demonstrations and Hands-On Learning.

Demonstrations and hands-on activities are extensively used throughout MC364. MC364 is one of the most conceptually challenging courses in the engineering curriculum as you begin to discuss material behavior under different combinations of loading. The idea of a stress element (infinitesimally small point) and its performance under a combined state of stress is difficult for cadets to comprehend. It is critical that we have visualizations that help simplify complex concepts. A complete list of hands-on demonstration, supplies and activities used in MC364 is available on Castle. These demonstrations include foam beams, tension and torsion machines, pool noodles, aluminum beams, stress elements, chalkboard compass and protractor, thin-walled pressure vessels, and a tower of torque demonstration.

C. Student Outcome Embedded Indicators.

The MC364 Labs are a great embedded indicator to assess Communication (#3) and Team (#5). The format of the laboratory reports is a short one-two page executive summary, where cadets must practice the ability to concisely communicate the results of their laboratory to their audience. This also for four labs to be conducted and for continuous feedback to be provided to the cadets to improve their technical writing skills. The laboratory submissions also include additional enclosure requirements to evaluate the cadet's ability to apply the concepts they are learning to the lab product. Having teams of three cadets allows for all cadets to contribute to the success of the team. The cadet groups do not change for the entire semester.

MC364 Course Assessment Package, AY 22

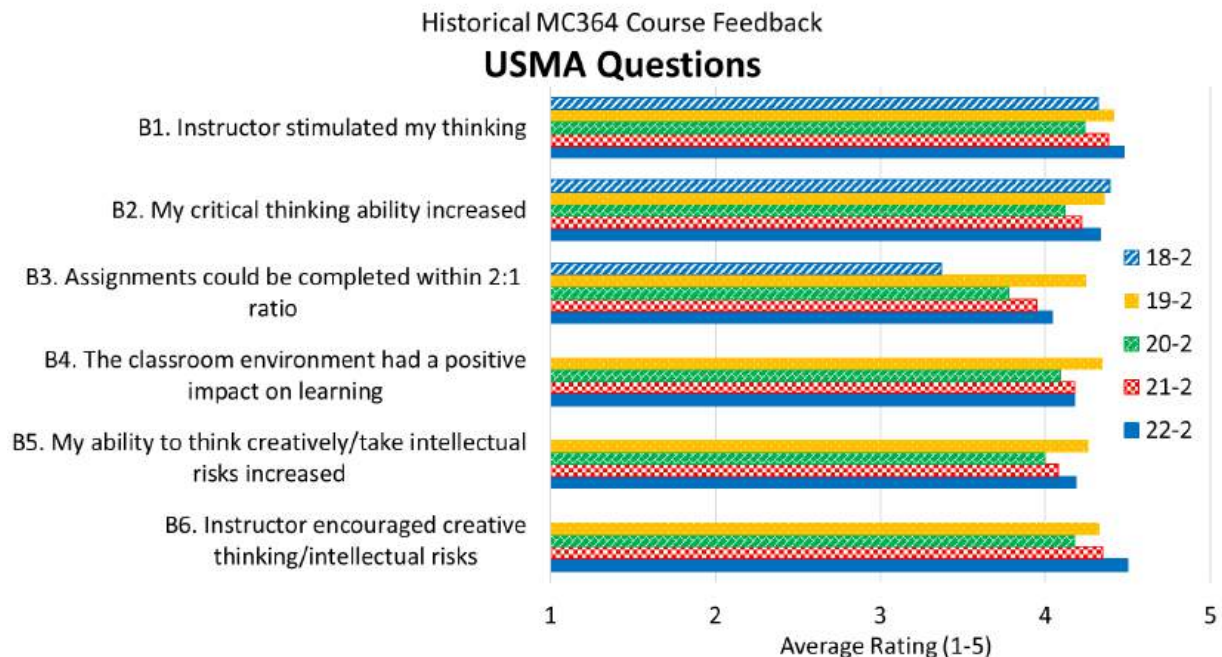
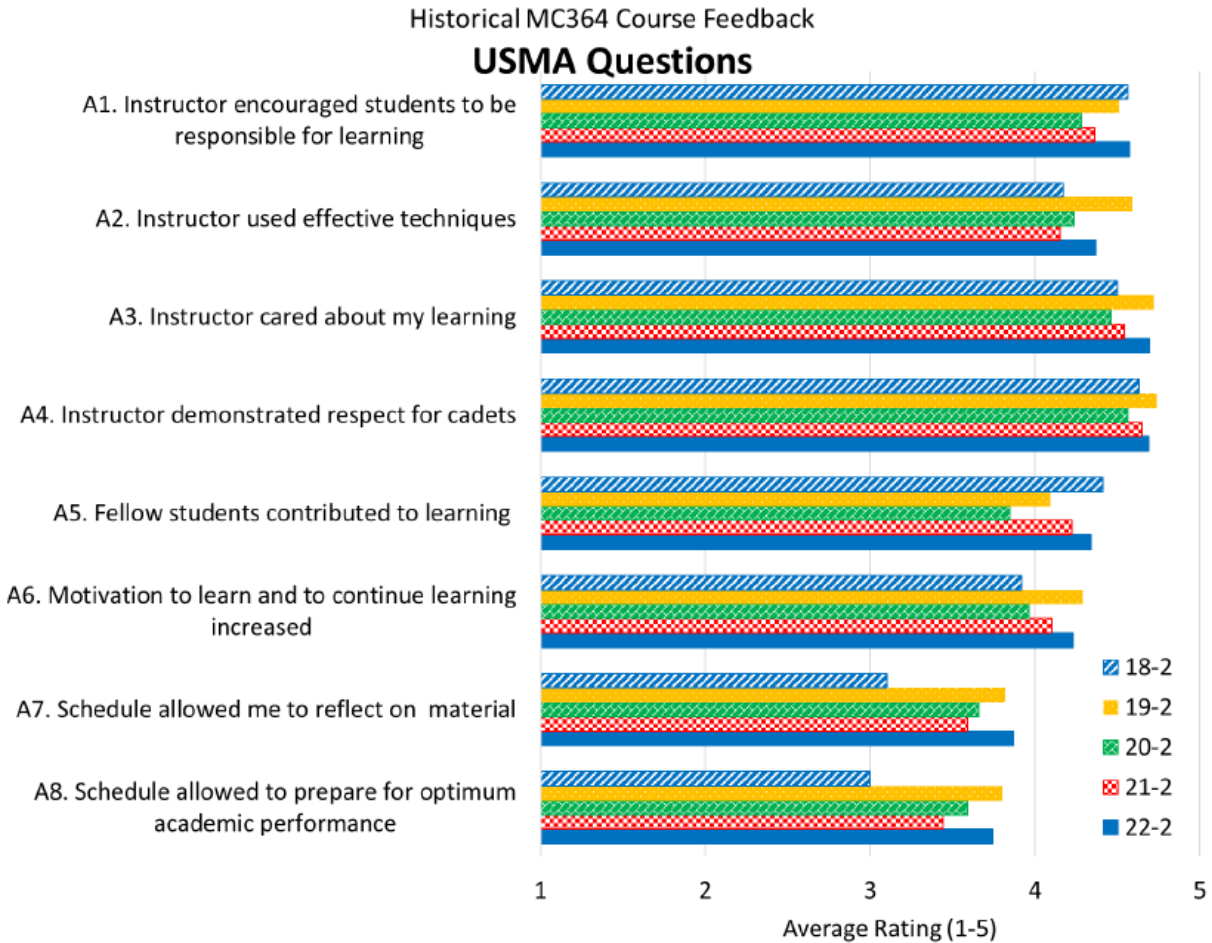
Enclosures

- 1. Historical Course Feedback on USMA Questions**
- 2. Term 22-2 Course Feedback on USMA Questions**
- 3. Historical Course Feedback on C&ME Questions**
- 4. Term 22-2 Course Feedback on C&ME Questions**
- 5. Course Narrative Feedback**
- 6. Course Numeric Feedback**
- 7. Other Feedback**

MC364 Course Assessment Package, AY 22

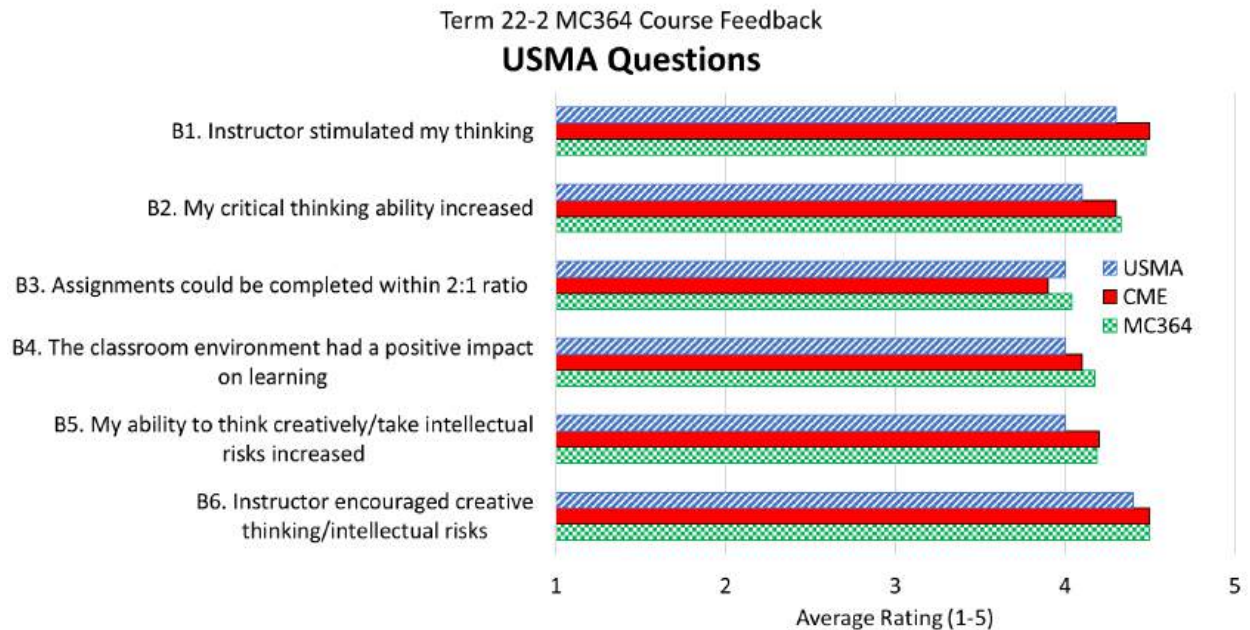
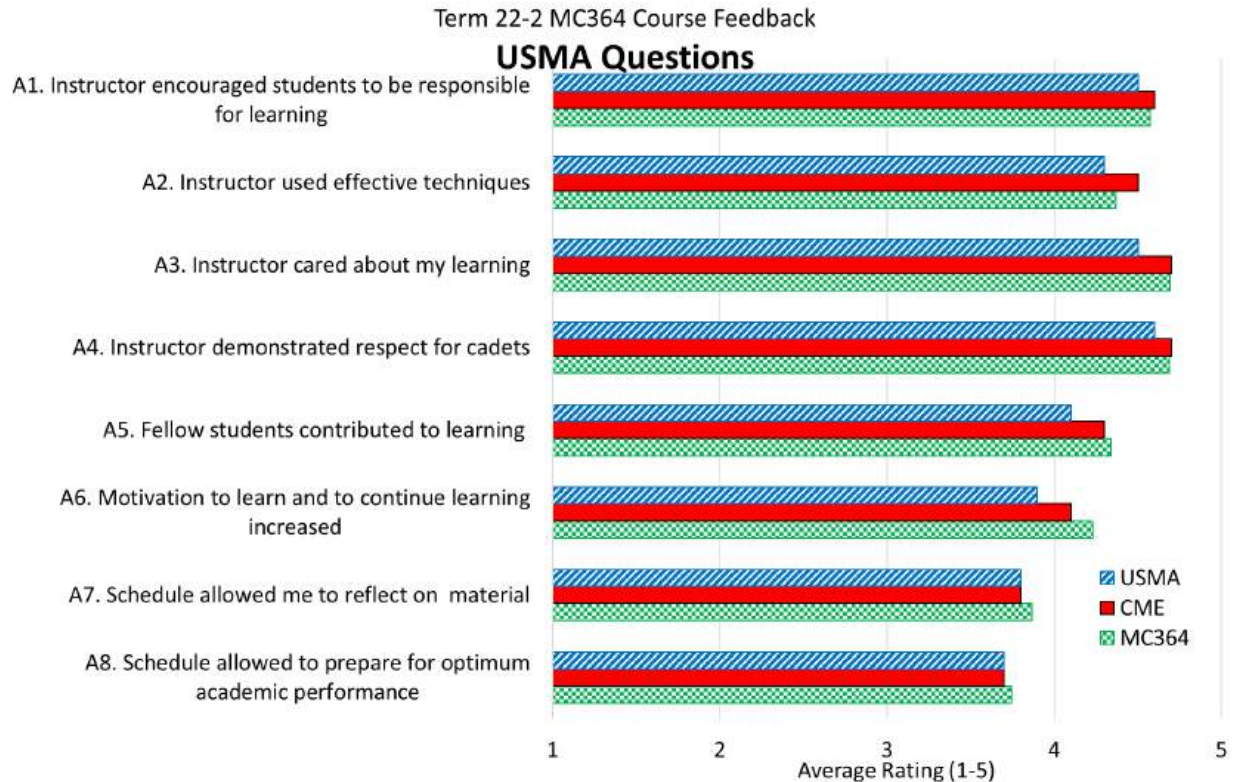
ENCLOSURES

1. Historical Course Feedback on USMA Questions



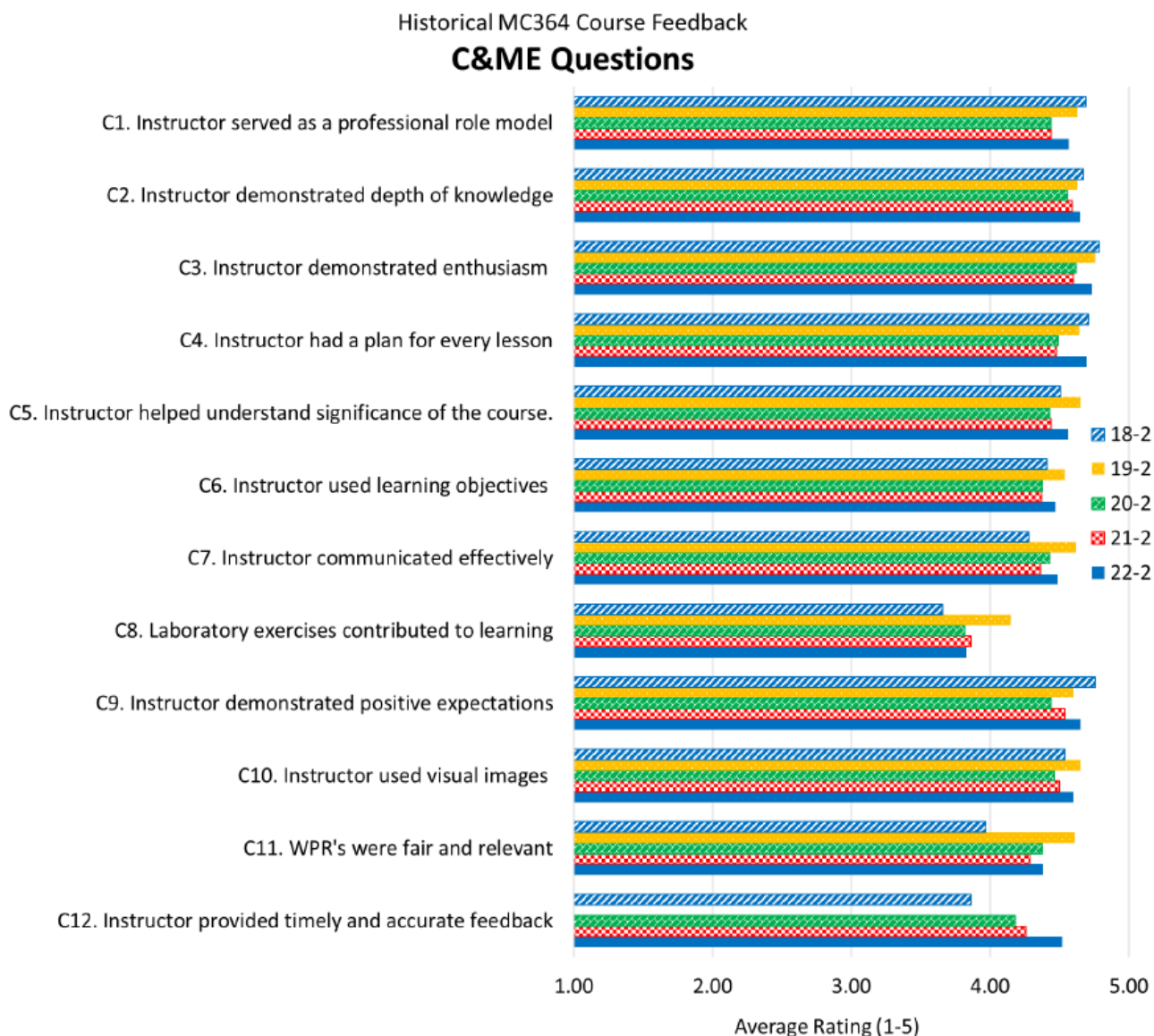
MC364 Course Assessment Package, AY 22

2. Term 22-2 Course Feedback on USMA Questions



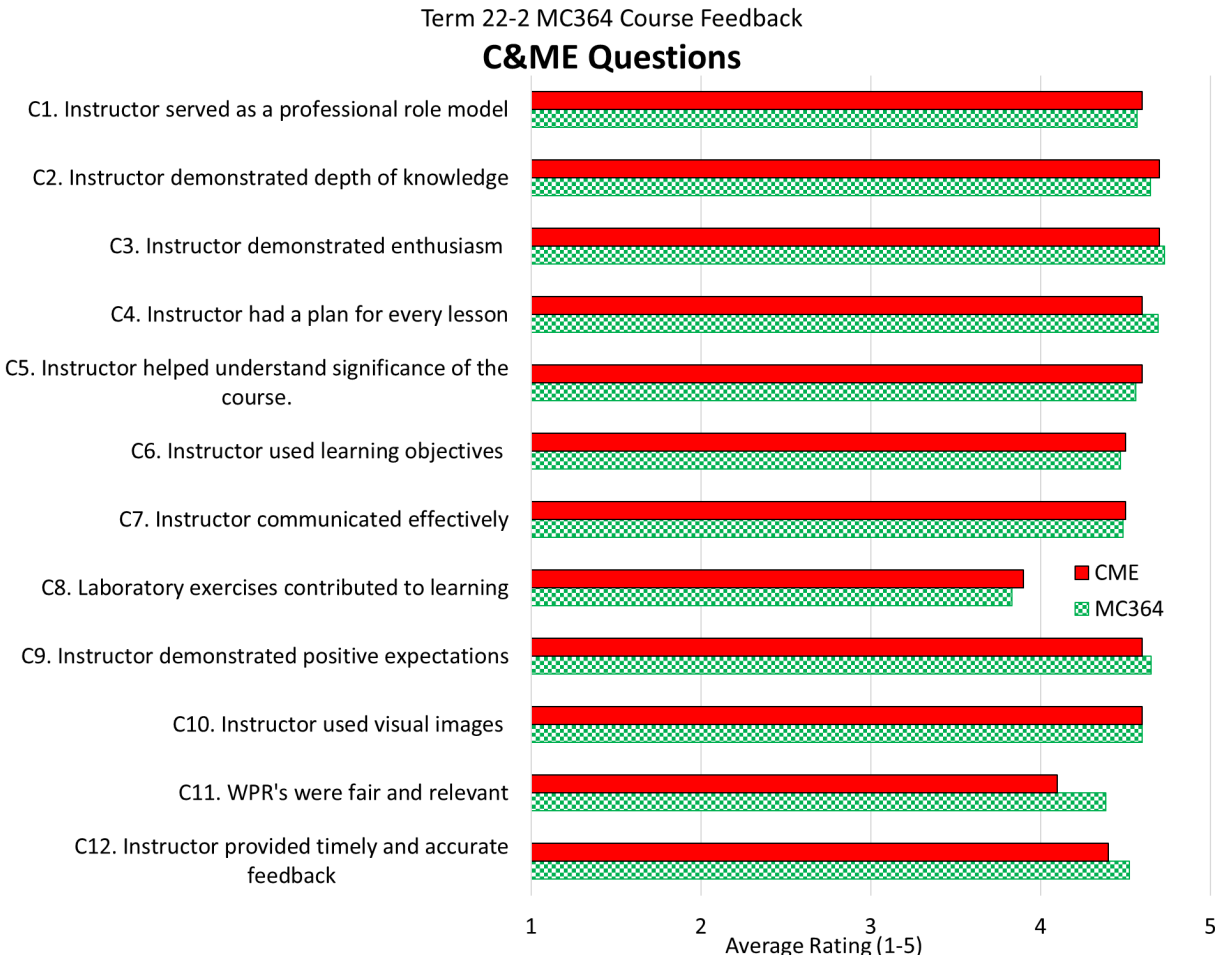
MC364 Course Assessment Package, AY 22

3. Historical Course Feedback on C&ME Questions



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4. Term 22-2 Course Feedback on C&ME Questions



5. All Course Narrative Feedback for the current period of assessment.

A. Additional Comments about the Course

- Overall I really enjoyed this course. I greatly appreciated the lesson videos, they helped me learn the course material easily.
- I was a fan of the flipped classroom. It made it so I knew what was going to be going on in class before I got there. Then I was able to ask questions on parts I might not have understood instead of trying to keep up for all of class.
- I like board work. I wish the board problems had **written out solutions**
- The **reading quizzes have a lot of points** associated with them when i don't think there should be as much. I would recommend making the homework's worth more points since they usually take at least an hour, while a HW quiz is only 8 minutes. I think it would be a better split of point if it was more based on the amount of work it takes to complete the assignments.
- The flipped schedule had its ups and downs. I feel sometimes it was more appropriate to do a problem together as a class, then transition to working boards.
- The flipped classroom was okay, but instead of reading quizzes, make **small 10 point homeworks** due at the beginning of every class on the material covered in the lesson videos. That way we can learn the material beforehand.

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- The method of watching videos before class then doing only problems in class was not helpful for me. I learn much better when the instructor walks through the material, and then goes through a problem with the class. We did that in MC300, which was so much more productive. In the future, I would recommend that the instructor goes through the material and practice problem each lesson.
- This was a great course following MC300. It was rewarding to have the theories and formulas that were left unexplained in MC300 be unpacked in this course. I also liked the flipped classroom, where there were intro videos and problem videos we could watch before class to get the simple learning (definitions and formulas) out of the way and focus on the complex problem solving in class.
- The **reading quizzes are worth way too many points**. They are useful but they should be worth less points. It took too long to adjust and learn how to prepare well for the reading quizzes so my grade suffered
- A little more time to work on the lab reports would have been nice.
- Reading quizzes did not accurately reflect my knowledge of the material.
- **Group activities were horrible**. Mostly that is group dependent but I put forth the majority of the work and most of the deliverables and my members put forth little and were rewarded the same for little to no production. Most of the assignments were of my doing and 10-25% would be others.
- Challenging but rewarding course. The reading quizzes seemed to do nothing to benefit my learning, only lowered my grade. Some were unfair.
- This course is difficult. It is very easy to fall behind even if you miss one lesson or have a rough night. It takes a lot of time to prepare for class and later internalize the material. I do not like the quizzes. If the format were such that the reading quizzes were on the prior lesson's material. Or if there were clear objectives to prepare for the quizzes, that would be beneficial. The flipped classroom is okay. Sometimes it is good, other times not so much. Sometimes the videos are lengthy and other times they are not.
- I think the **HWs should be weighted more than the Quizzes** due to the time spent on them.
- The first WPR was poorly set up, but the course recognized the flaws and were more fair on than the first.
- I liked the flipped classroom, however, I wish that the reading quizzes were held at the end of class after we had already had a chance to go over the material.
- I strongly was not a fan of the way this course was run. The format of having lesson videos with reading quizzes and then just doing board problems in class was not helpful to me.
- I highly disliked the flipped classroom and group work.
- More in class examples worked through with the teacher.
- Our knowledge was only tested in homeworks and WPRs but the homeworks were mostly lower level questions so while it helped develop my initial understanding, any extra work to become proficient had to be done in any extra time I had but wasn't reflected pointwise anywhere so I wasn't sure if I was understanding concepts until the WPR which was high risk.
- Everything is great about the course but I would go back to doing daily homeworks rather than reading quizzes.
- I felt that the WPRs and homeworks weren't proof read. A handful of issues made it feel like no one had looked over them other than the person that made them.
- I liked the videos for each lesson but sometimes the videos were way too long.
- I did not like how the reading quizzes were on topics we had not covered yet. If they were always on topics we had gone over already, it would be more fair.
- More focus should be made on sign rules, whether or not to write positive or negative into an equation.
- The quizzes and homeworks having the same point values was weird. Also, there was not enough time to complete many of the quizzes, and if I was given just a couple more minutes on each quiz, I feel my grades would have increased on many of them.
- I really enjoyed this course and the flipped classroom. The only thing I found ineffective was when the instructor walked us through the problems directly.

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- I disliked the flipped classrooms, I would have liked more focus on teaching the specific concepts in a place where you can instantly ask questions.
- The Microsoft Stream videos were helpful to my learning, however nothing of that sort can compete with an instructor working a the problem on the board and explaining the solving procedure. More so, 20% of the grade being attributed to reading quizzes/homework is extreme, specifically in regards to reading quizzes. I would rather have an increase in homework assignments, where I am not pressed to 8 minutes to display my knowledge of MC364. Reading quizzes were much more time/ability restraining than WPRs.
- I absolutely loved the lesson videos provided - the combination of the videos and lesson powerpoints provided prior to each lesson were instrumental to my success.
- I know I am good at this class, my WPR grades reflect it, but the weight of the reading quizzes has greatly brought down my grade to something I don't believe I deserve. **I think they are important for ensuring reading ahead is done but I think the weight and difficulty aren't fair.**
- **The reading quizzes were brutal for my grade.** Constant 17 or 18 out of 20s make it hard to recover.

B. Additional comments about the instructor (Removed specific comments)

- Always got my assignments back quickly with a lot of feedback. Always played music while working on problems, which made the process more enjoyable where I could practice and ask questions but it felt more like I was able to just practice what i needed to.
- The most professional, fun, and effective instructor I've had at West Point. He is what I think of when I think of a professional engineer.
- Always open for AI and willing to get back to you over many different communication platforms. They graded assignments and returned them very quickly.
- Extremely speedy responses whenever I approach them for additional help on assignments when outside of class.
- Inspirational, down to earth.
- Awesome teacher who keep students engage with humor but still keeps classroom expectations high. Makes students want to work hard for them

C. Do you believe the reading quizzes effectively assessed your learning of the course content? Why or why not?

52 Yes/47 No

- Yes they were a decent reputation of what they learned.
- Yes. It forces you to learn and retain whatever that was taught in class.
- Yes, they were fairly well attuned to my ability and represented a good snapshot of my performance on a topic.
- Yes because it assess the the past two lessons and this helps in my learning.
- I think the content on all of the reading quizzes was helpful and fair. Sometimes, however, I think a longer time limit should have been given because, oftentimes, I knew how to solve a question but ran out of time.
- I believe they did because they also helped to encourage doing the reading/watching the videos before class.
- They did. They were a good check on whether or not the student had watched the videos, which helped me prepare for class.
- Yes, I think the WPR simulated environment helped me during the exams.
- I believe that the reading quizzes were effective, however there was **too many** of them. I think a happy medium would be maybe 10 reading quizzes with 2 dropped at the end of the semester instead of the 15.

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- Yes, for the most part I believe the reading quizzes were fair. Sometimes I was not exactly sure if the material on quiz would be more on what we did last lesson or the homework for that night so that was the only difficult thing.
- I like the reading quizzes better than nightly homework assignments. It allowed me to get something out of the material and also allowing me time to complete other homework. Also the MS Streams were very effective and better than the textbook
- I think the reading quizzes were effective because they made me prepare for class knowing there were points connected to my preparation for class.
- Yes I thought reading quizzes were fair even though they were difficult. The only thing that was difficult to manage was time.
- Yes, they force us to check the course content prior to class, but they are 'high stakes' for a getting a good grade. They measure whether you understood the material rather than trying to complete the reading before that lesson. I think it is a little unreasonable that the cadets are able to complete a problem under an exams time constraints without proper examples they themselves complete or the ability to ask questions to a teacher unless they worked a lesson ahead.
- Yes they helped test what I learned, but I would much rather prefer them as daily homework assignments than pop quizzes because I would still benefit from them but my grade would not suffer
- Yes, because it tested whether or not we understood the last lesson and at least watched videos for the new lesson so we would not get lost and have some understanding of the new topic being discussed.
- Yes they forced me to read and learn the material every day before class. There was not enough time to take them though
- Yes, except sometimes the timeframe on it was difficult and I got caught up doing calculations that take time.
- Reading quizzes were very effective because they forced me to review the material carefully before each class, rather than just skimming.
- Yes, the forced me to prepare well in class although there is so much information that can be asked I wish it was open note instead of one page.
- Yes, because the main feature of the reading quiz was questions regarding the previous lesson. Which I had the pre-class work, in class-work and if needed AI after the class to obtain a solid understanding of the material. Also, the one question about that days class helped me be motivated to do the pre-class work and take notes. When I got something wrong it was pointing out a gap in my understanding which following that I could immediately acknowledge and remedy.
- They were effective; however, I enjoyed the ones that were 10 minutes long rather than the 8 minutes.
- Yes, but they would have been more effective if we took them at the end of class instead of at the beginning of class.
- They did, however I think they could have been 10 points instead of 20. 20 Points per quiz adds up very very fast and ended up being weighted more than 2 WPRs which in my opinion is a bit too much. Also, a lot of the time, we didn't really have enough time to complete them.
- Cadets don't have the time to get multiple lessons ahead on work. Nor does the academic program expect us to. However, when you quiz in class on something we learned the night before, it doesn't give us the ability to ask questions to our instructor about it, essentially punishing us for not understanding the material without help.
- No, the reading quizzes did not effectively assess my learning of the course. I would say the reading quizzes in this class discouraged me and diminished my confidence to succeed. I knew that even if I prepared for class, there was still a large possibility that I would not preform well on the reading quiz. Having a quiz on something that you have not been taught yet is not a fair assessment of understanding. While videos were meant to teach students on the subject matter. The artificial interaction in a video is not conducive for true learning. I understand the idea behind the flipped classroom method and I am perfectly fine with it. However, being tested on this material for a large portion of your grade is unfair and does

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promote true learning. If the quizzes had a retake option for partial credit, this would provide students with the opportunity to learn from their mistakes and perhaps level their grade to reflect their effort in the course. Or, reading quizzes could function more as a check on learning, having easier questions and being lower stakes in terms of points. The reading quizzes in this course were not fair in assessing knowledge nor in terms of grading scale. Even after preparing for class and participating in the flipped classroom, I often felt unprepared and unsatisfied during the quiz. I can understand if the intention behind this was to challenge us and encourage better problem solving skills, but then it is unfair to make reading quizzes worth such a large portion of our grade.

- The reading quizzes were a terrible assessment of learning. I always did the lesson prep, and they were not well-correlated to the videos. Additionally, taking the quizzes at the end of the course would be a better use because then they could test our guided knowledge.
- No. The flipped classroom didn't work, and taking the quiz at the beginning of class made it impossible to know if I had taught myself the material correctly.
- No. The reading quizzes were made with a good intent behind them, but they did not assess learning accurately. **An eight minute quiz carrying the same value as a homework that is available for a week does not make sense.** And if you're curious on the data, just look at how quiz grades compare to homework grades, then how overall grades relate to both. I would bet homework grades are significantly higher than quiz grades and closer to reflecting overall grades. Homework is a good assessment of ability because it shows what you actually can produce when given enough time. For improvement: Do the quizzes at the end of class. Make them worth less points. Make them take-home quizzes. Make them either all on the day's lesson or on the previous lesson. Divide them into blocks and only take the best few grades from each block.
- No. I do not learn well on my own, and I believe it was unfair to grade us on new material each lesson. Fortunately, the quizzes were not heavily weighted, but they were not effective. I did poorly on the quizzes, but well on the WPRs. In the future, I would suggest either eliminating the quizzes, or drastically lowering their weight, **such as to 5 points.** that way, they can be a check on learning without dragging our grades down.
- No, they were not a good assessment of knowledge. I enjoyed the MC300 class style with large homework assignments and no reading quizzes.
- I do not. Although I understand the importance of the reading quizzes to hold students accountable for learning, a lot of the reading quizzes seemed to be designed to trick students. I did not do super well on the reading quizzes even though I got an A+ on every WPR. This could be because I need to study each night more or that the questions might have been assessing the wrong content.
- No. They were always very rushed and especially when working with newly learned content, I need time to make sure I get the small things right, I needed time to think through conceptual problems.
- No, they were sometimes unpredictable and could catch cadets off guard even if they study for 2 hours before class. This is because cadets are expected to watch long videos, sometimes poorly produced, to learn new content that will only be a 2 pt question on the test while the majority of the RQ is generally a question based on the last lesson (except importantly the question is usually different from the questions we do in class forcing cadets to look at the rest of the problems for that class to try and figure out which aspect of the prior lesson is needed). **Also, the RQ's were almost entirely the same as last year I came to find out at the end of the course which means that if a cadet comes across an older RQ while studying they now, even if without knowing, have a huge advantage over others.** Lesson format should be one problem with the teacher at the beginning, then go to the boards, then a 20pt homework due by the next lesson. Every class we weren't assigned a RQ this is generally how it was structured since the teacher knew no one prepared and I felt like I learned much more in a shorter amount of time then sitting in my room watching the videos.
- I personally did not like the stress of reading quizzes. I think at a certain point I would have preferred take home quizzes, so that I could reflect on the material. Additionally, I think I

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came into every reading quiz prepared, but losing one or two points every class was annoying.

- No. **8 minutes to think/solve/report an answer** for several questions from preparing for over an hour is not a representation of my learning. If anything, reading quizzes detracted from my learning of the course content.
- No, there is minimal reflection done prior to taking them and there is minimal practice before hand. At the point of the quizzes, a very basic understanding of the material is had and with the time limit it made them very difficult. I had an almost full page of notes for each and was usually getting a 17 or 18 on them.
- No. They were often too much of a time crunch to think clearly and, in my opinion, did not resemble the amount of time available when solving these problems on a WPR. They were not very straightforward in the expectations; often times, I did the videos and entire reading and a concept from the previous lesson would be quizzed. Whenever I did not anticipate this by putting relevant equations/problems on my notesheet, I was still penalized despite doing the reading. The point distribution did not make sense to me either. The homework assignments were the same point value as reading quizzes, when reading quizzes are expected to be completed in 8 minutes and homework is slow and controlled work. I recognize the reasoning behind implementing the flipped class. However, I ultimately did not learn the course content since focusing on prep for the reading quizzes (having the tools I need on my notesheet) took away from actually learning any of the content. I think this class would have been excellent if taught in the same style as MC300 was.
- Yes and No. Yes I think they were helpful to highlight areas that you may need more work in, However, I think more time should have been allowed when taking them. The time restraint took much more points from me than points that I lost because I didn't understand something.
- When the reading quizzes were on the content from the last lesson, yes. It's hard to do a problem when we haven't practiced in class. The content only questions were okay to have before the actual lesson.
- I think that some of them did, but some of them were too long to do in time based on our current knowledge of the material. It was all on relatively new material, and some were harder than I thought some of the WPR questions were. I think maybe just a little more consistency with how they are structured.

D. Do you believe the homework assignments were effective in helping you learn the course content? Why or why not?

119 Yes / 8 No

- I thought the homework assignments were effective because they were challenging in that they tested us on new material but the problems were not overly difficult to the point where we could not solve them on our own.
- Yes, they were effective but not in learning course content more in studying it before the tests. The videos on Teams were the best tool to learn content and study for the reading quizzes.
- Yes, the homework assignments were effective in helping me learn course content because they forced me to work through topics I did not fully understand. This allowed me to gain a better of the topics.
- They were far more effective than the daily quizzes. It allowed time to actually problem solve and learn the material more effectively than just reading or watching videos.
- I really liked the 7 HW assignments. This gave me time to solve specific problems and gave me the time to reflect on the why of it all.
- Yes, they were worth enough points that they couldn't be skipped and forced us to do the problems correctly, each problem testing a different lesson. It forced us to keep up with the content and stay on pace.
- Yes. They were fair and helped me guide my studying as I used them as mini WPR preps for each section of the course.

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- Yes because they gave a preview of what the WPR questions would look like and allowed me to have to solve them without having just seen it in class.
 - Yes, because we were allowed to complete them using MS streams and any notes for help. However, they took much longer to complete compared to reading quizzes, so the point value doesn't make much sense.
 - Yes, relevant problem structure, complex enough to require thought and effort but not so much to require excessive time.
 - Yes, the homework was a good review of everything that we had learned for that block and was a good study tool to understand what kind of questions we would be asked on the test.
 - Yes because although there was more homework than MC300, they seemed more manageable and they helped prepare more for WPRs.
 - I believe that the videos for homework really helped you take more control for your learning and helped increase engagement with the content.
 - I like the homework assignments much better than the problem sets in MC300. They were just as effective and they didn't take nearly as much time. They also reflected the types of problems that would be on the WPRs very well
 - Yes the homework assignments were applicable but took way too long to complete for only 20 points.
 - Yes, the hw assignments reflected the type of problems that were given in the WPRs. It was just weird that they would take a significantly longer time than RQs, yet they were worth the same amount of points.
 - Yes. They were summaries of each main unit and covered majority of the problem solving skills we needed for that unit. They were manageable 1-2 hour assignments with ample time to complete facilitating AI and peer assistance. They are also similar to if not a slight step above WPR questions which prepared me well.
 - The videos were helpful most of the time, but this eliminated the need to buy the \$200 textbook.
 - I think the homeworks were incredibly effective. I got to try to figure it out on my own using example problems and videos. And if I couldn't, I could ask someone to explain it to me and cite it. That way I was learning.
 - They had the same point values as reading quizzes. They should be more points than the quizzes. Have the quizzes be 10 points and the homework as 40
 - The homework assignments were not the best bang for their buck. Except for a few of them, most of them took a substantial amount of time on top of the usual class prep time, yet they were only worth 20 points. Making them worth 40 points might make them more effective.
 - Yes and no. Yes because the homework assignments provided built-in points directly related to the material in class that would be seen on the WPR; however, oftentimes I found myself somewhat lost on the homework assignments resulting in a loss of points or requirement of AI in order to complete fully.
- E. Do you believe working problems at the boards in class was effectively in helping you learn how to solve problems in the course? Why or why not?

91 Yes / 16 No or Kinda

- Yes, board problems were extremely effective in helping me learn the topics of this course. This is because I was able to work through problems mainly by myself, so I got good practice, and because my teacher was able to be resourced quickly if I had a question. Board problems allowed me to practice problems and better understand the topics.
- yes, **however it often felt as though they took more time than they were worth.** I would rather have more hw, less quizzes, and more going over hw problems in class. Perhaps the hw being due at the end of the class makes the most sense. Not enough time to complete the whole thing in class but enough time to ask key questions and go over the harder problems. That way people have to show up with at least part of the HW complete.
- Yes, there was always **immediate feedback available**

MC364 Course Assessment Package, AY 22

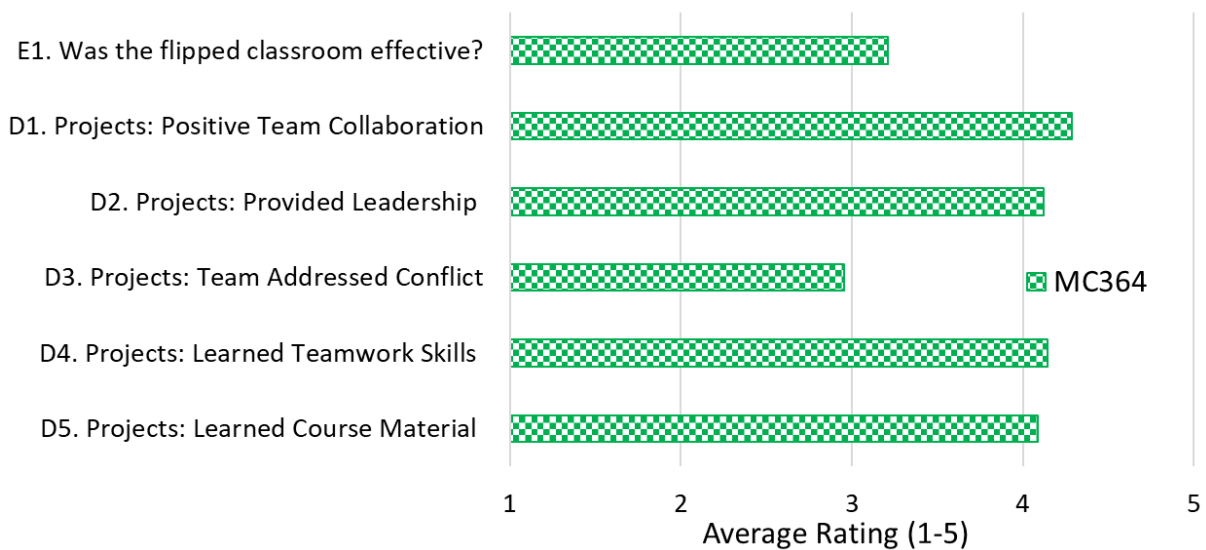
- This was helpful gave us time to work through the problems with teachers assistance if needed
- Yes I do. I believe it is important for getting reps at problems and also everyone can stay engaged and up to pace as they can learn from one another's work
- Yes, getting feedback from a teacher is really beneficial
- Yes, I loved doing board work. I was able to communicate and collaborate with my peers while simultaneously ensuring that I actually understand the material.
- Yes because it allowed us to practice the material we learned before class and if we messed anything up or had questions in class, we could ask.
- Yes it was effective because I could collaborate with my classmates and ask the instructor for help.
- I really liked working the problems on the board as it allowed you to get **instant feedback** on your work - you did not have to wait a week for an instructor to grade the problem like you would have with a problem set.
- Yes, because it forced us to either seek guidance when we had questions with problems or forced us to demonstrate that we could do the new problems.
- Yes, working problems in class was very helpful to my understanding. The best teaching method for problem solving classes is to try problems on your own and have the teacher review your work and give you feedback. That portion of the class is very important.
- I only want to do board problems. I hate lectures and if i already learned it the night before for the quiz then i don't need to be lectured for hours
- I do as they allowed us to make mistakes and get them fixed by the instructors so that we made sure to do them right.
- Working problems in class are very effective, however the method of **using boards is not helpful** to me because I like to have my notes and problems in my notebook for studying. Additionally, the problems we do in class need to be simpler and easier. Most of the time, we were given in class problems that were so complex that they took almost the whole class period. While complex problems are good for practice, they also keep me from understanding the main concept, and long problems usually bog me down before I can complete it.
- Yes. Going to the board was just wonderful for my understanding and confidence on homework and WPRs. Also, when I made slight mistakes or hit a roadblock my instructor was there to point out where I went wrong and correct my thought process before going all the way down the wrong path. Board problems held me accountable to solve or attempt to solve the problems as an active participant as opposed to a passive participant watching and copying my teacher solving a problem. Also having pictures of all my board problems was helpful for studying.
- **Yes, this was the only way that I learned the material. I didn't watch any videos or read the textbook.**
- No. Even after the videos/reading, I was extremely lost on most board problems. I would have preferred to have the content taught before going to the boards. This was sometimes remedied by doing a problem together before sending us to the boards.
- No. The flipped classroom meant that instructors were not required to go over the concepts in class. Most of my classmates spent most of 75 min block doing one (maybe two) problem because they were still confused about the topic. The instructors can't help everyone at once.
- No, as it required you to already know how to solve the problem, which is unrealistic given the conditions at west point.
- Every lesson, there would be several cadets who would get stumped on a particular part of the board problem. With a student: instructor ratio of ~15:1, it is not possible for the instructor to attend to everyone's questions.
- Having the teacher do an example problem first and then going to boards was helpful. Going to boards first did not contribute to my learning.

MC364 Course Assessment Package, AY 22

6. All Course Numeric Feedback

Term 22-2 MC364 Course Feedback

MC364 Questions





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TAB E AY2021 COURSE ASSESSMENT

MC364 Course Assessment Package, AY 21

I. Course Administration

A. Redbook Description

1. Current Redbook Description:

MC364 (Version: 2020 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
MC364	MECHANICS OF MATERIALS	2020	1	Civil and Mechanical Engineering	3.5 (BS=0.0, ET=3.5, MA=0.0)
SCOPE					
This course studies the behavior of a variety of materials under normal, shear, torsional, bending and combined loads. The concepts of stress, strain, failure theory and failure mechanisms are explored. The loading, geometry, functional environment and material properties of machine or structural parts are used to relate the forces applied to a body to the resulting internal forces and deformations so that performance can be evaluated. Practical applications involving the design and adequacy of mechanical and structural elements under various loading and environmental conditions are emphasized. The course includes several laboratory experiences that require students to develop and conduct experiments, analyze and interpret data, and use engineering judgement to draw conclusions.					
LESSONS: 25 @ 75 min (1.560 Att/wk) LABS: 10 @ 75 min					
SPECIAL REQUIREMENTS:					
None					

MC364 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
CO REQUISITE					
	MA205	2017	1	1	Y
	MA255	2016	2	2	Y
	MA366	2018	2	3	Y
DISQUALIFIER					
	CE364	2006	1	1	Y
PRE REQUISITE					
	MC300	2017	1	1	Y

Terms offered and enrollments.

AY 12	12-1	144		AY17	17-1	169		AY22	22-1	
	12-2				17-2	149			22-2	172
AY 13	13-1	175		AY18	18-1			AY23	22-8	1
	13-2				18-2	112			23-1	
AY14	14-1	197		AY19	19-1			AY24	23-2	26
	14-2				19-2	144			24-1	
AY15	15-1	162		AY20	20-1				24-2	3
	15-2				20-2	128				
AY16	16-1	109		AY21	21-1					
	16-2				21-2	118				

2. Enrollment Issues: Currently our highest projected enrollment in 5 years. 18 cadets/section x 3 rooms/section x 4 sections = 216; therefore, there no issues anticipated from a capacity standpoint under normal operations. Potential points of discussion: if section sizes are decreased due to continued social distancing requirements, we may need a significant number of additional sections. Increasing the number of simultaneous sections may not be feasible due to restrictions on available demonstration and laboratory equipment. 12 cadets/section x 3 rooms/section x 5 section = 180; would have to teach during K1 hour.

3. Proposed Changes to Redbook Description: No changes are proposed.

MC364 Course Assessment Package, AY 21

B. Course Objectives

1. Current Course Objectives:

- a. Calculate the internal forces, internal stresses, and deformations of axially loaded members, circular members in torsion, thin-walled pressure vessels, prismatic beams in bending, columns, and members subjected to combined loading and/or thermal effects.
- b. Apply compatibility of deformations to analyze and design members of a statically indeterminate structure subjected to loading and/or thermal effects.
- c. Apply appropriate theories of failure to analyze and design thin-walled pressure vessels and members subjected to loading and/or thermal effects.
- d. Given strain data from a strain rosette, determine the state of strain and the state of stress.
- e. Given a state of stress at a point, determine the principal stresses, the maximum in-plane shear stress, the angle to the principal plane, and the state of stress on any plane.
- f. Apply deformation—strain, strain—stress, and stress—force relationships to analyze and design structural members and machine components.
- g. Conduct laboratory experiments to verify and apply methods, theories and scientific laws learned throughout the course, and prepare proper technical reports to clearly communicate the conduct and results of those experiments.

2. Proposed changes to Course Objectives:

- d. Given strain data from a strain gage, determine the applied force, moment, or pressure on a structure.

C. Textbook(s)

1. Current Textbook: Mechanics of Materials, 4th Edition by Timothy A. Philpot. ISBN: 978-1-118-08347-5
2. Textbook Changes: Mechanics of Materials, 5th Edition by Timothy A. Philpot and Jeffery S. Thomas. ISBN: 978-1-119-60301-6

MC364 Course Assessment Package, AY 21

II. Course Content

A. Outline of Lessons and Labs as planned

1. Course Schedule for Term 21-2

Lesson	Title/Topic	Supported Course Objectives						
		a	b	c	d	e	f	g
1R	Course Intro & MC300 Review	a						
2R	Uniaxial Tension Lab	a					f	g
3R	Statically Indeterminate Axially Loaded Members I	a	b				f	
4R	Statically Indeterminate Axially Loaded Members II	a	b				f	
5R	Thermal Effects on Axially Loaded Members	a	b				f	
6	Torsion Lab	a					f	g
7R	Introduction to Torsion	a					f	
8	Solving Torsion Problems	a					f	
9	Gears and Power Transmission	a					f	
10	WPR 1 Review Lesson	a	b				f	
11	WPR 1 (Axial and Torsion)	a	b				f	
12	Beam Lab	a					f	g
13R	Shear and Moment Diagrams for Beams	a						
14	Flexural Normal Stress	a					f	
15	Flexural Shear Stress	a					f	
16	Deflections of Beams	a					f	
17	Statically Indeterminate Bending Members I	a	b				f	
18	Statically Indeterminate Bending Members II	a	b				f	
19	WPR 2 Review Lesson	a	b				f	
20	WPR2 (Flexural)	a	b				f	
21	Stress Elements	a					f	
22	Stress Transformation	a					f	
23	Principal Stresses and Max Shear Stress	a				e	f	
24	Mohr's Circle	a				e	f	
25	Strain Transformation	a			d	e	f	
26	TWPV (Strain Gage Mounting)	a			d	e	f	
27	TWPV Lab	a			d	e	f	g
28	Combined Loading (Axial and Torsion)	a			d	e	f	
29	Principal Stresses in Flexural Members	a			d	e	f	
30	WPR 3 Review Lesson	a			d	e	f	
31	WPR 3 (Stress & Strain Transformation and TWPV)	a			d	e	f	
32	General Combined Loading	a			d	e	f	
33	Theories of Failure - Ductile Material	a		c		e	f	
34	Theories of Failure - Brittle Material	a		c		e	f	
35	Course Review and Wrap-up	a	b	c	d	e	f	

*R. Indicates the class was remote – either for COVID-19 protocol or Code Red

*One Lab was removed per Dean's Guidance in response to COVID-19

MC364 Course Assessment Package, AY 21

2. Proposed Course Schedule for AY22-2

Lesson	Title/Topic	Supported Course Objectives							Comments
		a	b	c	d	e	f	g	
1	Course Intro & MC300 Review	a							
2	Uniaxial Tension/ Compression Lab	a					f	g	*Will require Sect Prop
3	Axial Deformation	a					f		
4	Buckling II	a					f		*Will require Sect Prop
5	Statically Indet. Axial Members I	a	b				f		
6	Statically Indet. Axial Members II	a	b				f		
7	Thermal Effects on Axial Members	a	b				f		
8	Torsion Lab	a					f	g	*Will require Sect Prop
9	Introduction to Torsion	a					f		
10	Solving Torsion Problems	a					f		
11	WPR 1 (Axial and Torsion)	a	b				f		
12	Geometric Section Properties	a							
13	Beam Lab	a					f	g	
14	Flexural Normal Stress	a					f		
15	Flexural Shear Stress	a					f		
16	Deflections of Beams	a					f		
17	Statically Indet. Bending Members I	a	b				f		
18	Statically Indet. Bending Members II	a	b				f		
19	Beam Analysis	a					f		
20	Beam Design	a					f	?	*Sup Lab5 with an EDP?
21	WPR2 (Flexural)	a	b				f		
22	Stress Elements	a					f		
23	Stress Transformation	a					f		
24	Principal Stresses and Max Shear Stress	a				e	f		
25	Mohr's Circle	a				e	f		
26	Strain Transformation	a			d	e	f		
27	TWPV (Strain Gage Mounting)	a			d	e	f		
28	TWPV Lab	a			d	e	f	g	
29	Combined Loading (Axial and Torsion)	a			d	e	f		
30	Principal Stresses in Flexural Members	a			d	e	f		
31	WPR 3 (Stress & Strain Trans. & TWPV)	a			d	e	f		
32	General Combined Loading	a			d	e	f	?	*Tower of Torque Lab?
33	Theories of Failure - Ductile Material	a		c		e	f		
34	Theories of Failure - Brittle Material	a		c		e	f		
35	Course Review and Wrap-up	a	b	c	d	e	f		

MC364 Course Assessment Package, AY 21

B. Proposed changes to Lessons and Labs.

1. Remove three WPR Review Lessons. These may be supplemented with out-of-class homeworks. If switch is made from reading quizzes to homeworks, approximately 230 minutes of in-person contact time will be recovered.
2. Remove Gears and Power Transmission and Shear and Moment Diagrams. Gears will be covered in ME403 and Shear and Moment Diagrams will be covered extensively in MC300.
3. Add Axial Deformation and Buckling II class based on changes from MC300
4. Add Section Properties lesson focused exclusively on how to solve for moments of inertia and first moment of outward area, etc. (Add to LSN2? Or LSN12?)
5. Add Beam Analysis and Beam Design classes
6. Include 5 Labs
 - i. Axial: Add Buckling –Test same material in Tension and Compression.
 - ii. Torsion
 - iii. Beam – would like to test different geometries, this hands-on demo was removed from MC300
 - iv. TWPV – Use strain rosette – more expensive ~3x\$
 - v. Tower of Torque? Beam EDP? Separate Axial & Buckling Labs?

C. Graded Requirements

1. Summary of graded requirements for AY21-2

Requirement	Qty	Value	Total	% of Grade
Lab (Group)	4	100	400	20%
Reading Quizzes (25 each – 3 lowest)	22	20	440	22%
WPRs	3	200	600	30%
TEE	1	500	500	25%
Instructor Points		60	60	3%
			2000	100%

2. Proposed changes to graded requirements

Requirement	Qty	Value	Total	% of Grade
Lab (Group)	5	100	500	25%
HW/RQ (23 each – 3 lowest)	20	20	400	20%
WPRs	3	200	600	30%
TEE	1	500	500	25%
			2000	100%

- a. Add fifth lab back into the course. Lab was removed due to COVID-19 restrictions in AY20-2 and AY21-2.
- b. Switch Reading Quizzes to Homeworks? Or half and half? The grade distribution for the course was good. However, cadets still managed to pass the course even with poor performance on major graded events. Need accurate assessment of individual performance, scores are inflated by labs (Group work).

MC364 Course Assessment Package, AY 21

III. Course Assessment

A. Course Director Assessment of Course Objectives

1= Objective Not Met. Objective clearly not met, most (75%) of the students did not achieve it.

2= Objective Marginally Met. Objective met by half the students or minimally by most.

3= Objective Satisfactorily Met. Objective clearly met by a solid majority (70%) of students.

4= Objective Solidly Met. Objective clearly met by the vast majority (90%) of students.

5= Objective Clearly Met. All students have achieved the objective and can demonstrate it.

The definition of “meeting a course objective” is achieving a “C” level (70%) on the task.

Course Objective	Value	How Evaluated and Remarks
a. Calculate internal forces ...	4	Evaluated through RQs, WPRs, and TEE. 94% of cadets earned a C-Level or higher on the TEE
b. Apply compatibility of deformations to analyze and design members...	3	Evaluated through RQs, WPRs, and TEE. 77% of cadets earned a C-Level or higher on the TEE
c. Apply appropriate theories of failure to analyze and design ...	3↓	Evaluated through RQs, WPRs, and TEE. 88% of cadets earned a C-Level or higher on the TEE
d. Given strain data from a strain rosette, determine the state ...	1↓	Evaluated through Labs Exercises. No strain rosettes were available.
e. Given a state of stress at a point, determine the principal stresses...	4	Evaluated through RQs, WPRs, and TEE. 92% of cadets earned a C-level or higher on the TEE
f. Apply deformation-strain-stress-force relationships ...	3↓	Evaluated through RQs, WPRs, and TEE. 88% of cadets earned a C-Level or higher on the TEE
g. Conduct laboratory experiments to verify and apply....	4	Evaluated through Lab Exercises. 98% of cadets earned a C-Level or higher on four group lab submissions

B. Program Outcomes Directly Supported By This Course

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors;
3. **an ability to communicate effectively with a range of audiences;**
 - Lab Report 4: Written communication sub-grade
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts;
5. **an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives;**
 - Lab Group Surveys

MC364 Course Assessment Package, AY 21

6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;

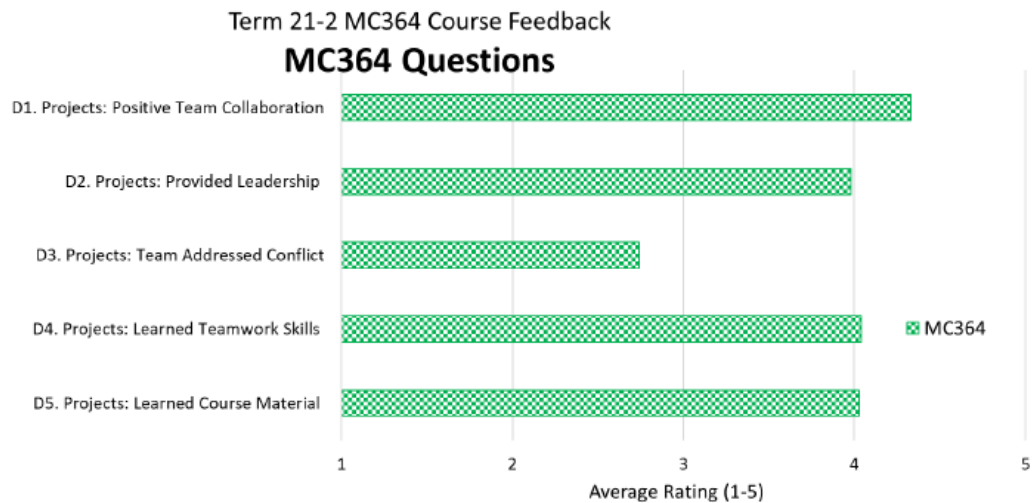
- Lab Report 4: Conduct of Experiment sub-grade

7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

C. Assessment of Embedded Indicators.

Graded Event	Pts	Average	A	B	C	D-F
Lab Report 4: Written Expression Sub-Score	50	45.3	25	10	0	0
Lab Report 4: Conduct of Experiment Sub-Score	50	45.8	23	12	0	0

Note – the data listed above is only for civil engineering majors.



MC364 Course Assessment Package, AY 21

D. Assessment of Changes Made this Semester.

1. *Change Textbook.* The textbook was changed from the 3rd edition of Philpot to the 4th edition. This was a good switch. The new edition had little changes to the content and provided good example problems. With the removal of Adobe Flashplayer MecMovies can no longer be accessed from MS&T. Cadet's can only access them via the Ebook. In the future, will highly encourage all cadets to purchase and register their ebook.
2. *Addition of 5th laboratory exercise (Tower of Torque):* unable to assess this year due to COVID restrictions. This was the assignment and content removed per Dean's guidance.
3. *Transition away from textbook problems for WPRs.* This was a sustain. The problems created for the WPRs were similar to textbook problems and often used similar graphics, but were original questions which targeted course and lesson objectives.
4. *Two Lesson turn around for lab report submissions.* This is a sustain. It may not be the perfect response, but cadets had sufficient time to coordinate and complete their laboratory assignment without dragging it out to far. It provided them with opportunity to ask instructors questions. It also gave them the flexibility to deconflict their schedules.
5. *Fully Flipped MC364.* This is a sustain. The fully flipped MC364 gives the cadets the opportunity to learn independently. It allows the class period to be spent solving complex problems and answering thought provoking questions.

E. Further Recommended Changes for Future Semesters.

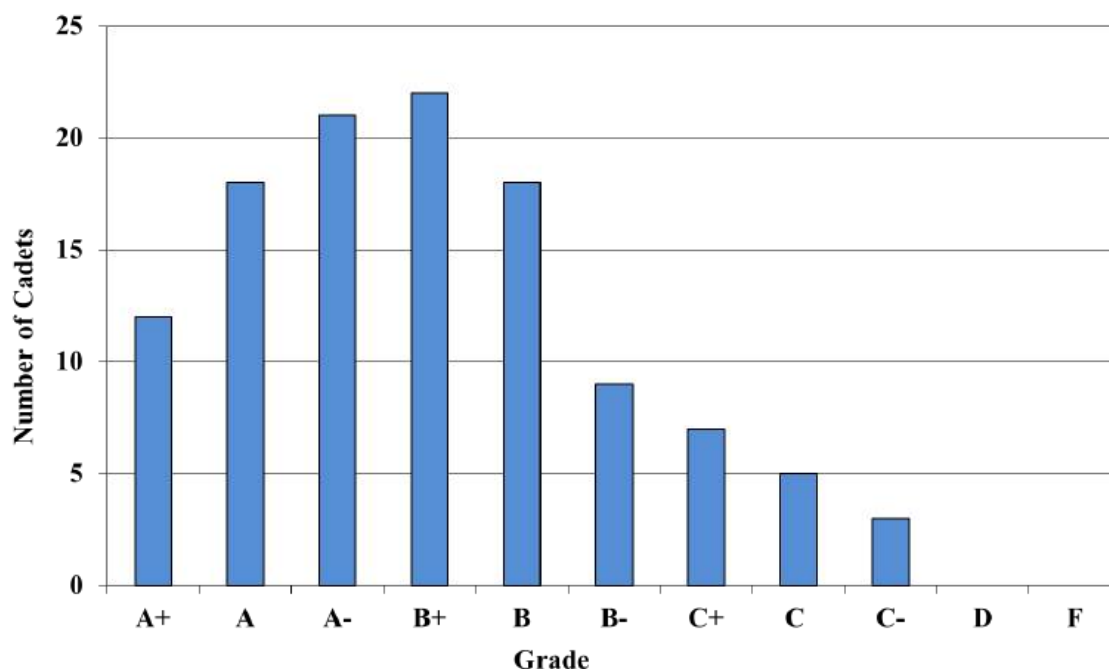
1. *Transitioning Reading Quizzes to Homework Assignments.* As the semester weened the cadets tended to stop preparing for class. Some instructors were forced to review lesson material in class taking away from time for cadets to practice problem solving. Some cadets struggled with reading quizzes, not knowing what was important to prepare or study. Recommend in future semesters either supplementing reading quizzes with homeworks or completely transitioning to homeworks. These homeworks should be targeted at testing the cadets conceptual understanding of the material and should encourage them to engage with the textbook and the lesson videos. They can also test their problem solving ability, but should not be the sole focus as the class period is spent solving problems.
2. *Using a Strain Rosette for the TWPV.* This will require purchasing rosettes, which are approximately 3x more expensive than uniaxial strain gages. This will allow us to achieve the current course objective d in its current form. It will also help illustrate that there is no plane shear stress on the outer surface of a thin wall pressure vessel. It will also allow cadets to see the relationship between hoop and longitudinal stress.
3. *Add a Fifth Laboratory Exercise.* If the axial stress lab is combined with the buckling lab, there is still a need for a fifth lab. This could either be a Tower of Torque lab or an EDP. I am also open to other options.

MC364 Course Assessment Package, AY 21

F. Assessment of Resources and Impacts.

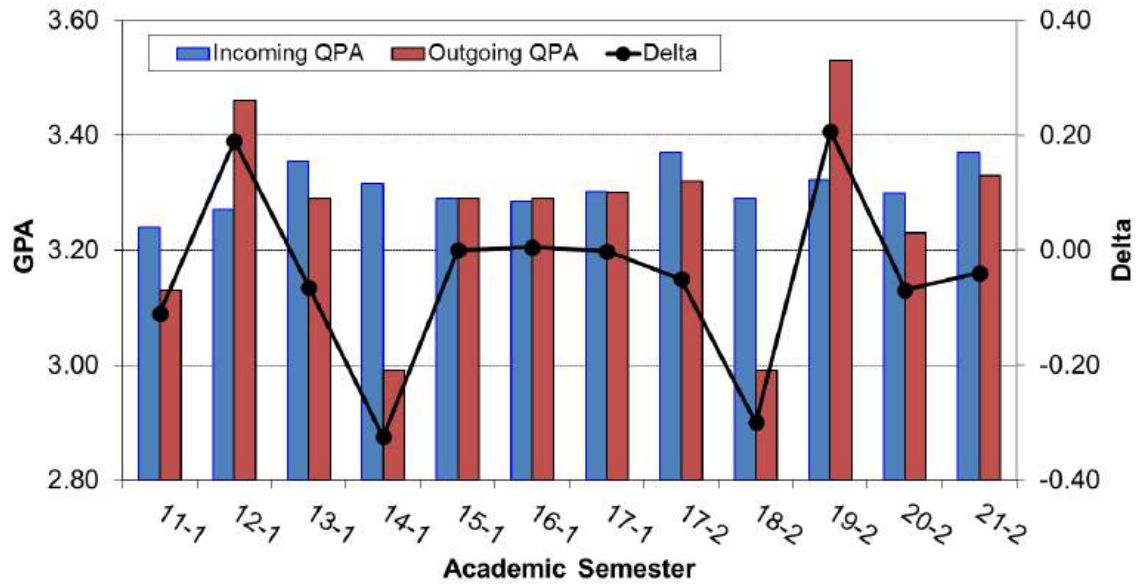
1. *Remaining issue from AY20-2:* During the Torsion lab exercise, the conduct of the lab was greatly constrained by a lack of sufficient torsion testing machines. The machines would not fit well into the classrooms, so we were limited to six machines, the number we could easily fit in the hallway. This required lab groups to work so quickly to collect data that some understanding, and synthesis were probably lost. Previous versions of this course allowed each cadet team to work on their own torsion testing machine and allocated a separate lab space for this to occur. This arrangement would be preferable to the current arrangement.
2. *Need for strain gage rosettes.* Strain gage rosettes are needed for the TWPV Lab in order to achieve Course Objective D. These instruments will allow cadets to achieve a better understanding of TWPV theory through the lab.
3. *Mechanics of Materials Box.* Would like to investigate the literature on providing cadets with hands on demo kits for the semester. Interesting in providing every cadets with hands on demos that they can use throughout the semester. Current items: Foam beam (bending), pool noodle (axial and torsion), work-out band (stress elements), balloons (TWPV), etc.

G. Grade Distribution

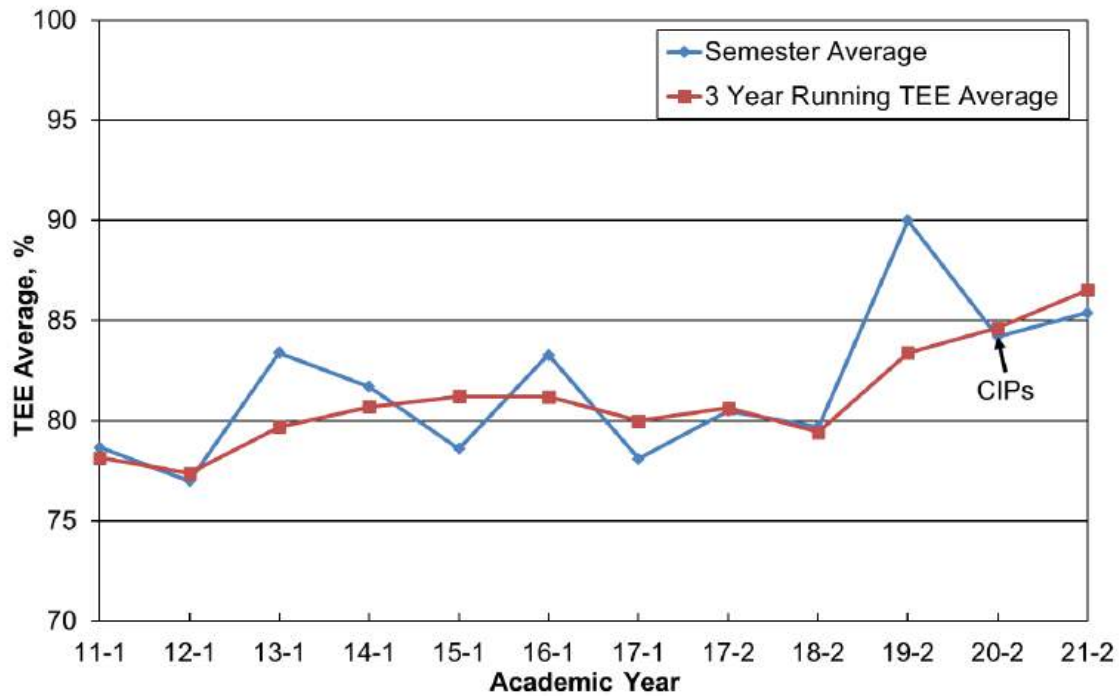


MC364 Course Assessment Package, AY 21

H. Historical GPA Comparison



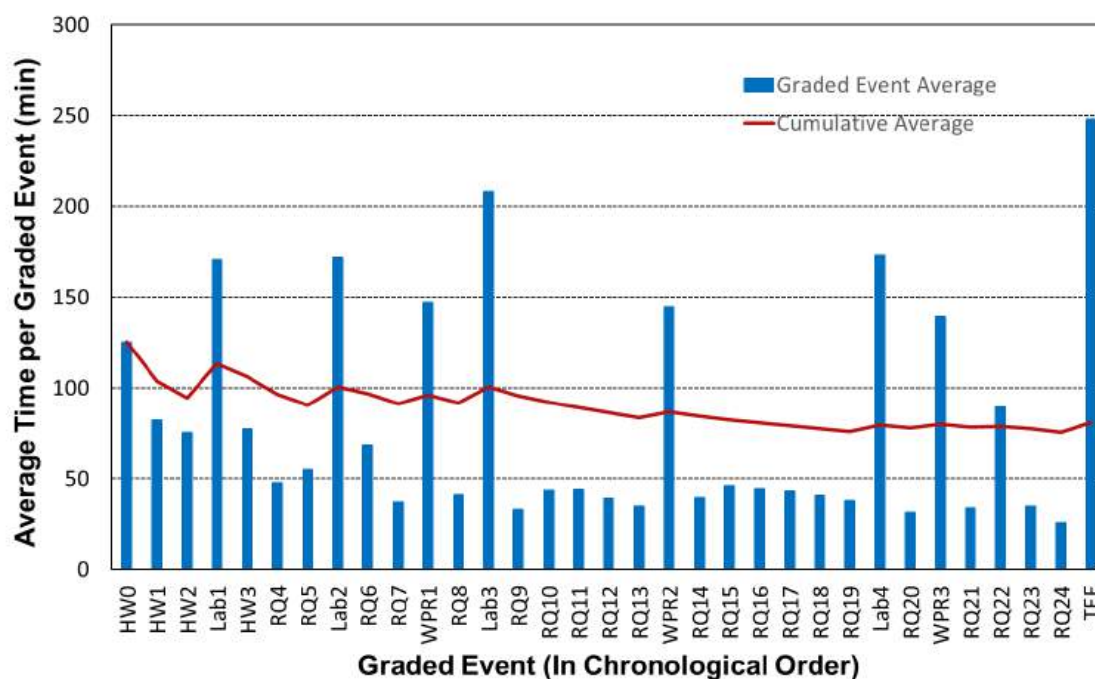
I. Historical TEE Averages



MC364 Course Assessment Package, AY 21

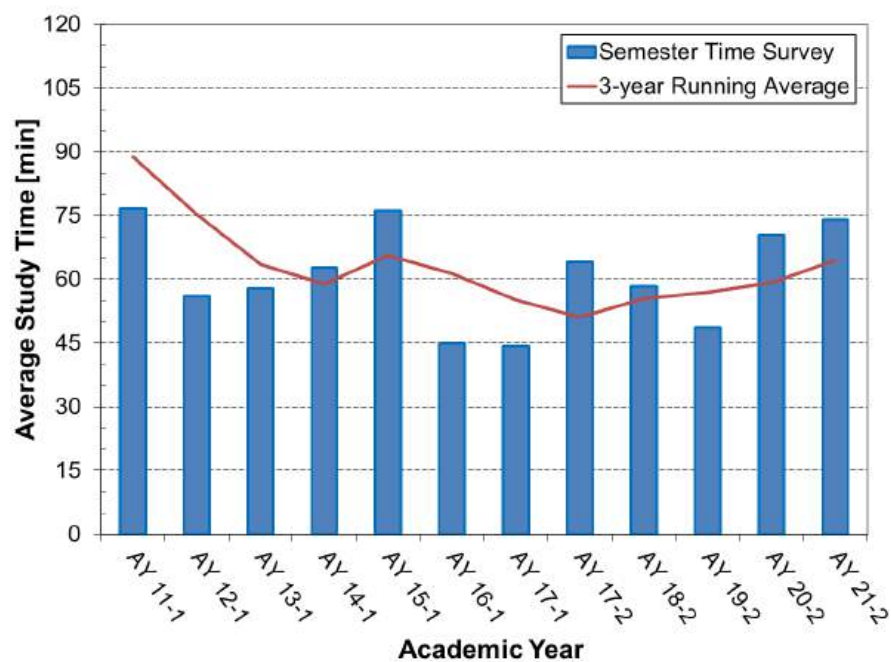
J. Course Time Study Graph

1. Current Semester time survey data.



Note: Time survey data was collected per graded event opposed to per lesson.
Average Time Per Assignment (33 Assignments): 81 minutes/assignment
Average Time per Lesson (35 Lessons + TEE): 74 minutes/lesson

2. Historical Time Survey Data



MC364 Course Assessment Package, AY 21

K. MC364 Specific Course End Survey Questions

Question	Question Type
I provided leadership in completing team assignments	Likert
My team had to address conflict to complete team assignments	Likert
My team worked in a positive collaborative manner	Likert
Teamwork activities were effective in learning course material	Likert
Teamwork activities were effective in learning teamwork skills	Likert
Do you believe the flipped class format was effective in helping you learn the course content?	Likert
Do you believe the reading quizzes effectively assessed your learning of the course content? Why or why not?	Open
Do you believe working problems at the boards in class was effective in helping you learn how to solve problems in the course? Why or why not?	Open
Do you believe you were able to effectively learn the course content without assigned homework? Why or why not?	Open

IV. CE Divisions Areas of Emphasis

A. Prerequisite Material

MC300 is the primary prerequisite course for MC364. A HW0 was assigned due on the first lesson to assess cadets knowledge on trusses, frames, axial stress, strain, and deformation, shear and moment diagrams, and flexural normal stress. Cadets struggle with beam analysis in MC364, which is extensively covered in MC300. This material will be removed from MC300 in AY22-1. Cadets struggle with visualizing 3D problems, this should be addressed by 3D equilibrium being introduced into MC300. One other area of concern is integration both multivariable calculus and solving for constants via boundary conditions.

B. Program Assessment – Embedded Indicators, Communication, and Teamwork

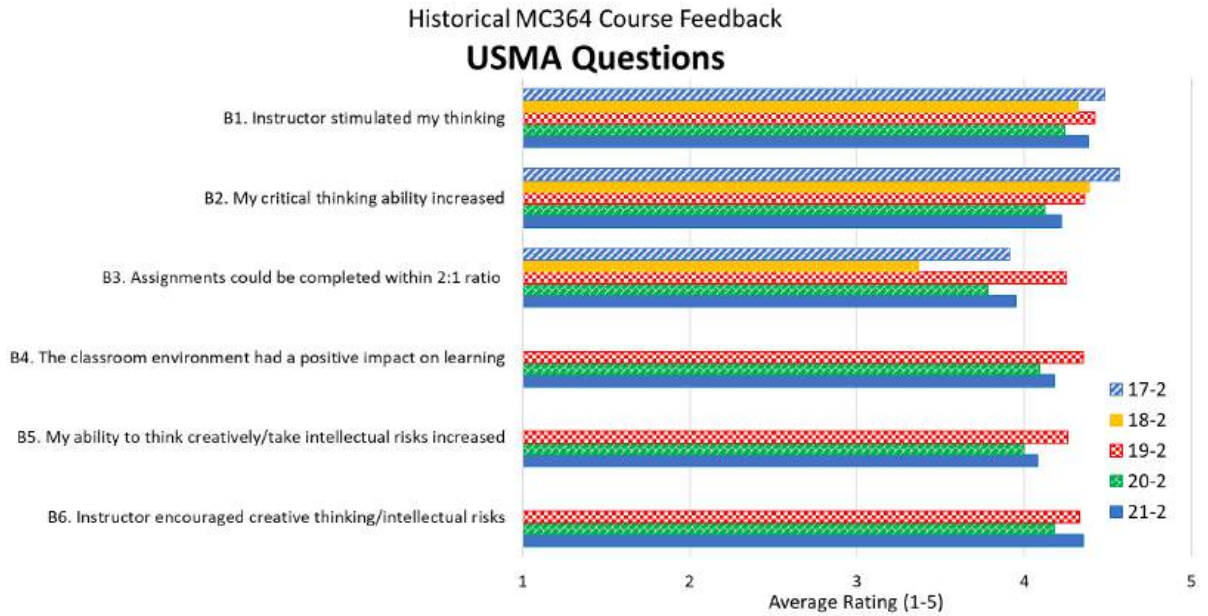
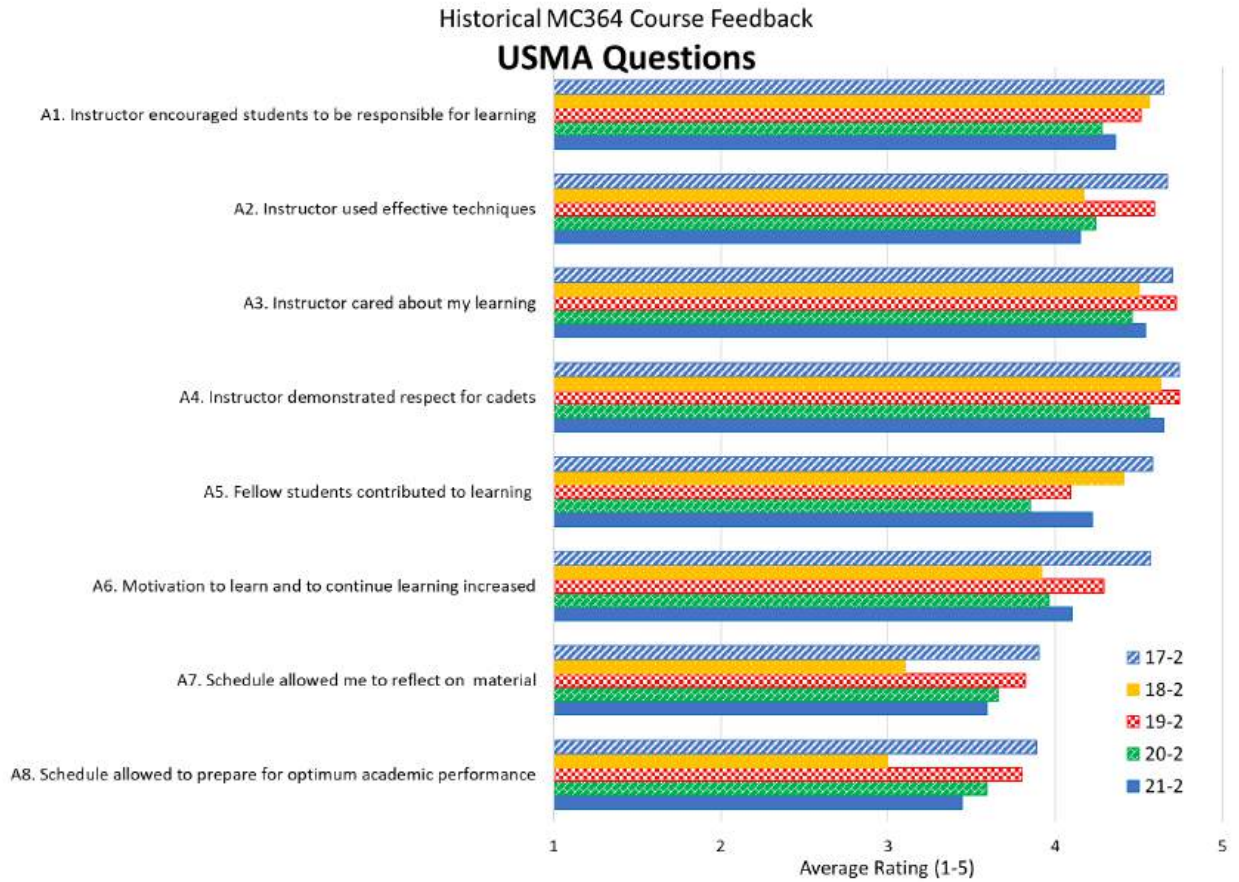
The thin-walled pressure vessel lab remains a good test of cadet's ability to design and conduct an experiment. Although they are not necessarily designing the experiment. The ExSum write-up is good written communication. They have completed three prior to the TWPV lab so it is a good assessment. The only issue is the writing and experimentation rubrics. This is a bit of a challenge for instructors to grade and cadets to focus on as directed rubrics are used for the first three labs. For the Team evaluation, CATME was an effective tool, should be executed across the course. May be implemented at the course director level.

V. Enclosures

1. Historical Course Feedback on USMA Questions
2. AY21-2 Term Course Feedback on USMA Questions
3. Historical Course Feedback on CME Questions
4. AY21-2 Term Course Feedback on CME Questions
5. AY21-2 Term Course Feedback on MC364 Questions
6. AY21-2 Qualitative Feedback on Instructor and Course

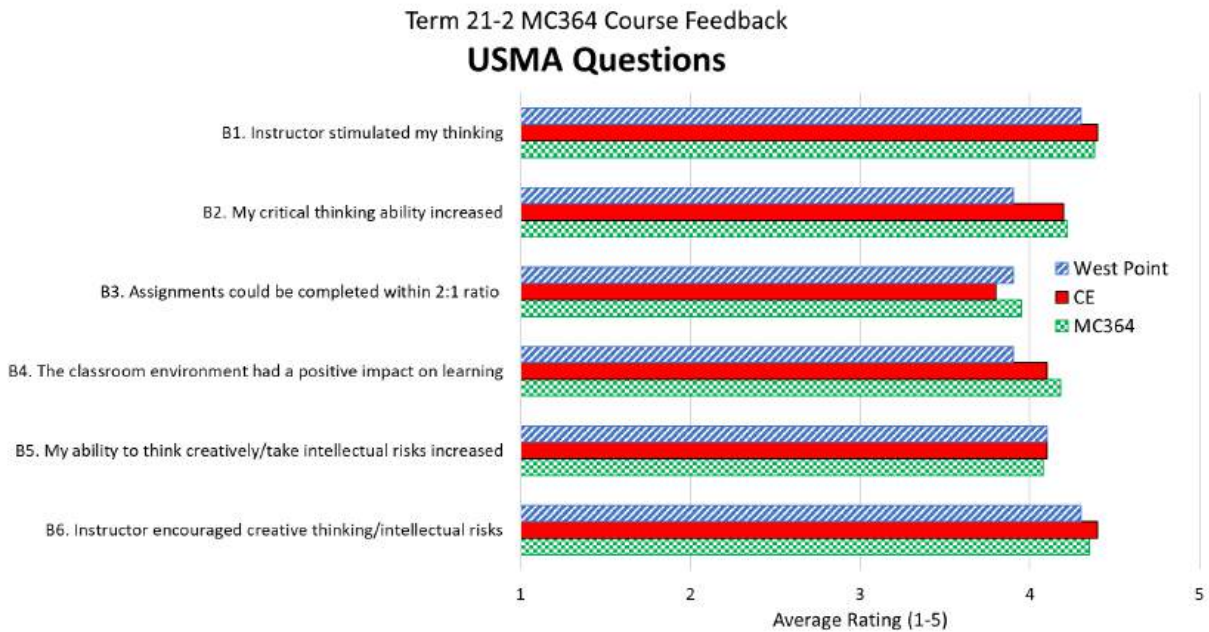
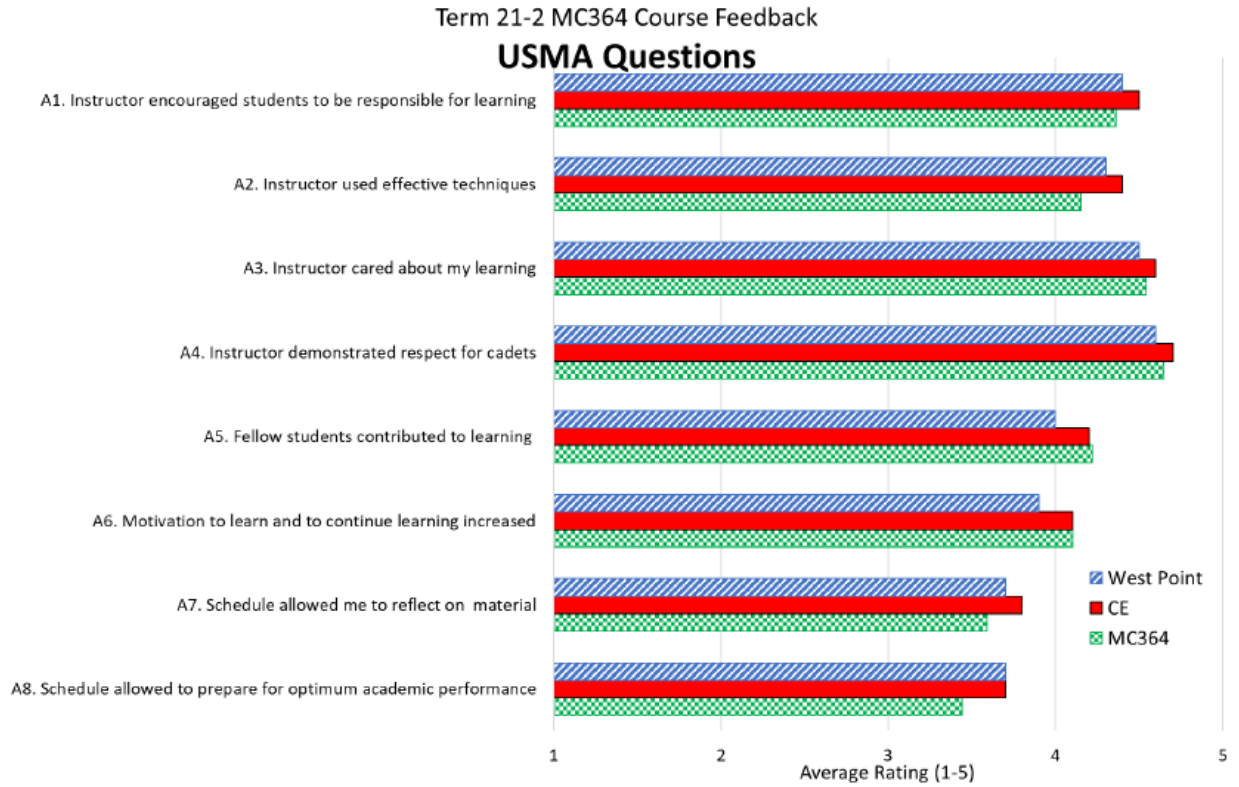
MC364 Course Assessment Package, AY 21

ENCL 1. Historical Course Feedback on USMA Questions



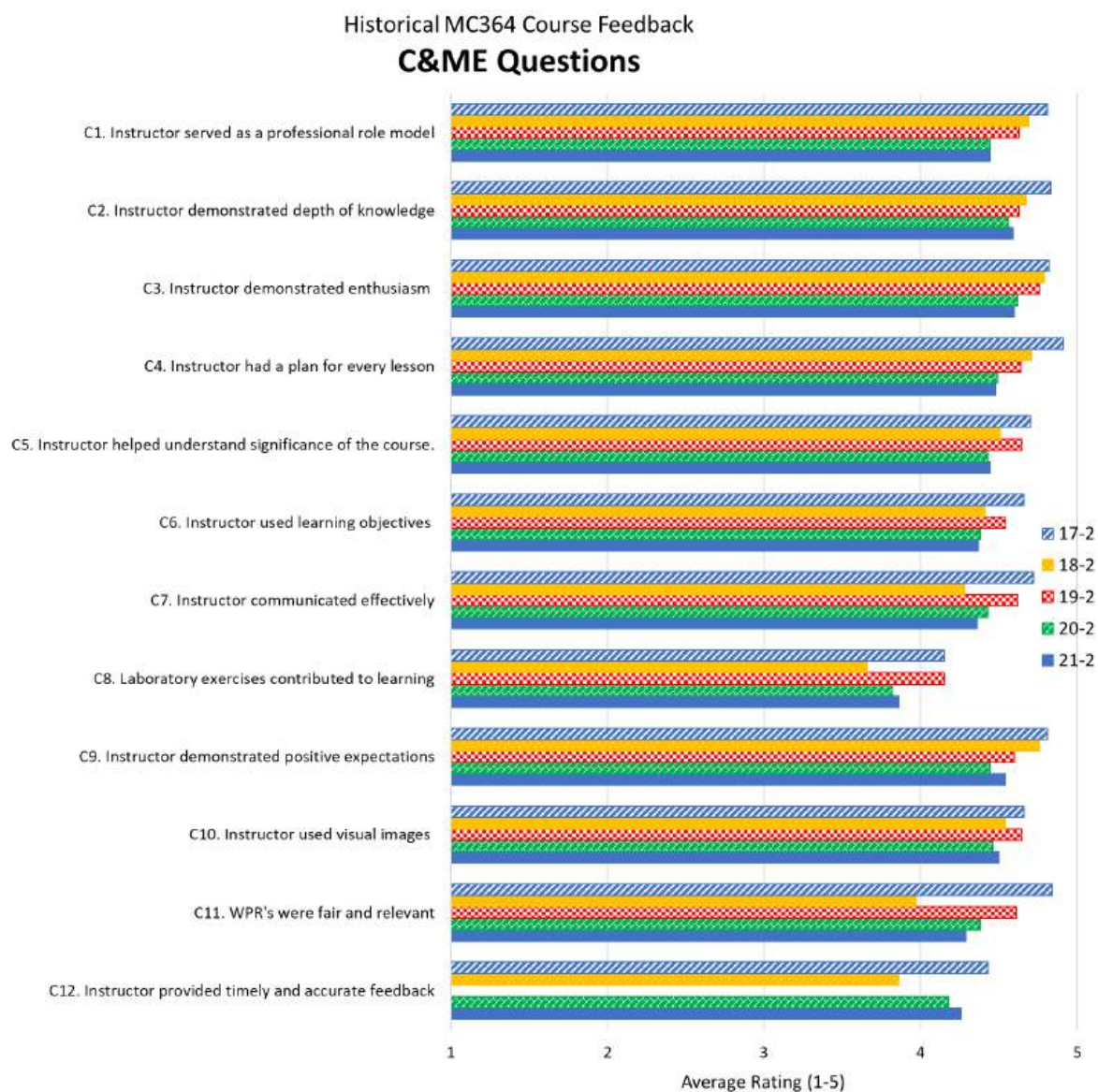
MC364 Course Assessment Package, AY 21

ENCL 2. AY21-2 Term Course Feedback on USMA Questions



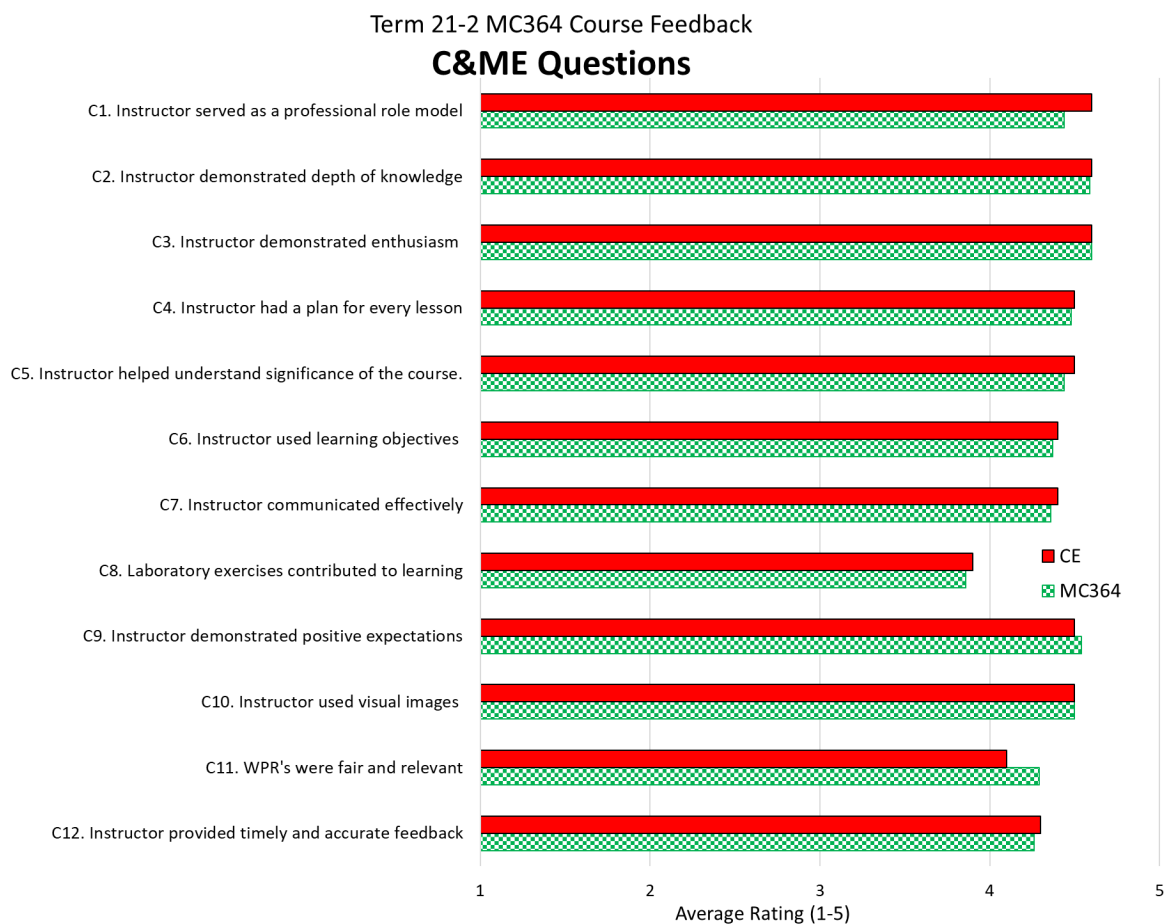
MC364 Course Assessment Package, AY 21

ENCL 3. Historical Course Feedback on C&ME Questions



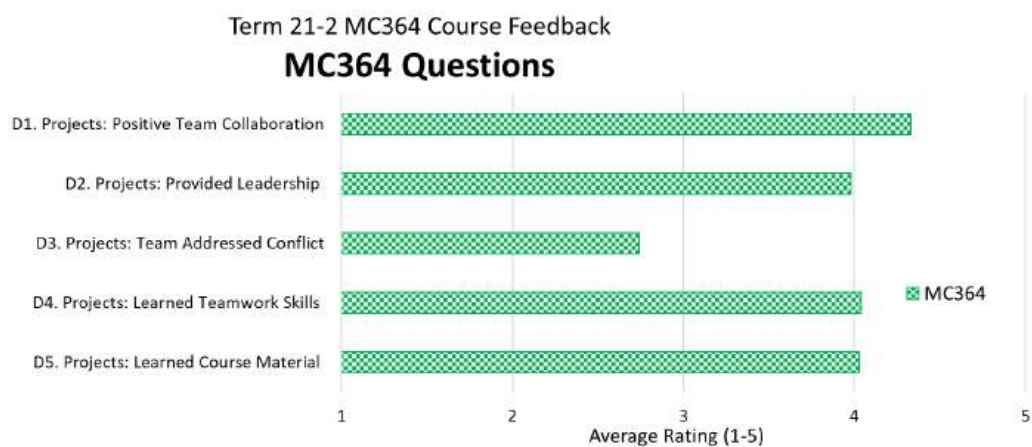
MC364 Course Assessment Package, AY 21

ENCL 4. AY21-2 Term Course Feedback on C&ME Questions

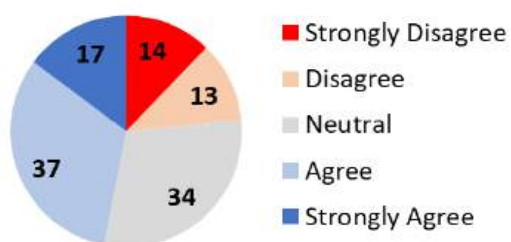


MC364 Course Assessment Package, AY 21

ENCL 5. AY21-2 Term Course Feedback on MC364 Questions



Do you believe the flipped class format was effective in helping you learn the course content?



MC364 Course Assessment Package, AY 21

ENCL 6. AY21-2 Qualitative Course-End Feedback Responses

1. Additional comments about the course

- I like the reading quizzes, but wish they were done differently, like **multiple choice, fill in the blank, etc.**, since having only 8 minutes for a problem that was new material made it difficult to complete without simple errors.
- I **enjoyed being able to have videos posted online** that I could preview and try to understand so that when I was in class I could apply it at board problems, or ask questions when I realized I didn't understand what was going on with some questions.
- The instructors and content produced by the instructors were very useful and helpful and enabled us to be successful in the course.
- The reading quizzes, while difficult, did force the **student to be responsible for his own learning.**
- This course helped me **increase my critical thinking skills**
- **More group problems** should be worked in class rather than just sending students to the boards to figure out the problems. I think this would allow students to see problems done properly and then do it again on the board. Yes, the lesson videos serve to run problems but students do not always have time to watch all the videos in depth.
- I don't like the flipped class room at all. I want to be taught the material **in person.** and I need to watch someone go through a problem in front of me in order to understand.
- Labs were useful, but I **hated** doing write-ups.
- I really appreciated the ability to take boards much more in this course. Last semester, in MC300, I would **fall asleep** in the class since there was little participation and I was not able to engage in the learning. With this semester, I thought the class was laid out perfectly in the course videos and then almost immediately taking boards.
- **This was a great course that took slow steps in getting understanding the big question of, Will it fail? Many of my classes did not revolve around a single concept and seemed to be distant. However, MC364 was a course that brought it full circle in understanding what it was that I was solving for.**
- **I feel that if we were allowed to have notes on the reading quizzes more people would watch and take notes on the videos.**
- Great follow up to MC300. The videos are extremely helpful and because of them I **barely had to open my textbook.**
- The labs were quite a bit of work with what felt like **no turnaround time.** In other words, we only had a few days from when we conducted the labs to when we had to turn the labs in.
- Flipping the class was not helpful. It would be more beneficial to do practice problems together as a class with the instructor taking the lead. This would allow students to take notes and have something to refer to after class. The **lack of notes I have from this class was not helpful at all.**
- I liked reading quizzes, but do **miss the traditional homework** style.
- Well organized course. The **video lessons are fantastic,** and being able to work through problems in class is ideal.
- I think the quizzes should be replaced by some other graded event. Spot knowledge doesn't actually ensure that we internalize the information.
- The material should be gone over in class and not just expected of cadets to know. Lecture-style teaching is much more effective for me (personally) than spending class time on practice problems.
- The flipped classroom made this course much more difficult. After a year of remote classes because of COVID, it would have been nice to get **in person** instruction rather than just doing example problems that we could do on our own.
- I would alter the **time constraint** on the lesson reading quizzes. Oftentimes, I felt I knew how to solve the problem I just did not have enough time to effectively do it or demonstrate I knew how to at least.
- The **flipped classroom is a terrible teaching strategy.** The efficacy of teachers comes from their ability to take complex topics from the textbook and explain them in easily understandable ways. By giving cadets videos, you lose this benefit and force them to grapple with complex concepts with inadequate explanation: the videos are not the same as an in-person lesson. I would recommend following a similar approach to MC300 in which cadets are given homework due before each lesson rather than reading quizzes

MC364 Course Assessment Package, AY 21

2. Do you believe the reading quizzes effectively assessed your learning of the course content? Why or why not?

- 58 – Yes, 35 – No
- absolutely! as stated above: the quizzes pushed to me really comprehend the material and were only 10 mins instead of an hour of homework.
- Although they were on the material learned from the videos, I do not believe they accurately assessed my learning of the course because of time constraints as well as we could not fully master the lesson the night before and take a quiz on it first thing in class.
- I do not. For me being able to sit down and digest the material with a homework assignment benefited me the most. The reading quizzes produced quite a bit of stress as well.
- I personally think I get more out of the homeworks and problem sets similar to what we did in MC300. Towards the end when motivation was slow I started to not prepare as much for reading quizzes but I know with homeworks I was always attentive.
- I think that they were not effective as it had not prepared me for learning and was instead a stress induced assignment that failure came as a time restraint rather than knowledge on the subject.
- I would say 60% of the reading quizzes effectively assessed my learning of the course content
- No because I felt rushed to try and repeat something I watched an example of one time and I don't have the time to rewatch until I understand because of other work
- No as they were labeled as being about the content we learned or watched videos of the night prior and sometimes they were, other times they were based on something we had learned the week or two prior.
- No because it is easy to make small mistakes that add up when on a time crunch and it does not really test the material well
- No. They were hard to study for because sometimes they directly related to the videos we needed to watch for class and other times they related to material from the prior lesson. If the prior lesson were on a Friday and the quiz was on a Monday, it was pretty ineffective and harder than necessary.
- The quizzes served the purpose well of forcing students to learn the course content, but I believe that a blend of some homework and quizzes would be more effective.
- The time limits for reading quizzes, at times, hinder my ability to successfully apply new content learned. More allotted time would allow me to be warier of careless mistakes.
- They did--but I wish they were more consistent in what they evaluated. Some were heavily weighted towards the last class' material, while others focused way more on the videos from the night before. It was hard to target my preparation, as I typically only have the time to prioritize one class of content.
- yes they did because it gave me a lot of practice in a testing environment under a good time constraint every day.
- Yes, because I believe that they were quick checks on learning on the previous/current lesson that fairly assessed if we watched the video.
- Yes, because they were mostly easy; I should have just prepared myself better.
- Yes, but the flipped class required you to know the material before you actually knew what you were doing. I believe the quizzes should be more focused on the previous lesson material than the current material.
- Yes, they served the purpose of problem sets but without the lumped time cost and were comparatively low risk. They could have been longer though.
- Yes, they were relevant to the lesson material and easy to do well on if you did the lesson prep.
- Yes, this kept cadets accountable for their learning and encouraged them to do the reading the night prior.
- Yes, unfortunately they did. When I prepared they were easy and when I did not, they were not.
- Yes. I do prefer the pre class homeworks from MC300 over the quizzes.
- Yes. It is obvious if I didn't do the reading, since I didn't do well on the quiz the next day.

MC364 Course Assessment Package, AY 21

3. Do you believe working problems at the boards in class was effectively in helping you learn how to solve problems in the course? Why or why not?
- 84 – yes, 8 no
 - Board problems were much more effective than they have been in any of my previous classes, most likely due to my **instructor's active role** in answering questions and making sure cadets were on the right path to the correct answer.
 - **Definitely, although I personally like a mixture of working problems at the boards and at my desk so I can walk away with more complete notes. I fixed this partially by just taking pictures of my board work and going over these problems later.**
 - I personally thought class went best when the teacher did **one review problem** before we went to boards for the harder material.
 - I think it would be better to make **board work optional**, especially for students who prefer to do work at their desks. I believe it is also easier to organize work when done at a desk rather than a board.
 - I think that they are very effective because I feel like I learn the best when I actually solve the problems as opposed to seeing how to do it.
 - No, i could not take notes and had **nothing to refer to** after class, the same thing can be accomplished by letting students do problems at their desks
 - No. There was not enough instructor student interaction for questions. Also, doing work on the boards with a partner allowed for one person to take over and they would only be asking questions or getting feedback and the **rest of the group was quiet.**
 - Yes, but I prefer to just do them in a notebook at my desk. Working problems is great but when we do them at the boards I don't get to review the process I used which is not always the same as the one used in the posted solutions.
 - Working board problem and especially receiving feedback in real time from my instructor and classmates made understanding how to solve problems much easier. Doing problems at the board made learning easier because corrections can be made quickly and peers can analyze each others' work.
 - Working board problems was the most effective way for me to learn in class, because it allowed me to work out problems and also get **immediate feedback** from my instructor.
 - Yes! Going through problems on the board is a great environment to work through the concepts and having students/instructor present to help was the ideal situation.
 - **Yes! Throw me into the deep end and I will learn how to swim! Especially, when the lifeguard is walking around.**
 - Yes, although I preferred to work in my notebook, but my instructor was fine with that.
 - Yes, but I wish we worked **more collaboratively.**
 - Yes, but the instructor should work a problem at the beginning of class first.
 - yes, especially when the teacher explained to us where we went wrong
 - Yes, if nothing else, this allowed me to **stay awake** and get something out of class. Additionally, it was good to be able to see others in this class either figure it out or get stuck somewhere. Sometimes being able to identify where others were struggling and helping them through that can be a better teacher than just learning how to do the problem.
 - Yes, the only downside is that I have to erase my work after.
 - Yes. It allowed my teacher to see what areas I was struggling in and work with my classmates.
 - Yes. This is the most effective technique and way better than watching the teacher solve problems at the board.
 - Yes; Allowed me to work with others and receive immediate feedback and help from the instructor and other students when needed

MC364 Course Assessment Package, AY 21

4. Do you believe you were able to effectively learn the course content without assigned homework? Why or why not?

- RQ – 55, HW – 31
- I learned this course really well from the board problems.
- Yes, because I knew I was expected to complete the review videos in preparation for the reading quizzes, and so I focused on learning the material rather than just finding ways to barely finish homework without really learning.
- I think I was, but I believe that assigned homework like MC300 had can be very effective in increasing confidence in the course material because it provides a way to practice the material before class.
- Yes because there were opportunities to learn and apply the course content during reading quizzes and board work. This was also helpful considering there is a high demand for completing work and learning the content in other courses, especially in other CME courses.
- I would have preferred homework to the reading quizzes. I felt like I often settled for mediocrity on the reading quizzes because I knew I didn't have a full understanding of the material. This hurt my grade.
- Yes. Especially for someone with a stronger background in MC300, it is not a hard course to learn without homework. The instructor videos make both the theoretical/scientific background and the problem solving application of the course material very easy to understand.
- Some content should have been homework, as i needed some reps with my textbook in hand before i could do a quiz
- I would have preferred homework to force myself to work on problems outside of class.
- I would prefer assigned homework because it gives me more of an incentive to schedule AI and better understand the material on each homework. I think that would lead to a higher understanding of the course for me.
- Yes, but honestly I would be in favor of more required problems (homework, problem sets, etc.) outside of class since they helped me a lot in MC300 last semester.
- Yes, I think the video problems into the reading quizzes were able to effectively capture my understanding of the lesson. I was able work through any areas of confusion through board work.
- Yes because I was able to be independent and focus on the videos. This lead to me to actually be able to focus on the videos.
- N o. The homeworks motivate me to work to get the correct answer before I turn it in. The reading quizzes do not, and I often end up turning in a reading quiz without fully understanding what's on it (which was not usually true of homework in MC300).
- No, as I stated in the previous question, I preferred the MC300 model because it incentivized me to take my time to sit down and practice the material and read the textbook more versus working in groups the last 30 minutes of class to solve one problem. I struggled in my learning in this class with the video/flipped format because I felt constrained or limited to one or two 15 minute lessons to absorb all the material before taking a RQ at the beginning of class. Last semester in MC300 (and even now) I think I had a better grasp on that material because I could watch my instructor go through a problem live with the opportunity to ask questions and then go to boards to solve more together. I thought with this being my only in-person class I would not need to spend as much time on the computer but I found I spent as much time if not more preparing for class than my fully virtual classes. I understand we live in a COVID environment but I have found my ability to focus and retain material has greatly decreased with constant use of a screen to communicate and learn all day. My recommendation for the future is to prioritize in-person learning/activities and to utilize green engineering paper for assignments that more closely resemble the concepts seen on the WPR's.
- No, the homework method as experienced in MC300 seemed to be much more effective. Homework allowed for me to do problems on my own and then ask questions in class where I got stuck. Watching

MC364 Course Assessment Package, AY 21

- videos before class did not allow me to fully understand the concepts and took too much time to be able to watch all of the videos and complete problems on my own.
- Yes, but half homework and half reading quizzes would be good.
 - Absolutely. The daily quizzes were plenty of motivation to actually prepare for class. The videos were perfect for learning the material, the quizzes were motivation to learn on our own, and class time was practice problems and clarification from instructor. Brilliant organization.
 - The quizzes served the purpose well of forcing students to learn the course content, but I believe homework would have been more effective in that cadets have to actively attempt the problems before class as opposed to just following along.
 - Actually more so! When I would do _homework_ before the class period, I had a hard time focusing because if I already succeeded it felt like a waste of time due to me struggling with a concept for an extra 2 hours. A quiz was something I studied for and on the spot addressed what I didn't understand.
 - Yes, I believe I was able to effectively learn but I also think assigned homeworks would actually increase my knowledge as I would have spent more time preparing for each lesson.
 - I feel I learned all of the concepts fairly well. The reading quizzes were motivation to take notes and write example problems, and working in class meant that I could ask questions if necessary rather than waiting until the next day.



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION II

EXAMINATIONS



TAB F – WPR1

DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

WRITTEN PARTIAL REVIEW I
CE364, 26-2

NAME: _____ SECTION _____

Left Section Room at: _____ (to be completed for early departure only)

Special Instructions:

1. Begin work when directed. You will have 75 minutes to complete the WPR
2. Write your name at the top of each page. Ensure you have a complete test (9 pages double sided, including this title page.)
3. This test is closed book and closed notes except for the CE364 reference document. Fundamentals of Engineering Exam approved calculators are authorized.
4. No cell phones or smart watches permitted in the classroom during the exam. Cadets may not collaborate or communicate with anybody else while taking the WPR.
5. You may use extra sheets of paper if you run out of room on the WPR. When completed, staple all pages together *in order*, including any additional sheets, and place the exam inside the folder on the exam proctor's desk in the hallway.
6. You are not authorized to discuss the content, structure, or any other information about the WPR until the WPR has been released from academic security.
7. SHOW ALL OF YOUR WORK! **Work Quickly but Legibly**

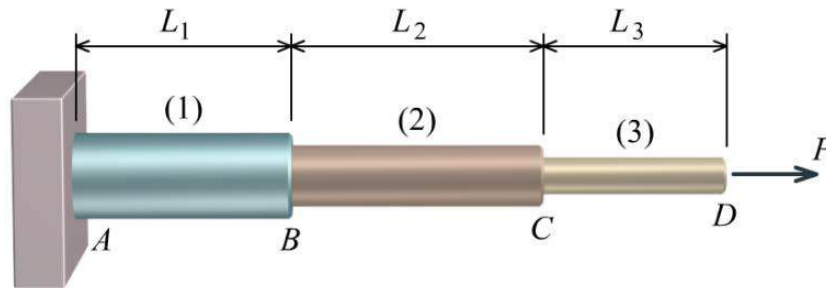
Question/Section	Points	Score
1. Axial Loading	50	
2. Buckling	25	
3. Indeterminate Axial Members	75	
4. Thermal Effects	50	
5. Torsion	50	
Total		

"I did not use any sources, nor did I receive any assistance while completing this WPR. I will not discuss this WPR with anyone except a CE364 instructor until it is released from academic security on **10FEB26**"

Signature: _____ Date Signed: _____

1. Three solid cylindrical rods are welded together to form the compound axial member shown in the figure below. The compound axial member is attached to a fixed support at A. Each rod has an elastic modulus of $E = 90 \text{ GPa}$. Use the below values for the rod lengths and areas.

Member	Length (mm)	Area (mm^2)
1	1660	275
2	1700	150
3	1150	70

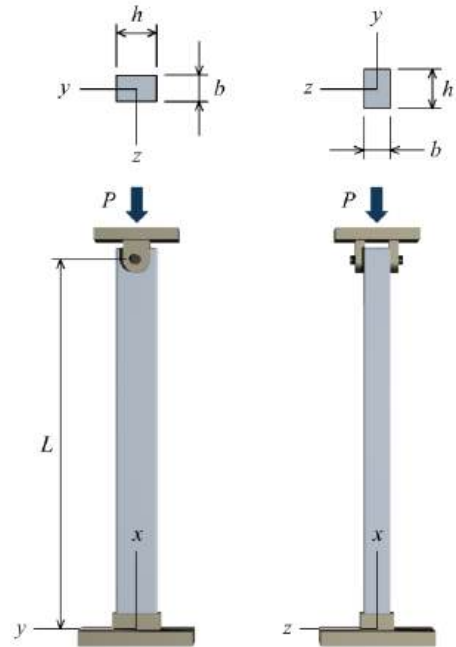


- a. **Calculate** the magnitude of load P required to displace end D (with respect to the fixed point at A) a distance of 75 mm to the right. **[35 pts]**

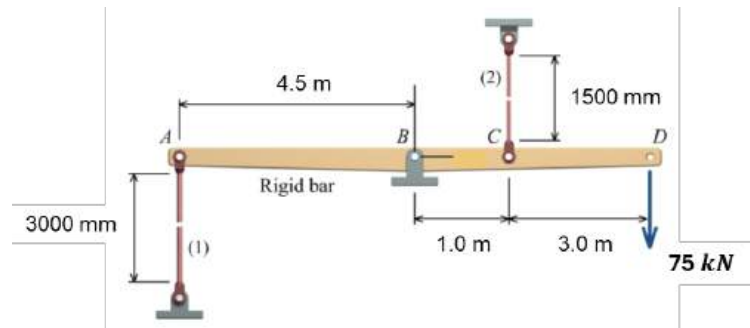
- b. **Calculate** the displacement of point C with respect to point B if the applied load P is 120 kN. **[15 pts]**

2. The column shown in the figure is made of 2014-T6 aluminum [$E = 10600 \text{ ksi}$] with a rectangular cross section and supports an axial load of P . The base of the column is fixed. The support at the top allows rotation of the column in the x - y plane (i.e., bending about the strong axis) but prevents rotation in the x - z plane (i.e., bending about the weak axis).

Determine the critical buckling load in the **weak direction** of the column for the following parameters: $L = 50 \text{ in}$, $b = 0.5 \text{ in}$, and $h = 0.875 \text{ in}$. You do not need to check the strong direction or the elastic assumption for Euler's buckling equation. **[25 pts]**



3. Rigid beam ABCD supports a load at point D by two axial members as shown. Both axial members are made of A992 steel ($E = 200 \text{ GPa}$). Assume Bars (1) and (2) do not buckle. Use the values in the table below for length and area.



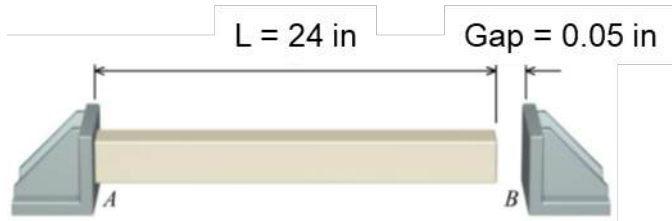
Member	Length (mm)	Area (mm ²)
1	3000	2850
2	1500	4420

- a. Solve for the force in Bar (2). Show all your work. [75 pts]

Continue work and place your final answer on the next page

$F_2 =$ _____

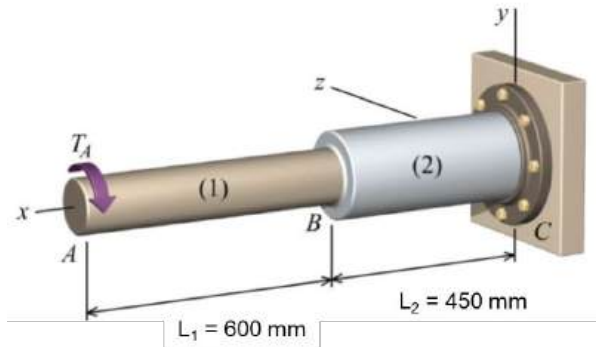
4. A block of C86100 bronze, with material properties listed below, is fixed at point A and has a gap of 0.05 in between the right end of the member and point B. The block has a length of $L = 24$ in and area of $A = 8.5$ in²



$$E = 15000 \text{ ksi} \quad \alpha_T = 9.6 \cdot 10^{-6} / ^\circ\text{F} \quad \sigma_y = 50 \text{ ksi}$$

- a. **Calculate** the change in temperature (ΔT) required to cause the block to reach its yield stress. **[25 pts]**
- b. **Solve** for the thermal deformation (δ_T) and thermal strain (ϵ_T) in the post if the change in temperature is equal to $\Delta T = +150$ °F **[20 pts]**
- c. If the deformation of the block is equal to 0.04 in, what is the thermal stress (σ_T)? **[5 pts]**

5. A solid compound shaft is made of two materials. Segment (1) is made of C83400 Red Brass and Segment (2) is made from 6061-T6 Aluminum, with material properties listed in the table below. The shaft is fixed at point C and is subjected to a torque of $T = 550 \text{ N} \cdot \text{m}$ at point A.



Member	Shear Modulus (GPa)	Length (mm)	Diameter (mm)
(1)	37	600	50
(2)	26	450	75

- a. **Determine** the maximum shear stress experienced by the compound shaft. [20 pts]

$\tau_{\max} = \underline{\hspace{2cm}}$

- b. **Draw** the shear stress distribution on the face of the cross-section on the image below. Clearly indicate the location of maximum shear stress found from part (a). [10 pts]



- c. **Calculate** the rotation angle of point A with respect to point C in **degrees**. [20 pts]

$\Phi_{A|C} =$ _____

Instructor Solution

DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

WRITTEN PARTIAL REVIEW I
CE364, 26-2

NAME: SOLUTION SECTION _____

Left Section Room at: _____ (to be completed for early departure only)

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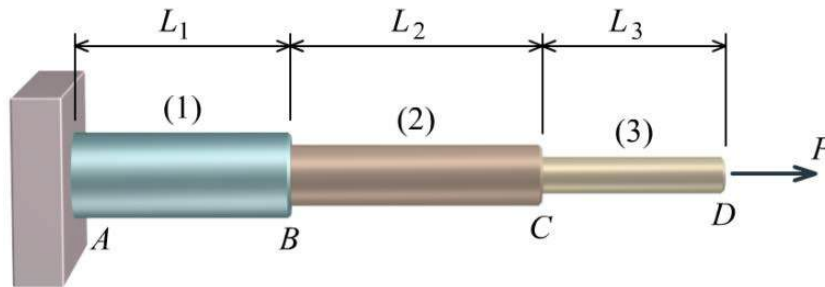
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Signature: _____ Date Signed: _____

1. Three solid cylindrical rods are welded together to form the compound axial member shown in the figure below. The compound axial member is attached to a fixed support at A. Each rod has an elastic modulus of $E = 90 \text{ GPa}$. Use the below values for the rod lengths and areas.

Member	Length (mm)	Area (mm^2)
1	1660	275
2	1700	150
3	1150	70



- a. **Calculate** the magnitude of load P required to displace end D (with respect to the fixed point at A) a distance of 75 mm to the right. [35 pts]

$$\delta_{D/A} = \frac{P L_1}{A_1 E} + \frac{P L_2}{A_2 E} + \frac{P L_3}{A_3 E}$$

$$\delta_{D/A} = \frac{P}{E} \left(\frac{L_1}{A_1} + \frac{L_2}{A_2} + \frac{L_3}{A_3} \right)$$

$$75 \text{ mm} = \frac{P}{90 \text{ kN/mm}^2} \left[\frac{1660 \text{ mm}}{275 \text{ mm}^2} + \frac{1700 \text{ mm}}{150 \text{ mm}^2} + \frac{1150 \text{ mm}}{70 \text{ mm}^2} \right]$$

$$75 \text{ mm} = \frac{P}{90 \text{ kN/mm}^2} \left(33.8 \frac{\text{mm}}{\text{mm}^2} \right)$$

$$\therefore \underline{\underline{P = 200 \text{ kN}}} \quad \text{ANS}$$

- b. **Calculate** the displacement of point C with respect to point B if the applied load P is 120 kN. [15 pts]

(10)

$$\delta_{C/B} = \frac{F_{BC} \cdot L_{BC}}{A_{BC} \cdot E_{BC}} = \frac{(120 \text{ kN})(1700 \text{ mm})}{(150 \text{ mm}^2)(90 \text{ kN/mm}^2)}$$

$$\delta_{C/B} = 15.1 \text{ mm} \rightarrow$$

ANS

-5 IF USED AB OR CD

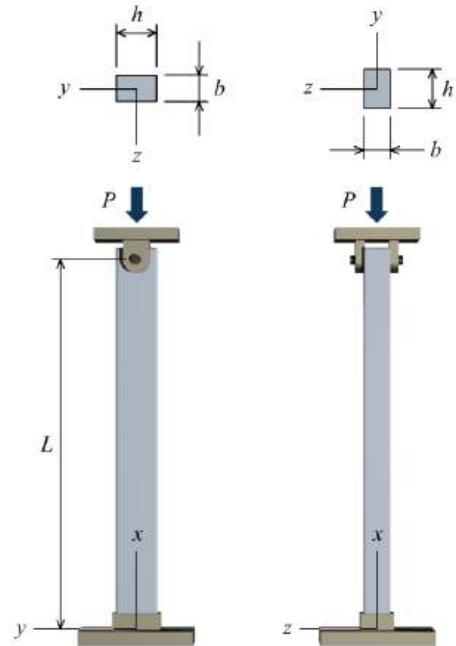
① UNITS

① DIRECTION

① SIG FIGS

2. The column shown in the figure is made of 2014-T6 aluminum [$E = 10600 \text{ ksi}$] with a rectangular cross section and supports an axial load of P . The base of the column is fixed. The support at the top allows rotation of the column in the x - y plane (i.e., bending about the strong axis) but prevents rotation in the x - z plane (i.e., bending about the weak axis).

Determine the critical buckling load in the **weak direction** of the column for the following parameters: $L = 50 \text{ in}$, $b = 0.5 \text{ in}$, and $h = 0.875 \text{ in}$. You do not need to check the strong direction or the elastic assumption for Euler's buckling equation. [25 pts]



$$\textcircled{5} \quad I_y = \frac{hb^3}{12} = \frac{(0.875 \text{ in})(0.5 \text{ in})^3}{12}$$

$$\underline{I_y = 0.009115 \text{ in}^4}$$

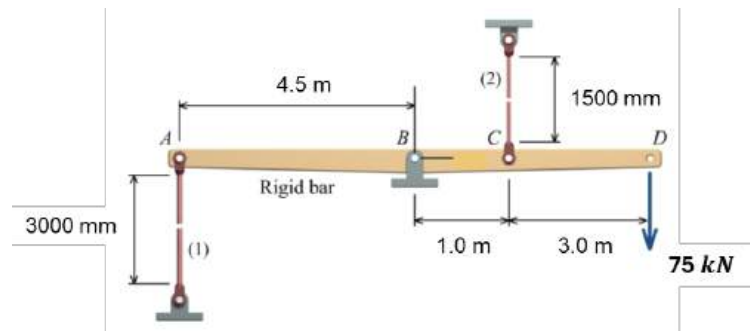
$$\textcircled{5} \quad k = 0.5 \quad (\text{FIX-FIX IN WEAK DIRECTION})$$

$$\textcircled{10} \quad P_{\text{cr},y} = \frac{\pi^2 EI_y}{(kL)^2} = \frac{\pi^2 (10600 \text{ ksi})(0.009115 \text{ in}^4)}{(0.5 \cdot 50 \text{ in})^2}$$

$$\underline{P_{\text{cr},y} = 1.53 \text{ k}} \quad \textcircled{5}$$

ANS

3. Rigid beam ABCD supports a load at point D by two axial members as shown. Both axial members are made of A992 steel ($E = 200 \text{ GPa}$). Assume Bars (1) and (2) do not buckle. Use the values in the table below for length and area.

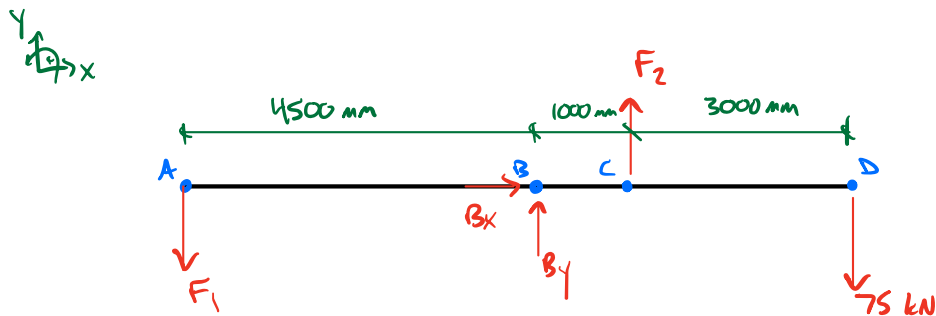


Member	Length (mm)	Area (mm^2)
1	3000	2850
2	1500	4420

- a. Solve for the force in Bar (2). Show all your work. [75 pts]

20

1) EQUILIBRIUM

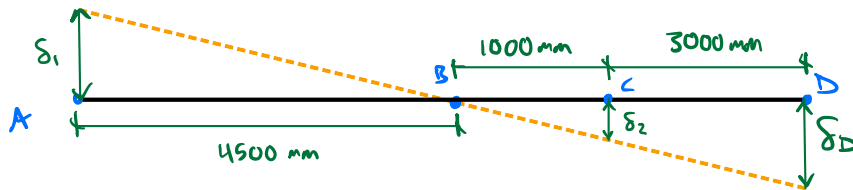


$$\sum M_B = 0$$

$$0 = F_1 (4500 \text{ mm}) + F_2 (1000 \text{ mm}) - 75 \text{ kN} (4000 \text{ mm})$$

$$\textcircled{I} \quad 4500 \text{ mm } F_1 + 1000 \text{ mm } F_2 = 300000 \text{ kN}\cdot\text{mm}$$

Continue work and place your final answer on the next page

20) 2) COMPATIBILITY

$$\frac{\delta_1}{4500 \text{ mm}} = \frac{\delta_2}{1000 \text{ mm}} \quad \therefore \delta_1 = 4.5 \delta_2$$

20) 3) LOAD-DISPLACEMENT RELATIONSHIP

$$\delta_1 = \frac{F_1 L_1}{A_1 E_1}$$

$$\delta_2 = \frac{F_2 L_2}{A_2 E_2}$$

$$\frac{F_1 L_1}{A_1 E_1} = \frac{4.5 F_2 L_2}{A_2 E_2}$$

$$\therefore \frac{F_1 (3000 \text{ mm})}{(2850 \text{ mm}^2)} = \frac{4.5 F_2 (1500 \text{ mm})}{(4420 \text{ mm}^2)}$$

$$\textcircled{\text{II}} \quad 1.053 F_1 - 1.527 F_2 = 0$$

10) 4) COMBINE & SOLVE

$$\left. \begin{array}{l} \textcircled{\text{I}} \quad 4500 \text{ mm } F_1 + 1000 \text{ mm } F_2 = 300000 \text{ kN}\cdot\text{mm} \\ \textcircled{\text{II}} \quad 1.053 F_1 - 1.527 F_2 = 0 \end{array} \right\} \text{2x2 SYS-SOLV}$$

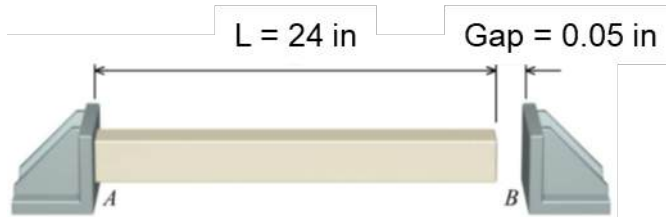
$$F_1 = 57.8 \text{ kN (T)}$$

$$F_2 = 39.9 \text{ kN (T)}$$

5

$$F_2 = \underline{39.9 \text{ kN (T)}}$$

4. A block of C86100 bronze, with material properties listed below, is fixed at point A and has a gap of 0.05 in between the right end of the member and point B. The block has a length of $L = 24$ in and area of $A = 8.5$ in²



$$E = 15000 \text{ ksi}$$

$$\alpha_T = 9.6 \cdot 10^{-6} / ^\circ\text{F}$$

$$\sigma_y = 50 \text{ ksi}$$

- a. **Calculate** the change in temperature (ΔT) required to cause the block to reach its yield stress. [25 pts]

$$\delta = \text{GAP}$$

$$\delta_T + \delta_F = \text{GAP}$$

$$\textcircled{5} \quad \alpha \Delta T L + \frac{F L}{A E} = \text{GAP} \textcircled{5}$$

$$\alpha \Delta T L + \sigma_y \frac{L}{E} = \text{GAP}$$

$$\Delta T = \frac{\left(\text{GAP} - \sigma_y \cdot \frac{L}{E} \right)}{\alpha \cdot L}$$

$$\Delta T = \frac{\left(0.05 \text{ in} - \frac{(-50 \text{ ksi}) \cdot 24 \text{ in}}{15000 \text{ ksi}} \right)}{\left(9.6 \times 10^{-6} / ^\circ\text{F} \cdot 24 \text{ in} \right)}$$

-2 IF NOT (-) FOR (+)

$$\textcircled{5} \quad \underline{\underline{\Delta T = 564 ^\circ\text{F}}}$$

-1 UNITS

-1 3 SIGFIGS

- b. **Solve** for the thermal deformation (δ_T) and thermal strain (ϵ_T) in the post if the change in temperature is equal to $\Delta T = +150$ °F [20 pts]

$$\textcircled{10} \quad \delta_T = \alpha \Delta T L = (9.6 \times 10^{-6} / ^\circ\text{F})(150 ^\circ\text{F})(24 \text{ in}) = \underline{\underline{0.0346 \text{ in}}}$$

ANS

$$\textcircled{10} \quad \epsilon_T = \alpha \Delta T = (9.6 \times 10^{-6} / ^\circ\text{F})(150 ^\circ\text{F}) = \underline{\underline{0.00144 \text{ in/in}}}$$

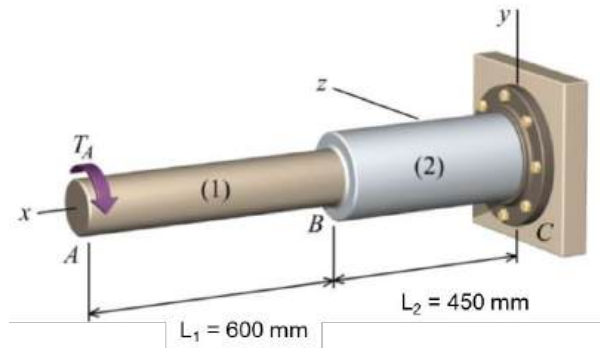
ANS

- c. If the deformation of the block is equal to 0.04 in, what is the thermal stress (σ_T)? [5 pts]

$$\textcircled{5} \quad \underline{\underline{\sigma_T = 0}}$$

ANS

5. A solid compound shaft is made of two materials. Segment (1) is made of C83400 Red Brass and Segment (2) is made from 6061-T6 Aluminum, with material properties listed in the table below. The shaft is fixed at point C and is subjected to a torque of $T = 550 \text{ N} \cdot \text{m}$ at point A.



Member	Shear Modulus (GPa)	Length (mm)	Diameter (mm)
(1)	37	600	50
(2)	26	450	75

- a. **Determine** the maximum shear stress experienced by the compound shaft. [20 pts]

(10)

SEGMENT 1

$$J_1 = \frac{\pi (50 \text{ mm})^4}{32} = 613592 \text{ mm}^4$$

$$\gamma_1 = \frac{(550 \text{ N} \cdot \text{m}) \left(1000 \frac{\text{mm}}{\text{m}}\right) (25 \text{ mm})}{(613592 \text{ mm}^4)}$$

$$\gamma_1 = 22.4 \text{ MPa}$$

(10)

SEGMENT 2

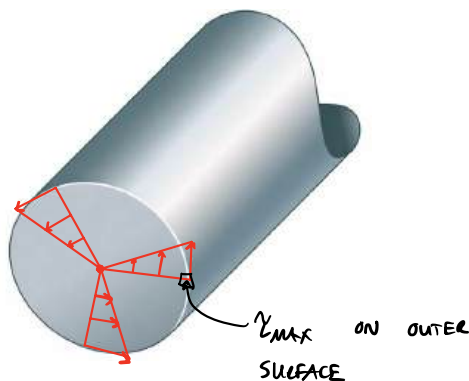
$$J_2 = \frac{\pi (75 \text{ mm})^4}{32} = 3106311 \text{ mm}^4$$

$$\gamma_2 = \frac{(550 \text{ N} \cdot \text{m}) \left(1000 \frac{\text{mm}}{\text{m}}\right) (37.5 \text{ mm})}{(3106311 \text{ mm}^4)}$$

$$\gamma_2 = 6.64 \text{ MPa}$$

$$\tau_{\max} = 22.4 \text{ MPa}$$

- b. **Draw** the shear stress distribution on the face of the cross-section on the image below. Clearly indicate the location of maximum shear stress found from part (a). [10 pts]



-5 IF NOT LINEAR

-5 IF $\gamma_{\max}$ NOT INDICATED / INCORRECT

c. **Calculate** the rotation angle of point A with respect to point C in **degrees**. [20 pts]

$$\phi_{C/A} = \frac{L_1 T_1}{J_1 G_1} + \frac{L_2 T_2}{J_2 G_2}$$

(10) (10)

$$= \frac{(-600 \text{ mm})(550 \text{ N}\cdot\text{m})(1000 \frac{\text{mm}}{\text{m}})}{(613592 \text{ mm}^4)(37 \text{ kN/mm}^2)(1000 \frac{\text{N}}{\text{kN}})} + \frac{(-450 \text{ mm})(550 \text{ N}\cdot\text{m})(1000 \frac{\text{mm}}{\text{m}})}{(3106311 \text{ mm}^4)(26 \text{ kN/mm}^2)(1000 \frac{\text{N}}{\text{kN}})}$$

$$= 0.0145 + 0.00306$$

$$\phi_{C/A} = 0.0176 \text{ RAD} \cdot \frac{180}{\pi} = 1.01^\circ$$

$\Phi_{A/C} = 1.01^\circ \quad \curvearrowright$

-2 IF IN
RADIANS

-1 IF NO
DIRECTION $\curvearrowright$
CCW



TAB G – WPR2

DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

WRITTEN PARTIAL REVIEW II
CE364, 26-2

NAME: _____ SECTION _____

Left Section Room at: _____ (to be completed for early departure only)

Question/Section	Points	Score
Problem 1: Flexural Analysis	100	
Problem 2: Deflections	100	
Problem 3: Combined Loading	50	
Total		

Special Instructions:

1. Begin work when directed. You will have 75 minutes to complete the WPR
2. Write your name at the top of each page. Ensure you have a complete test (8 pages double sided, including this cover sheet.)
3. This test is closed book and closed notes except for the CE364 authorized reference packet. Fundamentals of Engineering Exam-approved calculators are authorized.
4. No cell phones or smart watches permitted in the classroom during the exam. Cadets may not collaborate or communicate with anybody else while taking the WPR.
5. You may use extra sheets of paper if you run out of room on the WPR. When completed, staple all pages together *in order*, including any additional sheets, and place the exam inside the folder on the exam proctor's desk in the hallway.
6. You are not authorized to discuss the content, structure, or any other information about the WPR until the WPR has been released from academic security.
7. **SHOW ALL OF YOUR WORK!** Ensure that you report your final answer to 3 significant figures, with appropriate units, and direction (if applicable)

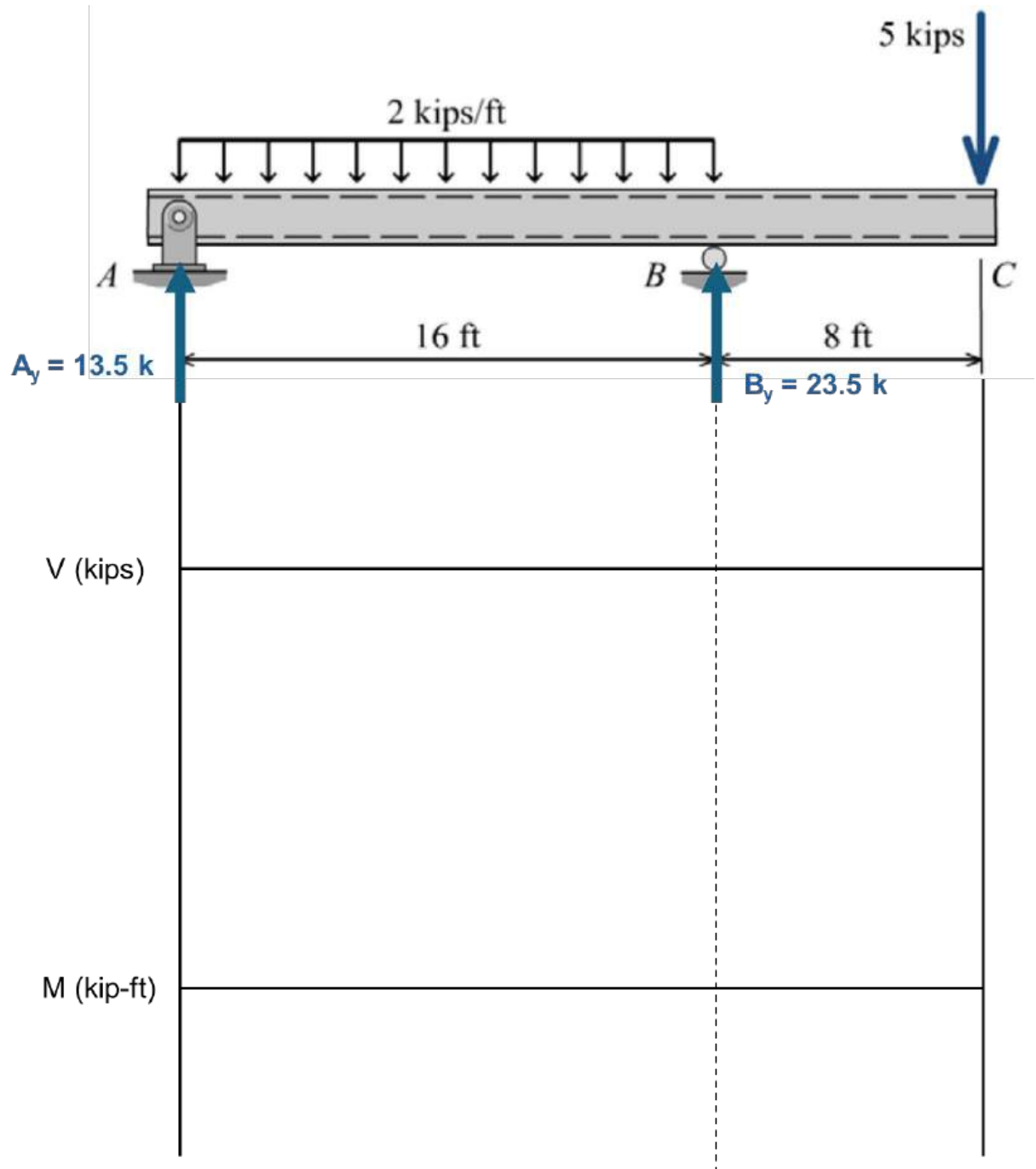
Work Quickly but Legibly

"I did not use any sources, nor did I receive any assistance while completing this WPR. I will not discuss this WPR with anyone except a CE364 instructor until it is released from academic security on _____"

Signature: _____

Problem 1 (100 pts):

- a. Construct a complete shear and moment diagram for the beam shown below. Label and report the locations of the maximum shear force and maximum internal bending moments. Show all work according to V&M Standards. Ensure to clearly label the shape of curves on your diagrams. The reactions are given at point A and point B (30 pts)



$V_{\max} =$ _____ @ $x =$ _____ $M_{\max} =$ _____ @ $x =$ _____

- b. Using a maximum internal moment of $M = 75 \text{ kip} \cdot \text{ft}$ (this is not the answer to part a). Check to see if a W10x45 is adequate with respect to flexural normal stress. Use a yield stress of $\sigma_y = 36 \text{ ksi}$ and a factor of safety of 1.5. Indicate whether the beam is safe or if it fails by filling in the bubble below. You may assume the beam is bending about the strong axis (30 pts)

Flexural Normal Stress:



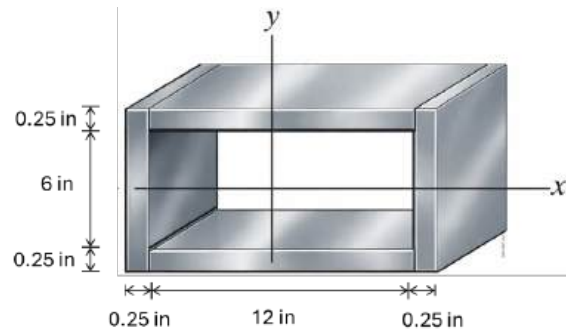
Safe



Fail

- c. Assuming a maximum internal shear force of $V = 25 \text{ kip}$ (this is not the answer to part a), determine the maximum flexural shear stress if you decided to use the HSS section below. (40 pts)

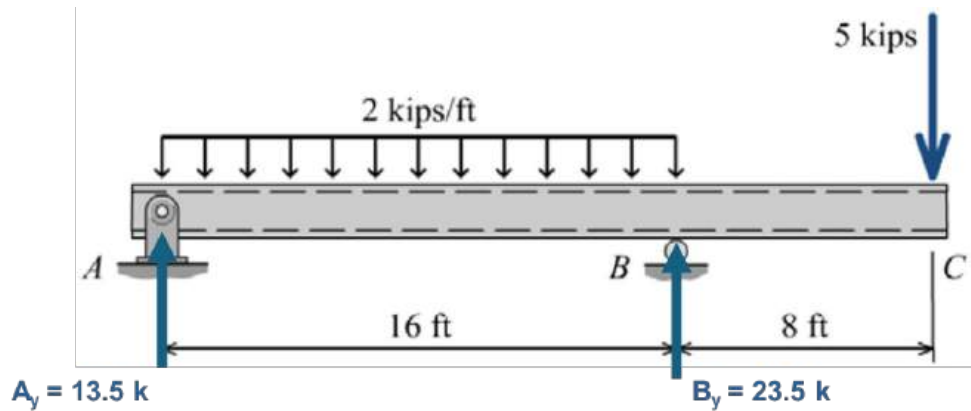
$$I_x = 70.1 \text{ in}^4$$



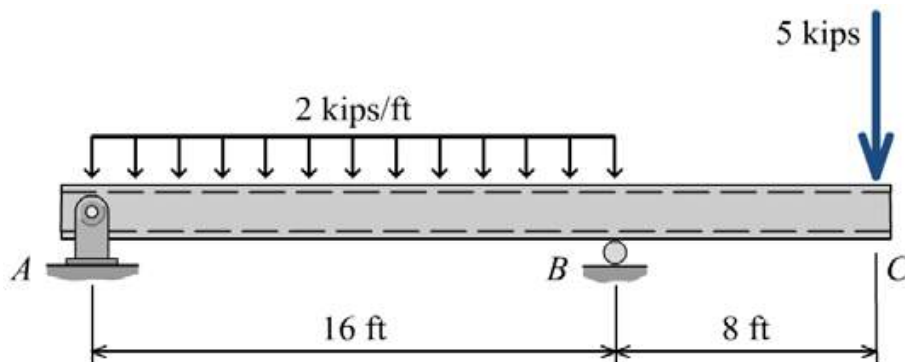
$$\tau_{max} =$$

Problem 2:

The same simply supported beam from problem 1 is shown below. Assume that EI is constant for the beam. Solve for the following: (100 pts)



- a. Draw the displaced shape of the beam ABC due to the applied loading on the picture of the beam below (10 pts):



- b. Identify the four boundary conditions needed to solve for the displacement at point C. Put each answer in the four blocks below (10 pts):

1.	2.
3.	4.

- c. Find the moment function between points A and B as a function of x . Draw a complete FBD with all appropriate labels. Write the moment function in the box below. (20 pts)

Moment function:

- d. Assuming the moment function is $M(x) = -6x^2 + 30x$ (this is not the answer to part c), use the integration method to find the slope and displacement functions between A and B. Solve for the two constants of integration and put the answers in the block below. Assume EI is constant (30 pts)

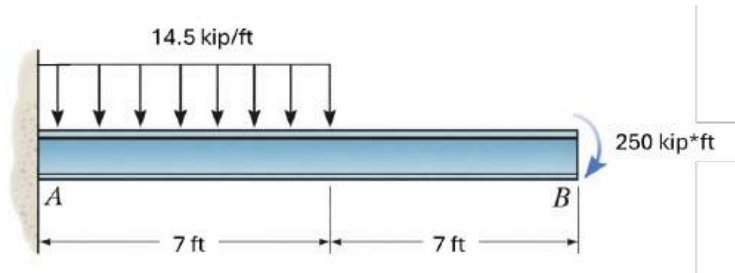
$c_1 =$

$c_2 =$

- e. Given the loading configuration below, determine the displacement at **point B** using the principle of superposition. Use the following properties for E and I (30 pts)

$$E = 29000 \text{ ksi}$$

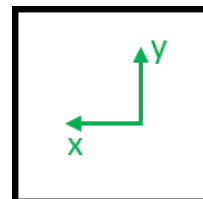
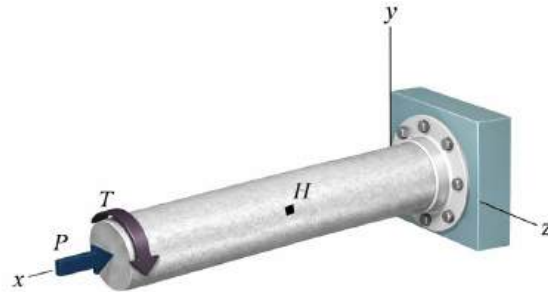
$$I_x = 984 \text{ in}^4$$



$$v_B =$$

Problem 3:

A solid shaft with a diameter of $d = 65 \text{ mm}$ is fixed to the wall and is subjected to a torque of $T = 650 \text{ N} \cdot \text{m}$ and an axial force of $P = 50 \text{ kN}$. Determine the state of stress at point H. Show the stress on the stress element below. Show all of your work! (50 pts)



Instructor Solution

DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

WRITTEN PARTIAL REVIEW II
CE364, 26-2

NAME: SOLUTION SECTION _____

Left Section Room at: _____ (to be completed for early departure only)

Question/Section	Points	Score
Problem 1: Flexural Analysis	100	
Problem 2: Deflections	100	
Problem 3: Combined Loading	50	
Total		

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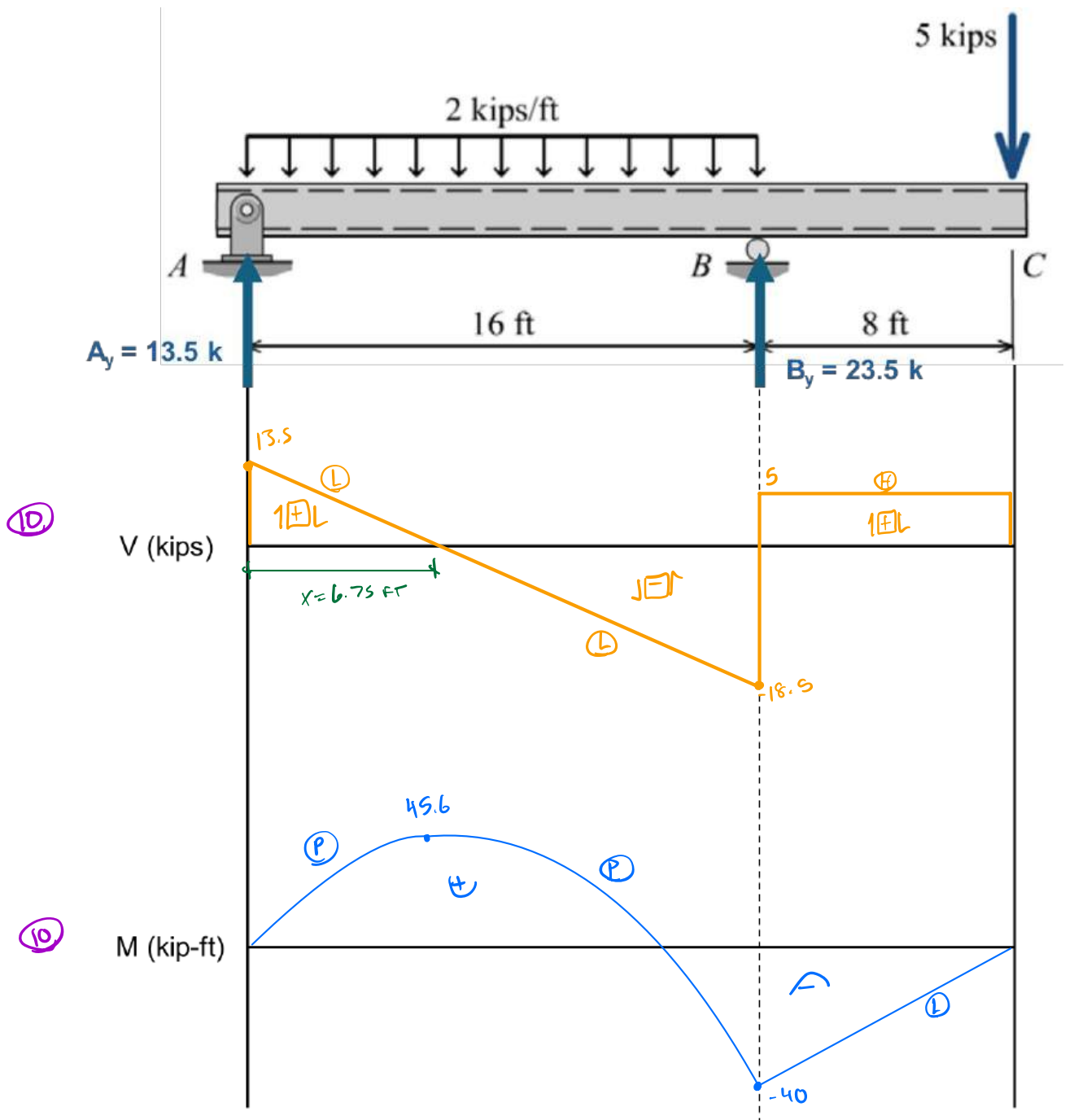
Work Quickly but Legibly

"I did not use any sources, nor did I receive any assistance while completing this WPR. I will not discuss this WPR with anyone except a CE364 instructor until it is released from academic security on _____"

Signature: _____

Problem 1 (100 pts):

- a. Construct a complete shear and moment diagram for the beam shown below. Label and report the locations of the maximum shear force and maximum internal bending moments. Show all work according to V&M Standards. Ensure to clearly label the shape of curves on your diagrams. The reactions are given at point A and point B (30 pts)



$$V_{\max} = 18.5 \text{ k} @ x = 16 \text{ ft}$$

$$M_{\max} = 45.6 \text{ k-ft} @ x = 6.75 \text{ ft}$$

- b. Using a maximum internal moment of $M = 75 \text{ kip} \cdot \text{ft}$ (this is not the answer to part a). Check to see if a W10x45 is adequate with respect to flexural normal stress. Use a yield stress of $\sigma_y = 36 \text{ ksi}$ and a factor of safety of 1.5. Indicate whether the beam is safe or if it fails by filling in the bubble below. (30 pts)

FROM F.E.R.M.

$$I_x = 248 \text{ in}^4$$

$$S_x = 49.1 \text{ in}^3$$

(5)

$$\sigma_{ACT} \leq \sigma_{ALL}$$

$$\frac{M}{S_x} \leq \frac{\sigma_y}{F.S.}$$

$$\frac{(75 \text{ k} \cdot \text{ft})(12 \text{ in/ft})}{49.1 \text{ in}^3} \leq \frac{36 \text{ ksi}}{1.5}$$

$$\underline{18.3 \text{ ksi}} \leq \underline{24 \text{ ksi}}$$

✓ SAFE

(5)

Flexural Normal Stress:



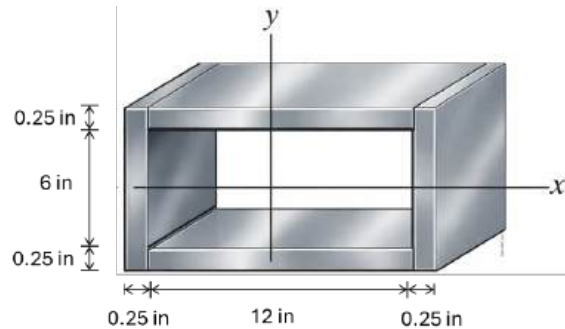
Safe



Fail

- c. Assuming a maximum internal shear force of $V = 25 \text{ kip}$ (this is not the answer to part a), determine the maximum flexural shear stress if you decided to use the HSS section below. (40 pts)

$$I_x = 70.1 \text{ in}^4$$



$$\tau_{max} = \frac{V \cdot Q}{I_x \cdot t}$$

$$\textcircled{20} \quad Q = Q_{out} - Q_{in} = [(3.25 \text{ in})(12.5 \text{ in})(1.625 \text{ in})] - [(3 \text{ in})(12 \text{ in})(1.5 \text{ in})]$$

$$Q = 12.02 \text{ in}^3$$

$$\textcircled{5} \quad t = 0.5 \text{ in} \quad (\text{TWO WALL THICKNESSES})$$

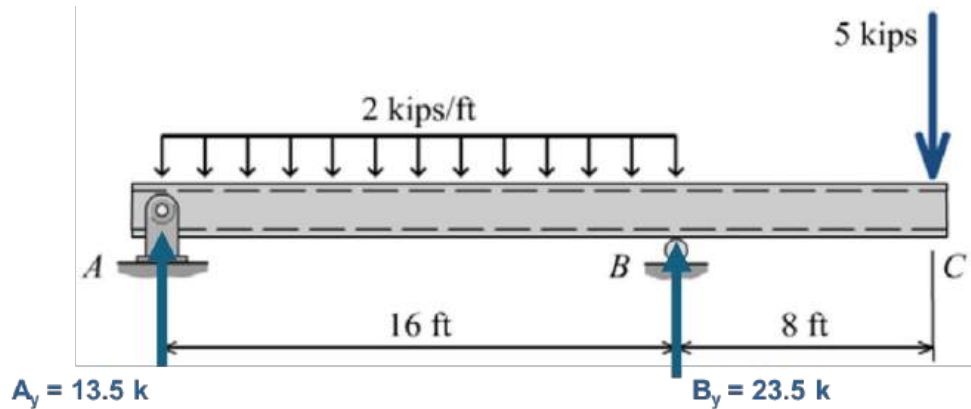
$$\textcircled{10} \quad \tau_{max} = \frac{(25 \text{ kip})(12.02 \text{ in}^3)}{(70.1 \text{ in}^4)(0.5 \text{ in})} = 8.57 \text{ ksi}$$

$\textcircled{5}$

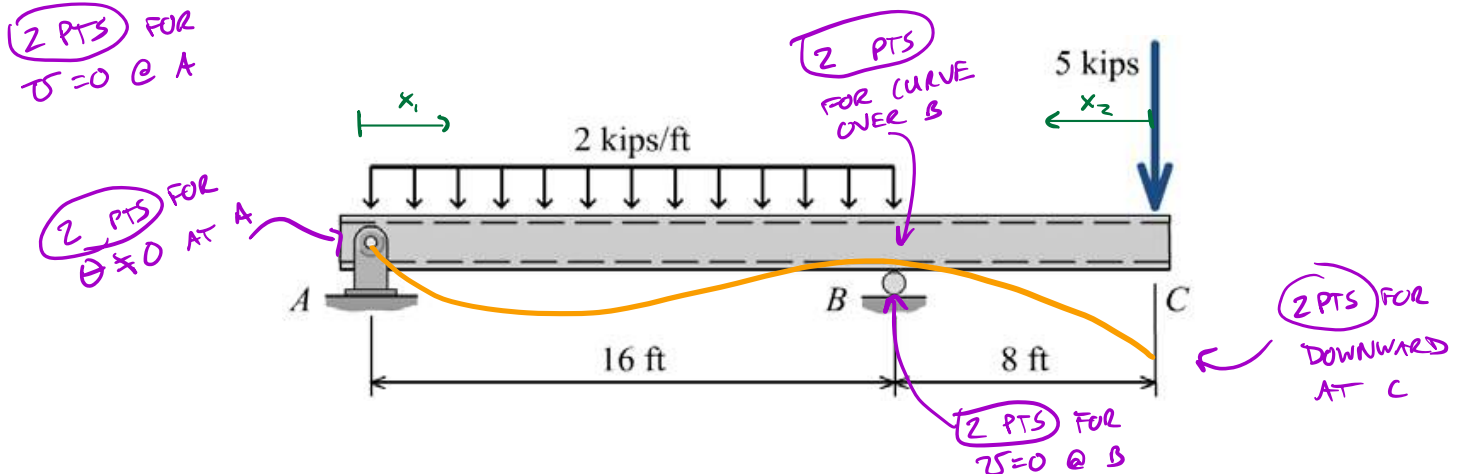
$$\tau_{max} = 8.57 \text{ ksi}$$

Problem 2:

The same simply supported beam from problem 1 is shown below. Assume that EI is constant for the beam. Solve for the following: (100 pts)



- a. Draw the displaced shape of the beam ABC due to the applied loading on the picture of the beam below (10 pts):



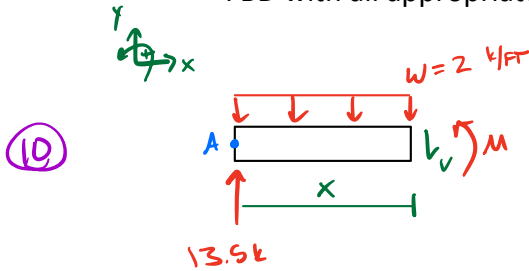
- b. Identify the four boundary conditions needed to solve for the displacement at point C. Put each answer in the four blocks below (10 pts):

2.5 EACH

1. $v_{A1}(0 \text{ FT}) = 0$	2. $v_{B1}(16 \text{ FT}) = 0$
3. $v_{B2}(8 \text{ FT}) = 0$	4. $\theta_{B1}(16 \text{ FT}) = \theta_{B2}(8 \text{ FT})$

*NOTE - THE DIRECTION OF x IS NOT SPECIFIED FOR BEAM 1 & 2

- c. Find the moment function between points A and B as a function of x . Draw a complete FBD with all appropriate labels. Write the moment function in the box below. (20 pts)



$$\sum M_{cur} = 0$$

$$0 = M(x) - 13.5x + (2 \text{ k/ft})x \cdot \frac{x}{2}$$

$$M(x) = -x^2 + 13.5x$$

(10)

Moment function:

$$M(x) = -x^2 + 13.5x$$

- d. Assuming the moment function is $M_x = -6x^2 + 30x$ (this is not the answer to part c), use the integration method to find the slope and displacement functions between A and B. Solve for the two constants of integration and put the answers in the block below.

Assume EI is constant (30 pts)

$$R = \frac{M(x)}{EI} = \frac{1}{EI} (-6x^2 + 30x)$$

(10)

$$\theta(x) EI = -2x^3 + 15x^2 + C_1$$

(10)

$$v(x) EI = -\frac{x^4}{2} + 5x^3 + C_1x + C_2$$

BOUNDARY CONDITIONS

(5)

$$v(0 \text{ ft}) = 0$$

$$0 = -\frac{1}{2}(0 \text{ ft})^4 + 5(0 \text{ ft})^3 + C_1(0 \text{ ft}) + C_2$$

$$\therefore \underline{C_2 = 0}$$

(5)

$$v(16 \text{ ft}) = 0$$

$$0 = -\frac{1}{2}(16 \text{ ft})^4 + 5(16 \text{ ft})^3 + C_1(16 \text{ ft})$$

$$0 = -32768 + 20480 + 16C_1$$

$$\therefore \underline{C_1 = 768}$$

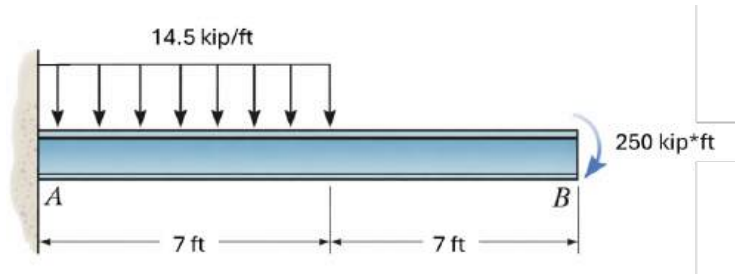
$$C_1 = 768$$

$$C_2 = 0$$

- e. Given the loading configuration below, determine the displacement at **point B** using the principle of superposition. Use the following properties for E and I (30 pts)

$$E = 29000 \text{ ksi}$$

$$I_x = 984 \text{ in}^4$$



$$v_B = v_{WB} + v_{MB}$$

$$\textcircled{12} \quad v_{WB} = \frac{-7 w L^4}{384 EI} = \frac{-7 (14.5 \text{ k/ft}) (\frac{1}{12} \text{ ft}) (14 \text{ ft} \cdot 12 \text{ in/ft})^4}{384 (29000 \text{ ksi}) (984 \text{ in}^4)} = \underline{0.615 \text{ in} \downarrow}$$

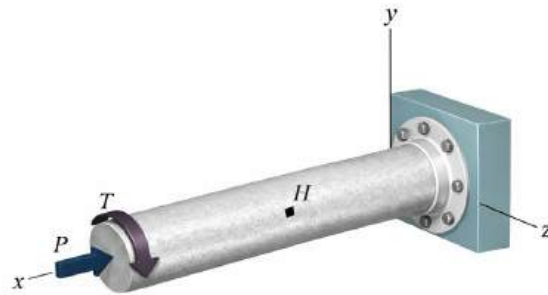
$$\textcircled{12} \quad v_{MB} = \frac{-M_0 L^2}{2 EI} = \frac{-(250 \text{ k-ft}) (12 \text{ in/ft}) (14 \text{ ft} \cdot 12 \text{ in/ft})^2}{2 (29000 \text{ ksi}) (984 \text{ in}^4)} = \underline{1.48 \text{ in} \downarrow}$$

$$\textcircled{6} \quad v_B = -0.615 \text{ in} - 1.48 \text{ in} = \underline{2.10 \text{ in} \downarrow}$$

$$v_B = 2.10 \text{ in} \downarrow$$

Problem 3:

A solid shaft with a diameter of $d = 65 \text{ mm}$ is fixed to the wall and is subjected to a torque of $T = 650 \text{ N} \cdot \text{m}$ and an axial force of $P = 50 \text{ kN}$. Determine the state of stress at point H. Show the stress on the stress element below. Show all of your work! (50 pts)



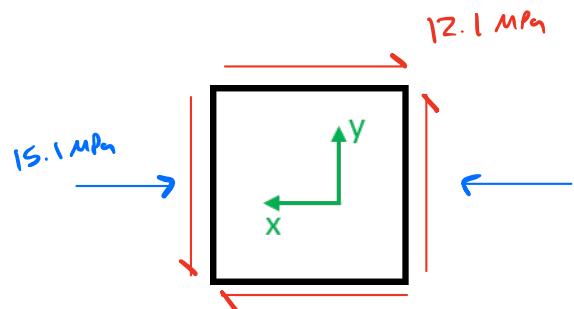
(20)

$$\sigma_H = \frac{P}{A} = \frac{(50 \text{ kN}) \cdot (1000 \text{ N/kN})}{\frac{\pi}{4} (65 \text{ mm})^2} = \underline{15.1 \text{ MPa (C)}}$$

(20)

$$\tau_H = \frac{T_c}{J} = \frac{(650 \text{ N} \cdot \text{m}) (1000 \text{ mm/m}) (32.5 \text{ mm})}{\frac{\pi}{2} (32.5 \text{ mm})^4} = \underline{12.1 \text{ MPa}}$$

(10)





TAB H – WPR3

DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

**WRITTEN PARTIAL REVIEW III
CE364, 26-2**

NAME: _____ SECTION _____

Left Section Room at: _____ (to be completed for early departure only)

Question/Section	Points	Score
Problem 1: Stress Transformation	50	
Problem 2: Mohr's Circle & Principal Stresses	75	
Problem 3: Strain Transformation and Generalized Hooke's Law	75	
Problem 4: TWPV & Failure Theory	50	
Total		

Special Instructions:

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Work Quickly but Legibly

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Signature: _____

Equation sheet**Normal and Shear Stress Transformation:**

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\theta - \tau_{xy} \sin 2\theta$$

$$\tau'_{xy} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

Strain Transformation:

$$\varepsilon_a = \varepsilon_x \cos^2 \theta_a + \varepsilon_y \sin^2 \theta_a + \gamma_{xy} \sin \theta_a \cos \theta_a$$

Maximum Shear Stress Failure Theory:

$$|\sigma_1| \geq \sigma_y \text{ if } \sigma_1 \text{ \& } \sigma_2 \text{ are positive}$$

$$|\sigma_2| \geq \sigma_y \text{ if } \sigma_1 \text{ \& } \sigma_2 \text{ are negative}$$

$$|\sigma_1 - \sigma_2| \geq \sigma_y \text{ if } \sigma_1 \text{ is positive \& } \sigma_2 \text{ is negative}$$

Maximum Distortion Energy Failure Theory:

$$\sigma_M \geq \sigma_y$$

$$\sigma_M = \sqrt{\sigma_1^2 - \sigma_1 \sigma_2 + \sigma_2^2}$$

Maximum Normal Stress Failure Theory:

$$|\sigma_1| \geq \sigma_{UT}$$

$$|\sigma_2| \geq \sigma_{UC}$$

Mohr's Failure Criterion:

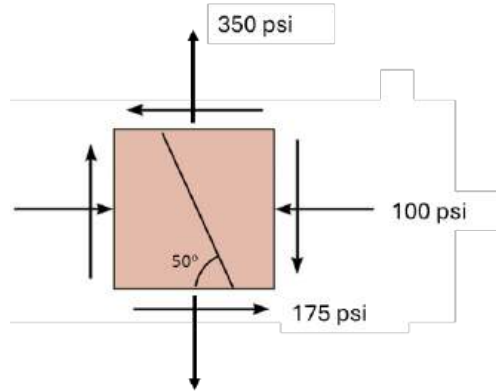
$$\sigma_1 \geq \sigma_{UT} \text{ if } \sigma_1 \text{ \& } \sigma_2 \text{ are positive}$$

$$\sigma_2 \leq -\sigma_{UC} \text{ if } \sigma_1 \text{ \& } \sigma_2 \text{ are negative}$$

$$\frac{\sigma_1}{\sigma_{UT}} - \frac{\sigma_2}{\sigma_{UC}} \geq 1 \text{ if } \sigma_1 \text{ is positive \& } \sigma_2 \text{ is negative}$$

Problem 1 (50 points):

A hollow circular gas pipeline with a spiral weld is subjected to an axial load, an internal gas pressure, and direct shear. The pipeline is welded at a 50-degree angle measured from horizontal. After conducting a combined loading analysis, you found the state of stress below. Determine the stresses that are normal and tangential to the weld and the shear stress parallel to the weld (50 pts)

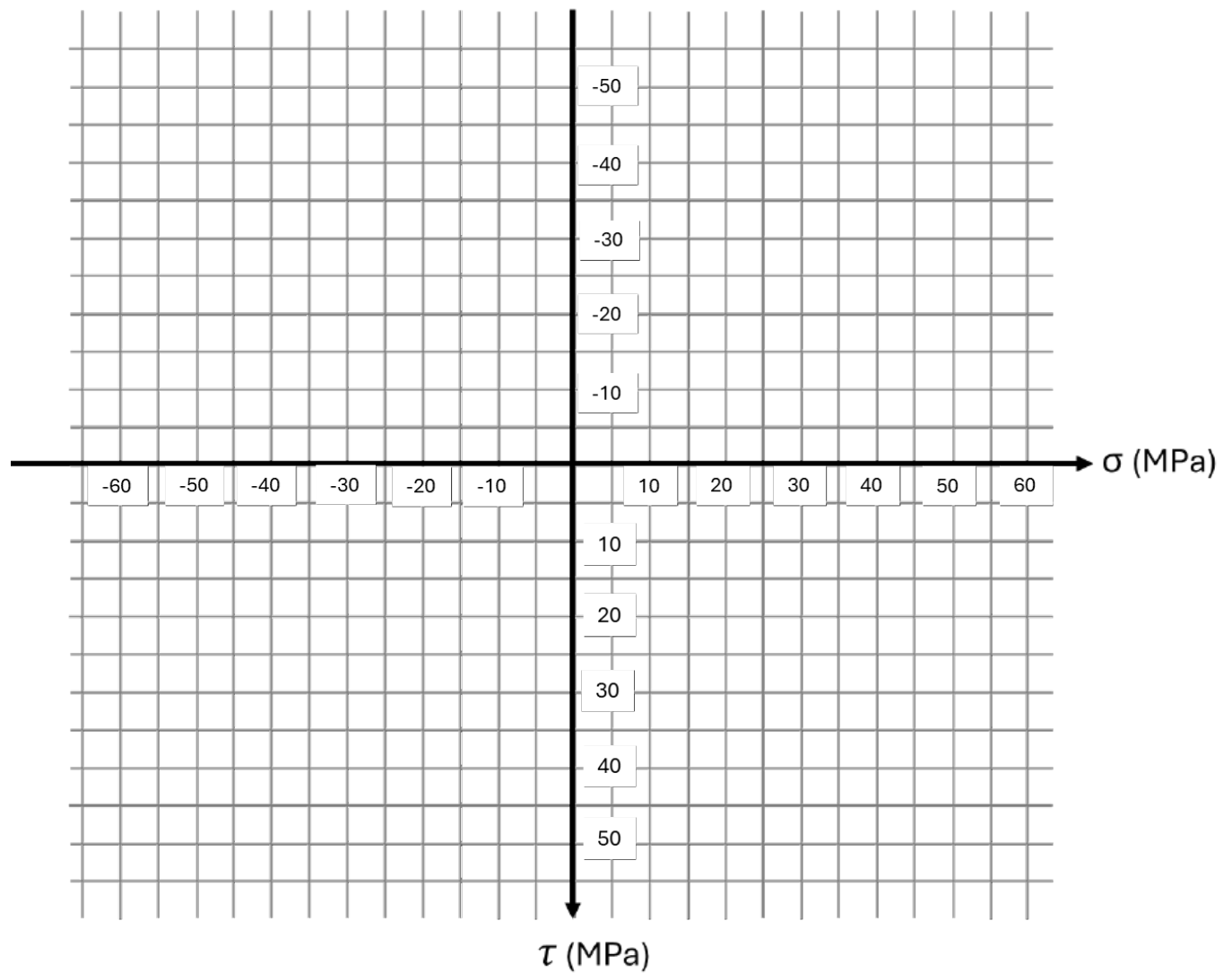
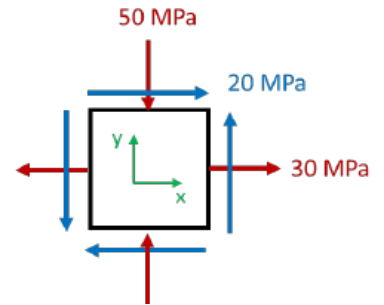


$\sigma_{x'} =$ _____ $\sigma_{y'} =$ _____ $\tau_{x'y'} =$ _____

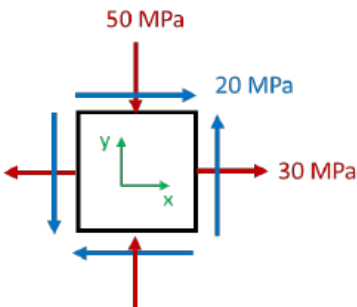
Problem 2 (75 points):

a) Construct a complete Mohr's Circle for the state of stress shown below and label the following properties on the circle. **Show your work!** (40 pts)

- 1) Reference Point, **A**
- 2) Average Normal Stress, σ_{avg}
- 3) Maximum In-Plane Shear Stress, τ_{max}
- 4) Maximum Principal Normal Stress, σ_1
- 5) Minimum Principal Normal Stress, σ_2
- 6) Orientation of principal planes, $2\theta_p$
- 7) Orientation of maximum shear plane, $2\theta_s$



- b) Given the same state of stress from part a, record the principal stresses, maximum in-plane shear stress, average normal stress, and principal stress orientation angle in the answer blocks below. Show these stresses on an appropriate sketch (i.e., the two square stress element method or the wedge-shaped stress element method). (35 pts)



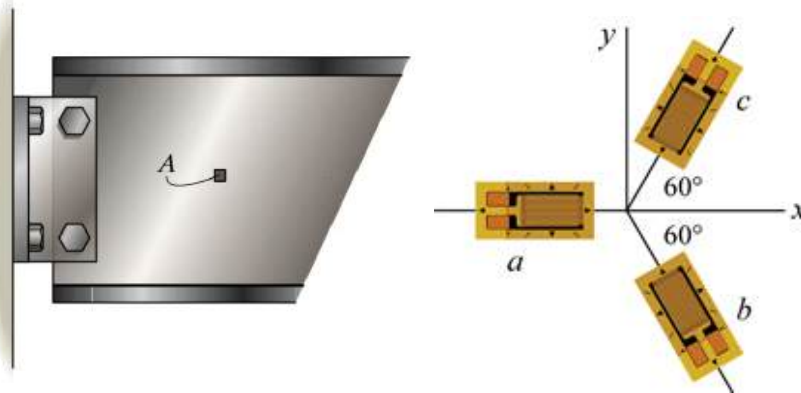
$\sigma_1 =$ _____ $\sigma_2 =$ _____ $\theta_p =$ _____

$\tau_{\max} =$ _____ $\sigma_{\text{AVG}} =$ _____

Problem 3 (75 points):

A strain rosette is mounted to the web of a beam ($E = 200 \text{ GPa}$, $\nu = 0.27$) at point A and records the following strains.

$$\varepsilon_a = 450 \mu\varepsilon; \varepsilon_b = 200 \mu\varepsilon; \varepsilon_c = -150 \mu\varepsilon$$



- a. Find the corresponding normal strains and shear strain in the x-y plane (35 pts):

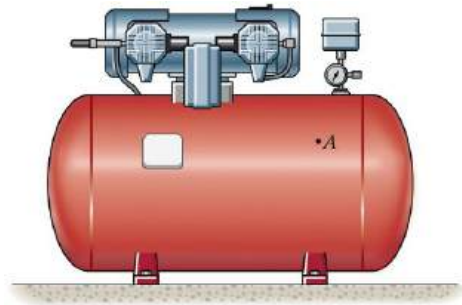
$\varepsilon_x =$ _____ $\varepsilon_y =$ _____ $\gamma_{xy} =$ _____

- b. Find the normal stresses and shear stress ($\sigma_x, \sigma_y, \tau_{xy}$) which produced these strains. You may assume plane stress conditions (ie. $\sigma_z = 0$) (40 pts):

$\sigma_x =$ _____ $\sigma_y =$ _____ $\tau_{xy} =$ _____

Problem 4 (50 points):

You are designing an air compressor that needs to be filled to a maximum internal pressure of 200 psi. The air compressor must have an outer diameter of 30 inches and be constructed of Structural A36 Steel. Determine the following:



- a) Using the Maximum Shear Stress Failure Theory (Tresca) and a factor of safety of 2, determine the minimum wall thickness required so that it does not fail. Steel shells are limited to thicknesses of 0.125-inch increments (40 pts)

Minimum safe wall thickness, $t =$ _____ in

- b) Does your wall thickness pass the criteria for a thin-walled pressure vessel? Show all of your work and circle your answer below. (10 pts)

Pass TWPV Criteria: YES / NO

Instructor Solution

DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

**WRITTEN PARTIAL REVIEW III
CE364, 26-2**

NAME: SOLUTION SECTION _____

Left Section Room at: _____ (to be completed for early departure only)

Question/Section	Points	Score
Problem 1: Stress Transformation	50	
Problem 2: Mohr's Circle & Principal Stresses	75	
Problem 3: Strain Transformation and Generalized Hooke's Law	75	
Problem 4: TWPV & Failure Theory	50	
Total		

Special Instructions:

1. Begin work when directed. You will have 75 minutes to complete the WPR
2. Write your name at the top of each page. Ensure you have a complete test (8 pages double sided, including this cover sheet.)
3. This test is closed book and closed notes except for the CE364 authorized reference packet. Fundamentals of Engineering Exam-approved calculators are authorized.
4. No cell phones, smart watches, or smart glasses permitted in the classroom during the exam. Cadets may not collaborate or communicate with anybody else while taking the WPR.
5. You may use extra sheets of paper if you run out of room on the WPR. When completed, staple all pages together *in order*, including any additional sheets, and place the exam inside the folder on the exam proctor's desk in the hallway.
6. You are not authorized to discuss the content, structure, or any other information about the WPR until the WPR has been released from academic security.
7. SHOW ALL OF YOUR WORK! Ensure that you report your final answer to 3 significant figures, with appropriate units, and direction (if applicable)

Work Quickly but Legibly

"I did not use any sources, nor did I receive any assistance while completing this WPR. I will not discuss this WPR with anyone except a CE364 instructor until it is released from academic security on _____"

Signature: _____

Equation sheet**Normal and Shear Stress Transformation:**

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\theta - \tau_{xy} \sin 2\theta$$

$$\tau'_{xy} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

Strain Transformation:

$$\varepsilon_a = \varepsilon_x \cos^2 \theta_a + \varepsilon_y \sin^2 \theta_a + \gamma_{xy} \sin \theta_a \cos \theta_a$$

Maximum Shear Stress Failure Theory:

$$|\sigma_1| \geq \sigma_y \text{ if } \sigma_1 \text{ \& } \sigma_2 \text{ are positive}$$

$$|\sigma_2| \geq \sigma_y \text{ if } \sigma_1 \text{ \& } \sigma_2 \text{ are negative}$$

$$|\sigma_1 - \sigma_2| \geq \sigma_y \text{ if } \sigma_1 \text{ is positive \& } \sigma_2 \text{ is negative}$$

Maximum Distortion Energy Failure Theory:

$$\sigma_M \geq \sigma_y$$

$$\sigma_M = \sqrt{\sigma_1^2 - \sigma_1 \sigma_2 + \sigma_2^2}$$

Maximum Normal Stress Failure Theory:

$$|\sigma_1| \geq \sigma_{UT}$$

$$|\sigma_2| \geq \sigma_{UC}$$

Mohr's Failure Criterion:

$$\sigma_1 \geq \sigma_{UT} \text{ if } \sigma_1 \text{ \& } \sigma_2 \text{ are positive}$$

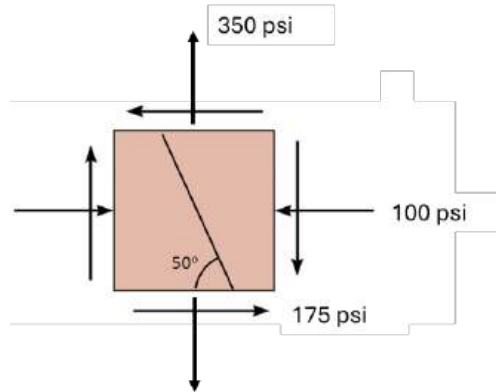
$$\sigma_2 \leq -\sigma_{UC} \text{ if } \sigma_1 \text{ \& } \sigma_2 \text{ are negative}$$

$$\frac{\sigma_1}{\sigma_{UT}} - \frac{\sigma_2}{\sigma_{UC}} \geq 1 \text{ if } \sigma_1 \text{ is positive \& } \sigma_2 \text{ is negative}$$

Problem 1 (50 points):

A hollow circular gas pipeline with a spiral weld is subjected to an axial load, an internal gas pressure, and direct shear. The pipeline is welded at a 50-degree angle measured from horizontal. After conducting a combined loading analysis, you found the state of stress below. Determine the stresses that are normal and tangential to the weld and the shear stress parallel to the weld (50 pts)

⑤ $\theta = +40^\circ$ (ccw)



$$\sigma_x = -100 \text{ psi}$$

$$\sigma_y = 350 \text{ psi}$$

$$\tau_{xy} = -175 \text{ psi}$$

⑬
$$\sigma_x' = \frac{-100 \text{ psi} + 350 \text{ psi}}{2} + \frac{-100 \text{ psi} - 350 \text{ psi}}{2} \cos(2 \cdot 40^\circ) + (-175 \text{ psi}) \sin(2 \cdot 40^\circ)$$

$$125 \text{ psi} + (-39.1 \text{ psi}) + (-172.3 \text{ psi}) = \underline{\underline{-86.4 \text{ psi}}} \text{ANS}$$

⑭
$$\sigma_y' = 125 \text{ psi} - (-39.1 \text{ psi}) - (-172.3 \text{ psi}) = \underline{\underline{336 \text{ psi}}} \text{ANS}$$

⑮
$$\tau_{x'y'} = -\left(\frac{-100 \text{ psi} - 350 \text{ psi}}{2} \sin(2 \cdot 40^\circ)\right) + (-175 \text{ psi} \cos(2 \cdot 40^\circ))$$

$$= 221.6 \text{ psi} - 30.4 \text{ psi} = \underline{\underline{191 \text{ psi}}} \text{ANS}$$

$\sigma_{x'} = \underline{-86.4 \text{ psi}} \quad \sigma_{y'} = \underline{336 \text{ psi}} \quad \tau_{x'y'} = \underline{191 \text{ psi}}$

-1 MISSING
SIGN

-1 UNITS
-1 SIG FIGS

Problem 2 (75 points):

a) Construct a complete Mohr's Circle for the state of stress shown below and label the following properties on the circle. **Show your work!** (40 pts)

- 1) Reference Point, **A**
- 2) Average Normal Stress, σ_{avg}
- 3) Maximum In-Plane Shear Stress, τ_{max}
- 4) Maximum Principal Normal Stress, σ_1
- 5) Minimum Principal Normal Stress, σ_2
- 6) Orientation of principal planes, $2\theta_p$
- 7) Orientation of maximum shear plane, $2\theta_s$

i) $A = (30, 20)$ (5)

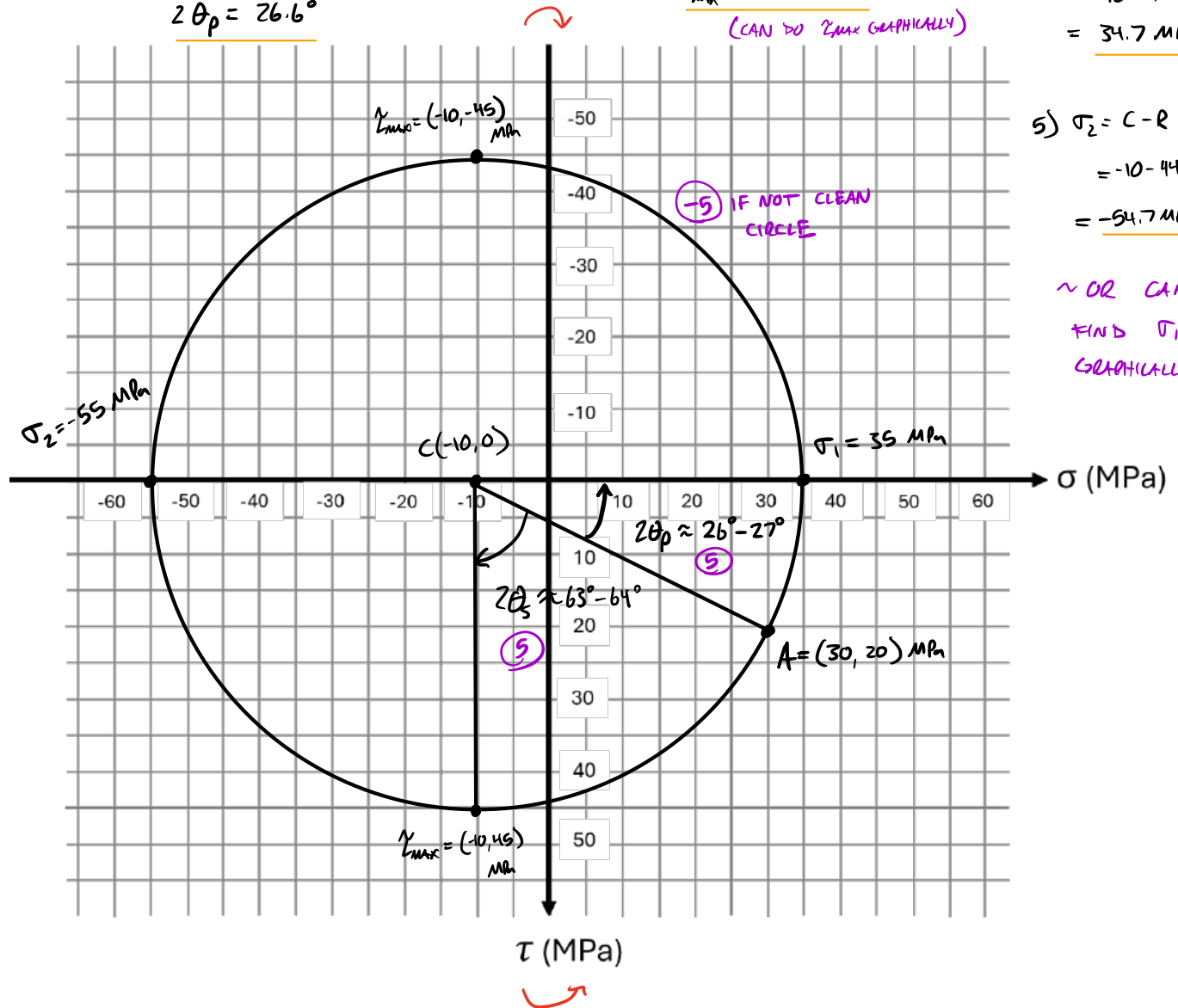
2) $\sigma_{avg} = C = \frac{\sigma_x + \sigma_y}{2}$ (5)
 $= \frac{30 - 50}{2} = -10 \text{ MPa}$

3) $\tau_{max} = \sqrt{\left(\frac{30 - (-50)}{2}\right)^2 + 20^2}$ (5)
 $\tau_{max} = 44.7 \text{ MPa}$ (5)
 (CAN DO τ_{max} GRAPHICALLY)

4) $\sigma_1 = C + R$ (5)
 $= -10 + 44.7 \text{ MPa}$
 $= 34.7 \text{ MPa}$

5) $\sigma_2 = C - R$ (5)
 $= -10 - 44.7 \text{ MPa}$
 $= -54.7 \text{ MPa}$

~ OR CAN
FIND σ_1 & σ_2
GRAPHICALLY

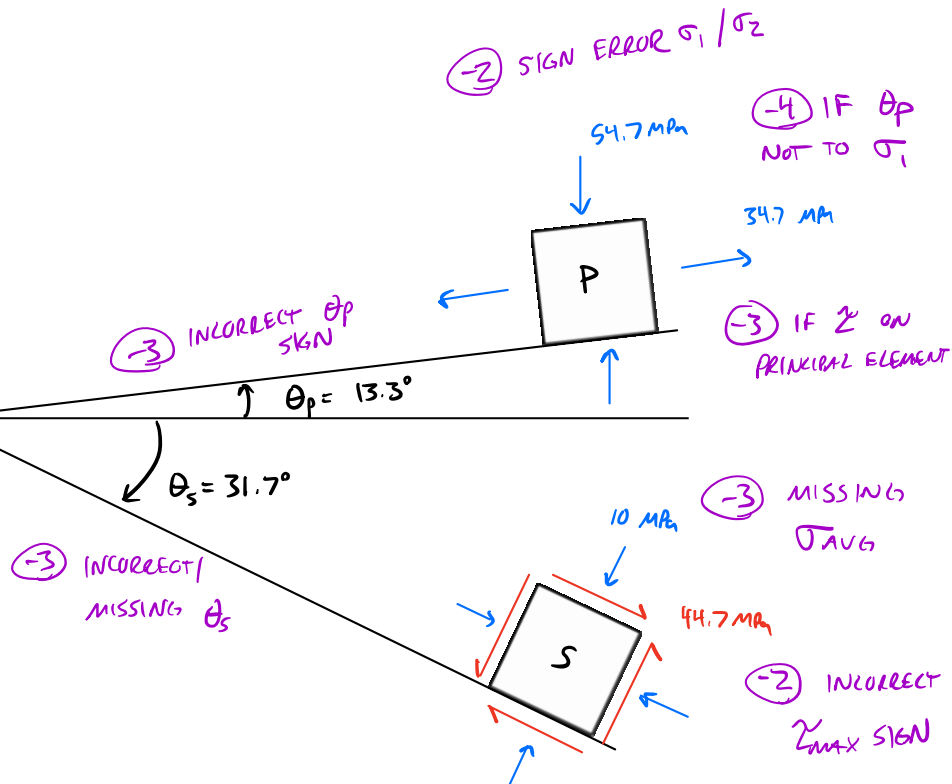
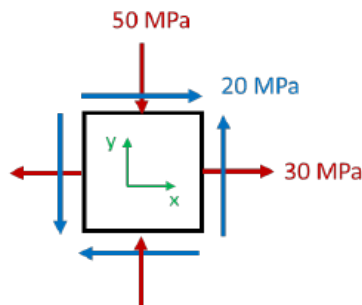


- b) Given the same state of stress from part a, record the principal stresses, maximum in-plane shear stress, average normal stress, and principal stress orientation angle in the answer blocks below. Show these stresses on an appropriate sketch (i.e., the two square stress element method or the wedge-shaped stress element method). (35 pts)

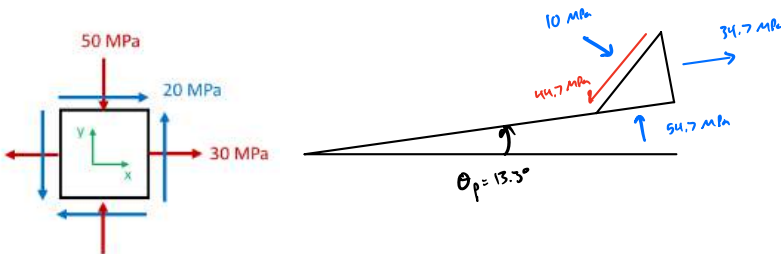
OVERALL

(20) SKETCH

(15) σ_1/σ_2 , $\tau_{\max}$, θ_p , σ_{AVG}



~OR~

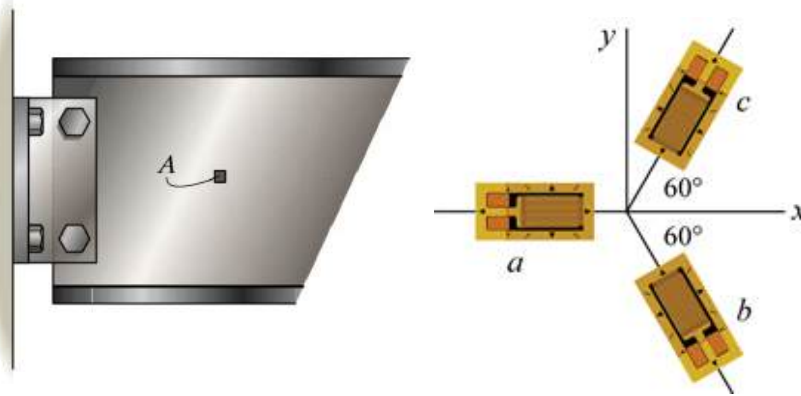


$\sigma_1 = 34.7 \text{ MPa}$ $\sigma_2 = -54.7 \text{ MPa}$ $\theta_p = 13.3^\circ$
 $\tau_{\max} = 44.7 \text{ MPa}$ $\sigma_{\text{AVG}} = -10 \text{ MPa}$

Problem 3 (75 points):

A strain rosette is mounted to the web of a beam ($E = 200 \text{ GPa}$, $\nu = 0.27$) at point A and records the following strains.

$$\varepsilon_a = 450 \mu\varepsilon; \varepsilon_b = 200 \mu\varepsilon; \varepsilon_c = -150 \mu\varepsilon$$



(-4) FOR EACH INCORRECT ANGLE

$$\theta_a = 180^\circ (0^\circ)$$

$$\theta_b = 300^\circ (-60^\circ)$$

$$\theta_c = 60^\circ$$

- a. Find the corresponding normal strains and shear strain in the x-y plane (35 pts):

$$450 \mu\varepsilon = \varepsilon_x \cos^2(180^\circ) + \varepsilon_y \sin^2(180^\circ) + \gamma_{xy} \sin(180^\circ) \cos(180^\circ)$$

$$200 \mu\varepsilon = \varepsilon_x \cos^2(300^\circ) + \varepsilon_y \sin^2(300^\circ) + \gamma_{xy} \sin(300^\circ) \cos(300^\circ)$$

$$-150 \mu\varepsilon = \varepsilon_x \cos^2(60^\circ) + \varepsilon_y \sin^2(60^\circ) + \gamma_{xy} \sin(60^\circ) \cos(60^\circ)$$

↳ 3x3 SYS-SOLVE

$$\varepsilon_x = 450 \mu\varepsilon$$

$$\varepsilon_y = -117 \mu\varepsilon$$

$$\gamma_{xy} = -404 \mu\text{RAD}$$

(5) MATH

-1 FOR INCORRECT γ_{xy} UNITS

$$\varepsilon_x = 450 \mu\varepsilon \quad \varepsilon_y = -117 \mu\varepsilon \quad \gamma_{xy} = -404 \mu\text{RAD}$$

- b. Find the normal stresses and shear stress (σ_x , σ_y , τ_{xy}) which produced these strains. You may assume plane stress conditions (ie. $\sigma_z = 0$) (40 pts):

$$E = 200 \text{ GPa}$$

$$\nu = 0.27$$

OPTION 1

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = \frac{200,000 \text{ MPa}}{1 - 0.27^2} \begin{bmatrix} 1 & 0.27 & 0 \\ 0.27 & 1 & 0 \\ 0 & 0 & \frac{1-0.27}{2} \end{bmatrix} \begin{bmatrix} 0.000450 \text{ } \epsilon \\ -0.000117 \text{ } \epsilon \\ -0.000404 \text{ rad} \end{bmatrix}$$

↳ MATRIX FUNCTION IN TI-36X

norm

OPTION 2

$$\textcircled{10} \quad \sigma_x = \frac{E}{1-\nu^2} (\epsilon_x + \nu \epsilon_y) = \frac{200,000 \text{ MPa}}{1-0.27^2} (0.000450 + 0.27(-0.000117)) = \underline{\underline{90.3 \text{ MPa}}} \text{ Ans}$$

$$\textcircled{10} \quad \sigma_y = \frac{E}{1-\nu^2} (\epsilon_y + \nu \epsilon_x) = \frac{200,000 \text{ MPa}}{1-0.27^2} (-0.000117 + 0.27(0.00045)) = \underline{\underline{0.971 \text{ MPa}}} \text{ Ans}$$

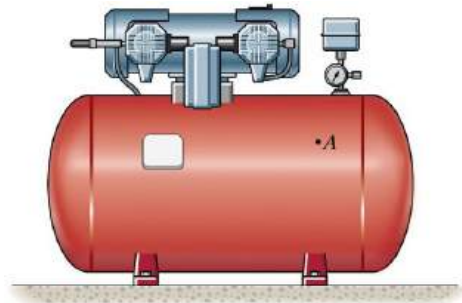
$$\textcircled{10} \quad \tau_{xy} = \frac{E}{1-\nu^2} \left(\frac{1-\nu}{2} \right) \gamma_{xy} = \frac{200,000 \text{ MPa}}{1-0.27^2} \left(\frac{1-0.27}{2} \right) (-0.000404 \text{ rad}) = \underline{\underline{-31.8 \text{ MPa}}} \text{ Ans}$$

$$\sigma_x = \underline{90.3 \text{ MPa}} \quad \sigma_y = \underline{0.971 \text{ MPa}} \quad \tau_{xy} = \underline{-31.8 \text{ MPa}}$$

Problem 4 (50 points):

You are designing an air compressor that needs to be filled to a maximum internal pressure of 200 psi. The air compressor must have an outer diameter of 30 inches and be constructed of Structural A36 Steel. Determine the following:

$$\sigma_y = 36 \text{ ksi (FERM)}$$



- a) Using the Maximum Shear Stress Failure Theory (Tresca) and a factor of safety of 2, determine the minimum wall thickness required so that it does not fail. Steel shells are limited to thicknesses of 0.125-inch increments (40 pts)

(10) TRESCA

$$\sigma_1 \leq \frac{\sigma_y}{2} = \frac{36 \text{ ksi}}{2}$$

(-5) IF NO F.S.

(-5) IF DID NOT USE r_i

$$\sigma_1 = \frac{p \cdot r_i}{t}$$

(10) HOOP STRESS

(-5) IF USED $\frac{p r_i}{2t}$

$$\frac{200 \text{ psi} (15 \text{ in} - t_w)}{t_w} \leq 18000 \text{ psi}$$

$$3000 - 200 t_w \leq 18000 \text{ psi}$$

$$18200 t_w \geq 3000 \quad \therefore t_w \geq 0.165 \text{ in} \quad \nabla \quad \underline{\underline{0.25 \text{ in}}} \quad \text{ANS}$$

Minimum safe wall thickness, $t = \underline{\underline{0.25}}$ in

(-4) IF DID NOT ROUND UP BY 0.125 IN INCREMENTS

- b) Does your wall thickness pass the criteria for a thin-walled pressure vessel? Show all of your work and circle your answer below. (10 pts)

(-5) FOR INCORRECT RATIO

$$\frac{r_i}{t} \geq 10$$

$$\frac{14.75 \text{ in}}{0.25 \text{ in}} = 59 \geq 10$$

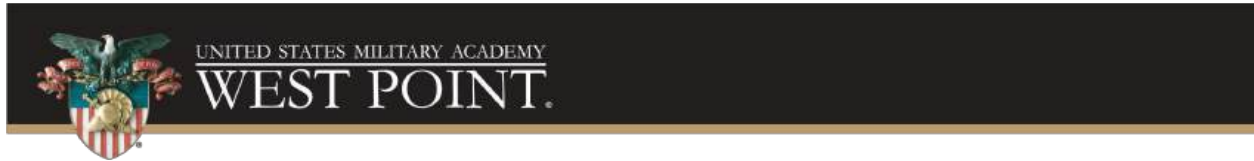
Pass TWPV Criteria: YES / NO

-5 IF NOT SELECTED



TAB I – TERM END EXAMINATION (TEE)

Note: Term End Examinations are kept in a secure location and can be accessed by request to the Curriculum Coordinator



TAB J – FINAL COURSE GRADES

STUDENT	YEAR	MAJOR	HW's (200 pts)	Group Work (550 pts)	WPRs (750 pts)	TEE (500 pts)	Course Average	Final Grade	Proposed Final Grade
ZUNHA, DANIEL M	2028	MEN3	99.0%	98.5%	97.7%	99.1%	98.4%	A+	A+
CAIRNEY, JAMES L	2028	MEN3	99.5%	92.2%	97.7%	98.4%	96.5%	A	A+
MURPHY, SHAUN R	2028	MEN3	99.2%	96.5%	94.3%	98.6%	96.5%	A	A+
KITTINGER, JACKSON D	2028	MEN3	99.0%	93.1%	96.9%	97.6%	96.3%	A	A+
WENDELL, CONRAD Q	2028	MEN3	98.5%	95.8%	96.9%	94.8%	96.2%	A	A+
GILLAND, BROD A	2026	CEN2	97.3%	94.0%	97.6%	93.4%	95.5%	A	A
LEHMAN, JAMES H	2028	MEN3	98.8%	95.5%	96.5%	92.7%	95.5%	A	A
SCHANG, NATHAN S	2028	MEN3	98.7%	93.8%	97.6%	92.6%	95.4%	A	A
AGRES, JOSH T	2028	MEN3	87.3%	98.5%	94.9%	95.4%	95.3%	A	A
ELLSWORTH, MATTHEW A	2026	CEN2	99.2%	93.4%	93.2%	98.0%	95.0%	A	A
JUD, KEVIN M	2028	MEN3	98.2%	90.5%	97.5%	94.2%	94.8%	A	A
BUSECK, LINNORA Z	2028	CVN3	96.5%	93.1%	94.4%	95.8%	94.6%	A	A
FANASWALA, RAJ	2028	MEN3	97.0%	92.5%	95.1%	95.0%	94.6%	A	A
CHABOT, JACK D	2028	MEN3	86.2%	93.5%	96.7%	95.0%	94.3%	A	A
MAYBURY, LAYTON R	2028	MEN3	97.7%	94.2%	94.0%	93.6%	94.3%	A	A
FUERTES, AARON A	2028	MEN3	99.3%	92.8%	92.5%	96.6%	94.3%	A	A
THEIN, RYAN A	2028	MEN3	99.7%	95.8%	93.7%	91.2%	94.2%	A	A
PUTNAM, OWEN C	2028	MEN3	96.5%	92.4%	94.7%	93.4%	93.9%	A	A
ROWE, PATRICK C	2028	MEN3	97.5%	87.8%	96.3%	95.4%	93.9%	A	A
KIM, EULI H	2028	MEN3	99.0%	96.5%	94.1%	88.4%	93.8%	A	A
CLARKE, MICHAEL E	2028	MEN3	97.2%	89.1%	96.2%	93.8%	93.7%	A	A
RUFFING, CHASE E	2027	NEN1	97.5%	89.2%	95.9%	93.9%	93.7%	A	A
HUNTER, LUKE C	2028	CVN3	98.5%	89.0%	93.4%	97.4%	93.7%	A	A
RAPP, ANGELA C	2028	CVN3	98.2%	92.0%	93.7%	93.2%	93.6%	A	A
BULL, NICHOLAS R	2028	CVN3	98.0%	90.7%	92.6%	94.2%	93.0%	A	A
MELTON, ELIZABETH J	2028	MEN3	97.7%	92.5%	91.4%	93.0%	92.7%	A-	A
MORRIS, ANDREW R	2028	MEN3	96.0%	92.2%	91.7%	93.4%	92.7%	A-	A
GEORGE, CHARLIE S	2028	MEN3	99.0%	90.7%	93.5%	89.8%	92.4%	A-	A-
JOSE, JACOB L	2028	NEN1	99.5%	92.5%	91.5%	90.0%	92.2%	A-	A-
LEANDRES, ANTONIO G	2028	CVN3	88.8%	93.4%	90.1%	94.0%	91.9%	A-	A-
CLANCY, JAMESON P	2028	CVN3	98.7%	95.5%	89.5%	88.4%	91.8%	A-	A-
BOLIVAR, GABRIEL A	2028	MEN3	95.2%	93.8%	92.3%	86.8%	91.6%	A-	A-
FIACCO, DOMINIC J	2028	MEN3	98.5%	93.6%	93.0%	82.7%	91.2%	A-	A-
HOLTMAN, ETHAN M	2028	CVN3	98.0%	88.7%	91.4%	90.6%	91.1%	A-	A-
EMERY, GABRIEL J	2028	CVN3	98.3%	94.5%	88.9%	87.2%	91.0%	A-	A-
SHAW, HARRISON C	2028	CVN3	94.7%	92.9%	91.1%	87.0%	90.9%	A-	A-
CENZAL, ALEXANDER J	2028	MEN3	98.8%	90.5%	91.9%	86.4%	90.9%	A-	A-
BEVILACQUE, VINCENT T	2027	NEN1	97.0%	91.8%	88.1%	91.2%	90.8%	A-	A-
ENMAN, JAKE W	2027	NEN1	90.3%	90.5%	93.2%	85.4%	90.2%	A-	A-
MATSON, RYAN J	2028	CVN3	99.2%	93.3%	90.8%	81.7%	90.0%	A-	A-
HAMPTON, JOSEPH R	2028	MEN3	94.8%	95.4%	91.0%	79.8%	89.8%	B+	B+
BRANCAMP, JOSHUA W	2028	CVN3	96.7%	89.1%	87.2%	91.3%	89.7%	B+	B+
ALLEN, QUANG DAN S	2028	CVN3	97.8%	93.4%	83.4%	91.8%	89.7%	B+	B+
GRIFFIN, ELI Q	2028	CVN3	96.3%	93.3%	85.8%	88.4%	89.6%	B+	B+
CONRAD, ANDREW J	2027	NEN1	92.2%	89.2%	92.6%	84.0%	89.5%	B+	B+
AMBROSIO-VELEGAS, OMAR	2028	CVN3	95.3%	93.0%	84.0%	90.4%	89.2%	B+	B+
LICEA, JUAN P	2028	MEN3	99.5%	97.6%	85.3%	81.0%	89.0%	B+	B+
RYGH, MADELEINE G	2027	NEN1	99.5%	95.6%	84.4%	83.9%	88.9%	B+	B+
VESS, ETHAN M	2028	MEN3	95.0%	91.8%	85.9%	87.6%	88.9%	B+	B+
CIESLA, JOSHUA R	2028	MEN3	93.0%	94.7%	86.9%	83.5%	88.8%	B+	B+
MERRIMAN, TRENTON C	2028	MEN3	97.7%	91.1%	88.7%	81.4%	88.4%	B+	B+
ACKER, AIDAN J	2028	MEN3	99.0%	91.6%	84.7%	84.0%	87.9%	B+	B+
CASWELL, JOHN PAUL I	2027	NEN1	98.5%	90.4%	84.4%	85.9%	87.8%	B+	B+
POWELL, STEVEN C	2028	MEN3	96.0%	89.3%	86.5%	83.6%	87.5%	B+	B+
FLANDERS, KATELYN A	2027	NEN1	100.0%	94.8%	83.4%	80.3%	87.4%	B+	B+
SUAREZ, NICHOLAS	2028	MEN3	93.8%	94.0%	82.7%	84.6%	87.4%	B+	B+
MORROW, RYAN S	2028	CVN3	94.8%	93.2%	83.9%	83.3%	87.4%	B+	B+
SULLIVAN, PATRICK T	2028	MEN3	91.2%	92.7%	80.7%	89.3%	87.2%	B+	B+
HUANG, JONATHAN Y	2028	MEN3	93.2%	92.5%	86.5%	79.6%	87.1%	B+	B+
KIDWELL, JACKSON T	2028	CVN3	92.1%	90.7%	83.9%	85.6%	87.0%	B+	B+
ONI, TOLUWANI O	2028	CVN3	84.5%	88.1%	82.4%	92.2%	86.6%	B	B+
GANSNIRT, MASON B	2028	MEN3	94.3%	94.4%	84.6%	77.4%	86.5%	B	B
MAALA, FRANCES ANGELO L	2027	NEN1	98.5%	95.4%	80.9%	80.0%	86.4%	B	B
MARSIGLIANO, MICHAEL P	2027	NEN1	98.0%	93.2%	87.7%	71.7%	86.3%	B	B

STUDENT	YEAR	MAJOR	HW's (200 pts)	Group Work (550 pts)	WPRs (750 pts)	TEE (500 pts)	Course Average	Final Grade	Proposed Final Grade
BAIK, RAYMOND I	2028	CVN3	90.2%	89.1%	79.3%	91.6%	86.1%	B	B
BENCHOFF, CALEB R	2028	CVN3	93.5%	91.8%	86.1%	76.8%	86.1%	B	B
VAIR, ELIANA E	2028	MEN3	99.5%	96.3%	82.3%	75.2%	86.1%	B	B
ADAMS, NICHOLAS W	2028	MEN3	98.0%	93.8%	83.1%	77.1%	86.0%	B	B
KELLY, BRENDAN R	2028	CVN3	82.2%	92.2%	83.8%	83.0%	85.7%	B	B
BOLON, COLE T	2028	MEN3	96.3%	95.1%	76.2%	85.2%	85.7%	B	B
RYDER, PAUL J	2028	CVN3	96.5%	92.7%	83.6%	76.6%	85.7%	B	B
MCFADDEN, CARSON D	2028	MEN3	94.1%	93.5%	83.5%	76.5%	85.6%	B	B
MARESCA, ANTHONY J	2027	NEN1	89.0%	93.8%	81.9%	80.5%	85.5%	B	B
TANNER, SAM T	2028	CVN3	96.2%	95.1%	81.2%	76.6%	85.4%	B	B
MOREHOUSE, SEAN W	2028	MEN3	96.5%	92.2%	85.7%	72.7%	85.3%	B	B
INCE, NATHANIEL R	2028	MEN3	98.0%	93.0%	76.8%	84.0%	85.2%	B	B
BRIDGFORTH, ROSS W	2028	CVN3	98.7%	89.1%	88.6%	70.2%	85.1%	B	B
HOBBAUGH, JOSHUA N	2028	MEN3	96.2%	93.8%	87.5%	66.9%	85.0%	B	B
SUMMERLIN, MARISSA L	2028	MEN3	96.8%	95.1%	81.8%	72.4%	84.6%	B	B
READ, DANIEL F	2028	CVN3	97.0%	89.1%	82.4%	77.6%	84.5%	B	B
SHAH, VEEBHU T	2028	CVN3	96.0%	93.8%	76.7%	80.6%	84.3%	B	B
MUMMA, LUKE H	2028	MEN3	93.0%	89.3%	84.7%	74.3%	84.2%	B	B
CHAMPLIN, EVERETT B	2028	CVN3	98.5%	95.5%	76.9%	76.3%	84.0%	B	B
WILLIS, KATHERINE M	2028	MEN3	95.5%	92.9%	84.1%	68.6%	83.8%	B	B
PASULKA, ALEXANDER M	2028	CVN3	88.9%	92.2%	82.5%	74.1%	83.7%	B	B
KOWALCZYK, EMMA G	2027	NEN1	98.7%	89.5%	86.1%	66.8%	83.4%	B	B
BAKER, GRAYSON F	2028	CVN3	87.8%	92.2%	78.3%	79.5%	83.4%	B	B
MALIK, MOHAMMAD E	2028	CVN3	95.8%	86.9%	83.2%	74.6%	83.3%	B	B
EGAN, ANDREW J	2028	MEN3	93.8%	92.9%	70.5%	86.9%	83.1%	B	B
POIRIER, PRESTON H	2027	NEN1	70.9%	94.8%	81.7%	76.4%	82.9%	B-	B-
SANDIE, DEREK W	2027	NEN1	94.0%	85.1%	84.7%	73.0%	82.8%	B-	B-
LANE, CONNOR M	2028	MEN3	98.7%	87.8%	78.9%	75.6%	82.5%	B-	B-
CURTIS, SAMUEL S	2028	MEN3	96.3%	89.1%	79.2%	74.4%	82.4%	B-	B-
BOSARGE, JARET S	2028	CVN3	97.5%	92.7%	78.8%	69.4%	82.2%	B-	B-
CAMPBELL, CHARLOTTE S	2028	CVN3	98.5%	86.0%	81.1%	72.6%	82.1%	B-	B-
BANZUELA, JOSHUA M	2028	CVN3	97.0%	92.0%	76.1%	73.5%	81.9%	B-	B-
BUSCH, NOAH K	2028	MEN3	96.2%	87.3%	82.5%	68.9%	81.8%	B-	B-
MAWBAY, LUKE P	2028	MEN3	88.2%	76.7%	86.5%	77.0%	81.6%	B-	B-
SOUHRADA, ANDREW W	2028	MEN3	96.5%	70.0%	78.3%	91.2%	81.1%	B-	B-
TROUT, NICHOLSON B	2028	CVN3	90.5%	92.8%	75.2%	73.0%	81.0%	B-	B-
RUDISILL, CALEB R	2028	CVN3	96.5%	89.3%	76.8%	71.5%	80.9%	B-	B-
KISSELL, GUNTHER E	2027	MEN3	99.0%	92.9%	75.3%	68.2%	80.8%	B-	B-
STOKES, LIA C	2026	CEN2	89.5%	86.2%	75.9%	78.3%	80.7%	B-	B-
STAUNTON, JAMES C	2028	CVN3	98.7%	97.6%	78.5%	57.5%	80.5%	B-	B-
CONNORS, JAMESON P	2026	NEN1	59.4%	91.0%	87.7%	66.4%	80.5%	B-	B-
GILFOIL, JAKOB A	2028	MEN3	96.5%	91.3%	74.8%	66.7%	79.5%	C+	C+
GREEN, ISAAC	2027	NEN1	94.5%	90.0%	72.3%	72.2%	79.4%	C+	C+
REILLY, SAMUEL J	2028	CVN3	95.5%	89.3%	69.8%	76.2%	79.3%	C+	C+
LOWE, MACLAIN J	2027	NEN1	98.2%	82.8%	79.0%	68.4%	79.3%	C+	C+
VANSCHOY, RYAN C	2028	CVN3	98.0%	90.9%	77.9%	60.9%	79.3%	C+	C+
TAYLOR, JORDAN A	2028	MEN3	99.2%	90.7%	74.9%	63.9%	78.9%	C+	C+
CIBES-LEE, HECTOR R	2027	NEN1	97.5%	95.2%	64.3%	75.2%	78.9%	C+	C+
WEINER, JOSHUA J	2028	CVN3	95.0%	86.0%	74.7%	70.2%	78.7%	C+	C+
MANGIN, JUSTIN M	2027	NEN1	93.9%	81.9%	77.4%	70.8%	78.6%	C+	C+
BRIGGS, RODNEY L	2028	MEN3	94.8%	90.5%	68.9%	73.1%	78.5%	C+	C+
DAMODAR, SIVIN V	2028	CVN3	81.3%	90.5%	76.3%	66.7%	78.3%	C+	C+
SYKES, JOHN E	2027	NEN1	87.3%	84.7%	76.8%	68.8%	78.0%	C+	C+
LIU, HARRY J	2027	NEN1	95.5%	83.5%	68.7%	78.3%	77.9%	C+	C+
DILONE DIAZ, WALDO R	2027	NEN1	81.8%	82.8%	72.8%	75.5%	77.1%	C+	C+
BALAJI, ADITYA	2028	MEN3	92.7%	69.6%	83.7%	68.6%	76.9%	C	NC
AUCI, AARON R	2028	MEN3	88.2%	90.7%	73.5%	60.5%	76.4%	C	C
ZIMMERMANN, JAY P	2028	MEN3	75.5%	93.3%	69.6%	67.6%	76.2%	C	C

STUDENT	YEAR	MAJOR	HW's (200 pts)	Group Work (550 pts)	WPRs (750 pts)	TEE (500 pts)	Course Average	Final Grade	Proposed Final Grade
HEKIMIAN, MARY RAY M	2028	CVN3	94.8%	87.7%	65.7%	70.1%	75.8%	C	C
GEDLEKIRSTOS, ZERAYACOE	2028	CVN3	90.3%	92.9%	66.9%	62.5%	75.3%	C	C
MITCHELL, ELIJAH W	2027	NEN1	86.8%	89.3%	59.7%	75.2%	74.4%	C	C
WHITE, ZACHARY D	2027	NEN1	78.5%	95.2%	63.9%	65.4%	74.3%	C	C
DIOP, KHADIJA	2028	CVN3	96.5%	90.9%	60.9%	64.5%	73.6%	C	C
MASON, AIDEN C	2027	NEN1	86.2%	90.4%	67.7%	55.1%	72.6%	C-	C-
GRANGER, PEYTON L	2027	NEN1	91.2%	92.2%	59.5%	61.2%	72.1%	C-	C-
LOSCH, DIESEL L	2027	NEN1	77.9%	88.8%	61.3%	66.1%	71.7%	C-	C-
OSGOOD, TRISTA L	2028	CVN3	78.3%	88.8%	61.5%	61.4%	70.7%	C-	C-
CLARK, HAROLD A	2028	CVN3	84.9%	87.2%	59.8%	62.5%	70.5%	C-	C-
BEVERLY, NEHEMIAH M	2028	MEN3	71.5%	89.1%	59.9%	58.0%	68.6%	D	D
MCNALLY, KYLE J	2028	CVN3	84.5%	87.5%	65.5%	43.9%	68.0%	D	F
MEAN			94.0%	91.2%	83.1%	80.0%	85.7%		

Retain


DANIEL L. MYERS
MAJ, EN
CE364 Course Director AY26-2


WILLIAM C. GRAVES
LTC, FA47
Civil Engineering Program Director


DR. ELIZABETH BRISTOW
Associate Professor
CE Curriculum Coordinator


PHILIP J. DACUNTO
COL, Professor, USMA
Head, Department of Civil & Environmental Engineering

REQUEST FOR No Credit (NC) Grade

A request for a No Credit (NC) grade is a request to complete specific course-related requirements on an amended schedule with due dates extended past the end of the current term. You will receive no credit for the course in the current term, and you will use this form to specify a timeline for completion of the remaining course requirements. Once you have completed all course requirements, the supervising Department Head will notify Registrar Services to post the appropriate grade. "NC" status does not affect APS or QPA calculations. The subsequent award of a grade counts as the first award of a grade in the course. Cadets should clear "NC" grades as soon as practical by working closely with their instructors to complete the remaining course requirements by the date specified. To request a No Credit "NC" grade, use this form. Please consult with your instructor in completing this form.

When completed (with cadet and instructor signatures), the instructor will coordinate with the appropriate Course Director and Department Head for adjudication. The department will send the completed form to the Registrar's Office.

Cadet Name: BALAJI, ADITYA USMA ID# (C#): C50153830 GY 2028

Instructor Name: CPT Pace Murray Department: CEE

Course Number: CE364 Course Title: Mechanics of Materials Section: G2 Term: 26-2

Cadet's Current Grade (percent and letter grade): 77.32 / C+ Current % of Course Complete 95%

Reason Code & Detailed Explanation for the No Credit (NC) grade:

ADMIN = Administrative Exception EMER LV = Emergency Leave EXCHG = Exchange (SAEP/SAP) MEDEX = Medically Excused
PENDR = Pending Review of Honor Case* Reason Code: **PENDR**

Pending honor case on Torsion Lab Report. Worth 100 pts.

Work which must be done to remove the grade of No Credit (be specific):

Torsion Lab Report

*If Honor Case is not found, what would be the final course grade? C

**If Honor Case, state what specific graded event is involved and the % of total course points for that event:

Torsion Lab Report, 5%

Deadline for completion of work (provide a specific date): _____

* Suggested deadline is four weeks from the end of the current term, but Department Heads may approve later dates.

If the work is not completed by the established deadline, what is the default grade? _____

Student signature: BALAJI.ADITYA.1634563497 Digitally signed by BALAJI.ADITYA.1634563497 Date: 2026.05.14 20:42:48 -04'00' Date: 2026-05-14

Instructor signature: MURRAY.PACE.WARNER.1455085612 Digitally signed by MURRAY.PACE.WARNER.1455085612 Date: 2026.05.14 20:56:42 -04'00' Date: 2026-05-12

Department Head signature:  Date: 18 May 26

☒ Approved

☐ Disapproved



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION III

ASSIGNMENTS



TAB K PROBLEM SETS



UNITED STATES MILITARY ACADEMY
WEST POINT.

HOMEWORK 1 – REVIEW & AXIAL STRESS/STRAIN

● HW1 - Review; Axial Stress/Strain

[Edit](#)
[Overview](#)
[Diagnostics](#)
[Print View with Answers](#)

HW1 - Review; Axial Stress/Strain

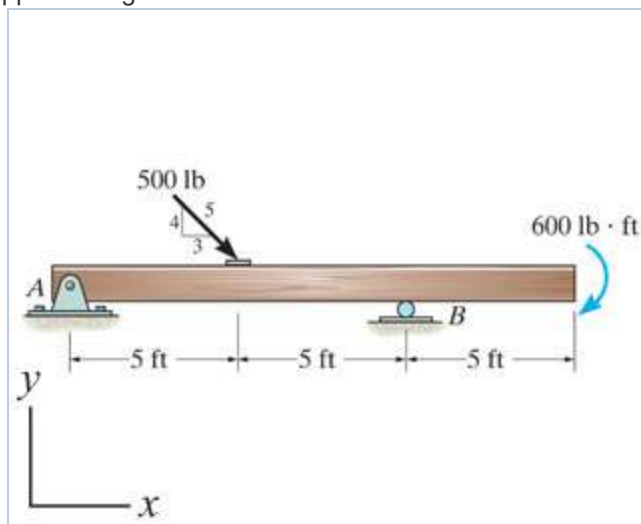
Due: 11:59pm on Friday, January 9, 2026

You will receive no credit for items you complete after the assignment is due. [Grading Policy](#)

Fundamental Problem 4.1 - Enhanced - with Hints and Feedback

Description: Determine the horizontal and vertical components of reaction at the supports. Neglect the thickness of the beam. (a) Determine the horizontal component of reaction at A. (b) Determine the vertical component of reaction at A. (c) Determine ...

Determine the horizontal and vertical components of reaction at the supports. Neglect the thickness of the beam.



Part A

Determine the horizontal component of reaction at A.

Express your answer in pounds to three significant figures. Enter positive value if the force is exerted in the positive x direction and negative value if the force is exerted in the negative x direction.

Hint 1. How to determine A_x

The unknown reactions can be determined from the equations of equilibrium. To determine the horizontal component of reaction at A, apply the force equilibrium equation along the x axis, $\Sigma F_x = 0$.

ANSWER:

$$A_x = -300 \text{ lb}$$

Part B

Determine the vertical component of reaction at A .

Express your answer in pounds to three significant figures. Enter positive value if the force is exerted in the positive y direction and negative value if the force is exerted in the negative y direction.

Hint 1. How to determine A_y

The unknown reactions can be determined from the equations of equilibrium. To determine the vertical component of reaction at A , apply the force equilibrium equation along the y axis, $\Sigma F_y = 0$. Note that you already need to know B_y .

Alternatively, you can apply the moment equilibrium equation about point B , $\Sigma M_B = 0$.

ANSWER:

$$A_y = 140 \text{ lb}$$

Part C

Determine the vertical component of reaction at B .

Express your answer in pounds to three significant figures. Enter positive value if the force is exerted in the positive y direction and negative value if the force is exerted in the negative y direction.

Hint 1. How to determine B_y

The unknown reactions can be determined from the equations of equilibrium. To determine the vertical component of reaction at B , apply the moment equilibrium equation about point A , $\Sigma M_A = 0$.

Alternatively, you can apply the force equilibrium equation along the y axis, $\Sigma F_y = 0$, using A_y from the previous part.

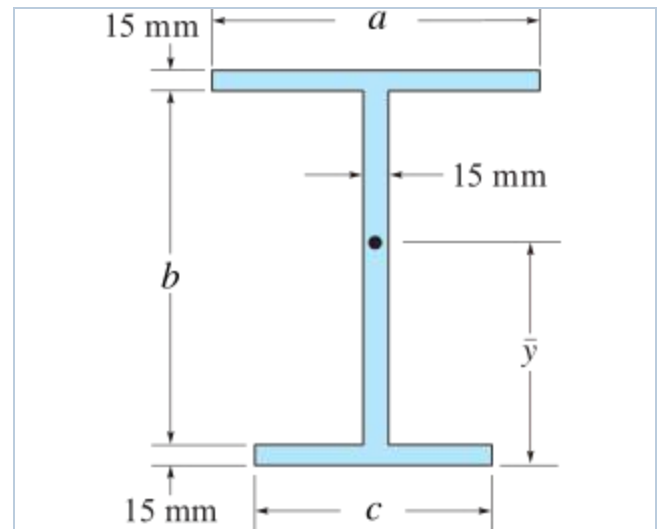
ANSWER:

$$B_y = 260 \text{ lb}$$

Problem 6.34

Description: Consider the beam's cross-sectional area shown in . Suppose that $a=##$ mm, $b=##$ mm, and $c=##$ mm. (a) Determine the location $\bar{y}$ of the centroid for the beam's cross-sectional area.

Consider the beam's cross-sectional area shown in . Suppose that $a = 360$ mm, $b = 480$ mm, and $c = 240$ mm.



Part A

Determine the location $\bar{y}$ of the centroid for the beam's cross-sectional area.

Express your answer to three significant figures and include the appropriate units.

ANSWER:

$$\bar{y} = \frac{a \left(15 + b + \frac{15}{2} \right) + b \left(15 + \frac{b}{2} \right) + c \frac{15}{2}}{a + b + c} = 283 \text{ mm}$$

Allowable Stress and Factor of Safety

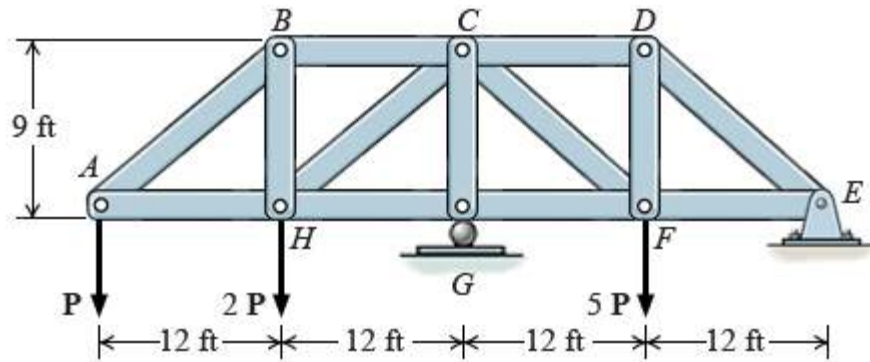
Description: In this tutorial, students will calculate an allowable stress, then, based on it, they will calculate the normal (or axial) forces in two truss members. Using these forces and the normal stress equation, the maximum allowable load P will be determined.

Learning Goal:

To calculate allowable stress by applying a factor of safety to the yield stress and to determine the maximum allowable load that can be applied based on the loads in individual truss members.

A pin-connected truss is loaded as shown. All members are two-force members. Consider the weights of each member to be negligible when compared to the loads applied to the truss. Members CH , BC , and GH are fabricated out of structural steel, which has a yield tensile strength of 36 ksi. The cross-sectional area of each member is 1.50 in^2 . Using a factor of safety of

2.5, calculate the maximum load P that can be applied to the truss so that the allowable stress will not be exceeded in



members BC and CH .

Part A - Find the normal forces in members BC and CH

Use the method of sections to determine the forces in members BC and CH in terms of P . Enter the force in BC , then the force in CH separated by a comma.

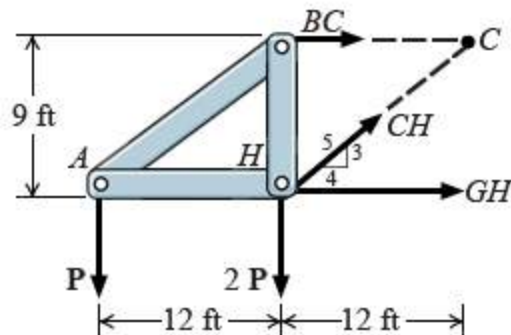
Express your answers in P to three significant figures.

Hint 1. How to approach the problem

Using the method of sections, cut vertically through the truss at members BC , CH , and GH ; take the left side as a free-body diagram. By taking the left side as a free-body diagram, you do not have to solve for any support reactions, thereby, simplifying this problem. Using the equations of equilibrium, solve for the forces in members BC and CH .

Hint 2. Find the force in member BC

Using the section free-body diagram shown below, sum moments about joint H to find the force in BC . This is the joint where the other two unknowns, CH and GH , intersect, so the moment equation will not involve either of these



unknowns.

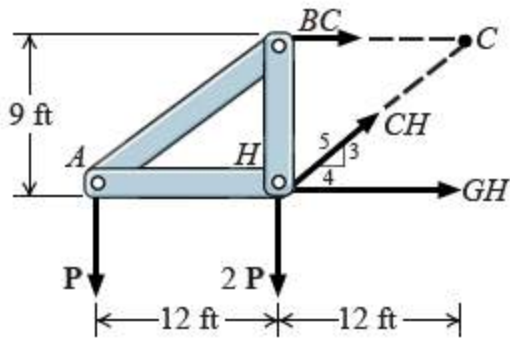
Express your answer in P to three significant figures.

ANSWER:

1.33 P

Hint 3. Find the force in member CH

Using the section free-body diagram, sum forces in the y direction to find the force in CH in terms of P .



Express your answer in P to three significant figures.

ANSWER:

5.00 P

ANSWER:

1.33, 5.00 P

Part B - Calculate the maximum allowable load, P , that can be applied to the truss

Apply the factor of safety to the yield stress to calculate the allowable stress, then use this allowable stress and the normal forces in members BC and CH to calculate the maximum allowable load, P .

Express your answer in kips to three significant figures.

Hint 1. Calculate the allowable stress

Apply the factor of safety to the yield stress to calculate the allowable stress.

Express your answer in ksi to three significant figures.

ANSWER:

14.4 ksi

Hint 2. Calculate the maximum allowable force in member BC

Using the axial stress equation and the allowable stress, calculate the maximum allowable axial force in member BC .

Express your answer in kips to three significant figures.

ANSWER:

21.6 kips

Note that the maximum allowable force in member CH will be the same as the maximum allowable force in BC because the cross-sectional area and allowable stress is the same for each truss member.

Hint 3. Calculate the allowable load, P , based on the maximum allowable force in member BC

Using the maximum allowable load in member BC , calculate the load, P , that can be applied to the truss.

Express your answer in kips to three significant figures.

ANSWER:

16.2 kips

Hint 4. Calculate the allowable load, P , based on the maximum allowable force in member CH

The maximum allowable force in member CH equals the maximum allowable force in member BC because they have the same material properties and the same cross-sectional area. Using the maximum allowable load in member CH , calculate the maximum allowable load, P , that can be applied to the truss.

Express your answer in kips to three significant figures.

ANSWER:

4.32 kips

Hint 5. Determine the maximum allowable load P that can be applied to the truss

Based on your calculations for members BC and CH , calculate the correct value of P such that the allowable stress is not exceeded.

Express your answer in kips to three significant figures.

ANSWER:

4.32 kips

In this situation, the smallest value of P , calculated based on the maximum allowable forces in each truss member, will be the correct answer so that the allowable stress is not exceeded.

ANSWER:

4.32 kips

Recall that an allowable stress is a value of stress that cannot be exceeded if a structure is to be considered safe. So, while the allowable stress may be reached in one member, another member may be subjected to a stress that is much lower than the allowable stress.

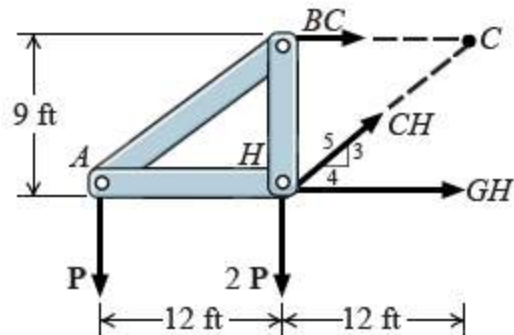
Part C - Find P based on the force in member GH

Using the allowable stress and the normal force in member GH , calculate the maximum allowable load, P . Note that, for structural steel, the yield strength is the same for tension and compression.

Express your answer in kips to three significant figures.

Hint 1. Find the axial force in member GH

Using the section free-body diagram, sum the moments about point C or sum forces in the x direction to find the force in member GH in terms of P . By summing moments about point C , the forces in members BC and CH are



eliminated, simplifying the equation.

Express your answer in P to three significant figures.

ANSWER:

-5.33 P

Hint 2. Calculate the allowable load, P , based on the normal stress in member GH

Calculate the allowable force P that can be applied to the truss based on the normal stress in member GH .

Express your answer in kips to three significant figures.

ANSWER:

4.05 kips

Note that 4.05 kips is actually less than the maximum load P that was calculated in Part B. Therefore, so that the maximum allowable normal stress of 14.4 ksi will not be exceeded, the maximum allowable load P is now 4.05 kips. An engineering designer must check *all* truss members to be sure that the axial stress does not exceed the allowable stress.

ANSWER:

4.05 kips

The force in member GH is in compression, but for steel the yield strength, and therefore the allowable stress, is the same in tension as in compression.

Hooke's Law

Description: In this tutorial, students will use the definitions of stress and strain, as well as Hooke's law, to calculate deformation, the modulus of elasticity, and the applied load.

Learning Goal:

To use Hooke's law and the definitions of stress and strain to calculate unknown values for linear elastic deformation of a rod.

A material undergoing uniaxial elastic deformation obeys Hooke's law. This law states that the stress is proportional to the strain, or $\sigma = E\epsilon$. The proportionality constant is called Young's modulus, or the modulus of elasticity.

Stress is a measure of force per unit area. When the stress is uniform over a region, it can be calculated using $\sigma = \frac{F}{A}$, where F is the load supported by the region, and A is the area of the region.

Strain is a measure of change in length per unit length, or $\epsilon = \frac{\Delta L}{L}$.

Part A - Change in length

A circular rod with a gage length of 3.2 m and a diameter of 3 cm is subjected to an axial load of 67 kN . If the modulus of elasticity is 200 GPa , what is the change in length?

Express your answer with appropriate units to three significant figures.

Hint 1. How to approach the problem

Use the given information to calculate the stress $\sigma = \frac{F}{A}$. Then use Hooke's law $\sigma = E\epsilon$ to calculate the strain. How is the strain related to the change in length?

Hint 2. Calculate the stress

What is the stress of a 67 kN load applied to a circular area with a diameter 3 cm ?

Express your answer with appropriate units to three significant figures.

ANSWER:

$$\sigma = \frac{4P1}{\pi D1D1} \cdot 10 = 94.8 \text{ MPa}$$

Hint 3. Calculate the strain

The stress and strain are related by $\sigma = E\epsilon$. What is the strain in the rod? Recall that the modulus of elasticity is $E = 200 \text{ GPa}$.

Express your answer to three significant figures.

ANSWER:

$$\epsilon = \frac{\frac{4P1}{E1\pi D1D1}}{100} = 4.74 \times 10^{-4}$$

ANSWER:

$$\Delta L = \frac{4P1L1}{E1\pi D1D1} \cdot 10 = 1.52 \text{ mm}$$

Part B - Modulus of elasticity

A circular rod with a length of 3.2 m and a diameter of 2.7 cm is subjected to an axial load of 67 kN . If the resulting change in length is 3.3 mm , what is the modulus of elasticity?

Express your answer with appropriate units to three significant figures.

Hint 1. How to approach the problem

Use the given information to calculate the stress $\sigma = \frac{F}{A}$ and strain $\epsilon = \frac{\Delta L}{L}$. Then use Hooke's law $\sigma = E\epsilon$ to calculate the modulus of elasticity.

Hint 2. Calculate the stress

What is the stress of a 67 kN load applied to a circular area with a diameter 2.7 cm ?

Express your answer with appropriate units to three significant figures.

ANSWER:

$$\sigma = \frac{4P^2}{\pi D^2 D^2} \cdot 10 = 117 \text{ MPa}$$

Hint 3. Calculate the strain

What is the strain in the rod if the original length is 3.2 m and the change in length is 3.3 mm ?

Express your answer to three significant figures.

ANSWER:

$$\epsilon = \frac{DL2}{L2 \cdot 1000} = 1.03 \times 10^{-3}$$

The stress and strain are related by the modulus of elasticity by $\sigma = E\epsilon$.

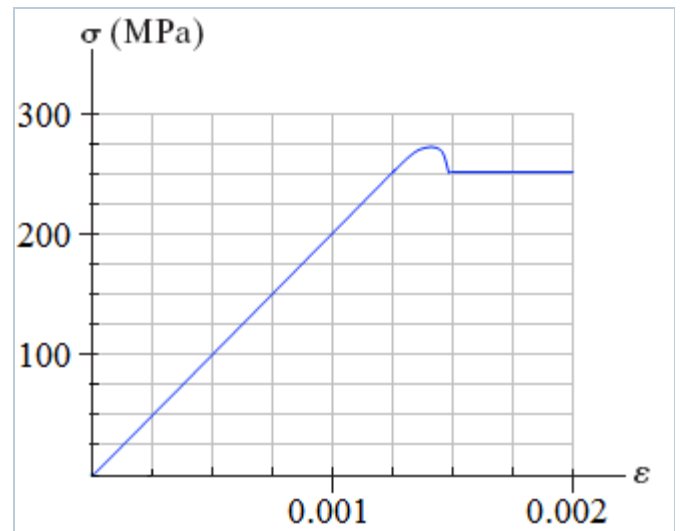
ANSWER:

$$E = \frac{4P^2 L2}{\pi D^2 D^2 DL2} \cdot 10 = 113 \text{ GPa}$$

Part C - Applied load

A circular rod with a length of 3.7 m and a diameter of 2.4 cm is subjected to an axial load and increases in length by 4 mm . The initial portion of the stress-strain curve of the material is given . What is the applied load?

Express your answer with appropriate units to three significant figures.

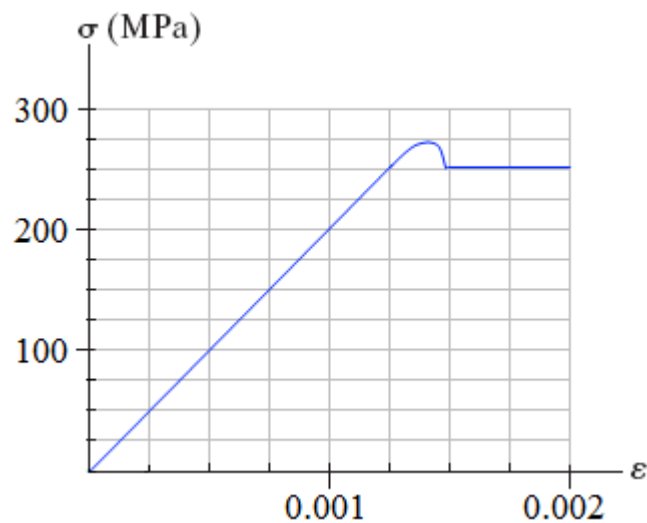


Hint 1. How to approach the problem

Use the stress-strain curve to determine the modulus of elasticity. Use the deformation and length to calculate the strain $\epsilon = \frac{\Delta L}{L}$. Then use Hooke's law to calculate the stress $\sigma = E\epsilon$. How is the stress related to the applied load?

Hint 2. Calculate the modulus of elasticity

Use the stress-strain curve to determine the modulus of elasticity for the material.



Express your answer with appropriate units to three significant figures.

ANSWER:

$$E = 200 \text{ GPa}$$

Hint 3. Calculate the strain

What is the strain in the rod if the original length is 3.7 m and the change in length is 4 mm ?

Express your answer to three significant figures.

ANSWER:

$$\epsilon = \frac{DL3}{L3 \cdot 1000} = 1.08 \times 10^{-3}$$

The stress and strain are related by the modulus of elasticity by $\sigma = E\epsilon$.

Hint 4. Calculate the stress

The stress and strain are related by $\sigma = E\epsilon$. What is the stress in the rod? Recall that the modulus of elasticity is $E = 200 \text{ GPa}$.

Express your answer with appropriate units to three significant figures.

ANSWER:

$$\sigma = \frac{E3DL3}{L3} = 216 \text{ MPa}$$

ANSWER:

$$F = \frac{E3DL3 \pi D3D3}{4L3} = 97.8 \text{ kN}$$

Poisson's Ratio

Description: In this tutorial, students will use the definition of Poisson's ratio and Hooke's law to relate the dimensional changes of square bars to axial forces acting on them.

Learning Goal:

To use Poisson's ratio and Hooke's law to examine the relationship between axial and lateral deformation by considering the deformation of an axially loaded bar.

When a force is applied to an object, it produces a deformation along the axis of the applied force. The object will also deform along axes perpendicular to the applied force. As long as the response of the material remains elastic, the relationship between the strains in the different directions can be described by a material property called Poisson's ratio, ν .

For the common case of a uniaxial load, the longitudinal (or axial) strain ϵ_{long} is related to the lateral strain ϵ_{lat} by

$\nu = -\frac{\epsilon_{\text{lat}}}{\epsilon_{\text{long}}}$. The negative sign indicates that axial stretching leads to lateral contraction while allowing Poisson's ratio to be a positive number.

Part A - Calculate Poisson's ratio

A square steel bar has a length of 6.7 ft and a 2.9 in by 2.9 in cross section and is subjected to axial tension. The final length is 6.70235 ft. The final side length is 2.89975 in. What is Poisson's ratio for the material?

Express your answer to three significant figures.

Hint 1. How to approach the problem

The given information can be used to calculate both the lateral and longitudinal strain. Once those are known, Poisson's ratio can be used.

Hint 2. Calculate the longitudinal strain

The original length is 6.7 ft and the final length is 6.70235 ft . What is the longitudinal strain?

Express your answer to three significant figures.

ANSWER:

$$\epsilon_{\text{long}} = \frac{L_{2b} - L_{2a}}{L_{2a}} = 3.51 \times 10^{-4}$$

Hint 3. Calculate the lateral strain

The original side length is 2.9 in and the final side length is 2.89975 in . What is the lateral strain?

Express your answer to three significant figures.

ANSWER:

$$\epsilon_{\text{lat}} = \frac{D_{2b} - D_{2a}}{D_{2a}} = -8.62 \times 10^{-5}$$

ANSWER:

$$\nu = \frac{-(D_{2b} - D_{2a}) L_{2a}}{D_{2a} (L_{2b} - L_{2a})} = 0.246$$

Part B - Calculate the load

A square steel bar has a length of 3.8 ft and a 1.7 in by 1.7 in cross section and is subjected to axial compression. Under load, the thickness becomes 1.70025 in . If the modulus of elasticity is $E = 2.9 \times 10^4$ ksi and Poisson's ratio is $\nu = 0.26$; what is the magnitude of the load that was applied to the bar?

Express your answer with appropriate units to three significant figures.

Hint 1. How to approach the problem

Use the given information to calculate the lateral strain in the bar. Then use Poisson's ratio to calculate the longitudinal strain. How is the longitudinal strain related to the load?

Hint 2. Calculate the lateral strain

The bar is square with an original thickness of 1.7 in . The final thickness is 1.70025 in . What is the lateral strain?

Express your answer to three significant figures.

ANSWER:

$$\epsilon_{\text{lat}} = \frac{D_{1b} - D_{1a}}{D_{1a}} = 1.47 \times 10^{-4}$$

Hint 3. Calculate the longitudinal strain

Once the lateral strain is known, use it to calculate the longitudinal strain. Recall that Poisson's ratio is given as $\nu = 0.26$. What is the longitudinal strain under the given load?

Express your answer to three significant figures.

ANSWER:

$$\epsilon_{\text{long}} = \frac{\frac{-(D_{1b} - D_{1a})}{D_{1a}}}{\nu} = -5.66 \times 10^{-4}$$

Hint 4. Calculate the axial stress

The axial stress is related to the longitudinal strain using Hooke's law. Use the calculated longitudinal strain to calculate the stress in the bar. Recall that $E = 2.9 \times 10^4$ ksi .

Express your answer with appropriate units to three significant figures.

ANSWER:

$$\sigma = \frac{\frac{-(D_{1b} - D_{1a})}{D_{1a}}}{\nu} E = -16.4 \text{ ksi}$$

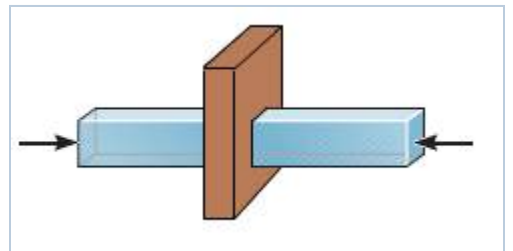
ANSWER:

$$F = \frac{D_{1b} - D_{1a}}{\nu} E D_{1a} = 47.4 \text{ kip}$$

Part C - Maximum load

A slider with a square hole is mounted on the bar described in part B . The largest load the bar ever carries is 116 kip . What is the minimum side length of the square hole so that it will not bind on the bar?

Express your answer with appropriate units to six significant figures.



Hint 1. How to approach the problem

Under compression, the bar gets slightly thicker. Use the maximum load and Poisson's ratio to find the largest thickness that the bar will have under load.

Hint 2. Calculate the longitudinal strain

The bar is 1.7 in thick. If the applied compressive force is 116 kip , what is the strain? Recall that $E = 2.9 \times 10^4$ ksi .

Express your answer to three significant figures.

Hint 1. Calculate the axial stress

The axial stress is the applied load divided by the area. Use the thickness 1.7 in and 116 kip to first calculate the stress. Be sure to use the correct sign.

Express your answer with appropriate units to three significant figures.

ANSWER:

$$\sigma = \frac{-P1}{D1a^2} = -40.1 \text{ ksi}$$

It is plausible that this stress would still be in the elastic region, as is being assumed. It is below the yield stress for A992 structural steel ($\sigma_Y = 50$ ksi) and L2 tool steel ($\sigma_Y = 102$ ksi).

ANSWER:

$$\epsilon_{\text{long}} = \frac{-P1}{D1a^2 E1} = -1.38 \times 10^{-3}$$

Hint 3. Calculate the lateral strain

Use the longitudinal strain and Poisson's ratio $\nu = 0.26$ to calculate the lateral strain.

Express your answer to three significant figures.

ANSWER:

$$\epsilon_{\text{lat}} = \frac{P1}{D1a^2 E1} \nu 1 = 3.60 \times 10^{-4}$$

ANSWER:

$$s_{\text{min}} = D1a \left(1 + \frac{P1}{D1a^2 E1} \nu 1 \right) = 1.70061 \text{ in}$$

The change in the side length is very small. In fact, it is significantly smaller than typical tolerances (permissible variations) used in machining and production. For example, machined parts without strict tolerances often use 0.01 or 0.005 in.

Statics and Mechanics of Materials, 6e

Hibbeler

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Ends: 05/31/26



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HOMEWORK #2 – AXIAL DEFORMATION & BUCKLING

● HW2 - Axial Deformation; Buckling

[Edit](#)
[Overview](#)
[Diagnostics](#)
[Print View with Answers](#)

HW2 - Axial Deformation; Buckling

Due: 11:59pm on Wednesday, January 21, 2026

You will receive no credit for items you complete after the assignment is due. [Grading Policy](#)

Columns Having Various Types of Supports

Description: In this tutorial, students will calculate maximum axial loads for a column based on yielding and bending about two axes with lateral support about one axis and including factors of safety.

Learning Goal:

To calculate the maximum axial loads for a column based on yielding and buckling about both axes, and allowing for factors of safety.

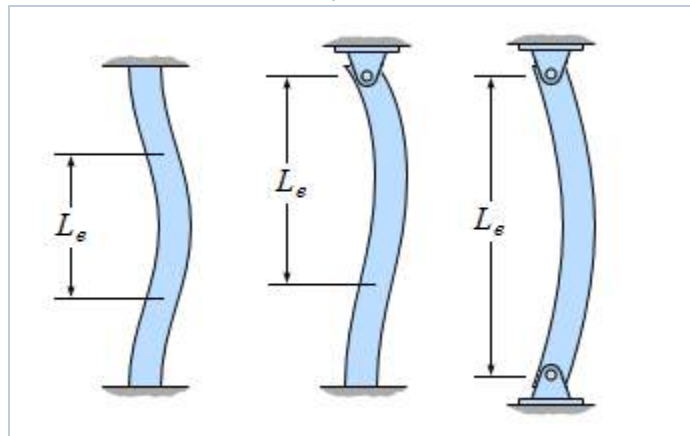
The critical load P_{cr} for buckling of a column depends on the vertical distance between supports and how the column is connected to those supports. The length L in the formula $P_{cr} = \frac{\pi^2 EI}{L^2}$ is the distance between points of zero moment. For a pin-supported column, this is the full height of the column.

For a column that is not pinned at both ends, the points of zero moment are not at the supports. The equation for the critical load can be modified to use an effective length $L_e = KL$, where K depends on the support conditions. For a column fixed at both ends, $K = 0.5$. For a column fixed at one end and pinned at the other, $K = 0.7$. The equation for the critical load then becomes

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

A column can also have lateral supports along its height. A lateral support prevents lateral movement in one direction, which reduces the effective length, L_e .

Many column sections are not circular. Since the critical load depends on the moment of inertia, it also depends on the axis about which the bending occurs. The axis with the smallest moment of inertia is called the weak axis, and the axis with the largest moment of inertia is called the strong axis.



A factor of safety is used to account for uncertainties in material properties and dimensions. The factor of safety is the ratio of the failure load to the allowable load, or the ratio of the maximum design load to the maximum expected load.

A column has a 5.1 cm by 8.7 cm rectangular cross section and a height 4 m. The column is fixed at both ends and has a lateral support at mid-height for the weak axis. The material has $E = 200 \text{ GPa}$ and $\sigma_Y = 250 \text{ MPa}$.

Part A - Yielding

The factor of safety for the column yielding is $FS = 1.5$. What is the maximum load that can be applied based on the column yielding due to the normal stress? Assume the load is applied at the centroid of the section.

Express your answer with appropriate units to three significant figures.

Hint 1. How to approach the problem

For a column with a load applied at the centroid, the normal stress is the average normal stress based on the load and the cross sectional area.

Hint 2. Calculate the load to cause yielding

What load will cause yielding in the column, assuming that it does not buckle?

Express your answer with appropriate units to three significant figures.

ANSWER:

$$P = \frac{\sigma Y wh}{10} = 1110 \text{ kN}$$

ANSWER:

$$P_{\max} = \frac{\frac{\sigma Y wh}{FSI}}{10} = 740 \text{ kN}$$

Part B - Strong axis buckling

The design is to have a factor of safety for buckling $FS = 2.5$. What is the maximum load based on buckling in the strong axis of the column?

Express your answer with appropriate units to three significant figures.

Hint 1. How to approach the problem

Determine the appropriate effective length and the moment of inertia for bending about the strong axis and use those values in the equation for the critical load.

Hint 2. Determine the effective length

What is the effective length for buckling about the strong axis for the column?

Express your answer in m to three significant figures.

ANSWER:

$$L_e = 0.5L = 2.00 \text{ m}$$

Hint 3. Calculate the moment of inertia

What is the strong axis bending moment of inertia for a rectangular section that is 5.1 cm by 8.7 cm ?

Express your answer with appropriate units to three significant figures.

ANSWER:

$$I = \frac{wh^3}{12} = 280 \text{ cm}^4$$

Hint 4. Calculate the critical load

Calculate the critical load for strong axis buckling.

Express your answer with appropriate units to three significant figures.

ANSWER:

$$P_{cr} = \frac{\frac{\pi^2 E (wh^3)}{12}}{\frac{0.25L^2}{100}} = 1380 \text{ kN}$$

Now use the factor of safety to determine the maximum load that can be applied considering strong-axis buckling.

ANSWER:

$$P_{strong} = \frac{\frac{\pi^2 E (wh^3)}{12}}{\frac{(0.5L)^2}{100}} = 552 \text{ kN}$$

Part C - Weak axis buckling

The design is to have a factor of safety for buckling $FS = 2.5$. What is the maximum load based on buckling in the weak axis of the column?

Express your answer with appropriate units to three significant figures.

Hint 1. How to approach the problem

Determine the appropriate effective length and the moment of inertia for bending about the weak axis and use those values in the equation for the critical load.

Hint 2. Determine the effective length

The lateral support at the mid-height will cause the buckling to be symmetric about that point. In this case, the support acts like a pin, so each half of the column will buckle like a pin/fixed column. What is the effective length for buckling about the weak axis for the column?

Express your answer in m to three significant figures.

ANSWER:

$$KL = 0.5 \cdot 0.7L = 1.40 \text{ m}$$

Hint 3. Calculate the moment of inertia

What is the weak axis bending moment of inertia for a rectangular section that is 5.1 cm by 8.7 cm ?

Express your answer with appropriate units to three significant figures.

ANSWER:

$$I = \frac{hw^3}{12} = 96.2 \text{ cm}^4$$

Hint 4. Calculate the critical load

Calculate the critical load for weak axis buckling.

Express your answer with appropriate units to three significant figures.

ANSWER:

$$P_{cr} = \frac{\frac{\pi^2 E (hw^3)}{12}}{(0.5 \cdot 0.7L)^2} = 969 \text{ kN}$$

Now use the factor of safety to determine the maximum load that can be applied considering strong-axis buckling.

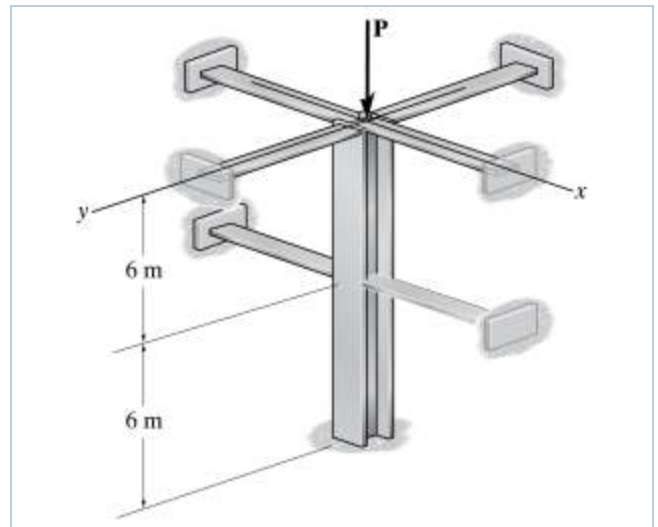
ANSWER:

$$P_{weak} = \frac{\frac{\pi^2 E (hw^3)}{12}}{\frac{FS^2}{100}} = 387 \text{ kN}$$

Fundamental Problem 17.3

Description: Assume the A992 steel column is pinned at its top and bottom and braced against its weak axis at the mid-height in . Take $A = 7.4(10^{-3}) \text{ m}^2$, $I_x = 87.3(10^{-6}) \text{ m}^4$, and $I_y = [\dots]$. Apply a F.S. = 2 against buckling. (a) Determine the maximum...

Assume the A992 steel column is pinned at its top and bottom and braced against its weak axis at the mid-height in . Take $A = 7.4(10^{-3}) \text{ m}^2$, $I_x = 87.3(10^{-6}) \text{ m}^4$, and $I_y = 15.8(10^{-6}) \text{ m}^4$. Apply a F. S. = 2 against buckling.



Part A

Determine the maximum allowable force P that the column can support without buckling.

Express your answer to three significant figures and include the appropriate units.

ANSWER:

$$P_{\text{allow}} = \frac{\frac{\pi^2 \cdot 200 \cdot 10^3 I_y}{36}}{2000} = 433 \text{ kN}$$

Fundamental Problem 17.4

Description: A steel pipe is fixed supported at its ends. It is 5 m long and has an outer diameter of 50 mm and a thickness of 10 mm. Take $E_{\text{st}} = 300 \text{ GPa}$, $\sigma_Y = 250 \text{ MPa}$. (a) Determine the maximum axial load P that it can carry without buckling.

A steel pipe is fixed supported at its ends. It is 5 m long and has an outer diameter of 50 mm and a thickness of 10 mm. Take $E_{\text{st}} = 300 \text{ GPa}$, $\sigma_Y = 250 \text{ MPa}$.

Part A

Determine the maximum axial load P that it can carry without buckling.

Express your answer to three significant figures and include the appropriate units.

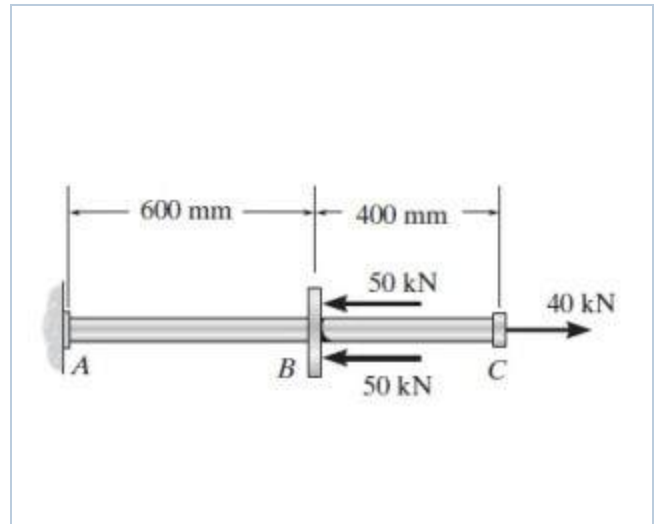
ANSWER:

$$P = \frac{\pi^2 E_{\text{st}} \cdot 267.04}{0.25 \cdot 25} = 1.27 \times 10^5 \text{ N}$$

Fundamental Problem 9.1

Description: The ##-mm-diameter A-36 steel rod is subjected to the axial forces shown. (a) Determine the displacement of end C with respect to the fixed support at A.

The 17-mm-diameter A-36 steel rod is subjected to the axial forces shown.



Part A

Determine the displacement of end C with respect to the fixed support at A .

Express your answer to three significant figures and include the appropriate units.

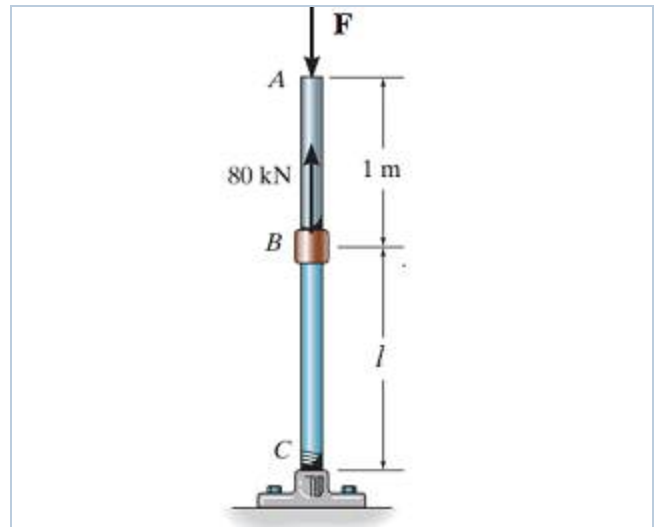
ANSWER:

$$\delta_C = \frac{-20 \cdot 1000 \cdot 10^6 \cdot 4 \cdot 1000}{200 \cdot 10^9 d^2 \pi} = -0.441 \text{ mm}$$

Problem 9.2

Description: The assembly consists of an A-36 steel rod CB and an a 2014-T6 aluminum rod BA, each having a diameter of 20 mm. The rod is subjected to the axial loadings at A and at the coupling B. Take $l = \text{##} \text{ m}$ and $F = \text{##} \text{ kN}$. Neglect the size of the connections at B...

The assembly consists of an A-36 steel rod CB and an a 2014-T6 aluminum rod BA , each having a diameter of 20 mm. The rod is subjected to the axial loadings at A and at the coupling B . Take $l = 1.7 \text{ m}$ and $F = 50 \text{ kN}$. Neglect the size of the connections at B and C .



Part A

Determine the displacement of the coupling B .

Express your answer to three significant figures and include appropriate units.

ANSWER:

$$\delta_B = \frac{(80 - F) \cdot 1000l}{0.1\pi \cdot 0.001 \cdot 200 \cdot 10^9} \cdot 1000 = 0.812 \text{ mm}$$

Part B

Determine the displacement of the end A .

Express your answer to three significant figures and include appropriate units.

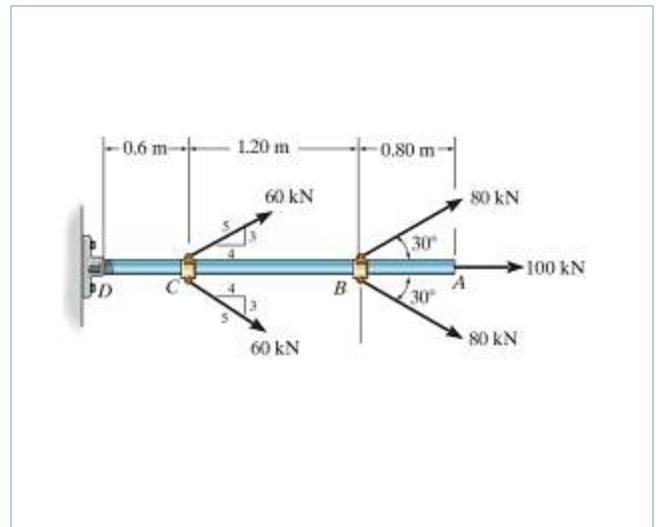
ANSWER:

$$\delta_A = \left(\frac{(80 - F) \cdot 1000l}{0.1\pi \cdot 0.001 \cdot 200 \cdot 10^9} + \frac{-F \cdot 1000}{0.1\pi \cdot 0.001 \cdot 73.1 \cdot 10^9} \right) \cdot 1000 = -1.37 \text{ mm}$$

Problem 9.5

Description: The A992 steel rod is subjected to the loading shown. The diameter of the rod is ## mm. Neglect the size of the couplings at B , C , and D . (a) Determine the displacement of B . (b) Determine the displacement of A .

The A992 steel rod is subjected to the loading shown. The diameter of the rod is 20 mm. Neglect the size of the couplings at B , C , and D .



Part A

Determine the displacement of B .

Express your answer to three significant figures and include appropriate units.

ANSWER:

$$\delta_B = \left(\frac{334.56 \cdot 0.6 \cdot 1000}{\frac{\pi}{4} \left(\frac{d}{1000} \right)^2 \cdot 200 \cdot 10^9} + \frac{238.5 \cdot 1000 \cdot 1.2}{\frac{\pi}{4} \left(\frac{d}{1000} \right)^2 \cdot 200 \cdot 10^9} \right) \cdot 1000 = 7.75 \text{ mm}$$

Part B

Determine the displacement of A .

Express your answer to three significant figures and include appropriate units.

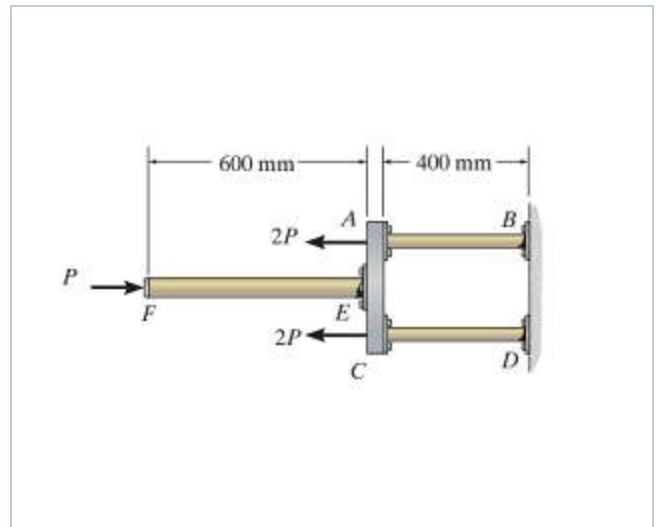
ANSWER:

$$\delta_A = \left(\frac{334.56 \cdot 0.6 \cdot 1000}{\frac{\pi}{4} \left(\frac{d}{1000} \right)^2 \cdot 200 \cdot 10^9} + \frac{238.5 \cdot 1000 \cdot 1.2}{\frac{\pi}{4} \left(\frac{d}{1000} \right)^2 \cdot 200 \cdot 10^9} + \frac{100 \cdot 1000 \cdot 0.8}{\frac{\pi}{4} \left(\frac{d}{1000} \right)^2 \cdot 200 \cdot 10^9} \right) \cdot 1000 = 9.02 \text{ mm}$$

Problem 9.7

Description: The assembly consists of two 12-mm-diameter A992 steel rods AB and CD , a 20-mm-diameter 6061-T6 aluminum rod EF , and a rigid bar AEC . (a) If $P = \#\# \text{ kN}$, determine the displacement of F .

The assembly consists of two 12-mm-diameter A992 steel rods AB and CD , a 20-mm-diameter 6061-T6 aluminum rod EF , and a rigid bar AEC .



Part A

If $P = 30 \text{ kN}$, determine the displacement of F .

Express your answer to three significant figures and include appropriate units.

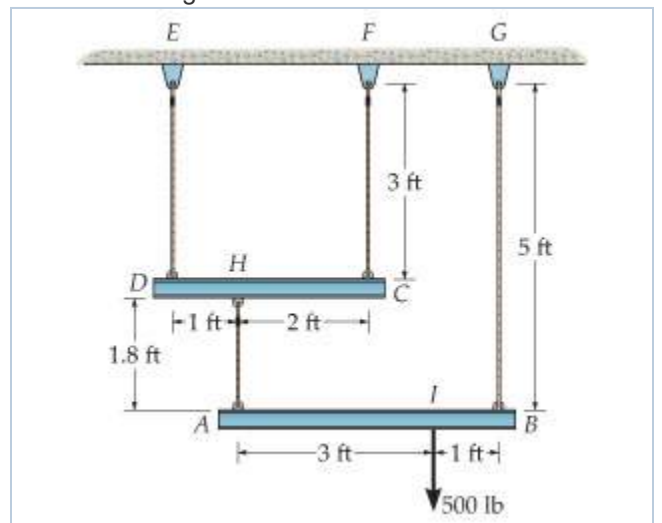
ANSWER:

$$\delta_F = \left(\frac{-P \cdot 1000 \cdot 0.6}{68.9 \cdot 10^9 \cdot 0.1 \pi \cdot 0.001} + \frac{\frac{-P + (4P)}{2} \cdot 1000 \cdot 0.4}{36 \cdot 10^{-6} \pi \cdot 200 \cdot 10^9} \right) \cdot 1000 = -3.58 \times 10^{-2} \text{ mm}$$

Problem 9.11

Description: The load is supported by the four 304 stainless steel wires that are connected to the rigid members AB and DC. (a) Determine the vertical displacement of the 500-lb load if the members are originally horizontal when the load is applied. Each wire has ...

The load is supported by the four 304 stainless steel wires that are connected to the rigid members AB and DC.



Part A

Determine the vertical displacement of the 500-lb load if the members are originally horizontal when the load is applied. Each wire has a cross-sectional area of 0.025 in^2 .

Express your answer in inches to three significant figures.

ANSWER:

$$\delta_l = 2.60 \times 10^{-2} \text{ in.}$$

[◀ All Assignments](#)

Statics and Mechanics of Materials, 6e

Hibbeler

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Ends: 05/31/26



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HOMEWORK #3 – INDETERMINATE AXIAL MEMBERS

● HW3 - Indeterminate Axial Members

[Edit](#)
[Overview](#)
[Diagnostics](#)
[Print View with Answers](#)

HW3 - Indeterminate Axial Members

Due: 11:59pm on Thursday, January 29, 2026

You will receive no credit for items you complete after the assignment is due. [Grading Policy](#)

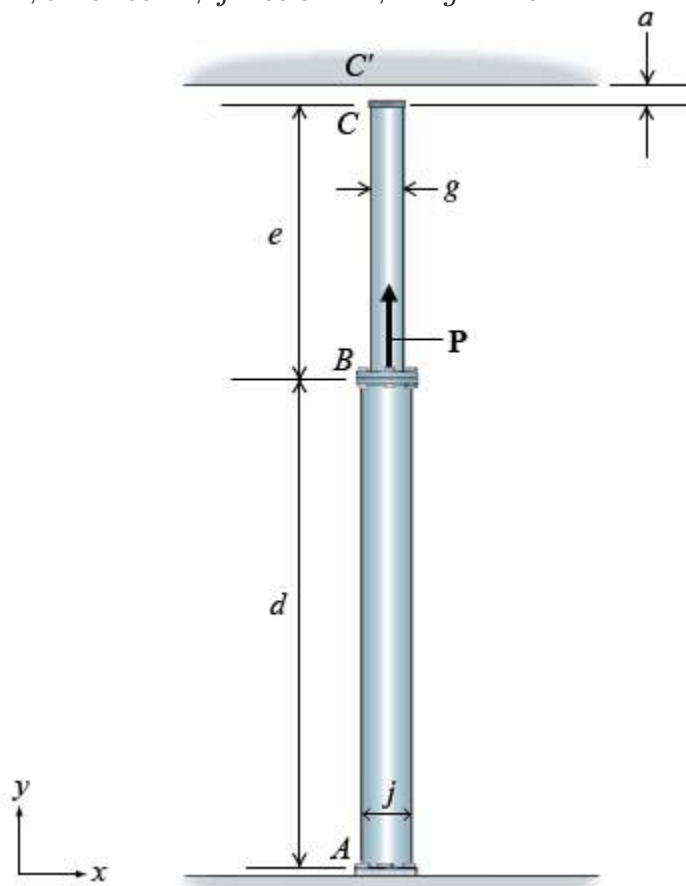
Principle of Superposition

Description: In this tutorial, students are asked to find the reaction forces in a rod. To solve this problem, the principle of superposition must be used.

Learning Goal:

To use the principle of superposition to determine the total deflection in a cylindrical rod due to a static loading.

A steel rod ($E = 200 \text{ GPa}$) is subjected to the load shown, where $P = 2800 \text{ kN}$. A gap $a = 1.80 \text{ mm}$ exists before the load is applied. The elongated rod contacts the top surface at C' . Assume the mass of the rod is negligible. The values for the figure below are $d = 0.700 \text{ m}$, $e = 0.200 \text{ m}$, $j = 60.0 \text{ mm}$, and $g = 44.0 \text{ mm}$.



Part A - Reaction force at C' after the load has been applied

Find the magnitude of the reaction force $\mathbf{F}_{C'}$ in the rod after the load $\mathbf{P}$ has been applied.

Express your answer to three significant figures and include appropriate units.

Hint 1. Analyzing the steel rod after deformation

After the loading is applied and assuming that the force is large enough to cause the rod's end C to contact the wall at C', the rod becomes statically indeterminate because we only have one equilibrium equation and two unknown reactions. Therefore, an additional compatibility equation is needed using $\delta_{C/A} = 1.80\text{mm}$.

Hint 2. Identify the equation for summation of forces in the y direction

To use the compatibility equation, the forces in each segment of the rod have to be determined. Select the correct equation for the sum of the forces in the y direction.

ANSWER:

- ☒ $\sum F_y = P - F_{C'} - F_A = 0$
- ☐ $\sum F_y = F_{C'} - F_A = 0$
- ☐ $\sum F_y = P - F_A = 0$
- ☐ $\sum F_y = P + F_{C'} + F_A = 0$

Hint 3. Find the force in segment AB in terms of the reaction force $F_{C'}$

To use the compatibility equation, the reaction force at A must be known. Find the magnitude of force F_A in the rod in terms of $F_{C'}$. Represent the magnitude of the force $F_{C'}$ with the variable F_R .

Express your answer in terms of the applied load P and the unknown reaction force at C', representing this force with the variable F_R .

ANSWER:

$$F_A = P - F_R$$

ANSWER:

$$F_{C'} = \frac{\left[\frac{a}{10^3} \cdot 200 \cdot 10^9 - \frac{b \cdot 10^3 d}{\pi \left(\frac{j}{2 \cdot 10^3} \right)^2} \right]}{\left[\frac{-d}{\pi \left(\frac{j}{2 \cdot 10^3} \right)^2} - \frac{e}{\pi \left(\frac{g}{2 \cdot 10^3} \right)^2} \right]} = 8.79 \times 10^5 \text{ N}$$

Part B - Reaction at A after the load has been applied

Determine the magnitude of the reaction force F_A in the steel rod after the load has been applied. Assume section AB of the rod is in tension.

Express your answer to three significant figures and include appropriate units.

Hint 1. How to find the reaction force in the rod after the loads have been applied

To determine the reaction force at A, draw a free-body diagram of the entire rod with all of the forces represented. Assume that the rod will elongate such that point C will contact the top wall. With the reaction force $\mathbf{F}_{C'}$ known, sum the forces in the y direction to calculate the reaction force at A.

Hint 2. Identify the equilibrium equation for the entire rod

To find the reaction force at A, the equilibrium equation must first be obtained from the free-body diagram of the entire rod. Select the correct equilibrium equation (solved for the reaction force at A) for the entire rod.

ANSWER:

- ☒ $\mathbf{F}_A = \mathbf{P} - \mathbf{F}_{C'}$
- ☐ $\mathbf{F}_A = \mathbf{P}$
- ☐ $\mathbf{F}_A = \mathbf{F}_{C'}$
- ☐ $\mathbf{F}_A = \mathbf{P} + \mathbf{F}_{C'}$

ANSWER:

$$F_A = (b \cdot 10^3) - \frac{\left[\frac{a}{10^3} \cdot 200 \cdot 10^9 - \frac{b \cdot 10^3 d}{\pi \left(\frac{j}{2 \cdot 10^3} \right)^2} \right]}{\left[\frac{-d}{\pi \left(\frac{j}{2 \cdot 10^3} \right)^2} - \frac{e}{\pi \left(\frac{g}{2 \cdot 10^3} \right)^2} \right]} = 1.92 \times 10^6 \text{ N}$$

Part C - Total deflection in the rod when the upper wall is removed

Determine the total deflection $\delta_{C/A}$ in the rod when the upper wall is removed.

Express your answer to three significant figures and include appropriate units.

Hint 1. How to approach the problem

To find the total deflection in the rod, all the forces in the rod must be known. Because the top wall is removed, the problem is now statically determinate with one equilibrium equation and one unknown reaction force. Draw a free-body diagram of the entire rod, then using Saint Venant's principle, make the necessary cuts to determine the axial forces acting on the rod in segments AB and BC.

Hint 2. Find the reaction force at A

With the wall removed, the problem is now statically determinate and the reaction force can be found directly from a free-body diagram of the entire rod. Find the magnitude of the reaction force $\mathbf{F}_A$ in the rod.

Express your answer to four significant figures and include appropriate units.

ANSWER:

$$F_A = b = 2800 \text{ kN}$$

Hint 3. Find the deflection in segment BC

Once the forces are known in each segment of the rod, the deflection due to the forces can be calculated. Find the deflection $\delta_{C/B}$ in the rod.

Express your answer to three significant figures and include appropriate units.

ANSWER:

$$\delta_{C/B} = 0.00 \text{ m}$$

ANSWER:

$$\delta_{C/A} = \frac{b \cdot 10^3 d}{\pi \left(\frac{j}{2 \cdot 10^3}\right)^2 \cdot 200 \cdot 10^9} = 3.47 \times 10^{-3} \text{ m}$$

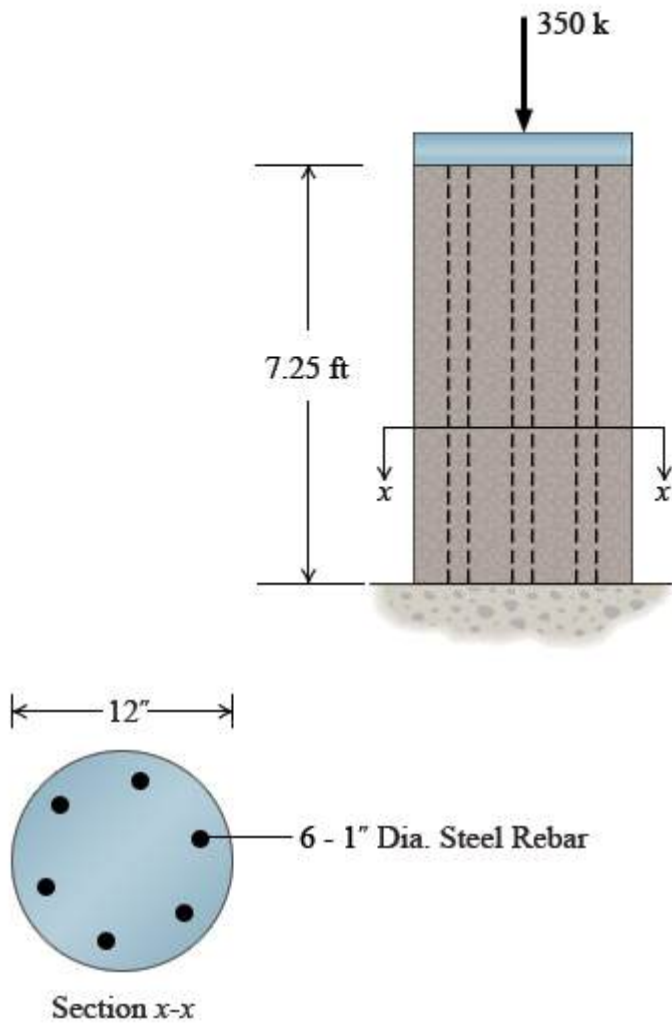
Statically Indeterminate Axial Members

Description: In this tutorial, students will develop an equation of static equilibrium from a free-body diagram, then use the deformation compatibility relationship to solve for unknown internal axial forces in two materials. Once the forces have been determined, students will find the axial stress developed in each material.

Learning Goal:

To learn how to find internal forces for a statically indeterminate axially loaded structure by using deformation compatibility relationships and static equilibrium equations.

A reinforced concrete pier supports a 350 kip axial compressive load as shown. The 12-in-diameter concrete pier is reinforced with 6 #8 (i.e., 1-in diameter) steel reinforcing bars (i.e., rebar). The steel and the concrete are assumed to be bonded together (i.e., there is no slip between them upon loading). The modulus of elasticity for the steel rebars is 29,000 ksi and the modulus of elasticity for the concrete is 4,200 ksi.



Part A - Find the axial stress in the concrete when the pier is loaded

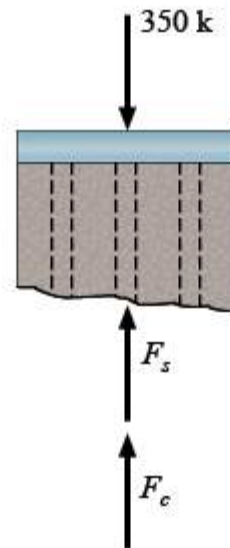
Find the axial stress in the concrete when the pier is loaded.

Express your answer in ksi to three significant figures. Indicate compressive stresses with a negative sign. A value entered with no sign is assumed to be a tensile stress.

Hint 1. How to approach the problem

Draw a free-body diagram of the pier to determine the internal forces in the steel and the concrete. Then, use $\sum F_y = 0$ to obtain a statics equation. Note that there are two unknowns in this equation: the internal force in the steel, F_S , and the internal force in the concrete, F_C . Therefore, this structure is statically indeterminate and you

must find a deformation compatibility relationship between the concrete and the steel to solve for the forces



developed in each material.

Hint 2. Relate the deformation of the steel to the deformation of the concrete

Relate the deformation of the steel to the deformation of the concrete when the 350 kip load is applied. Note that because the materials are bonded together, both will shorten by the same amount and $\delta_S = \delta_C$. Use $\delta = PL/AE$, substituting in the values for all known quantities.

Express your answer as the force in the steel in terms of the force in the concrete to three significant figures.

Hint 1. Calculate the area of the steel, A_S , and the area of the concrete, A_C

Calculate the area of the steel, A_S , and the area of the concrete, A_C .

Express your answers in square inches to three significant figures separated by a comma.

ANSWER:

4.71, 108 in²

ANSWER:

$F_S = 0.300 F_C$

Hint 3. Solve for the force in the concrete

Using your statics equation, $350 = F_S + F_C$, substitute in your compatibility relationship for F_S (obtained in Hint A2) and solve for the force in the concrete, F_C .

Express your answer in kips to three significant figures. If the force is compressive, enter your answer as a negative number. A value entered without a sign is assumed to be in tension.

ANSWER:

-269 kips

ANSWER:

-2.48 ksi

Part B - Find the axial stress in the steel when the pier is loaded

Find the axial stress in the steel when the pier is loaded.

Express your answer in ksi to three significant figures. Indicate compressive stresses with a negative sign. A value entered with no sign is assumed to be tensile.

Hint 1. Calculate the force in the steel

Use the statics equation and the known force in the concrete, F_C , to calculate the force in the steel

Express your answer in kips to three significant figures. If the force is compressive, enter it with a negative sign. A value entered without a sign is assumed to be tensile.

ANSWER:

-80.8 kips

Now that the axial force in the steel is known, apply the axial stress equation to calculate the stress in the steel when the pier is loaded.

ANSWER:

-17.1 ksi

Part C - Calculate the axial deformation of the reinforced concrete pier

Calculate the axial deformation of the reinforced concrete pier.

Express your answer in inches to three significant figures. If the pier gets shorter, be sure to include a negative sign. (Note that because they are equal, either the deformation of the concrete or the steel can be used. This was the deformation compatibility relationship upon which the problem solution was based.)

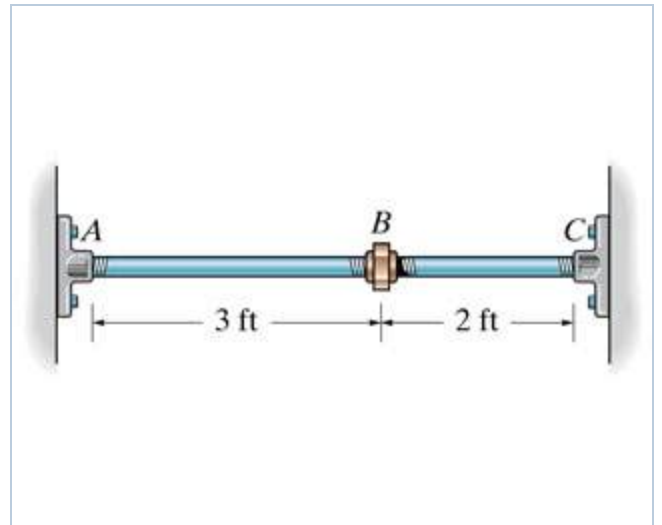
ANSWER:

-5.14×10^{-2} in

Review Problem 9.4

Description: Two A992 steel pipes, each having a cross-sectional area of 0.32 (in)^2 , are screwed together using a union at B as shown. Originally the assembly is adjusted so that no load is on the pipe. (a) If the union is then tightened so that its screw,...

Two A992 steel pipes, each having a cross-sectional area of 0.32 in^2 , are screwed together using a union at B as shown. Originally the assembly is adjusted so that no load is on the pipe.



Part A

If the union is then tightened so that its screw, having a lead of 0.18 in. , undergoes two full turns, determine the average normal stress developed in the pipe. Assume that the union and couplings at A and C are rigid. Neglect the size of the union. *Note:* The lead would cause the pipe, when *unloaded*, to shorten 0.18 in. when the union is rotated one revolution.

Express your answer in kilopounds per square inch to three significant figures.

ANSWER:

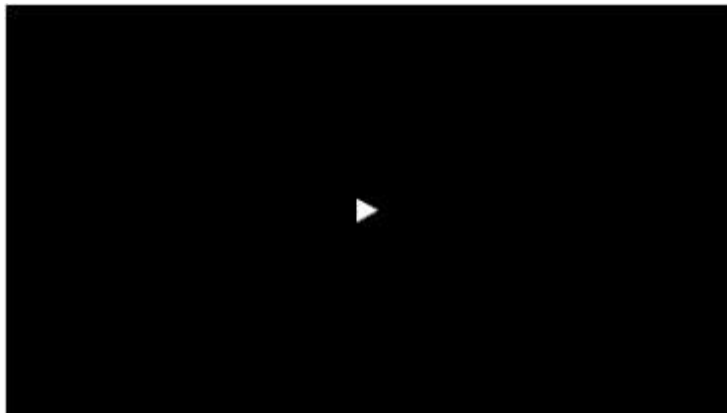
$$\sigma = \frac{2x \cdot 29}{60} \cdot 1000 = 174 \text{ ksi}$$

$$\text{Also accepted: } \frac{2x \cdot 29}{60 - 2x} \cdot 1000 = 175, \frac{2x \cdot 29}{60} \cdot 1000 = 174$$

Video Solution Problem for Chapter 9, Section 4 - Static Indeterminacy - Similar Triangles.

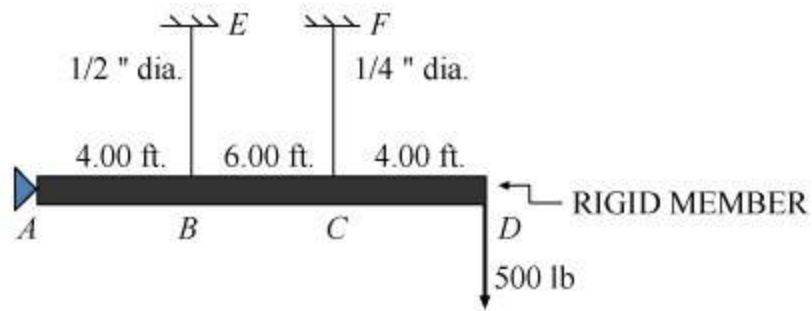
Description:

Watch the following video, then answer the question(s).



Part A

For the system shown below, BE and CF are steel wires having the same length and $ABCD$ is a **rigid** member pinned at A . The diameter of BE is $1/2$ in. and the diameter of CF is $1/4$ in.



Using similar triangles based on the displacement of the rigid member downward, determine the deformation compatibility equation with the force in CF expressed as a function of the force in BE .

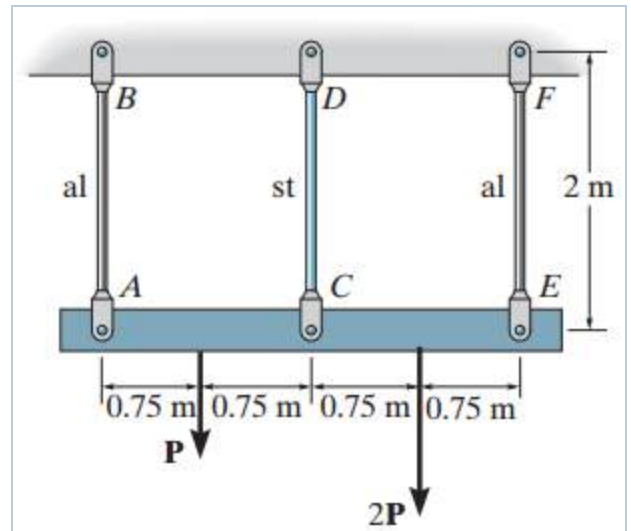
ANSWER:

- ☐ $F_{CF} = .400 F_{BE}$
- ☒ $F_{CF} = .625 F_{BE}$
- ☐ $F_{BE} = 1.60 F_{CF}$
- ☐ $F_{CF} = 2.50 F_{BE}$

Problem 9.31

Description: The rigid beam in is supported by the three suspender bars. Bars AB and EF are made of aluminum and bar CD is made of steel. (a) If each bar has a cross-sectional area of $## \text{ mm}^2$, determine the maximum value of P if the allowable stress is [...] for ...

The rigid beam in is supported by the three suspender bars. Bars AB and EF are made of aluminum and bar CD is made of steel.



Part A

If each bar has a cross-sectional area of 390 mm^2 , determine the maximum value of P if the allowable stress is $(\sigma_{\text{allow}})_{\text{st}} = 220 \text{ MPa}$ for the steel and $(\sigma_{\text{allow}})_{\text{al}} = 150 \text{ MPa}$ for the aluminum. $E_{\text{st}} = 200 \text{ GPa}$, $E_{\text{al}} = 70 \text{ GPa}$.

Express your answer to three significant figures and include the appropriate units.

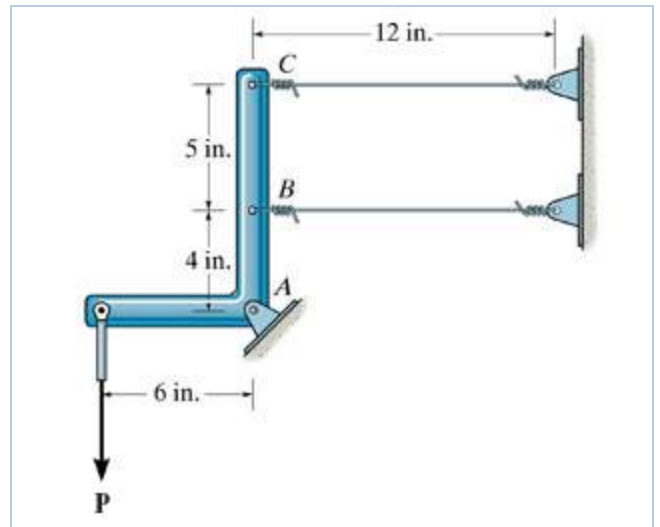
ANSWER:

$$P = \frac{\frac{SSTCSA}{1.7647}}{1000} = 48.6 \text{ kN}$$

Review Problem 9.7

Description: The rigid link is supported by a pin at A and two A-36 steel wires, each having an unstretched length of 12 in. and cross-sectional area of 0.0125 (in)^2 . (a) Determine the force developed in the wire B when the link supports the vertical load of...

The rigid link is supported by a pin at A and two A-36 steel wires, each having an unstretched length of 12 in. and cross-sectional area of 0.0125 in^2 .



Part A

Determine the force developed in the wire B when the link supports the vertical load of $P = 330$ lb.

Express your answer in pounds to three significant figures.

ANSWER:

$$F_B = \frac{P \cdot 6}{4 + \frac{81}{4}} = 81.6 \text{ lb}$$

Part B

Determine the force developed in the wire C when the link supports the vertical load of $P = 330$ lb.

Express your answer in pounds to three significant figures.

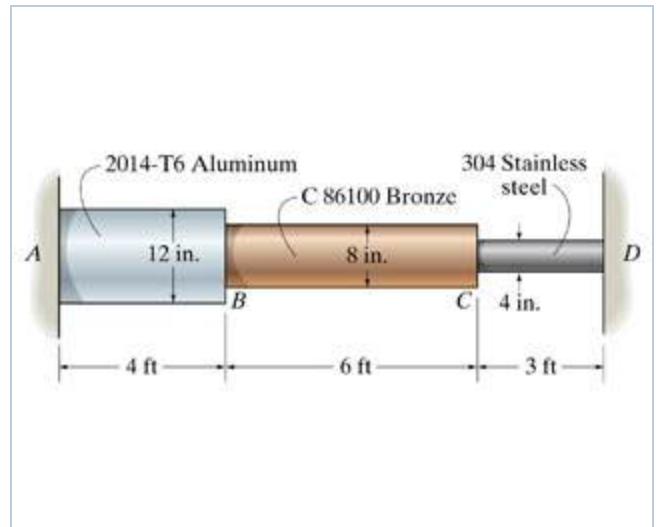
ANSWER:

$$F_C = \frac{\frac{P \cdot 6}{4 + \frac{81}{4}} \cdot 9}{4} = 184 \text{ lb}$$

Problem 9.44

Description: The assembly has the diameters and material indicated. It fits securely between its fixed supports when the temperature is $T_1 = 70$ degree(s) F. (a) Determine the average normal stress in the aluminum when the temperature reaches $T_2 = \text{##}$ degree(s) F. ...

The assembly has the diameters and material indicated. It fits securely between its fixed supports when the temperature is $T_1 = 70^\circ\text{F}$.



Part A

Determine the average normal stress in the aluminum when the temperature reaches $T_2 = 105^\circ\text{F}$.

Express your answer in kilopounds per square inch to three significant figures.

ANSWER:

$$\sigma_{\text{al}} = \frac{(T_2 - 70)(12.8 \cdot 4 + 9.6 \cdot 6 + 9.6 \cdot 3)}{(1000\pi \cdot 6^2) \left(\frac{4}{\pi \cdot 6^2 \cdot 10.6} + \frac{6}{\pi \cdot 4^2 \cdot 15} + \frac{3}{\pi \cdot 2^2 \cdot 28} \right)} = 2.15 \text{ ksi}$$

Part B

Determine the average normal stress in the bronze when the temperature reaches $T_2 = 105^\circ\text{F}$.

Express your answer in kilopounds per square inch to three significant figures.

ANSWER:

$$\sigma_{\text{br}} = \frac{(T_2 - 70)(12.8 \cdot 4 + 9.6 \cdot 6 + 9.6 \cdot 3)}{(1000\pi \cdot 4^2) \left(\frac{4}{\pi \cdot 6^2 \cdot 10.6} + \frac{6}{\pi \cdot 4^2 \cdot 15} + \frac{3}{\pi \cdot 2^2 \cdot 28} \right)} = 4.83 \text{ ksi}$$

Part C

Determine the average normal stress in the stainless steel when the temperature reaches $T_2 = 105^\circ\text{F}$.

Express your answer in kilopounds per square inch to three significant figures.

ANSWER:

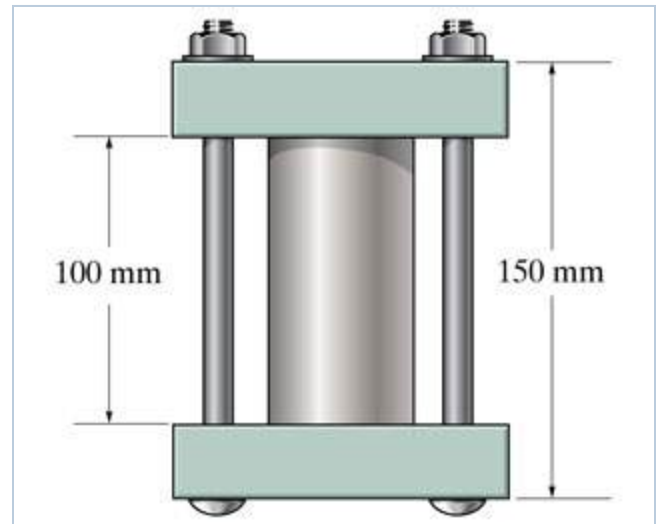
$$\sigma_{\text{st}} = \frac{(T_2 - 70)(12.8 \cdot 4 + 9.6 \cdot 6 + 9.6 \cdot 3)}{(1000\pi \cdot 2^2) \left(\frac{4}{\pi \cdot 6^2 \cdot 10.6} + \frac{6}{\pi \cdot 4^2 \cdot 15} + \frac{3}{\pi \cdot 2^2 \cdot 28} \right)} = 19.3 \text{ ksi}$$

Problem 9.56

Description: Consider . The D-mm-diameter cylinder is made from Am 1004-T61 magnesium and is placed in the clamp when the temperature is $T_1 = 20$ degree(s) C. (a) If the 304 stainless steel carriage bolts of the clamp each have a

diameter of d mm,...

Consider . The 50-mm-diameter cylinder is made from Am 1004-T61 magnesium and is placed in the clamp when the temperature is $T_1 = 20^\circ\text{C}$.



Part A

If the 304 stainless steel carriage bolts of the clamp each have a diameter of 14 mm, and they hold the cylinder snug with negligible force against the rigid jaws, determine the force in the cylinder when the temperature rises to $T_2 = 130^\circ\text{C}$.

Express your answer to three significant figures and include the appropriate units.

ANSWER:

$$F = \frac{10^{-6} (130 - 20) (26 \cdot 0.1 - 17 \cdot 0.15)}{\frac{44.7 \cdot 10^9 \cdot D^2 \pi}{4} + \frac{193 \cdot 10^9 \cdot 2d^2 \pi}{4}} = 1500 \text{ N}$$

[◀ All Assignments](#)

Statics and Mechanics of Materials, 6e

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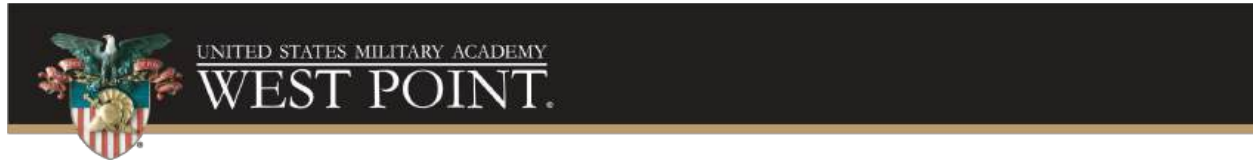
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HOMEWORK #4 – TORSION

● HW4 - Torsion

[Edit](#)[Overview](#)[Diagnostics](#)[Print View with Answers](#)

HW4 - Torsion

Due: 11:59pm on Thursday, February 5, 2026

You will receive no credit for items you complete after the assignment is due. [Grading Policy](#)

The Torsion Formula

Description: In this tutorial, students will use the torsion formula to relate the torque applied to a rod of circular cross section to the maximum shear stress in the rod.

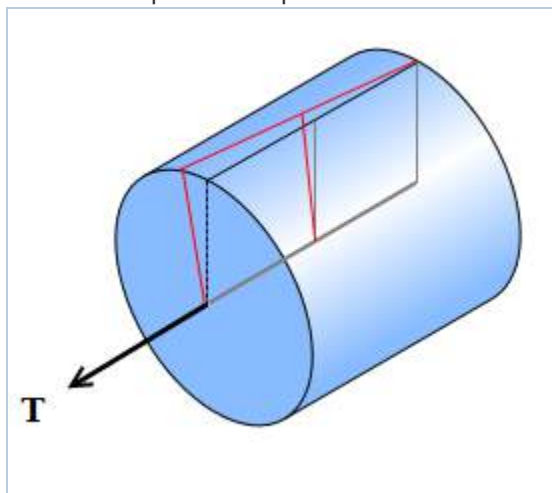
Learning Goal:

To use the torsion formula to relate the torque applied to a rod of circular cross section to the maximum shear stress in the rod.

When a torque T is applied to a shaft with a circular cross section, the shaft will deform by twisting about its longitudinal axis. For small deflections, the radial lines will remain straight and the cross sections will remain planar and parallel. The shear strain within the shaft will vary linearly along any radial line.

If the material is homogenous and elastic, then the shear stress will also vary linearly along any radial line as long as the maximum shear stress is no greater than the proportional limit for the material. Since the shear stress varies linearly, the maximum will occur on the outer boundary of the cross section.

The shear stress τ at a point on the cross section is related to the torque at the section by $\tau = \frac{T\rho}{J}$, where ρ is the distance of the point from the central axis and J is called the polar moment of inertia of the cross-sectional area. For a solid circular shaft $J = \frac{\pi}{2}c^4$, where c is the shaft radius. For a circular tube with inner radius c_i and outer radius c_o , the polar moment of inertia is $J = \frac{\pi}{2}(c_o^4 - c_i^4)$.



Part A - Maximum stress

A cross section of a solid circular rod is subject to a torque of $T = 3.7 \text{ kN} \cdot \text{m}$. If the diameter of the rod is $D = 6 \text{ cm}$, what is the maximum shear stress?

Express your answer with appropriate units to three significant figures.

Hint 1. How to approach the problem

The maximum stress will be at points on the rod's surface. Calculate the polar moment of inertia of the rod, and then use the torsion formula with the appropriate radial position ρ to calculate the stress.

Hint 2. Calculate J

What is the polar moment of inertia of a solid circular shaft with a diameter $D = 6 \text{ cm}$?

Express your answer with appropriate units to three significant figures.

ANSWER:

$$J = \frac{\pi}{2} \left(\frac{D1}{2} \right)^4 = 127 \text{ cm}^4$$

Hint 3. Identify the value for ρ

What is the radial distance for the point of maximum shear stress in the shaft? Recall that the diameter is $D = 6$ cm .

Express your answer in cm.

ANSWER:

$$\rho = \frac{D1}{2} = 3.00 \text{ cm}$$

ANSWER:

$$\tau_{\max} = \frac{T1 \frac{D1}{2}}{\frac{\pi}{2} \left(\frac{D1}{2} \right)^4} \cdot 1000 = 87.2 \text{ MPa}$$

Part B - Torque

The maximum stress in a section of a circular tube subject to a torque is $\tau_{\max} = 28$ MPa . If the inner diameter is $D_i = 4$ cm and the outer diameter is $D_o = 5.75$ cm , what is the torque on the section?

Express your answer with appropriate units to three significant figures.

Hint 1. How to approach the problem

In this case, the stress is given and the torque must be calculated. Begin by calculating the polar moment of inertia. Solve the torsion formula for the torque, and then use the appropriate radial position to calculate the value of the torque.

Hint 2. Calculate J

What is the polar moment of inertia of a circular tube with an outer diameter $D_o = 5.75$ cm and an inner diameter $D_i = 4$ cm ?

Express your answer with appropriate units to three significant figures.

ANSWER:

$$J = 0.5\pi \left((0.5D2b)^4 - (0.5D2a)^4 \right) = 82.2 \text{ cm}^4$$

Hint 3. Solve for T

The torsion formula is $\tau = \frac{T\rho}{J}$. Solve this equation for the torque.

Express your answer in terms of τ , ρ , and J .

ANSWER:

$$T = \frac{\tau J}{\rho}$$

The appropriate position for finding the maximum shear stress is at the outer surface. Use the radius at the outer surface for ρ .

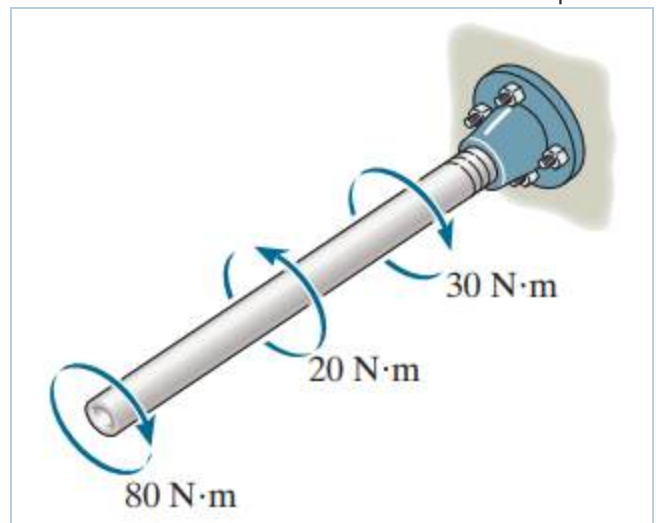
ANSWER:

$$T = \frac{0.5\pi ((0.5D2b)^4 - (0.5D2a)^4) \tau_2}{0.5D2b} \cdot 0.001 = 0.800 \text{ kN}\cdot\text{m}$$

Problem 10.4

Description: The copper pipe has an outer diameter of 40 mm and an inner diameter of d mm. It is fixed to the wall and three torques are applied to it as shown in the figure below. (a) Determine the absolute maximum shear stress developed in the pipe.

The copper pipe has an outer diameter of 40 mm and an inner diameter of 34 mm. It is fixed to the wall and three torques are applied to it as shown in the figure below.



Part A

Determine the absolute maximum shear stress developed in the pipe.

Express your answer to three significant figures and include the appropriate units.

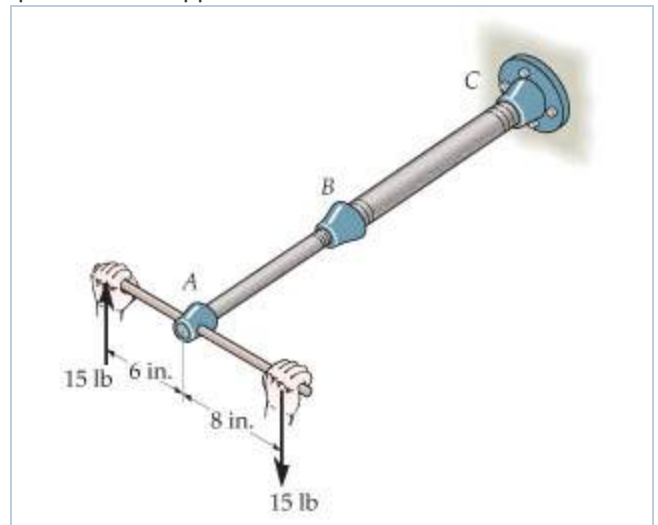
ANSWER:

$$\tau_{\max} = \frac{90 \cdot 0.02 \cdot 2}{\pi \left(0.02^4 - \left(\frac{d}{2000} \right)^4 \right)} = 1.50 \times 10^7 \text{ Pa}$$

Problem 10.11

Description: The assembly consists of two galvanized steel pipes connected together using a reducing coupling at *B*. The smaller pipe has an outer diameter of 0.75 in. and an inner diameter of d_1 in., whereas the larger pipe has an outer diameter of 1 in. and an inner diameter of 0.84 in. The pipe is secured to the wall at *C*. The couple shown is applied to the handles of the wrench.

The assembly consists of two galvanized steel pipes connected together using a reducing coupling at *B*. The smaller pipe has an outer diameter of 0.75 in. and an inner diameter of 0.67 in., whereas the larger pipe has an outer diameter of 1 in. and an inner diameter of 0.84 in. The pipe is secured to the wall at *C*. The couple shown is applied to the handles of the wrench.



Part A

Determine the maximum shear stress in pipe *AB*.

Express your answer in kilopounds per square inch to three significant figures.

ANSWER:

$$\tau_{AB} = \frac{210 \cdot 0.375}{\frac{\pi}{2} \left(0.375^4 - \left(\frac{d_1}{2} \right)^4 \right)} \cdot 10^{-3} = 6.98 \text{ ksi}$$

Part B

Determine the maximum shear stress in pipe *BC*.

Express your answer in kilopounds per square inch to three significant figures.

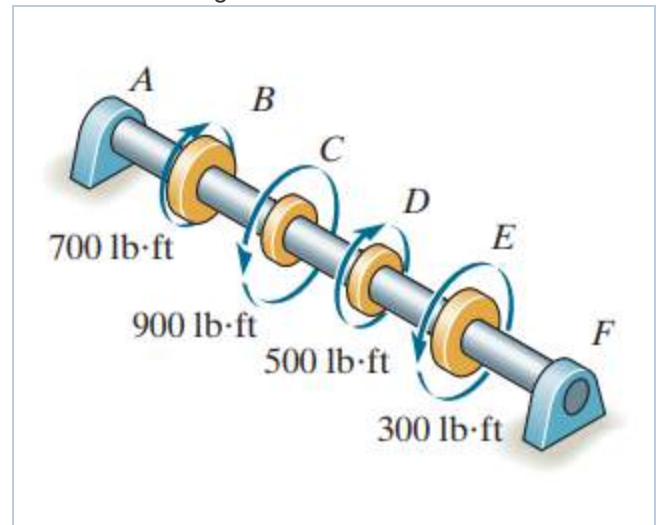
ANSWER:

$$\tau_{BC} = \frac{210 \cdot 0.5}{\frac{\pi}{2} \left(0.5^4 - \left(\frac{d_2}{2} \right)^4 \right)} \cdot 10^{-3} = 2.13 \text{ ksi}$$

Problem 10.12

Description: The solid shaft has a diameter of ## in. It is subjected to the torques shown. The bearings at A and F allow free rotation of the shaft. (a) Determine the maximum shear stress developed in region CD of the shaft. (b) Determine the maximum shear...

The solid shaft has a diameter of 1.4 in. It is subjected to the torques shown. The bearings at A and F allow free rotation of the shaft.



Part A

Determine the maximum shear stress developed in region *CD* of the shaft.

Express your answer in kilopounds per square inch to three significant figures.

ANSWER:

$$(\tau_{CD})_{\max} = \frac{200 \cdot 12}{\frac{\pi}{2} \left(\frac{d}{2} \right)^3} \cdot 10^{-3} = 4.45 \text{ ksi}$$

Part B

Determine the maximum shear stress developed in region *DE* of the shaft.

Express your answer in kilopounds per square inch to three significant figures.

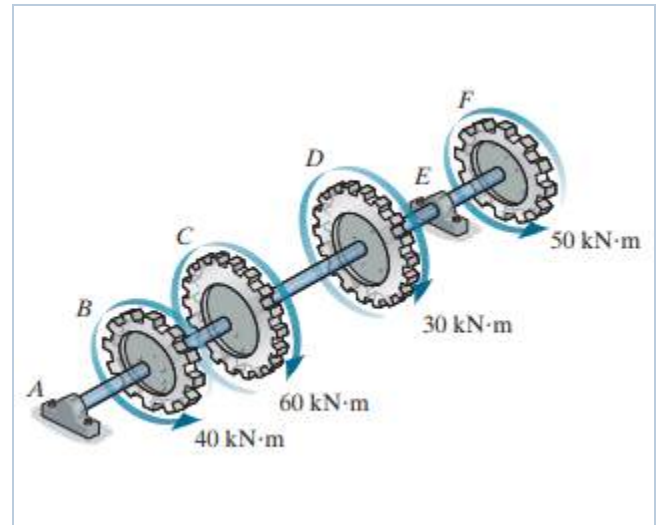
ANSWER:

$$(\tau_{DE})_{\max} = \frac{300 \cdot 12}{\frac{\pi}{2} \left(\frac{d}{2}\right)^3} \cdot 10^{-3} = 6.68 \text{ ksi}$$

Problem 10.28

Description: The tubular shaft is supported by bearings at A and E and has an outer diameter of ## mm. (a) If the tubular shaft is made from material having an allowable shear stress of $\tau_{\text{allow}} = 90 \text{ MPa}$, determine the required minimum wall thickness of the shaft ...

The tubular shaft is supported by bearings at A and E and has an outer diameter of 180 mm.



Part A

If the tubular shaft is made from material having an allowable shear stress of $\tau_{\text{allow}} = 90 \text{ MPa}$, determine the required minimum wall thickness of the shaft to the nearest millimeter.

Express your answer in millimeters as an integer.

ANSWER:

$$t = \text{ceil} \left(\left(\left(\frac{d_o}{2} \right)^4 - \frac{2 \cdot 50 \cdot 10^{-3} \frac{d_o}{2}}{\pi \cdot 90} \right)^{\frac{1}{4}} \cdot 1000 \right) = 14 \text{ mm}$$

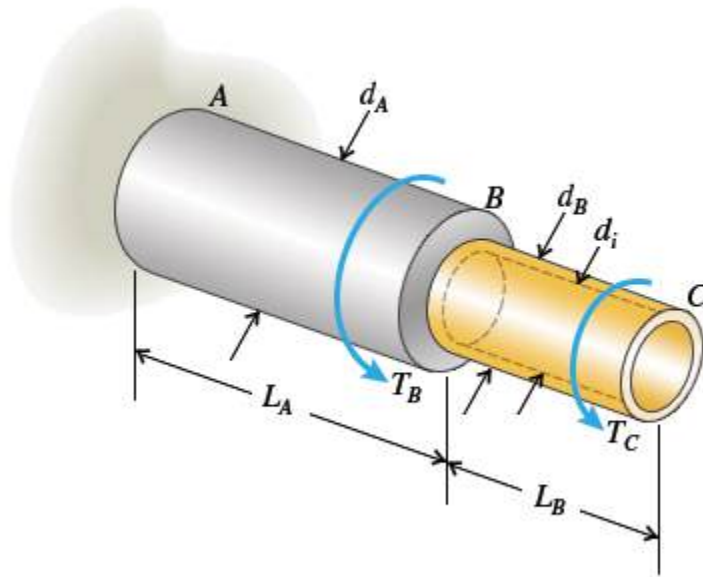
Angle of Twist

Description: In this tutorial, students will explore the concept of angle of twist while considering sign conventions of torque and angle of twist, as well as utilizing a torque diagram to aid in the completion of the problem involving a composite rod that is hollow in certain areas.

Learning Goal:

To determine the angle of twist on a composite rod given the geometry and externally applied torques, to properly apply a sign convention to determine the angle of twist, to use a torque diagram to aid in determining the angle of twist, and to determine the maximum applicable torque given a maximum allowable angle of twist.

The rod shown below is made of two different materials. Segment AB is made of aluminum ($G = 3.9 \times 10^6$ psi). Segment BC is bonded to segment AB and is made of brass ($G = 5.6 \times 10^6$ psi). Segment BC is hollow and has an inner diameter of $d_i = 0.9$ in. External torques are applied to the rod at point B and point C . Let $L_A = 60.0$ in, $L_B = 50.0$ in, $d_A = 3.5$ in, $d_B = 2.5$ in, $T_B = 8.0$ kip · in, and $T_C = 12.0$ kip · in.

**Part A - Angle of twist at point B**

Determine the angle of twist, ϕ_B , experienced by the rod at point B .

Express your answer to three significant figures and include appropriate units.

Hint 1. Internal torque in segment AB of the rod

The internal torque is found at a point on the axis of the rod by using the method of sections and the equation of moment equilibrium applied along the rod's axis.

Hint 2. Find the torque in segment AB

Using the right-hand rule and the established sign convention that positive torque is directed away from the sectioned end of the rod, the torque in segment AB can be determined. What is this torque?

Express your answer to two significant figures and include appropriate units.

Hint 1. How to approach the problem

Since the entire rod is subjected to two different torques twisting the rod in the same direction, remember to add both torques to get the total torque acting on the rod.

ANSWER:

$$T_{AB} = T_b + T_c = 20 \text{ kip}\cdot\text{in}$$

Hint 3. Find the polar moment of inertia of segment *AB*

The polar moment of inertia is an object's ability to resist torsion. This is a necessary parameter to determine the angle of twist in the rod. What is the polar moment of inertia for segment *AB*?

Express your answer to three significant figures and include appropriate units.

Hint 1. Equation for polar moment of inertia

The polar moment of inertia for a circular shaft is given by the equation

$$J = \frac{\pi}{2} c^4$$

where $c = (1/2)d$.

ANSWER:

$$J_{AB} = \frac{\pi}{32} d^4 = 14.7 \text{ in}^4$$

ANSWER:

$$\phi_B = \frac{5760 (T_b + T_c) L a}{\pi^2 d^4 \cdot 3900} = 1.20^\circ$$

Part B - Angle of twist at point C

Determine the angle of twist, ϕ_C , experienced by the rod at point C.

Express your answer to three significant figures and include appropriate units.

Hint 1. Principle of superposition

To determine the angle of twist at point C on the rod, the angle of twist of segment *AB* and segment *BC* must be added.

Hint 2. Find the internal torque for section *BC*

Using the right-hand rule and the established sign convention that positive torque is directed away from the sectioned end of the rod, the torque in segment *BC* of the rod can be determined. To obtain this number, cut through any point in segment *BC* and find the torque needed at that cross section for the rod to be at static equilibrium. Think about the reaction torque at the base of the rod (i.e., the internal torque for section *BC*). What is this torque?

Express your answer to two significant figures and include appropriate units.

ANSWER:

$$T_{BC} = T_c = 12 \text{ kip}\cdot\text{in}$$

Hint 3. Find the polar moment of inertia of segment BC

The polar moment of inertia is an object's ability to resist torsion. This is a necessary parameter to determine the angle of twist in the rod. What is the polar moment of inertia for segment BC ?

Express your answer to three significant figures and include appropriate units.

Hint 1. Equation for the polar moment of inertia

The polar moment of inertia for a hollow circular shaft is given by the equation

$$J = \frac{\pi}{2} (c_o^4 - c_i^4)$$

where $c = (1/2)d$.

ANSWER:

$$J_{BC} = \frac{\pi}{32} (db^4 - di^4) = 3.77 \text{ in}^4$$

ANSWER:

$$\phi_C = \frac{5760}{\pi^2} \left(\frac{(Tb + Tc) La}{3900 da^4} + \frac{TcLb}{5600 (db^4 - di^4)} \right) = 2.82^\circ$$

Part C - Angle of twist at point C if segment BC were solid instead of hollow

Determine the angle of twist, ϕ_C , experienced by the rod at point C if the segment BC is solid rather than hollow.

Express your answer to three significant figures and include appropriate units.

Hint 1. Comparing values for angle of twist

By removing the hollow portion of segment BC , the resistance to the torsion experienced by the rod is increased. This will result in a smaller angle of twist for the entire rod.

Hint 2. Find the polar moment of inertia of segment BC

Removing the hollow portion of segment BC changes its polar moment of inertia. What is the new polar moment of inertia for segment BC ?

Express your answer to three significant figures and include appropriate units.

Hint 1. How to look at the problem

By removing the hollow portion of the segment BC , the polar moment of inertia of the segment should increase in value.

ANSWER:

$$J_{BC} = \frac{\pi}{32} db^4 = 3.83 \text{ in}^4$$

Hint 3. Identify the effect of segment BC on segment AB

By removing the hollow portion of the rod, segment BC is affected. What effect does this action have on segment AB ?

ANSWER:

- ☐ The polar moment of inertia of segment AB is decreased.
- ☐ The internal torque in segment AB is increased.
- ☐ The polar moment of inertia of segment AB is increased.
- ☒ There is no effect on segment AB .

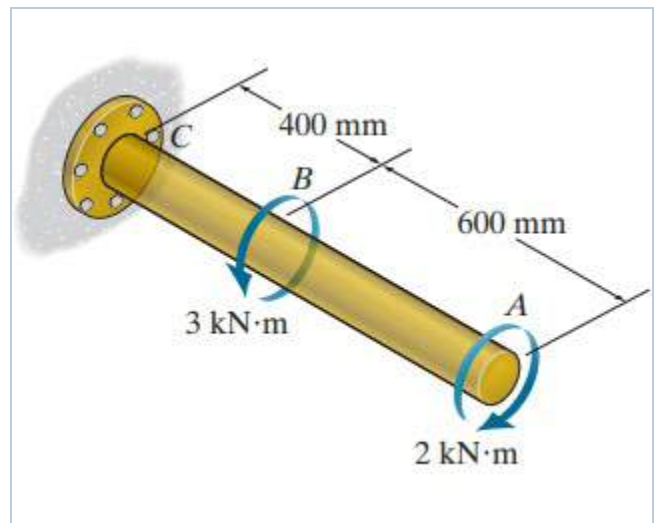
ANSWER:

$$\phi_C = \frac{5760}{\pi^2} \left(\frac{(Tb + Tc) La}{3900 da^4} + \frac{Tc Lb}{5600 db^4} \right) = 2.80^\circ$$

Fundamental Problem 10.9

Description: The ##-mm-diameter steel shaft is subjected to the torques shown. (a) Determine the magnitude of the angle of twist of end A with respect to C. Take $G = 75 \text{ GPa}$.

The 57-mm-diameter steel shaft is subjected to the torques shown.



Part A

Determine the magnitude of the angle of twist of end A with respect to C . Take $G = 75 \text{ GPa}$.

Express your answer in degrees to three significant figures.

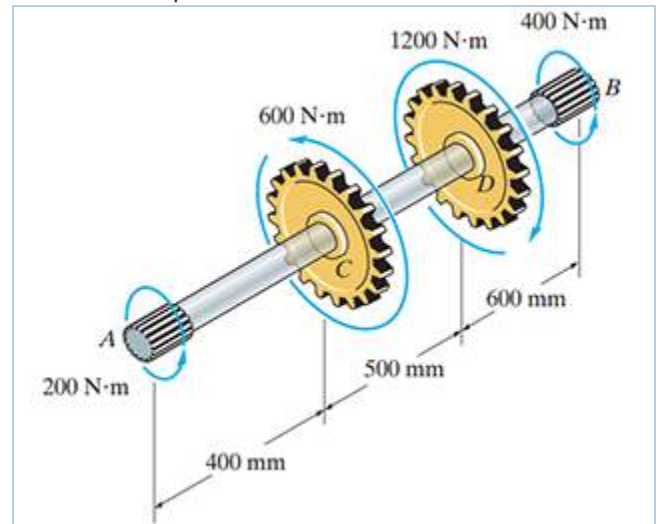
ANSWER:

$$\phi_{A/C} = \frac{800 \cdot 2}{\pi \cdot 75 \cdot 10^9 \left(\frac{d}{2000}\right)^4} \frac{180}{\pi} = 0.590^\circ$$

Problem 10.39

Description: The splined ends and gears attached to the A992 steel shaft are subjected to the torques shown in the . The shaft has a diameter of $d \text{ mm}$. (a) Determine the angle of twist of end B with respect to end A .

The splined ends and gears attached to the A992 steel shaft are subjected to the torques shown in the . The shaft has a diameter of 40 mm .



Part A

Determine the angle of twist of end B with respect to end A .

Express your answer in degrees to three significant figures.

ANSWER:

$$\phi_{B/A} = \frac{(200 \cdot 0.400 + (200 + 600) \cdot 0.500 + (200 + 600 - 1200) \cdot 0.600) \cdot 2 \cdot 2000^4 \cdot 180}{\pi d^4 \cdot 75 \cdot 10^9 \pi} = 0.730^\circ$$

Statics and Mechanics of Materials, 6e

Hibbeler

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Ends: 05/31/26



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HOMEWORK #5 – FLEXURAL STRESSES

● HW5 - Flexural Stresses

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HW5 - Flexural Stresses

Due: 11:59pm on Thursday, February 19, 2026

You will receive no credit for items you complete after the assignment is due. [Grading Policy](#)

The Flexure Formula

Description: In this tutorial, students apply the flexure formula to beams under load and find unknown stresses, moments, and forces.

Learning Goal:

To apply the flexure formula to beams under load and find unknown stresses, moments, and forces.

For straight members having a constant cross-section that is symmetrical with respect to an axis with a moment applied perpendicular to that axis, the maximum normal stress in the cross-section can be calculated using the flexure formula:

$$\sigma_{\max} = \frac{Mc}{I}$$

where M is the magnitude of the internal moment with respect to the neutral axis, c is the perpendicular distance from the neutral axis to the point farthest from the neutral axis, and I is the moment of inertia of the cross-section about the neutral axis. The maximum normal stress will always occur on the top or bottom surface of the beam; in fact, one of these surfaces will experience a maximal tensile stress while the other experiences the same magnitude of stress in compression.

For points not on a surface of the beam, we can use

$$\sigma_{\max}/c = -\sigma/y$$

to rewrite the flexure formula in the more general form

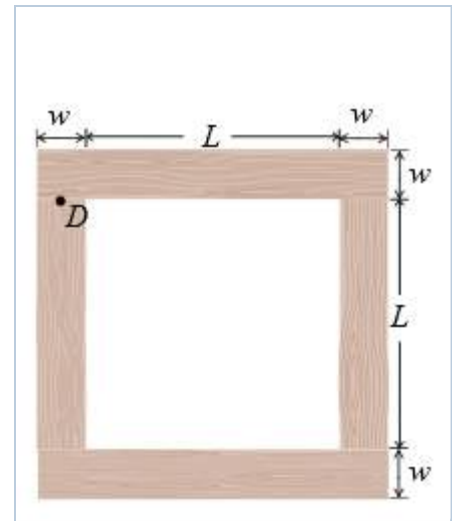
$$\sigma = -\frac{My}{I}.$$

Part A - Moment Required to Produce a Given Stress

The cross-section of a wooden, built-up beam is shown below. The dimensions are $L = 190$ mm and $w = 25$ mm.

Determine the magnitude of the moment $\mathbf{M}$ that must be applied to the beam to create a compressive stress of $\sigma_D = 22$ MPa at point D . Also calculate the maximum stress developed in the beam. The moment $\mathbf{M}$ is applied in the vertical plane about the geometric center of the beam.

Express your answers, separated by a comma, to three significant figures.



Hint 1. Applying the flexure formula

To apply the flexure formula to find an unknown moment, we need to know the stress, distance to the neutral axis, and the area moment of inertia. When the moment is applied about the center of the beam, the maximum stress in the beam will occur at a point farthest from the applied moment.

Hint 2. Calculate the area moment of inertia

What is the area moment of inertia of the beam?

Express your answer to three significant figures and include the appropriate units.

ANSWER:

$$I = \frac{\left(\frac{LA+2wA}{1000}\right)^4}{12} - \frac{\left(\frac{LA}{1000}\right)^4}{12} = 1.68 \times 10^{-4} \text{ m}^4$$

Hint 3. Determine distances from the neutral axis

What are the distances, y_D and c , from the neutral axis needed to calculate the required moment that will produce the 22 MPa stress at point D and the maximum stress in the beam, respectively?

Express your answers, separated by a comma, to three significant figures.

ANSWER:

$$y_D =, c = \frac{LA}{2}, \frac{LA}{2} + wA = 95.0, 120 \text{ mm, mm}$$

ANSWER:

$$M =, \sigma_{\max} = \frac{\sigma_{\max} DA \cdot 10^6 IA}{\frac{LA}{2}}, \frac{\sigma_{\max} DA (LA + 2wA)}{LA} = 38.9, 27.8 \text{ kN}\cdot\text{m, MPa}$$

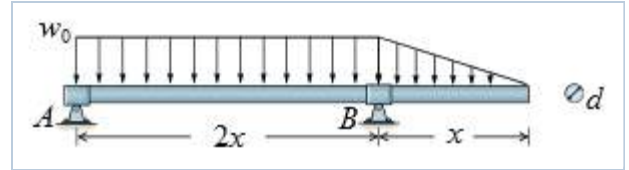
Notice how using both forms of the flexure formula simplifies this otherwise-lengthy calculation.

Part B - Minimum Allowable Cross-Section

The rod shown in the figure below is supported by smooth journal bearings at A and B that exert only vertical reactions on the shaft.

Determine the smallest allowable diameter of the rod, d , if $x = 2.0$ m, $w_0 = 9.2$ kN/m, and the maximum allowable bending stress is $\sigma_{\text{allow}} = 155$ MPa.

Express your answer to three significant figures and include the appropriate units.



Hint 1. How to determine the required diameter

Draw the free-body diagram and determine the support reactions. Use these to draw the shear diagram and determine the location on the rod where the maximum bending moment occurs. Once the bending moment is determined at that cross-section, the flexure formula can be used to determine the minimum diameter.

Hint 2. Determine the support reactions

What are the support reactions at A and B ?

Express your answers, separated by a comma, to three significant figures.

ANSWER:

$$A_y =, B_y = \frac{11xw_0}{12}, \frac{19xw_0}{12} = 16.9, 29.1 \text{ kN, kN}$$

Hint 3. Find the location on the rod where the shear is zero

Where, as measured from point A , is the shear in the rod equal to zero?

Express your answer to three decimal places.

ANSWER:

$$x_0 = \frac{11x}{12} = 1.833 \text{ m}$$

Because of the differential relationship between shear and moment, we know that this location corresponds to the maximum shear in the rod.

Hint 4. Calculate the maximum bending moment in the rod

What is the maximum bending moment in the bar?

Express your answer to three significant figures and include the appropriate units.

ANSWER:

$$M_{\max} = \frac{121xB^2w_0B}{288} = 15.5 \text{ kN}\cdot\text{m}$$

ANSWER:

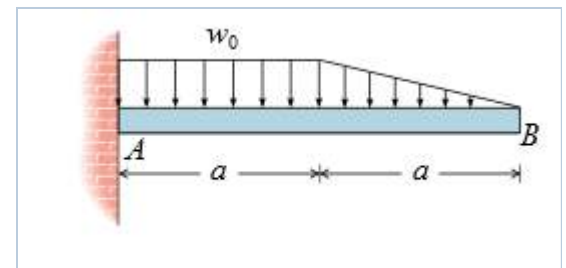
$$d = 2000 \sqrt[3]{\frac{\frac{121xB^2w_0B}{288}}{\frac{3.14159\sigma_{\max}B \cdot 10^3}{4}}} = 101 \text{ mm}$$

Part C - Absolute Maximum Bending Stress

Find the absolute maximum bending stress in the beam shown in the figure below.

The beam has a square cross-section of 6.5 in. on each side, is 20 ft long, and the initial value of the distributed load is $w_0 = 325 \text{ lb/ft}$.

Express your answer to three significant figures.



Hint 1. How to approach the problem

First, determine where along the beam the maximum stress will occur. Once this is known, apply the flexure formula at that point by finding the moment, centroid, and area moment of the beam.

Hint 2. Determine the location of the maximum stress

Where does the maximum stress occur in the beam?

ANSWER:

- ☒ Point A
- ☐ Point B
- ☐ $\frac{4}{3}(10 \text{ ft})$ to the right of Point A
- ☐ 10 ft to the right of Point A (where the distributed load changes from constant to decreasing)

Hint 3. Calculate the moment at Point A

What is the magnitude of the moment at Point A?

Express your answer to three significant figures.

ANSWER:

$$M_A = \frac{7w_0CaC^2}{6} = 3.79 \times 10^4 \text{ lb}\cdot\text{ft}$$

Hint 4. Area moment of inertia for a square beam

What is the formula used to calculate the area moment of inertia of a square beam whose sides have length s ?

ANSWER:

- ☐ $I = \frac{1}{2}s^4$
- ☒ $I = \frac{1}{12}s^4$
- ☐ $I = s^4 - \frac{1}{2}s^2$
- ☐ $I = s^2 + \frac{1}{2}s^2$

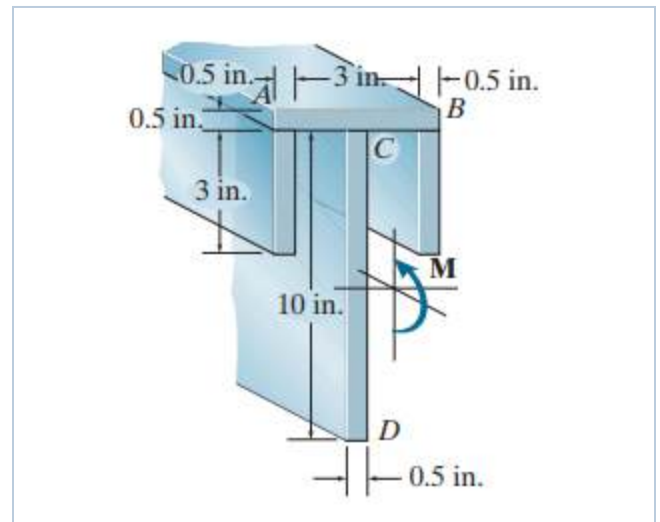
ANSWER:

$$\sigma_{\max} = \frac{84w_0CaC^2}{sC^3} = 9940 \text{ psi}$$

Problem 11.42

Description: Consider the cross section shown in . (a) Determine the moment M that will produce a maximum stress of $\#\#$ ksi on the cross section.

Consider the cross section shown in .



Part A

Determine the moment M that will produce a maximum stress of 14 ksi on the cross section.

Express your answer in kilopound-feet to three significant figures.

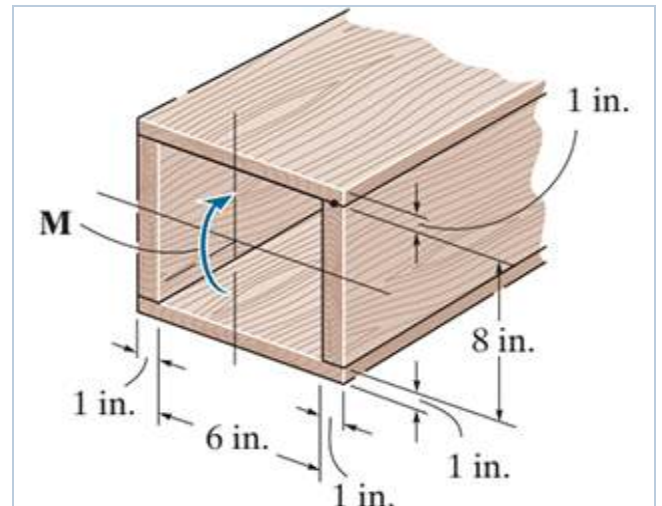
ANSWER:

$$M = \frac{\sigma \cdot I}{c} = 15.1 \text{ kip} \cdot \text{ft}$$

Problem 11.44

Description: The beam in is constructed from four pieces of wood, glued together as shown. (a) If $M = \text{##} \text{ kip} \cdot \text{ft}$, determine the maximum bending stress in the beam.

The beam in is constructed from four pieces of wood, glued together as shown.



Part A

If $M = 10 \text{ kip} \cdot \text{ft}$, determine the maximum bending stress in the beam.

Express your answer in kilopounds per square inch to three significant figures.

ANSWER:

$$\sigma_{\max} = \frac{M \cdot c}{I} = 1.46 \text{ ksi}$$

The Shear Formula

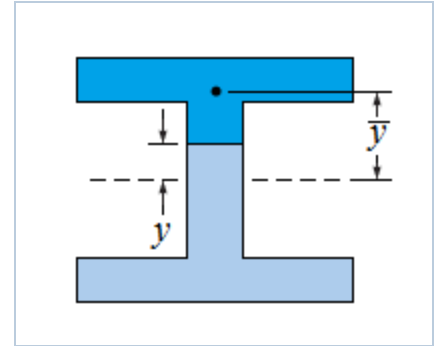
Description: In this tutorial, students will calculate the shear stress at a point in the web of an I-beam.

Learning Goal:

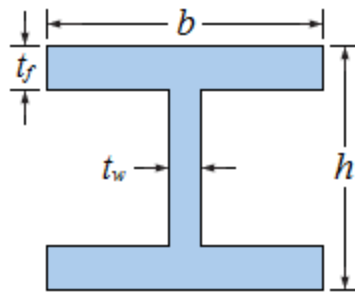
To calculate the shear stress at a point in the web of an I-beam section subjected to a shear force.

When a beam section is subjected to a shear load, a shear stress distribution is developed on the section. The distribution of the shear stress is not linear. Elasticity theory can be used to calculate the shear stress at any point.

However, a simpler method can be used to calculate the average shear stress across the width of the section, a distance y above or below the neutral axis. The average shear stress is given by $\tau = \frac{VQ}{It}$. Here V is the shear force on the section, I is the moment of inertia of the entire section about the neutral axis, and t is the width of the section at the distance y where the shear stress is being calculated. Q is the product of the area of the section above (or below) y and the distance from the neutral axis to the centroid of that area. In short, Q is the moment of the area about the neutral axis.



An I-beam has a flange width $b = 300$ mm, height $h = 300$ mm, web thickness $t_w = 10$ mm, and flange thickness $t_f = 15$ mm. Use the following steps to calculate the shear stress at a point 120 mm above the neutral axis.

**Part A - Moment of inertia**

The shear formula includes the moment of inertia of the whole cross section, I , about the neutral axis. Calculate the moment of inertia.

Express your answer with appropriate units to three significant figures.

Hint 1. How to approach the problem

Since the section has a horizontal axis of symmetry, it is the neutral axis. Divide the section into three rectangles: the web and the two flanges. Use the parallel axis theorem to calculate the moment of inertia for each.

Hint 2. Contribution from one flange

What is the contribution of one flange to the moment of inertia of the section?

Express your answer with appropriate units to three significant figures.

Hint 1. Calculate the flange moment about its own centroid

The moment of inertia of a rectangle about its own centroid is $I' = \frac{1}{12}bh^3$, where b is the width and h is the height. What is the moment of inertia of the flange about its own centroid? Recall that $b = 300$ mm and $t_f = 15$ mm .

Express your answer with appropriate units to three significant figures.

ANSWER:

$$I' = \frac{\frac{1}{12}bt_f^3}{10000} = 8.44 \text{ cm}^4$$

Now use the parallel axis theorem $I = I' + A\bar{y}^2$ to calculate the moment about the neutral axis of the beam.

Hint 2. Calculate the flange area

What is the area of the flange? Recall that $b = 300$ mm and $t_f = 15$ mm .

Express your answer with appropriate units to three significant figures.

ANSWER:

$$A = \frac{t_fb}{100} = 45.0 \text{ cm}^2$$

Hint 3. Calculate flange centroid position

What is the distance from the neutral axis to the centroid of the flange? Recall that $h = 300$ mm and $t_f = 15$ mm .

Express your answer with appropriate units to three significant figures.

ANSWER:

$$\bar{y} = \frac{0.5(h - t_f)}{10} = 14.3 \text{ cm}$$

ANSWER:

$$I_{\text{flange}} = I' + A\bar{y}^2 = 9150 \text{ cm}^4$$

Hint 3. Contribution from the web

What is the contribution of the web to the moment of inertia of the section?

Express your answer with appropriate units to three significant figures.

ANSWER:

$$I_{\text{web}} = I_w = 1640 \text{ cm}^4$$

ANSWER:

$$I = I_w + 2(I_f + A_f y_f^2) = 1.99 \times 10^4 \text{ cm}^4$$

Part B - Q for the given point

The shear stress 120 mm above the neutral axis depends on the value of Q for the area above that point. Calculate the value of Q .

Express your answer with appropriate units to three significant figures.

Hint 1. How to approach the problem

Q is the product of the area and the centroid of the area, $Q = A' \bar{y}'$. The calculation for the centroid can be written as $\bar{y} = \frac{\sum A' \bar{y}'}{A}$, where A' and $\bar{y}'$ are the area and centroid of the individual components of the area. Notice that these two expressions can be simplified to give $Q = \sum A' \bar{y}'$. The area for Q can be divided into two rectangles: the upper flange and a portion of the web.

Hint 2. Contribution from the flange

What is the product $A' \bar{y}'$ for the flange?

Express your answer with appropriate units to three significant figures.

ANSWER:

$$A' \bar{y}' = A_f y_f = 641 \text{ cm}^3$$

Hint 3. Contribution from the web

What is the product $A' \bar{y}'$ for the portion of the web more than 120 mm above the neutral axis?

Express your answer with appropriate units to three significant figures.

ANSWER:

$$A' \bar{y}' = \frac{tw(0.5h - tf - y)(y + 0.5(0.5h - tf - y))}{1000} = 19.1 \text{ cm}^3$$

ANSWER:

$$Q = A_f y_f + \frac{tw(0.5h - tf - y)(y + 0.5(0.5h - tf - y))}{1000} = 660 \text{ cm}^3$$

Part C - Shear stress

Use the results from Parts A and B to calculate the shear stress at a point 120 mm above the neutral axis if the shear force on the section is $V = 7.2 \text{ kN}$.

Express your answer with appropriate units to three significant figures.

Hint 1. How to approach the problem

The shear formula is $\tau = \frac{VQ}{It}$. The value for I was calculated in Part A and the value for Q was calculated in Part B.

Hint 2. Identify the section width

What is the width across the section 120 mm above the neutral axis?

Express your answer with appropriate units.

ANSWER:

$$t = tw = 10.0 \text{ mm}$$

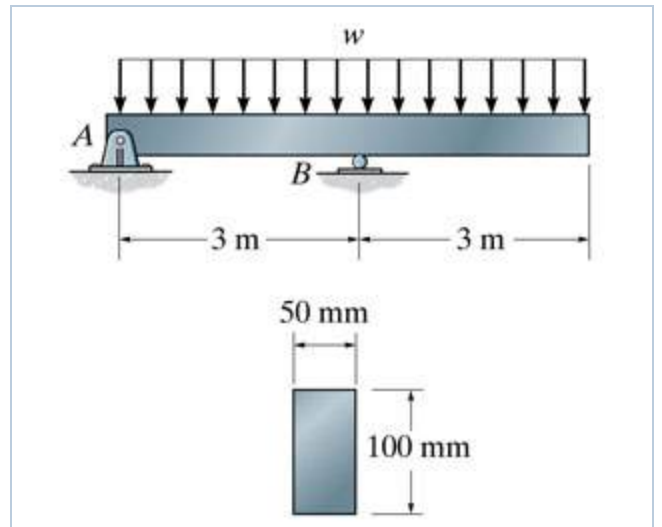
ANSWER:

$$\tau = \frac{VQ}{Itw} \cdot 100 = 2.39 \text{ MPa}$$

Problem 12.11

Description: The overhang beam is subjected to the uniform distributed load having an intensity of $w = \text{##} \text{ kN/m}$!. (a) Determine the maximum shear stress in the beam.

The overhang beam is subjected to the uniform distributed load having an intensity of $w = 25 \text{ kN/m}$.



Part A

Determine the maximum shear stress in the beam.

Express your answer to three significant figures and include the appropriate units.

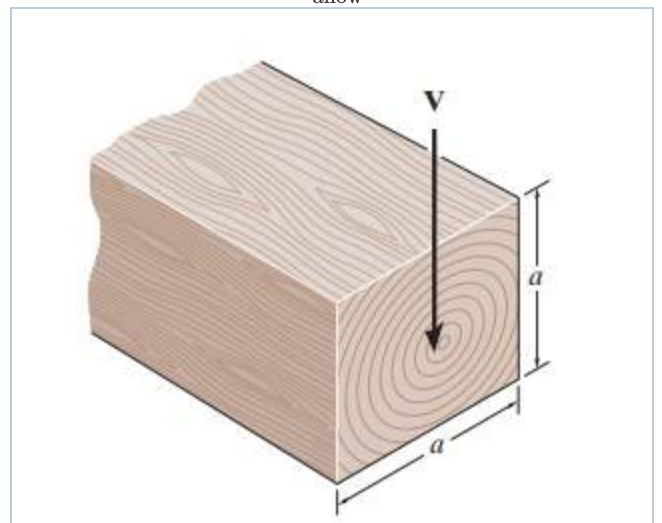
ANSWER:

$$\tau_{\max} = \frac{w \cdot 3 \cdot 1.5}{0.05 \cdot 0.1 \cdot 1000} = 22.5 \text{ MPa}$$

Problem 12.16

Description: The beam has a square cross section and is made of wood having an allowable shear stress of $\tau_{\text{allow}} = 0.70 \text{ ksi}$. (a) If it is subjected to a shear of $V = 10 \text{ kip}$, determine the smallest dimension a of its sides.

The beam has a square cross section and is made of wood having an allowable shear stress of $\tau_{\text{allow}} = 0.70 \text{ ksi}$.



Part A

If it is subjected to a shear of $V = 1.3$ kip, determine the smallest dimension a of its sides.

Express your answer in inches to three significant figures.

ANSWER:

$$a = \sqrt{\frac{3V}{\tau_{\text{allow}}}} = 1.67 \text{ in.}$$

◀ All Assignments

Statics and Mechanics of Materials, 6e

Hibbeler

myers59634

Ends: 05/31/26



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HOMEWORK #6 – DEFLECTIONS

● HW6 - Deflections

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HW6 - Deflections

Due: 11:59pm on Tuesday, March 3, 2026

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Slope and Displacement by Integration

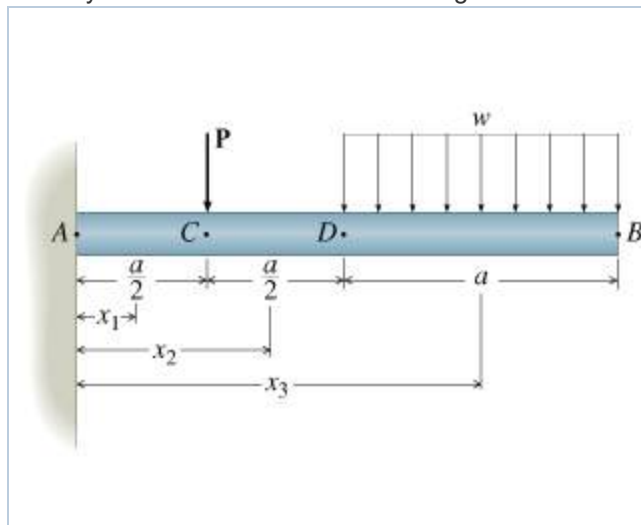
Description: In this tutorial, students will develop the general equations for the elastic curve of a cantilevered beam that is subjected to a distributed load and a concentrated load using three coordinate systems.

Learning Goal:

To determine the general equation for the elastic curve for a cantilevered beam that is subjected to a distributed load w and a concentrated load $\mathbf{P}$.

A cantilevered beam is loaded as shown. Because of the discontinuities from the different types of loading on the beam, three coordinate systems, x_1 , x_2 , and x_3 , are used to define the locations of the internal moments. Assume EI is constant.

Note: that the coordinate systems are at arbitrary distances and not necessarily at the locations shown in the figure.



Part A - Moment reaction at the wall

Before finding the equations of the moment, the moment reaction at A must be determined.

Express the moment reaction in terms of a , w , and $\mathbf{P}$ in simplified form.

Hint 1. How to determine the moment reaction at the wall

To find the moment reaction $\mathbf{M}_A$ at the wall, the equilibrium equation for the moment about the wall can be solved directly by taking into account the concentrated load $\mathbf{P}$ and the distributed load w .

Hint 2. Determine the moment due to the concentrated load $\mathbf{P}$

The concentrated load $\mathbf{P}$ causes a moment about the wall. Determine this moment.

Express the moment in terms of the force $\mathbf{P}$ and the distance a .

ANSWER:

$$\mathbf{M} = \frac{-Pa}{2}$$

Hint 3. Determine the moment due to the distributed load

The distributed load w causes a moment reaction at the wall. Determine this moment.

Express the moment in terms of w and a .

ANSWER:

$$\mathbf{M} = \frac{-3}{2}wa^2$$

ANSWER:

$$\mathbf{M}_A = \frac{-a}{2}(P + 3wa)$$

Part B - Equation of the elastic curve using x_1 over section AC

To find the elastic curve for the entire beam, the equation that uses the coordinate system x_1 must be determined for section AC of the beam. Write this equation in terms of EI , w , a , and $\mathbf{P}$. Let C_1 and C_2 denote the constants of integration.

Express your answer in terms of EI , w , a , $\mathbf{P}$, C_1 , and C_2 .

Hint 1. How to approach the problem

To find the elastic curve using the x_1 coordinate, the internal moment $M(x_1)$ must be determined by making a cut between the load $\mathbf{P}$ and the wall and solving the appropriate equilibrium equation for the moment. The reaction force at A will have to be determined in terms of w , a , and $\mathbf{P}$. Once the moment is determined, the elastic curve can then be solved by integrating $EI \frac{d^2v}{dx_1^2} = M(x_1)$ twice for $v(x_1)$.

Hint 2. Identify the internal moment $\{M\}(x_1)$ in the beam

To determine the equation for the elastic curve using x_1 , the moment $M(x_1)$ must be determined. Select the correct equation for $M(x_1)$.

Hint 1. How to approach the problem

To determine the elastic curve using x_1 , make a cut before the concentrated load then sum the moments about the point where the cut was made.

ANSWER:

- ☐ $M(x_1) = V(x_1)x_1 - \frac{Pa+3wa^2}{2}$
- ☐ $M(x_1) = (wa + P)x_1$
- ☒ $M(x_1) = (wa + P)x_1 - \frac{Pa+3wa^2}{2}$
- ☐ $M(x_1) = -\frac{Pa+3wa^2}{2}$

ANSWER:

$$v(x_1) = \frac{1}{EI} \left(\frac{(wa + P)x_1^3}{6} - \frac{(Pa + 3wa^2)x_1^2}{4} + C_1x_1 + C_2 \right)$$

Part C - Equation of the elastic curve using x_2 over section CD

To find the elastic curve for the entire beam, the equation that uses the coordinate system x_2 must be determined for section CD. Write this equation in terms of EI , w , and a . Let C_3 and C_4 denote the constants of integration.

Express your answer in terms of EI , w , a , C_3 , and C_4 .

Hint 1. How to approach the problem

To find the elastic curve using the x_2 coordinate, the internal moment $M(x_2)$ must be determined by making a cut at the distributed load and solving the appropriate equilibrium equation for the moment. The reaction force at A will have to be determined in terms of w , a , and P . Once the moment is determined, the elastic curve can then be solved by integrating $EI \frac{d^2v}{dx_2^2} = M(x_2)$ twice for $v(x_2)$.

Hint 2. Identify the internal moment $\{M\}(x_2)$ in the beam

To determine the equation for the elastic curve using x_2 , the moment $M(x_2)$ must be determined. Select the correct equation for $M(x_2)$.

ANSWER:

- ☐ $M(x_2) = wax_2 + \frac{Pa}{2}$
- ☒ $M(x_2) = wax_2 - \frac{3wa^2}{2}$
- ☐ $M(x_2) = wax_2 - \frac{Pa+3wa^2}{2}$
- ☐ $M(x_2) = -Px_2 - \frac{3wa^2}{2}$

ANSWER:

$$v(x_2) = \frac{1}{EI} \left(\frac{wax_2^3}{6} - \frac{3wa^2x_2^2}{4} + C_3x_2 + C_4 \right)$$

Part D - Equation of the elastic curve using x_3 over section DB

To find the elastic curve for the entire beam, the equation that uses the coordinate system x_3 must be determined for section DB . Write this equation in terms of EI , w , and a . Let C_5 and C_6 denote the constants of integration.

Express your answer in terms of EI , w , a , C_5 , and C_6 .

Hint 1. How to approach the problem

To find the elastic curve using the x_3 coordinate, the internal moment $M(x_3)$ must be determined by making a cut between the load $\mathbf{P}$ and the distributed load and solving the appropriate equilibrium equation for the moment. The reaction force at A will have to be determined in terms of w , a , and $\mathbf{P}$. Once the moment is determined, the elastic curve can then be solved by integrating $EI \frac{d^2v}{dx_3^2} = M(x_3)$ twice for $v(x_3)$.

Hint 2. Identify the internal moment $\{M\}(x_3)$ in the beam

To determine the equation for the elastic curve using x_3 , the moment $M(x_3)$ must be determined. Select the correct equation for $M(x_3)$.

Hint 1. How to approach the problem

To determine the elastic curve using x_3 , make a cut after the distributed load and then sum the moments about the point where the cut was made.

ANSWER:

- ☐ $M(x_3) = wax_3 - wx_3 + wa - \frac{3wa^2}{2}$
- ☐ $M(x_3) = -\frac{(x_3-a)^2}{2} - Px_3 - \frac{3wa^2}{2}$
- ☐ $M(x_3) = wax_3 + \frac{3wa^2}{2}$
- ☒ $M(x_3) = 2wax_3 - \frac{wx_3^2}{2} - 2wa^2$

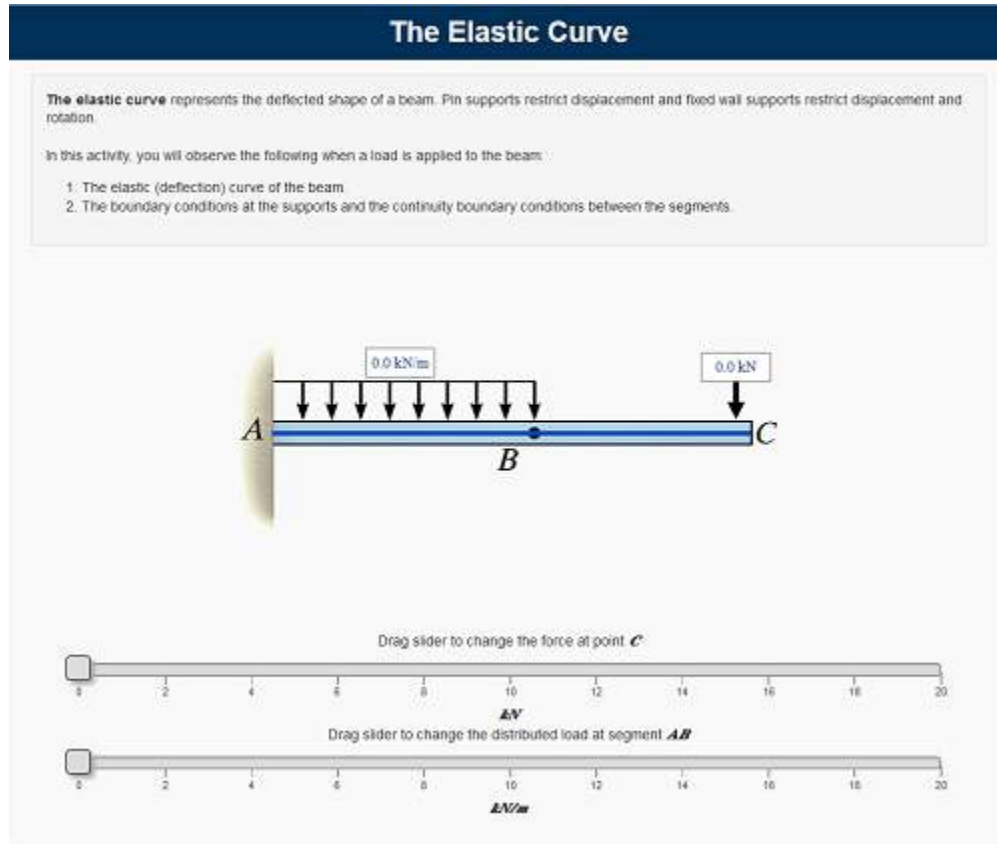
ANSWER:

$$v(x_3) = \frac{1}{EI} \left(\frac{wax_3^3}{3} - \frac{wx_3^4}{24} - wa^2x_3^2 + C_5x_3 + C_6 \right)$$

Interactive Figure: The Elastic Curve

Description: For the beam configuration in this animation, how does the elastic curve change as the loading increases?

Launch [the simulation](#), then answer the question.



Part A

For the beam configuration in this animation, how does the elastic curve change as the loading increases?

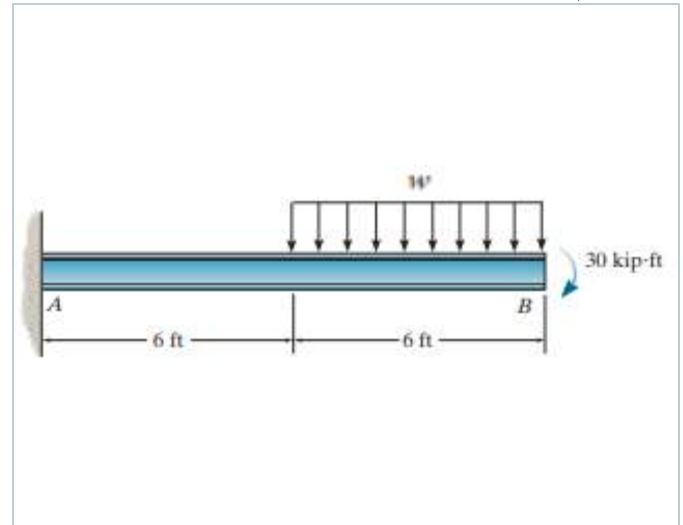
ANSWER:

- ☐ The slope of the curve becomes flatter in the distributed load region.
- ☐ The magnitude of the slope of the curve decreases.
- ☐ The slope of the curve becomes flatter in the point load region.
- ☒ The magnitude of the slope of the curve increases.

Problem 16.34

Description: The W14 * 30 cantilever beam is made of A-36 steel and is subjected to the loading shown in . Assume $w=##$ kip/ft. (a) Determine the slope at the end B of the beam, measured clockwise. (b) Determine the displacement at the end B of the beam.

The W14 × 30 cantilever beam is made of A-36 steel and is subjected to the loading shown in . Assume $w = 4.1$ kip/ft.



Part A

Determine the slope at the end B of the beam, measured clockwise.

Express your answer in radians to three significant figures.

ANSWER:

$$\theta_B = \frac{12^2}{291.29 \cdot 10^3} [(288w - 36w + 360)] = 2.38 \times 10^{-2} \text{ rad}$$

Part B

Determine the displacement at the end B of the beam.

Express your answer in inches to three significant figures. Enter positive value if the displacement is upward and negative value if the displacement is downward.

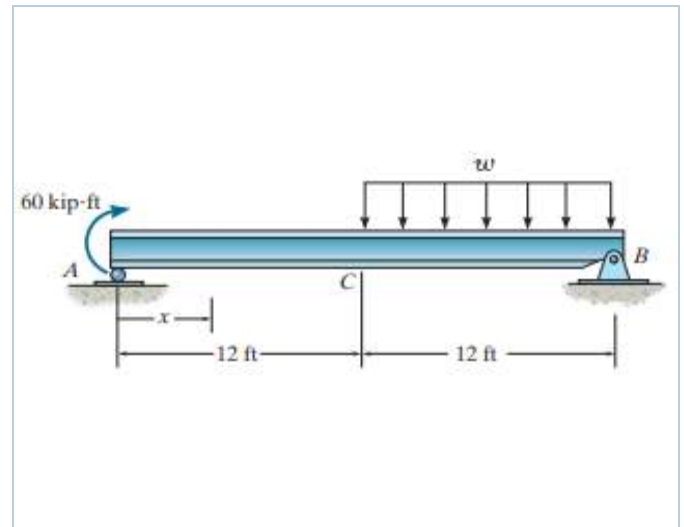
ANSWER:

$$v_B = -\frac{12^3}{291.29 \cdot 10^3} [2592w - 378w + 2160] = -2.30 \text{ in.}$$

Problem 16.35

Description: The W14 × 43 simply supported beam is made of A992 steel and is subjected to the loading shown in . Take $w = 1.8$ kip/ft. (a) Determine the displacement of center C of the beam.

The W14 × 43 simply supported beam is made of A992 steel and is subjected to the loading shown in . Take $w = 1.8$ kip/ft.



Part A

Determine the displacement of center C of the beam.

Express your answer in inches to three significant figures. Enter positive value if the displacement is upward and negative value if the displacement is downward.

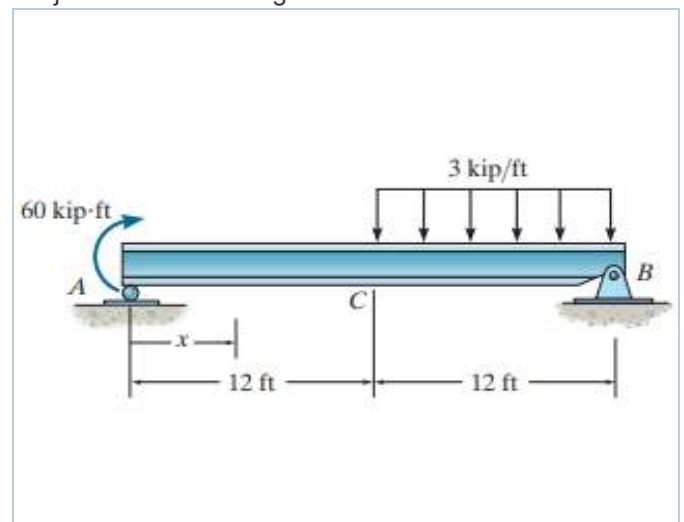
ANSWER:

$$v_C = -\frac{12^3}{428 \cdot 29 \cdot 10^3} [2160 (1 + w)] = -0.842 \text{ in.}$$

Problem 16.36

Description: The W14 * 43 simply supported beam is made of A992 steel and is subjected to the loading shown in . (a) Determine the slope at A, measured counterclockwise. (b) Determine the slope at B, measured counterclockwise.

The W14 × 43 simply supported beam is made of A992 steel and is subjected to the loading shown in .



Part A

Determine the slope at A , measured counterclockwise.

Express your answer in degrees to three significant figures.

ANSWER:

$\theta_A = -0.822^\circ$

Part B

Determine the slope at B , measured counterclockwise.

Express your answer in degrees to three significant figures.

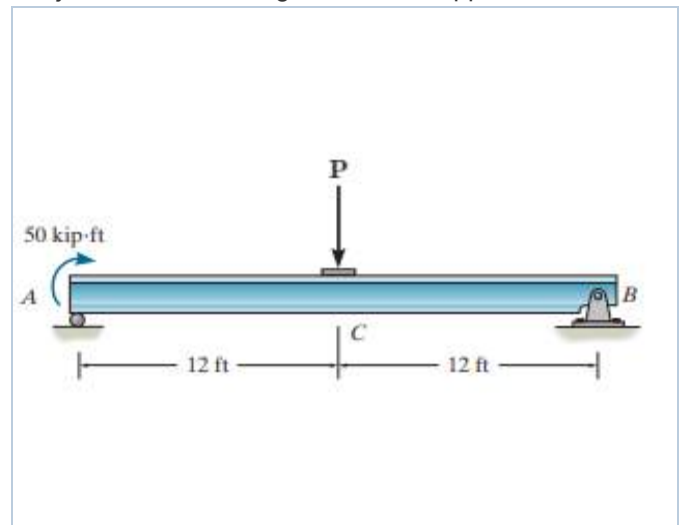
ANSWER:

$\theta_B = 0.806^\circ$

Problem 16.37

Description: The W12 * 45 simply supported beam is made of A-36 steel and is subjected to the loading shown in . Suppose that $P = \#\#$ kip. (a) Determine the displacement of the beam at its center C .

The W12 $\times$ 45 simply supported beam is made of A-36 steel and is subjected to the loading shown in . Suppose that $P = 5.0$ kip.



Part A

Determine the displacement of the beam at its center C .

Express your answer in inches to three significant figures. Enter positive value if the displacement is upward and negative value if the displacement is downward.

ANSWER:

$$v_C = -\frac{\left(\frac{P \cdot 24^3}{48} + 1800\right) \cdot 12^3}{29 \cdot 10^3 \cdot 350} = -0.552 \text{ in.}$$

◀ All Assignments

Statics and Mechanics of Materials, 6e

Hibbeler

myers59634

Ends: 05/31/26



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UNITED STATES MILITARY ACADEMY
WEST POINT.

HOMEWORK #7 – COMBINED LOADING

● HW7 - Combined Loading

[Edit](#)
[Overview](#)
[Diagnostics](#)
[Print View with Answers](#)

HW7 - Combined Loading

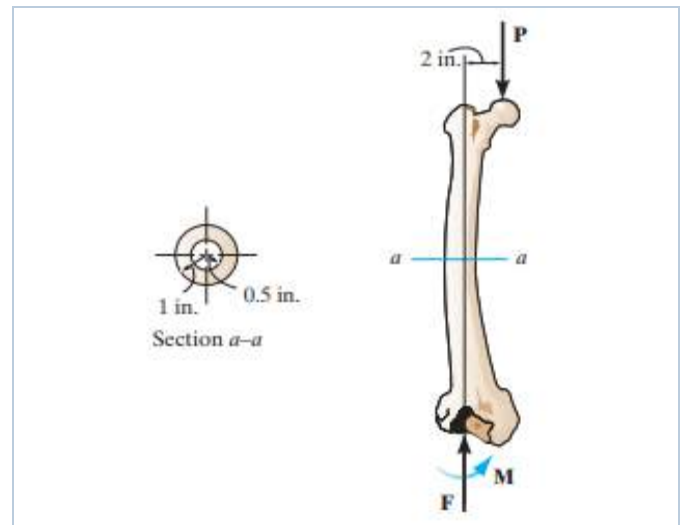
Due: 11:59pm on Thursday, March 12, 2026

You will receive no credit for items you complete after the assignment is due. [Grading Policy](#)

Review Problem 13.5

Description: The cross section of the femur at section $a-a$ can be approximated as a circular tube as shown. (a) Determine the maximum normal stress developed on the cross section at section $a-a$ due to the load of $P = 60$ lb.

The cross section of the femur at section $a-a$ can be approximated as a circular tube as shown.



Part A

Determine the maximum normal stress developed on the cross section at section $a-a$ due to the load of $P = 60$ lb.

Express your answer in pounds per square inch to three significant figures. Enter negative value in the case of compression and positive value in the case of tension.

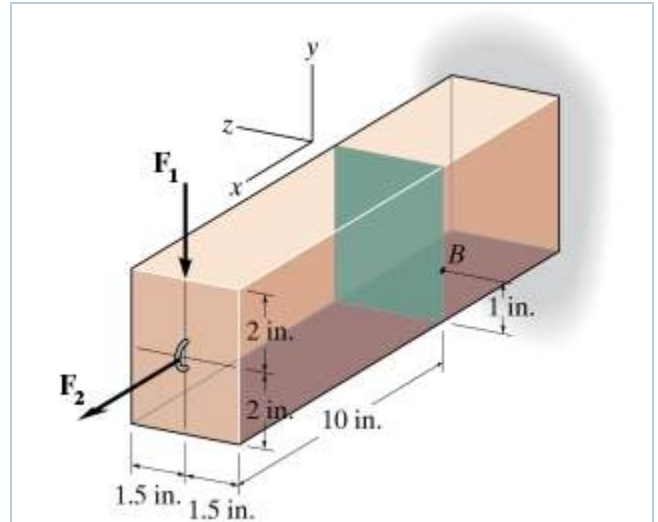
ANSWER:

$$\sigma_{\max} = \frac{-P}{\pi (1^2 - 0.5^2)} - \frac{P \cdot 2 \cdot 1}{\frac{\pi}{4} (1^4 - 0.5^4)} = -188 \text{ psi}$$

Fundamental Problem 13.5

Description: The beam has a rectangular cross section and is subjected to the loading shown. Determine the state of stress at point B. Take $F_1 = 250$ lb, $F_2 = 500$ lb. (a) Find σ_x . (b) Find σ_y . (c) Find τ_{xy} .

The beam has a rectangular cross section and is subjected to the loading shown. Determine the state of stress at point B. Take $F_1 = 250$ lb, $F_2 = 500$ lb.



Part A

Find σ_x .

Express your answer in pounds per square inch to three significant figures. Enter negative value in the case of compression and positive value in the case of tension.

ANSWER:

$$\sigma_x = \frac{500}{12} - \frac{10F_1}{16} = -115 \text{ psi}$$

Part B

Find σ_y .

Express your answer in pounds per square inch to three significant figures. Enter negative value in the case of compression and positive value in the case of tension.

ANSWER:

$$\sigma_y = 0 \text{ psi}$$

Part C

Find τ_{xy} .

Express your answer in pounds per square inch to three significant figures.

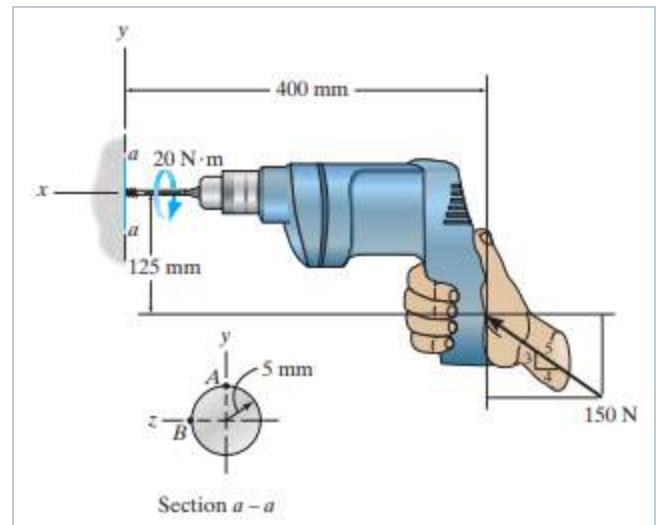
ANSWER:

$$\tau_{xy} = \frac{4.5F_1}{3 \cdot 16} = 23.4 \text{ psi}$$

Problem 13.36

Description: The drill is jammed in the wall and is subjected to the torque and force shown in . (a) Determine the normal stress at point A on the cross section of drill bit at section a-a. (b) Determine the transverse stress at point A on the cross section...

The drill is jammed in the wall and is subjected to the torque and force shown in .



Part A

Determine the normal stress at point A on the cross section of drill bit at section a-a.

Express your answer to three significant figures and include the appropriate units. Enter negative value in the case of compression and positive value in the case of tension.

ANSWER:

$$\sigma_A = -215 \text{ MPa}$$

Part B

Determine the transverse stress at point A on the cross section of drill bit at section a-a.

Express your answer to three significant figures and include the appropriate units.

ANSWER:

$$((\tau_{xy})_V)_A = 0 \text{ MPa}$$

Part C

Determine the torsional stress at point A on the cross section of drill bit at section $a-a$.

Express your answer to three significant figures and include the appropriate units.

ANSWER:

$$((\tau_{xz})_T)_A = 102 \text{ MPa}$$

[◀ All Assignments](#)

Statics and Mechanics of Materials, 6e

Hibbeler

myers59634

Ends: 05/31/26



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HOMEWORK #8 – STRESS TRANSFORMATION & MOHR'S CIRCLE

UNITED STATES MILITARY ACADEMY

CE364: MECHANICS OF MATERIALS

HOMEWORK 8

INSTRUCTOR: _____

CADET _____ SECTION ____

WEST POINT, NEW YORK

DATE: _____

Contents	Remarks	Value	Score
Cover Sheet	This Sheet	--	---
Problem 1	Stress Transformation	5	
Problem 2	Mohr's Circle	15	
Adjustments	---	Presentation is important	
Score	---	20	

_____ I CERTIFY THAT I HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT I USED TO COMPLETE THIS ASSIGNMENT AND THAT I ACKNOWLEDGED ALL ASSISTANCE I RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

_____ I CERTIFY THAT I DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

_____ I ACKNOWLEDGE THAT I USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

_____ I ACKNOWLEDGE THAT I DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

Signature: _____ Date: _____

Problem Set Organization

PDF (20 pts)

1. Stress Transformation
2. Mohr's Circle

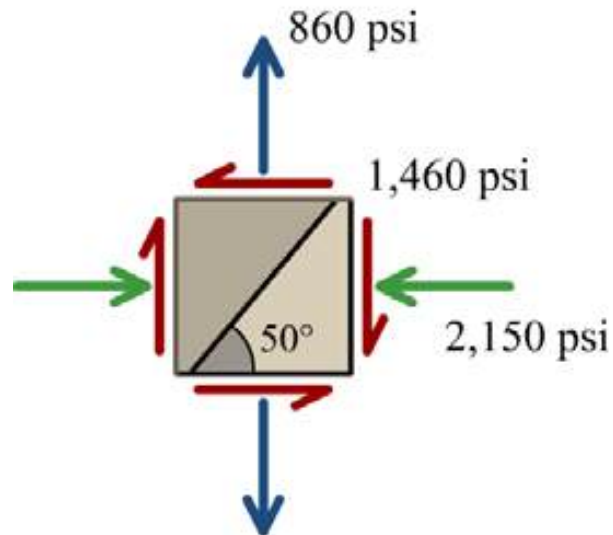
Note for Homework Submission

You are only required to turn in the cover page for the PDF portion. You do not need to turn in any other page from this homework document. Your "Given" information in your homework solution should include all relevant information provided in this document that was needed to solve each problem.

Citations and use of AI

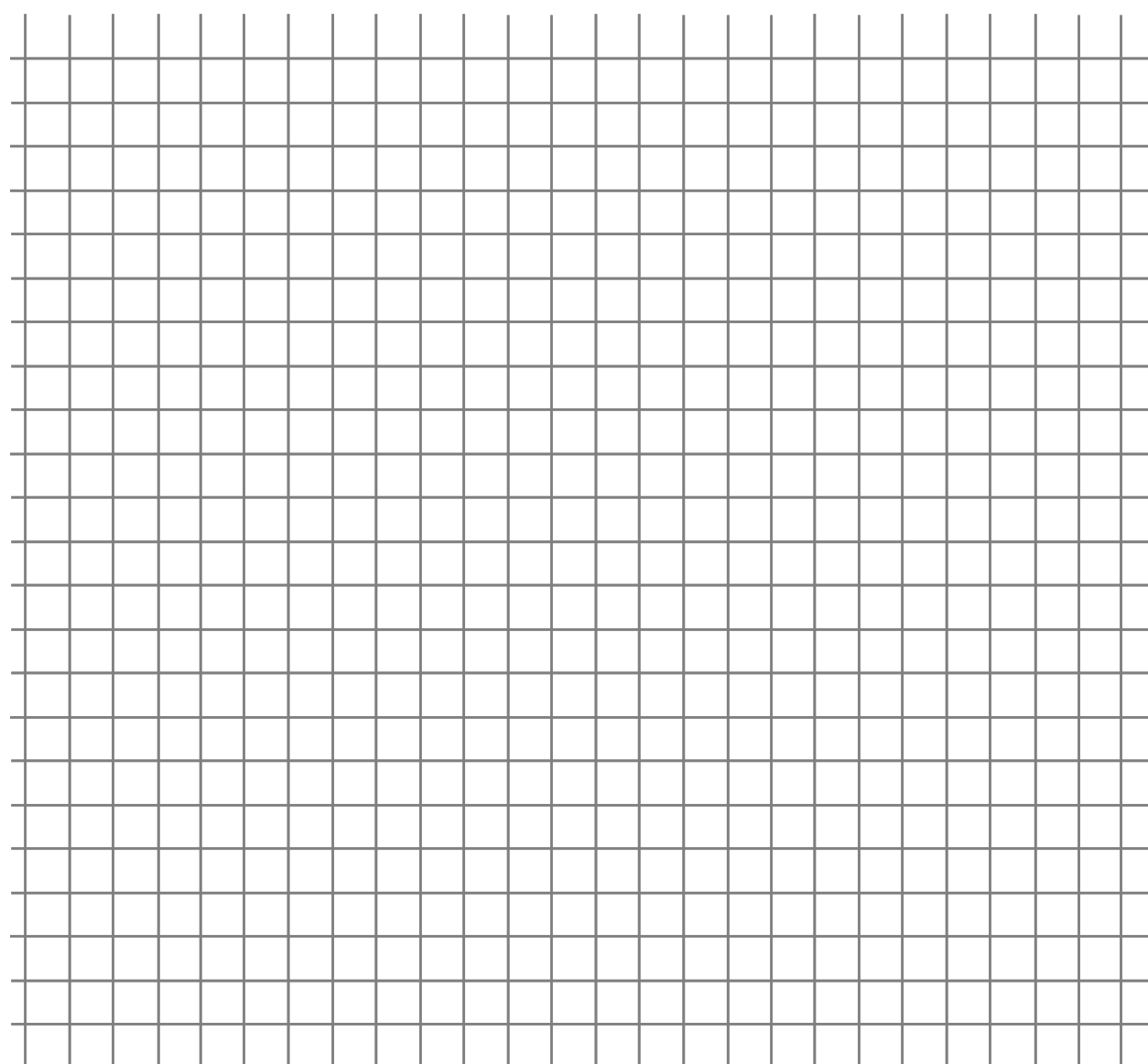
As described in the syllabus, all citations in the PDF portion must be parenthetical within your written work. Any collaborations or references used during the Mastering portion must be documented in a Work Cited page appended to the PDF submission. All uses of generative AI, e.g., ChatGPT, must be clearly documented using both i) parenthetical citations within the SOLVE section, and ii) screenshots of all prompts and ChatGPT responses as a Works Cited appendix, which must be part of your PDF submission. Solutions generated by ChatGPT that are used verbatim in your submission will be considered "copied work" and will receive a max of 50% credit.

1. Stress Transformation (5 pts) Consider a point in a structural member that is subjected to plane stress. Normal and shear stresses acting on horizontal and vertical planes at the point are shown. Determine the normal and shear stresses on the inclined plane. Show these stresses on an appropriate sketch.



2. Mohr's Circle (15 pts) Using the same state of stress from problem 1, determine the following:

- Draw Mohr's circle for this state of stress. Clearly label the reference point (A), Center (C), and Radius (R).
- Determine the principal stresses and the maximum in-plane shear stress acting at the point. Clearly label these on Mohr's circle.
- Show the principal stresses and maximum in-plane shear stress on an appropriate sketch (ie. wedge element or 2-square method)
- Check your work from problem 1 by plotting the coordinates of the transformed stresses on Mohr's circle. Label these points B ($\sigma_{x'}, \tau_{x'y'}$) and C ($\sigma_{y'}, \tau_{x'y'}$)



Instructor Solution

SOLUTION

UNITED STATES MILITARY ACADEMY

CE364: MECHANICS OF MATERIALS

HOMEWORK 8

INSTRUCTOR: MAJ THAYER

CADET SMART SECTION A1

WEST POINT, NEW YORK

DATE: DD MMM YY

Contents	Remarks	Value	Score
Cover Sheet	This Sheet	--	---
Problem 1	Stress Transformation	5	
Problem 2	Mohr's Circle	15	
Adjustments	---	Presentation is important	
Score	---	20	

_____ I CERTIFY THAT I HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT I USED TO COMPLETE THIS ASSIGNMENT AND THAT I ACKNOWLEDGED ALL ASSISTANCE I RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

TJS I CERTIFY THAT I DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

_____ I ACKNOWLEDGE THAT I USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

TJS I ACKNOWLEDGE THAT I DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

Signature: Smart Date: DD MMM YY



INSTRUCTOR MAJ THAYER

COURSE CE364

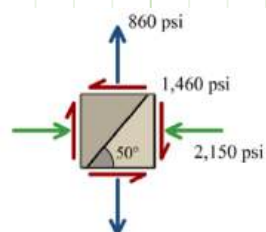
SECTION A1

PROBLEM 1

DATE AT 26-2

SHEET NO. 1 of 4

GIVEN:



$$\begin{aligned}\sigma_x &= -2,150 \text{ psi} \\ \sigma_y &= 860 \text{ psi} \\ \tau_{xy} &= -1,460 \text{ psi}\end{aligned}$$

FIND:

$\sigma_{x'}, \sigma_{y'}, \tau_{x'y'}$

SOLVE:

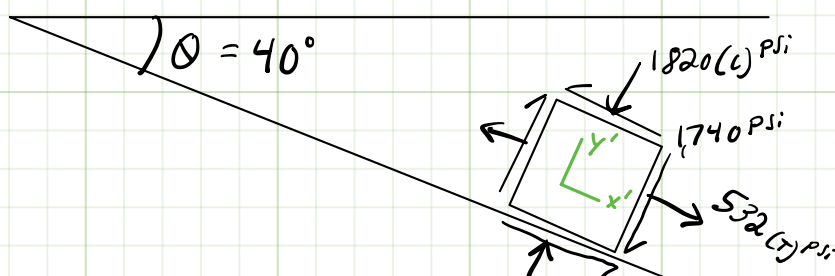
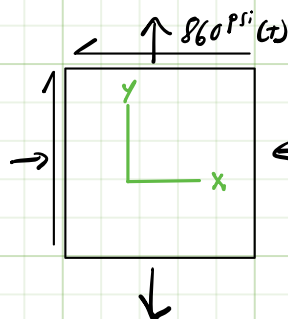


$\theta_p = -40^\circ$ OR 40° COUNTERCLOCKWISE

$$\begin{aligned}\sigma_{x'} &= \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos(2\theta) + \tau_{xy} \sin(2\theta) \quad [14-1] \\ &= \frac{(-2,150 + 860)}{2} + \frac{(-2,150 - 860)}{2} \cos(2 \cdot -40) + (-1,460) \sin(2 \cdot -40) \\ &= -645 + -2,61 + 1,438 \\ \sigma_{x'} &= 53 \text{ psi} = \underline{\underline{532 \text{ (T) psi}}} \quad \text{ANS}\end{aligned}$$

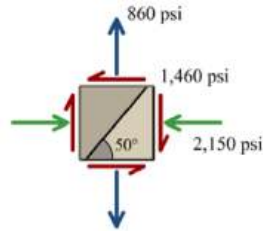
$$\begin{aligned}\sigma_{y'} &= \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos(2\theta) - \tau_{xy} \sin(2\theta) \quad [14-3] \\ &= -645 + 2,61 - 1,438 \\ \sigma_{y'} &= -1,822 \text{ psi} \approx -1,820 \text{ psi} = \underline{\underline{1,820 \text{ (C) psi}}} \quad \text{ANS}\end{aligned}$$

$$\begin{aligned}\tau_{x'y'} &= \frac{(\sigma_x - \sigma_y)}{2} \sin(2\theta) + \tau_{xy} \cos(2\theta) \quad [14-2] \\ &= \frac{(-2,150 - 860)}{2} \sin(-80) + (-1,460) \cos(-80) \\ \tau_{x'y'} &= -1,736 \text{ psi} \approx -1,740 \text{ psi} = \underline{\underline{1,740 \text{ (C) psi}}} \quad \text{ANS}\end{aligned}$$





GIVEN:



$$\begin{aligned}\sigma_x &= -2,150 \text{ psi} \\ \sigma_y &= 860 \text{ psi} \\ \tau_{xy} &= -1,460 \text{ psi}\end{aligned}$$

FIND: MOHR CIRCLE, σ_1 , σ_2 , τ_{max} , STRESS ELEMENT AT θ_p , POINT B, C.

SOLVE: PROBLEM 2A.

$$C = \frac{\sigma_x + \sigma_y}{2} = \frac{-2150 + 860}{2} = -645 \text{ psi}$$

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \sqrt{\left(\frac{-2150 - 860}{2}\right)^2 + (-1460)^2}$$

$$R = 2096.8 \approx 2100 \text{ psi}$$

PROBLEM 2B

SEE ENTIRE MOHR CIRCLE
ON FINAL PAGE OF THE
PROBLEM SET



INSTRUCTOR MAJ THAYER

COURSE CE364

SECTION A1

PROBLEM 2C-D

DATE AT 26-2

SHEET NO. 3 of 4

PROBLEM 2.C

CAN USE EQUATIONS (BELOW) OR CAREFUL INSPECTION OF MOHR'S CIRCLE.

$$\sigma_1, \sigma_2 = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{(-2,150 + 860)}{2} \pm \sqrt{\left(\frac{-2,150 - 860}{2}\right)^2 + (-1460)^2}$$

[14-5]

$$= -645 \pm 2096.8$$

$$\sigma_1 = 1452 \approx \underline{1450 \text{ PSI}} \quad \text{ANS}$$

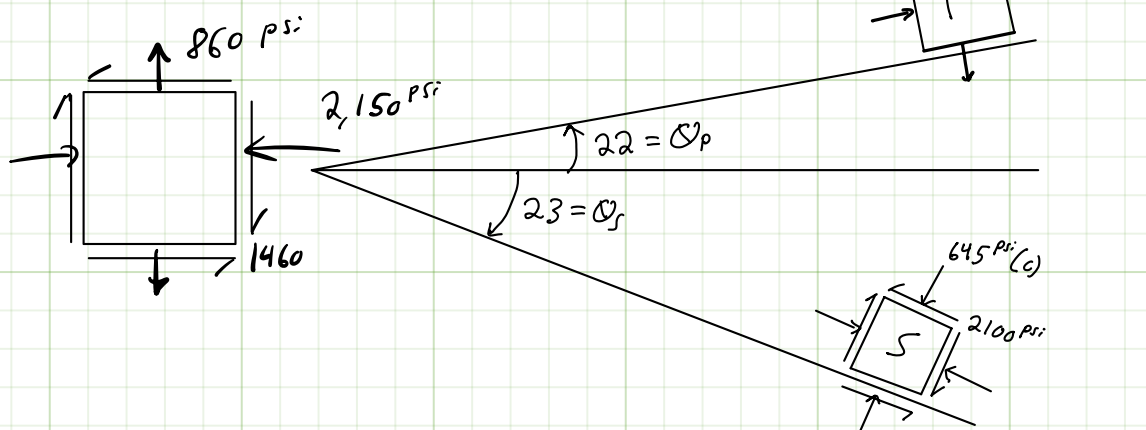
$$\sigma_2 = -2742 \approx \underline{2740 \text{ PSI (C)}} \quad \text{ANS}$$

$\sigma_x - \sigma_y = -2,150 - 860 = \text{NEGATIVE}$, so σ_2 ACTS OUT OF X-PLANE

$$\tau_{\max} = \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \sqrt{\left(\frac{-2,150 - 860}{2}\right)^2 + (-1460)^2}$$

[14-7]

$$\tau_{\max} = 2096.8 \approx \underline{2100 \text{ PSI}} \quad \text{ANS}$$



PROBLEM 2.d

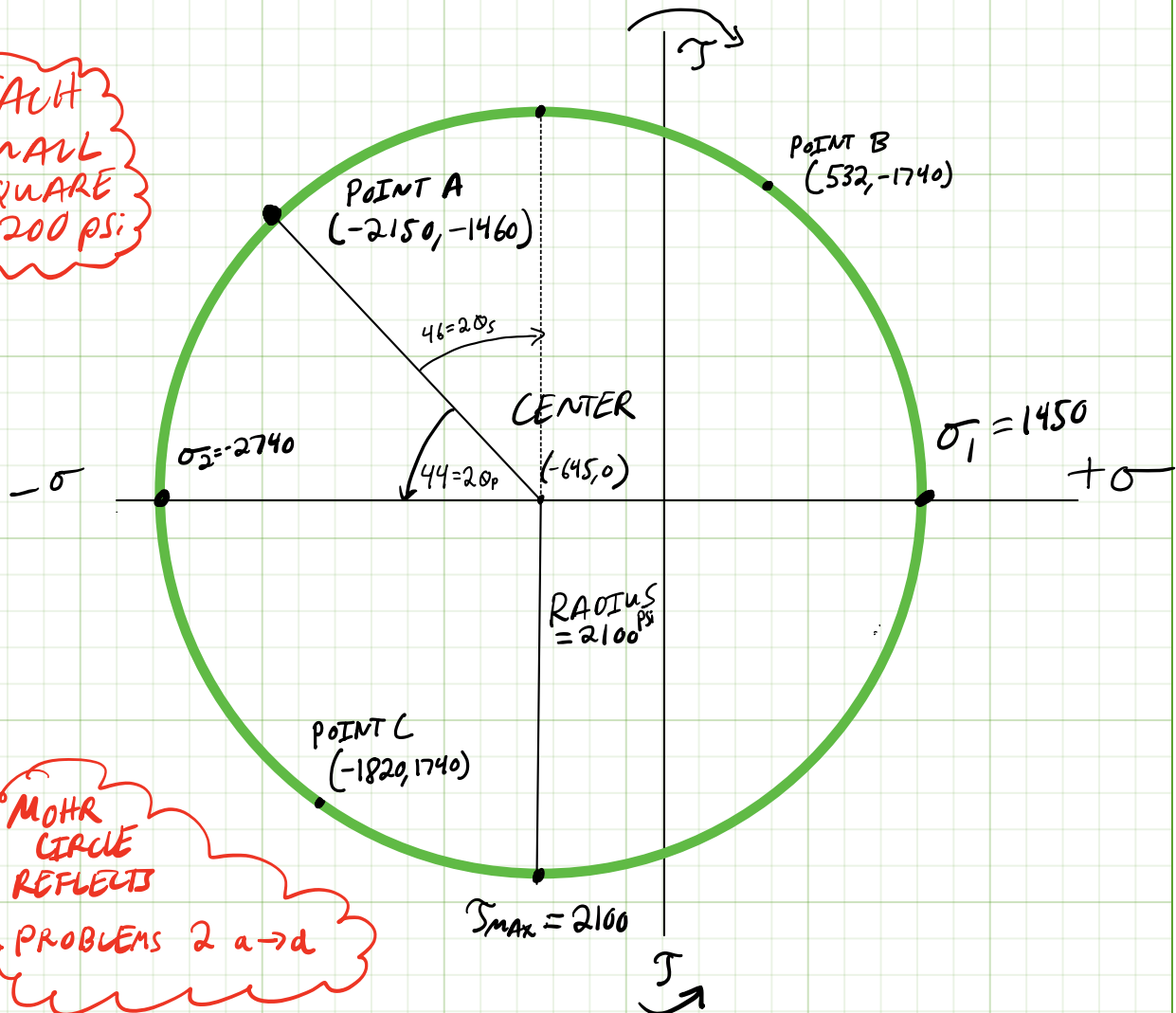
$$B (\sigma_{x'}, \tau_{xy'}) = (532, -1740)$$

$$C (\sigma_{y'}, \tau_{yx'}) = (-1820, 1740)$$

FROM PROBLEM #1



EACH
SMALL
SQUARE
= 200 PSI



MOHR
CIRCLE
REFLECTS
PROBLEMS 2 a → d



UNITED STATES MILITARY ACADEMY
WEST POINT.

HOMEWORK #9 – STRAIN TRANSFORMATION & THIN-WALLED PRESSURE VESSELS

● HW9 - Strain Transformation; TWPV

[Edit](#)[Overview](#)[Diagnostics](#)[Print View with Answers](#)

HW9 - Strain Transformation; TWPV

Due: 11:59pm on Tuesday, April 14, 2026

You will receive no credit for items you complete after the assignment is due. [Grading Policy](#)

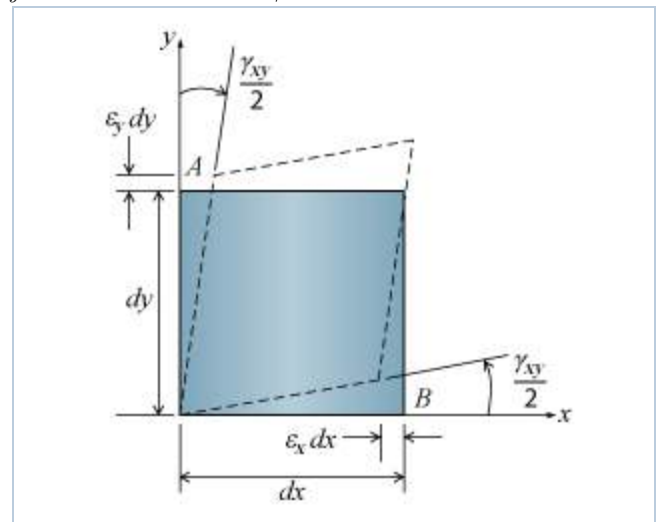
Plane Strain

Description: In this tutorial, students will analyze an element of material that is in a given state of normal and shear strain, determine the angles and values of in-plane principal strains, determine the angle and value of maximum in-plane shear strain, and determine the average normal strain.

Learning Goal:

To analyze an element of material that is in a given state of normal and shear strain, determine the angles and values of in-plane principal strains, determine the angle and value of maximum in-plane shear strain, and determine the average normal strain.

As shown, a differential element of material is distorted after it is subjected to stresses resulting in a state of plane strain given by $\epsilon_x = -390.0 \times 10^{-6}$ in/in, $\epsilon_y = 290.0 \times 10^{-6}$ in/in, and $\gamma_{xy} = 100.0 \times 10^{-6}$ in/in.



Part A - In-plane principal strains and their orientation

Determine the in-plane principal strains ϵ_1 and ϵ_2 and their orientations that are associated with the differential element of material.

Express your answers to four significant figures separated by commas.

Hint 1. How to determine the in-plane principal strains

The in-plane principal strains for the differential element of material can be determined via the following equation:

$$\epsilon_{1,2} = \frac{\epsilon_x + \epsilon_y}{2} \pm \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

Hint 2. Determine the orientation of the in-plane principal strains

The orientation of the in-plane principal strains can be determined from the geometry of the differential element of material and the state of plane stress. What is the equation for the orientation of the in-plane principal strains?

Hint 1. How to determine the orientation

The angle represented by the trigonometric identity is twice the desired angle.

ANSWER:

- ☐ $\tan 2\theta_p = \frac{\gamma_{xy}}{\epsilon_y - \epsilon_x}$
- ☐ $\tan \theta_p = \frac{\gamma_{xy}}{\epsilon_x - \epsilon_y}$
- ☐ $\tan \theta_p = \frac{\gamma_{xy}}{\epsilon_y - \epsilon_x}$
- ☒ $\tan 2\theta_p = \frac{\gamma_{xy}}{\epsilon_x - \epsilon_y}$

ANSWER:

$$\epsilon_1, \epsilon_2, \theta_p = \left(\frac{-x+y}{2} + \sqrt{\left(\left(\frac{-x-y}{2}\right)^2 + \left(\frac{t}{2}\right)^2}\right) \cdot 10^{-6}}, \right. \\ \left. \left(\frac{-x+y}{2} - \sqrt{\left(\left(\frac{-x-y}{2}\right)^2 + \left(\frac{t}{2}\right)^2}\right) \cdot 10^{-6}}, \frac{\text{atan}\left(\frac{t}{-x-y}\right)}{2} = 2.937 \times 10^{-4}, \right. \\ \left. -3.937 \times 10^{-4}, -7.301 \times 10^{-2} \right) \text{ in/in, in/in, rad}$$

Part B - Maximum in-plane shear strain and its orientation

Determine the magnitude of the maximum in-plane shear strain, $\gamma_{\text{in plane}}^{\text{max}}$, and its orientation relative to the differential element of material.

Express your answers to four significant figures separated by a comma.

Hint 1. How to determine the maximum in-plane shear strain

The maximum in-plane shear strain can be determined via the equation

$$\frac{\gamma_{\text{in plane}}^{\max}}{2} = \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

Hint 2. How to determine the proper sign of the maximum in-plane shear strain

The proper sign of the maximum in-plane shear strain (positive or negative) can be determined by solving the following equation for $\gamma_{x'y'}$: $\frac{\gamma_{x'y'}}{2} = -\left(\frac{\epsilon_x - \epsilon_y}{2}\right) \sin 2\theta + \frac{\gamma_{xy}}{2} \cos 2\theta$

Hint 3. Determine the orientation of the maximum in-plane shear strain

The orientation of the maximum in-plane shear strain can be determined from the geometry of the differential element of material and the state of plane stress. What is the equation for the orientation of the maximum in-plane shear strain?

Hint 1. How to determine the orientation

The angle represented by the trigonometric identity is twice the desired angle.

ANSWER:

- ☐ $\tan 2\theta_s = \left(\frac{\epsilon_x - \epsilon_y}{\gamma_{xy}}\right)$
- ☐ $\tan \theta_s = \left(\frac{\epsilon_x - \epsilon_y}{\gamma_{xy}}\right)$
- ☐ $\tan \theta_s = -\left(\frac{\epsilon_x - \epsilon_y}{\gamma_{xy}}\right)$
- ☒ $\tan 2\theta_s = -\left(\frac{\epsilon_x - \epsilon_y}{\gamma_{xy}}\right)$

ANSWER:

$$\gamma_{\text{in plane}}^{\max}, \theta_s = 2\sqrt{\left[\left(\frac{-x-y}{2}\right)^2 + \left(\frac{t}{2}\right)^2\right]} \cdot 10^{-6}, \frac{\text{atan}\left(\frac{y+x}{t}\right)}{2} = 6.873 \times 10^{-4}, 0.7124 \text{ in/in, rad}$$

Part C - Average normal strain in the differential element of material

Determine the average normal strain in the differential element of material.

Express your answer to four significant figures.

Hint 1. How to determine the average normal strain

To calculate the average normal strain, average the strains in the x and y directions by using the equation

$$\epsilon_{\text{avg}} = \frac{\epsilon_x + \epsilon_y}{2}$$

Hint 2. Find the nominal values of the normal strains

To properly implement the equation for the average normal strain, the nominal values (including their respective signs) must be known. What are the principal strains, ϵ_x and ϵ_y ?

Express your answers to four significant figures separated by a comma.

ANSWER:

$$\epsilon_x, \epsilon_y = -x(10^{-6}), y(10^{-6}) = -3.900 \times 10^{-4}, 2.900 \times 10^{-4} \text{ in/in, in/in}$$

ANSWER:

$$\epsilon_{\text{avg}} = \frac{y - x}{2} \cdot 10^{-6} = -5.000 \times 10^{-5} \text{ in/in}$$

General Equations of Plane Strain

Description: In this tutorial, students will determine the equivalent in-plane strains on an element oriented at an angle and determine the in-plane principal strains, maximum in-plane shear strain, and average normal strain.

Learning Goal:

To become familiar with the general equations of plane strain used for determining in-plane principal strain, maximum in-plane shear strain, and average normal strain.

The state of strain at a point has components of $\epsilon_x = 350.0 \times (10^{-6})$, $\epsilon_y = -200.0 \times (10^{-6})$, and $\gamma_{xy} = 190.0 \times (10^{-6})$.

Part A - Equivalent in-plane strains on the oriented element

Determine the equivalent in-plane strains on an element rotated counterclockwise at an angle of $\theta = 30.0^\circ$.

Express your answers, separated by commas, to three significant figures.

Hint 1. How to determine the equivalent in-plane normal strains

The equivalent in-plane normal strains on the oriented element can be determined from the equations

$$\epsilon_{x'} = \frac{\epsilon_x + \epsilon_y}{2} + \frac{\epsilon_x - \epsilon_y}{2} \cos 2\theta + \frac{\gamma_{xy}}{2} \sin 2\theta \quad \epsilon_{y'} = \frac{\epsilon_x + \epsilon_y}{2} - \frac{\epsilon_x - \epsilon_y}{2} \cos 2\theta - \frac{\gamma_{xy}}{2} \sin 2\theta$$

Hint 2. Identify the equation for the equivalent in-plane shear strain

Select the equation that is used to determine the equivalent in-plane shear strain on an element that is rotated through a counterclockwise angle.

Hint 1. How to approach the problem

The equation for the equivalent in-plane shear strain either can be referenced from a book or can be determined by considering the free-body diagram of a differential element subjected to the given state of strain.

ANSWER:

- ☐ $\frac{\gamma_{x'y'}}{2} = \left(\frac{\epsilon_x - \epsilon_y}{2}\right) \sin 2\theta + \frac{\gamma_{xy}}{2} \cos 2\theta$
- ☐ $\gamma_{x'y'} = -\left(\frac{\epsilon_x - \epsilon_y}{2}\right) \sin 2\theta + \frac{\gamma_{xy}}{2} \cos 2\theta$
- ☐ $\gamma_{x'y'} = \left(\frac{\epsilon_x - \epsilon_y}{2}\right) \sin 2\theta + \frac{\gamma_{xy}}{2} \cos 2\theta$
- ☒ $\frac{\gamma_{x'y'}}{2} = -\left(\frac{\epsilon_x - \epsilon_y}{2}\right) \sin 2\theta + \frac{\gamma_{xy}}{2} \cos 2\theta$

ANSWER:

$$\epsilon_{x'}, \epsilon_{y'}, \gamma_{x'y'} = \mathbf{a, b, c} = 2.95 \times 10^{-4}, -1.45 \times 10^{-4}, -3.81 \times 10^{-4}$$

Part B - In-plane principal strains on the oriented element

Determine the in-plane principal strains on the oriented element.

Express your answers, separated by a comma, to three significant figures.

Hint 1. How to determine the in-plane principal strains

The in-plane principal strains can be determined from the equation

$$\epsilon_{1,2} = \frac{\epsilon_x + \epsilon_y}{2} \pm \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

Hint 2. Identify the method for determining the in-plane principal strains

Select the statement below that describes the method used to determine the in-plane principal strains on the oriented element.

ANSWER:

- ☐ The in-plane principal strains on the oriented element can be found only by substituting the equivalent in-plane strains determined in Part A into the equation for in-plane principal strains. The strains on the nonoriented element could not be used.
- ☐ The in-plane principal strains on the oriented element can be found by determining the principal strains on the nonoriented element first. Then, the equivalent strains to these principal strains on the oriented element must be determined.
- ☐ The in-plane principal strains on the oriented element can be found only by substituting the in-plane strains on the nonoriented element into the equation for in-plane principal strains. The equivalent strains on the oriented element could not be used.
- ☒ The in-plane principal strains do not depend on the orientation of the element, so the strains on either oriented or nonoriented element can be substituted into the equation for in-plane principal strains. The results will be the same.

ANSWER:

$$\epsilon_1, \epsilon_2 = \frac{a+b}{2} \pm \sqrt{\left(\frac{a-b}{2}\right)^2 + \left(\frac{c}{2}\right)^2} = 3.66 \times 10^{-4}, -2.16 \times 10^{-4}$$

Part C - Maximum in-plane shear strain and average normal strain on the oriented element

Determine the maximum in-plane shear strain and the average normal strain on the oriented element.

Express your answers, separated by a comma, to three significant figures.

Hint 1. How to determine the average normal strain

The average normal strain on the oriented element can be determined from the equation $\epsilon_{\text{avg}} = \frac{\epsilon_x + \epsilon_y}{2}$

Hint 2. Identify the equation for determining the maximum in-plane shear strain

Select the equation that is used to determine the maximum in-plane shear strain on an element that is rotated through an angle.

Hint 1. How to approach the problem

The equation for the maximum in-plane shear strain either can be referenced from a book or can be determined by considering the free-body diagram of a differential element subjected to the given state of strain.

ANSWER:

- ☐ $\gamma_{in\ plane}^{max} = \sqrt{\left(\frac{\epsilon_x + \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$
- ☒ $\frac{\gamma_{in\ plane}^{max}}{2} = \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$
- ☐ $\frac{\gamma_{in\ plane}^{max}}{2} = \sqrt{\left(\frac{\epsilon_x + \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$
- ☐ $\gamma_{in\ plane}^{max} = \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$

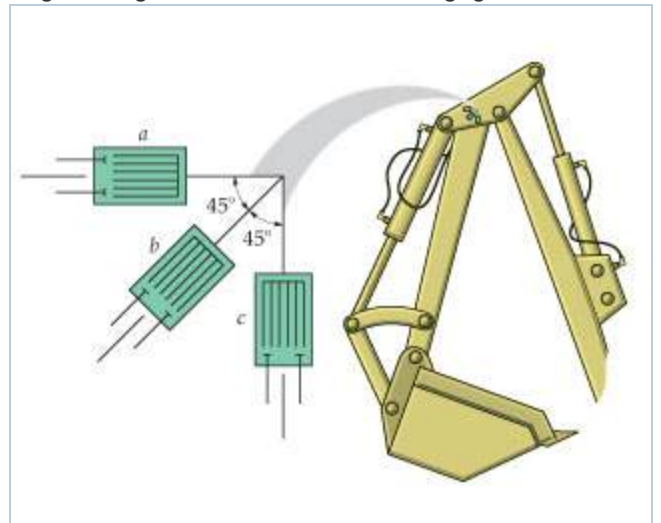
ANSWER:

$$\gamma_{in\ plane}^{max}, \epsilon_{avg} = 2\sqrt{\left(\frac{a-b}{2}\right)^2 + \left(\frac{c}{2}\right)^2}, \frac{a+b}{2} = 5.82 \times 10^{-4}, 7.50 \times 10^{-5}$$

Problem 14.70

Description: The 45 degree(s) strain rosette is mounted on the link of the backhoe. The following readings are obtained from each gage: [...], [...], [...]. (a) Determine the in-plane principal strains. (b) Determine the maximum in-plane shear strain. (c)...

The 45° strain rosette is mounted on the link of the backhoe. The following readings are obtained from each gage:
 $\epsilon_a = 650(10^{-6})$, $\epsilon_b = -230(10^{-6})$, $\epsilon_c = 480(10^{-6})$.



Determine the in-plane principal strains.

Enter your answers using three significant figures separated by a comma.

ANSWER:

$$\epsilon_1, \epsilon_2 = \frac{\epsilon_a + \epsilon_c}{2} + \sqrt{\left(\frac{\epsilon_a - \epsilon_c}{2}\right)^2 + \left(\epsilon_b - \frac{1}{2}(\epsilon_a + \epsilon_c)\right)^2}, \frac{\epsilon_a + \epsilon_c}{2} - \sqrt{\left(\frac{\epsilon_a - \epsilon_c}{2}\right)^2 + \left(\epsilon_b - \frac{1}{2}(\epsilon_a + \epsilon_c)\right)^2} = 1.36 \times 10^{-3}, -2.35 \times 10^{-4}$$

Part B

Determine the maximum in-plane shear strain.

Express your answer using three significant figures.

ANSWER:

$$\gamma_{\text{in-plane}}^{\text{max}} = 2\sqrt{\left(\frac{\epsilon_a - \epsilon_c}{2}\right)^2 + \left(\epsilon_b - \frac{1}{2}(\epsilon_a + \epsilon_c)\right)^2} = 1.60 \times 10^{-3}$$

Part C

Determine the associated average normal strain.

Express your answer using three significant figures.

ANSWER:

$$\epsilon_{\text{avg}} = \frac{\epsilon_a + \epsilon_c}{2} = 5.65 \times 10^{-4}$$

Problem 13.1

Description: A spherical gas tank has an inner radius of $r = 1.5$ m. (a) If it is subjected to an internal pressure of $p = \text{##}$ kPa, determine its minimum required thickness if the maximum normal stress is not to exceed 12 MPa.

A spherical gas tank has an inner radius of $r = 1.5$ m.

Part A

If it is subjected to an internal pressure of $p = 360$ kPa, determine its minimum required thickness if the maximum normal stress is not to exceed 12 MPa.

Express your answer to three significant figures and include the appropriate units.

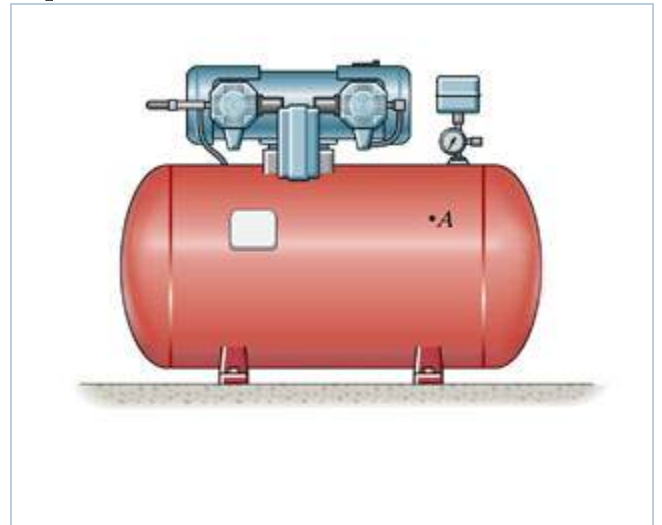
ANSWER:

$$t = \frac{p \cdot 1.5}{2} = 22.5 \text{ mm}$$

Problem 13.4

Description: The tank of the air compressor is subjected to an internal pressure of ## psi. The inner diameter of the tank is 22 in., and the wall thickness is 0.25 in. (a) Determine the hoop stress acting at point A. (b) Determine the longitudinal stress acting ...

The tank of the air compressor is subjected to an internal pressure of 110 psi . The inner diameter of the tank is 22 in., and the wall thickness is 0.25 in.



Part A

Determine the hoop stress acting at point A.

Express your answer in kilopounds per square inch to three significant figures.

ANSWER:

$$\sigma_1 = p \cdot 11.4 \cdot 10^{-3} = 4.84 \text{ ksi}$$

Part B

Determine the longitudinal stress acting at point A.

Express your answer in kilopounds per square inch to three significant figures.

ANSWER:

$$\sigma_2 = \frac{p \cdot 11.4}{2} \cdot 10^{-3} = 2.42 \text{ ksi}$$

Statics and Mechanics of Materials, 6e

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Ends: 05/31/26



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UNITED STATES MILITARY ACADEMY
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HOMEWORK #10 – FAILURE THEORIES

● HW10 - Failure Theories

[Edit](#)[Overview](#)[Diagnostics](#)[Print View with Answers](#)

HW10 - Failure Theories

Due: 11:59pm on Friday, April 24, 2026

You will receive no credit for items you complete after the assignment is due. [Grading Policy](#)

Problem 10.63

Description: A shaft is made of a material for which $\sigma_Y = \#\#$ ksi!. (a) Determine the torsional shear stress required to cause yielding using the maximum distortion energy theory.

A shaft is made of a material for which $\sigma_Y = 60$ ksi.

Part A

Determine the torsional shear stress required to cause yielding using the maximum distortion energy theory.

Express your answer in kilopounds per square inch to three significant figures.

ANSWER:

$$\tau = \frac{\sigma_Y}{\sqrt{3}} = 34.6 \text{ ksi}$$

Problem 10.66

Description: A bar with a square cross section is made of a material having a yield stress of $\#\#$ ksi. Suppose the bar is subjected to a bending moment of $\#\#$ kip·in. (a) Determine the required size of the bar according to the maximum distortion energy theory. Use ...

A bar with a square cross section is made of a material having a yield stress of 150 ksi. Suppose the bar is subjected to a bending moment of 40 kip·in.

Part A

Determine the required size of the bar according to the maximum distortion energy theory. Use a factor of safety of 1.5 with respect to yielding.

Express your answer in inches to three significant figures.

ANSWER:

$$a = \sqrt[3]{\frac{9M}{\sigma_Y}} = 1.34 \text{ in.}$$

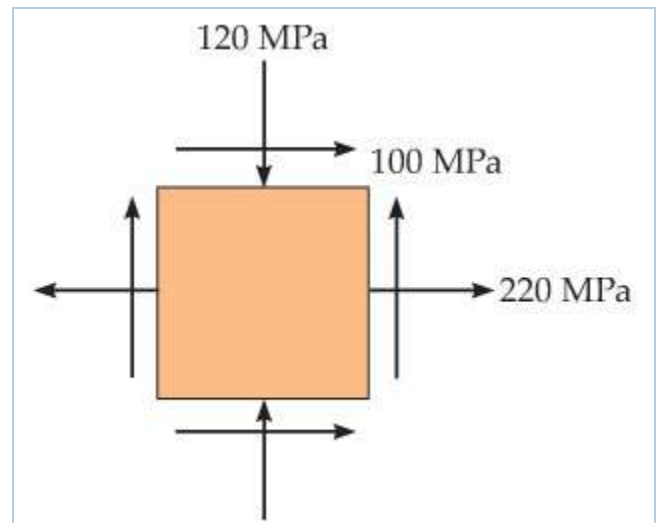
Problem 10.68

Description: Cast iron when tested in tension and compression has an ultimate strength of $(\sigma_{\text{ult}})_t = 280 \text{ MPa}$ and $(\sigma_{\text{ult}})_c = 420 \text{ MPa}$, respectively. Also, when subjected to pure torsion it can sustain an ultimate shear stress of $\tau_{\text{ult}} = 168 \text{ MPa}$.

Cast iron when tested in tension and compression has an ultimate strength of $(\sigma_{\text{ult}})_t = 280 \text{ MPa}$ and $(\sigma_{\text{ult}})_c = 420 \text{ MPa}$, respectively. Also, when subjected to pure torsion it can sustain an ultimate shear stress of $\tau_{\text{ult}} = 168 \text{ MPa}$.

Part A

If a part made of this material is subjected to the state of plane stress shown, determine if it fails according to Mohr's failure criterion.



ANSWER:

- ☒ Yes, it fails.
☐ No, it doesn't fail.

Part B

Determine the in-plane principal stresses.

Express your answers in megapascals to three significant figures separated by a comma.

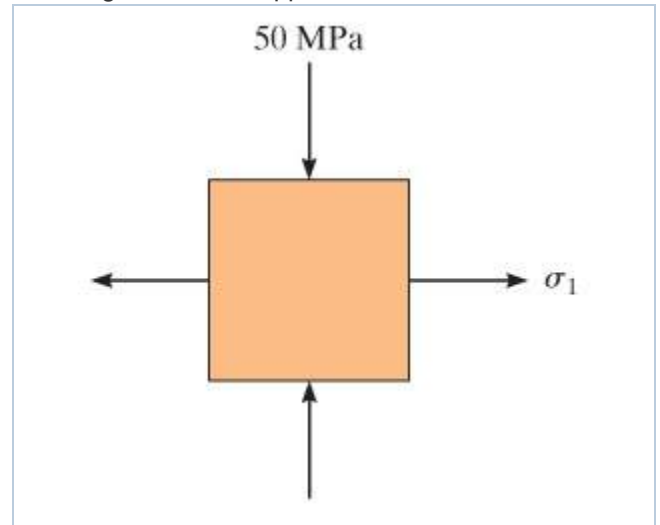
ANSWER:

$$\sigma_1, \sigma_2 = 247, -147 \text{ MPa}$$

Problem 10.69

Description: The principal plane stresses acting on a differential element are shown in the figure below. Suppose that $\sigma_1 = \text{##}$ MPa. The material is machine steel having a yield stress of $\sigma_Y = 700$ MPa. (a) Determine the factor of safety with respect to...

The principal plane stresses acting on a differential element are shown in the figure below. Suppose that $\sigma_1 = 40$ MPa. The material is machine steel having a yield stress of $\sigma_Y = 700$ MPa.



Part A

Determine the factor of safety with respect to yielding if the maximum shear stress theory is considered.

Express your answer using three significant figures.

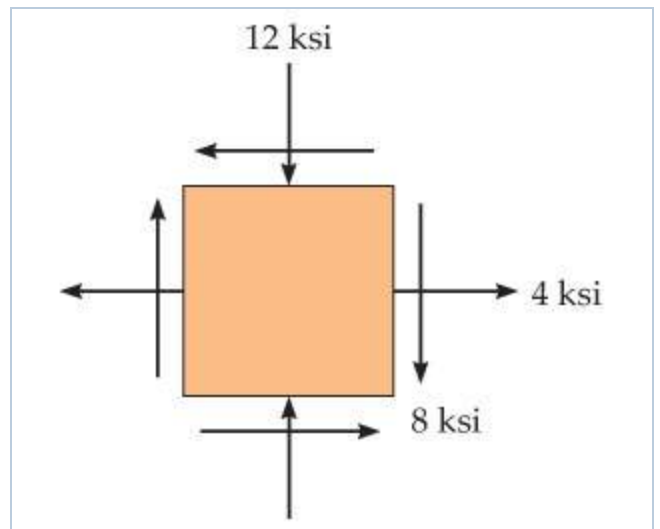
ANSWER:

$$\text{F. S.} = \frac{700}{\sigma_1 + 50} = 7.78$$

Problem 10.75

Description: The element is subjected to the stresses shown in . (a) If $\sigma_y = \text{##}$ ksi!, determine the factor of safety for the loading based on the maximum shear stress theory.

The element is subjected to the stresses shown in .



Part A

If $\sigma_Y = 52 \text{ ksi}$, determine the factor of safety for the loading based on the maximum shear stress theory.

Express your answer using three significant figures.

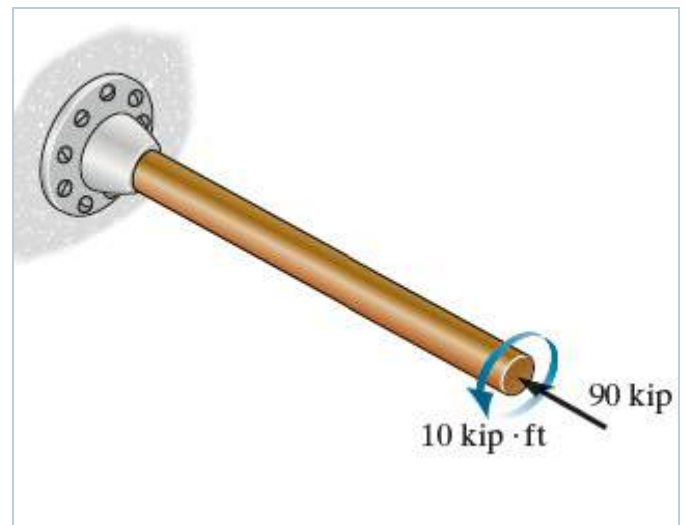
ANSWER:

$$F.S. = \frac{\sigma_y}{2 \cdot 11.314} = 2.30$$

Problem 10.77

Description: In , the ##-in.-diameter short rod is made from brittle material having an ultimate strength of $\sigma_{ult} = 60 \text{ ksi}$, for both tension and compression. Use a factor of safety of 2 against rupture. (a) Determine if the shaft fails according to the...

In , the 3.2-in.-diameter short rod is made from brittle material having an ultimate strength of $\sigma_{ult} = 60 \text{ ksi}$, for both tension and compression. Use a factor of safety of 2 against rupture.



Part A

Determine if the shaft fails according to the maximum normal stress theory.

ANSWER:

- ☒ No
☐ Yes

Part B

Determine the in-plane principal stresses.

Express your answers in kilopounds per square inch to three significant figures separated by a comma.

ANSWER:

$$\sigma_1, \sigma_2 = \frac{-90.4}{2\pi d^2} + \sqrt{\left(\frac{-90.4}{\pi d^2 \cdot 2}\right)^2 + \left(\frac{-10 \cdot 12 \cdot 2 \cdot 8}{\pi d^3}\right)^2}, \frac{-90.4}{\pi d^2 \cdot 2} - \sqrt{\left(\frac{-90.4}{\pi d^2 \cdot 2}\right)^2 + \left(\frac{-10 \cdot 12 \cdot 2 \cdot 8}{\pi d^3}\right)^2} = 13.9, -25.1 \text{ ksi}$$

[◀ All Assignments](#)

Statics and Mechanics of Materials, 6e

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Ends: 05/31/26



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TAB L LABORATORY ASSIGNMENTS



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LAB #1: UNIAXIAL TENSION LAB

CE364, 26-2, Tension Test Report (100 points) – DUE: 15JAN26 (Day 2) 16JAN26 (Day 1)

In this laboratory exercise, your team conducted a uniaxial tension test on two samples with the same cross-section but made of different materials. The purpose of this report is to present the results of your experimental testing clearly and concisely. You will need to compare and contrast the measured values from the experiment to theoretical values from equations and specified material properties. Follow the report format provided on Canvas. Include a Cover page IAW Dean's DAAW. Each cadet must annotate next to their signature on the lab cover page the time they spent working on the lab report outside of class. The main body of your report (Title, Introduction, Theory & Methods, Results, Conclusions) should be approximately 2 pages. Include figures, tables, and appendices after the main body. Be sure to include the following items in each of the sections.

Theory & Methods

Include a brief discussion and present relevant equations for the following items.

1. Relationship between Axial Force and Axial Stress
2. Relationship between Axial Deformation and Axial Strain
3. Relationship between Axial Stress and Axial Strain
4. Yield strength based on the 0.2% offset method
5. Resilience and Toughness – why are they important?

****Include assumptions needed to apply these equations****

Results

Answer the following questions in the results section. Be sure to reference the tables and figures.

1. Describe the materials' stress-strain graphs in words. You may reference your figures. In your description, report key material properties including: the proportional limit, yield strength, ultimate strength, fracture strength, Modulus of Elasticity, and the material resilience and toughness.
2. Identify the materials you tested based on the properties identified from the stress strain curve. Use specified material properties from the Hibbeler textbook or from other resources. Justify your selection and compare the theoretical values to experimental results.
3. Discuss any differences between your experimental and theoretical values. Explain why your answers make sense or why they are different. Report the percent differences for each value: yield strength, ultimate strength, and modulus of elasticity.
4. Report the averages and standard deviations for the ultimate strength values for the materials in tension. If you were designing a structure with each of these materials in tension, what ultimate strength value would you use for each and why?
5. Explain any sources of error which may have impacted the results. Identify any assumptions which may have impacted the validity of the results.

****Your response to the questions does not need to be in a specific order. Just include each item above.****

Figures

Include the following figures in your report

1. Figures of the experimental tests
2. Figures of the samples before and after testing
3. Create a stress-strain graph for all your data points with axial stress (in MPa) on the vertical axis and axial strain (in 10^{-6} mm/mm) on the horizontal axis. Label the key points to include ultimate stress (σ_u), fracture stress (σ_f), fracture strain (ϵ_f), and yield stress (σ_y). See Figure 1.
4. Create a stress-strain graph focusing only the elastic region with axial stress (in MPa) on the vertical axis and axial strain (in 10^{-6} mm/mm) on the horizontal axis. Label the key points to include yield stress (σ_y), yield strain (ϵ_y), and Modulus of Elasticity (E). See Figure 2.

****Add lines and textboxes to the graph in PowerPoint or Excel, then "group" the figure and paste special as an "Enhanced Metafile Image" in Word.****

Tables

Include the following tables of results

1. Table with experimental values and theoretical material specified values for yield strength, ultimate strength, and modulus of elasticity.
2. Table of tensile and ultimate strength for all groups in the section including the average and standard deviation.

Appendix

- A. Sample Calculations on Green/White Engineering Paper or Mathcad
 - a. Specimen cross sectional area and area moment of inertia
 - b. Stress and strain calculations for the fifth data point,
 - c. Modulus of Elasticity based on selecting two data points appropriately within the elastic region.
 - d. Modulus of Resilience by estimating the area under the elastic region.
 - e. Material Toughness by estimating the area under the entire stress-strain graph.
- B. Processed data spreadsheet includes a column for load (kN), deformation (mm), stress (MPa), and strain (10^{-6} mm/mm) for each data point. Ensure your units are correct.

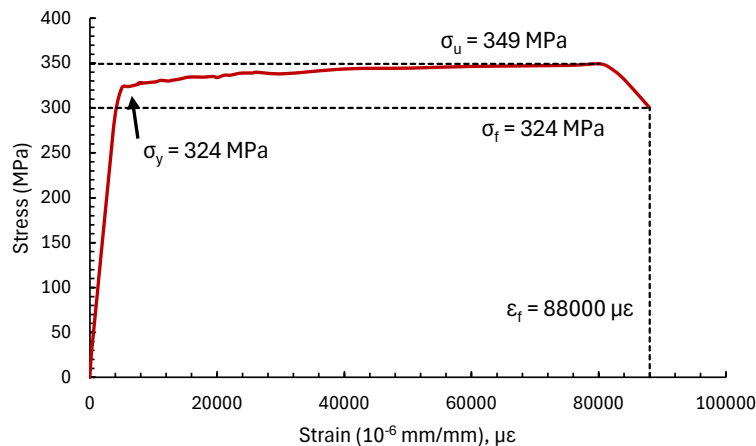


Figure 1: Axial Stress vs Axial Strain for Uniaxial Tension Test

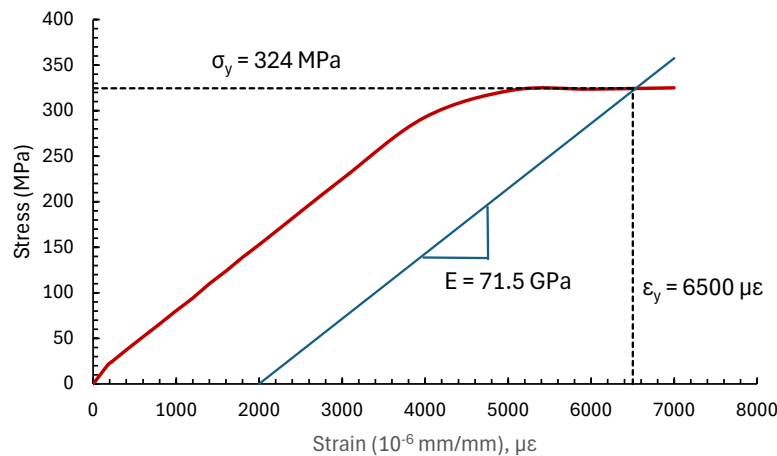


Figure 2: Elastic Region of Axial Stress vs Axial Strain for Uniaxial Tension Test

CE364, 26-2, Tension Test Report, Data Sheet

Uniaxial Tension Test

Diameter 1 = _____ mm
 Diameter 2 = _____ mm
 Diameter 3 = _____ mm
 Average Diameter = _____ mm
 Area = _____ mm²
 Initial Length L_0 = 50 (gage) mm
 Maximum Load = _____ kN
 (from Peak Reading)
 Rupture Length L_f = _____ mm
 (measured between gage length marks)

Maximum Stress = _____ MPa = N/mm²

Ultimate Stress from other Groups in Section

Group 1 =	MPa
Group 2 =	MPa
Group 3 =	MPa
Group 4 =	MPa
Group 5 =	MPa
Group 6 =	MPa

Rupture Load = _____ kN
 (as specimen fractured - watch carefully!)

Elongation	Load	Elongation	Load
mm	kN	mm	kN
0.02		2.50	
0.04		3.00	
0.06		3.50	
0.08		4.00	
0.10		4.50	
0.12		5.00	
0.14		5.50	
0.16		6.00	
0.18		6.50	
0.20		7.00	
0.30		7.50	
0.40		8.00	
0.50		8.50	
0.60		9.00	
0.70		9.50	
0.80		10.00	
0.90		10.50	
1.00		11.00	
1.50		11.50	
2.00		12.00	

At Rupture: At Rupture:

DO NOT REMOVE FROM THE LAB
UNIAXIAL TENSION TEST EXPERIMENTAL PROCEDURES
CE364, 26-2

Purpose: The uniaxial tension test provides information on the strength and ductility of a material.

Pre-Test Requirements

- a. **Safety glasses:** Put them on! (They look cool, why wouldn't you put them on?)
- b. **Universal Testing Machine (UTM):** Confirm the chucks on the UTM (see Figure 1) are screwed in all the way to avoid stripping the threads. Check rods supporting the hydraulic jack to ensure they are vertically straight and not twisted.

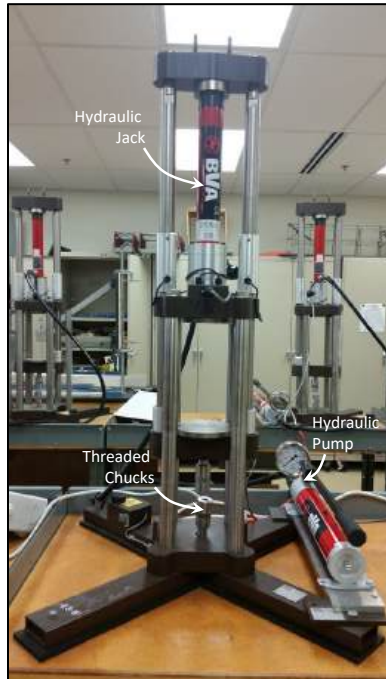


Figure 1 Uniaxial Tension Testing Machine

c. Specimen:

- (1) You are provided with two specimens for uniaxial tension testing. (Figure 2)



Figure 2 Uniaxial Tension Specimen

- (2) Use calipers (see Figure 3) to measure the diameter of each specimen at three locations. Record the diameter to an accuracy of 0.02 mm. Make sure your units are mm, not inches!

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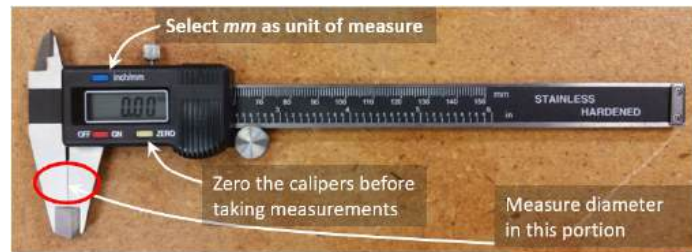


Figure 3 Digital Calipers

- (3) Mount the first specimen into the UTM. Test the specimen with the shinier surface first.
- Fully close the hydraulic release on the cylinder, near its gauge.
 - Use the hydraulic pump to raise the **top** chuck. Screw the specimen into the **top** chuck so that the threads are entirely within the chuck.
 - Gently lower the top chuck utilizing the hydraulic release, being careful not to jam the threads into the bottom chuck (or smash your fingers).
 - Close the release valve when the sample is close to the bottom chuck, grip the roller pivot above/below the chuck and screw the specimen into the bottom chuck.
 - You may have to lower the crosshead occasionally using the hydraulic release in order to maintain enough slack in the chuck. The specimen is properly placed when 2-3 threads are visible on both ends of the sample.
 - Take up the slack, using the manual pump, until all mechanical slack is removed. Do this slowly to avoid placing any appreciable load on the specimen.

d. Extensometer

- (1) You are provided one extensometer with a digital dial gauge. Ensure the extensometer is completely closed with the plunger on the bottom “diving board” as shown in the figure below.
- (2) Mount the extensometer on the specimen. Position the extensometer in the middle portion of the specimen. Avoid contacting the extensometer with the chuck.
- (3) Use the set screws to secure the extensometer to the specimen.
- (4) Make sure that the units are selected for millimeters, the gage is zeroed, and the resolution is three decimal places as shown.

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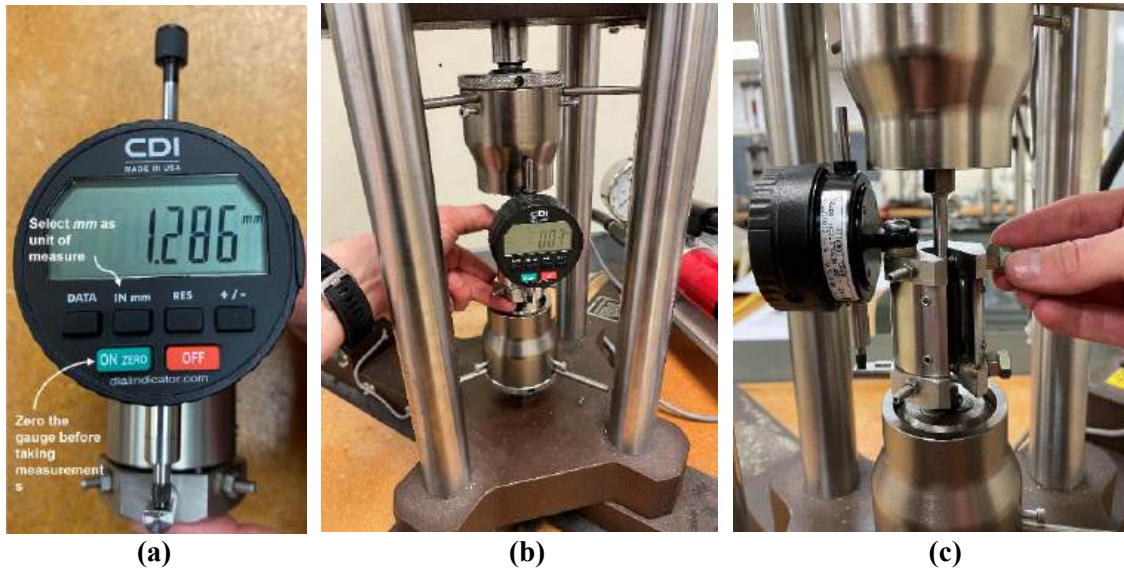


Figure 5 – (a) Digital Extensometer; (b) Mounting extensometer; (c) Tightening set screws

e. Digital Load Meter

(1) The digital load meter is used to measure the force on the specimen during testing. Make sure that the Digital Load Meter is plugged in and on.



Figure 6 Digital Load Meter

(2) After you have taken the slack out of the sample, press the “TARE” button to zero the meter. Press the “PEAK RESET” button in order to eliminate any previous peak readings. NOTE: The load is measured in kN.

f. **Safety:** Emplace the safety shield before loading the specimen. Do not stand behind the testing machine. Be careful when handling the specimen to remove it from the machine after the test since the samples might be hot or have jagged edges.

***** STOP *****
***** Have an instructor or lab technician check your set up before proceeding *****

Tension Test

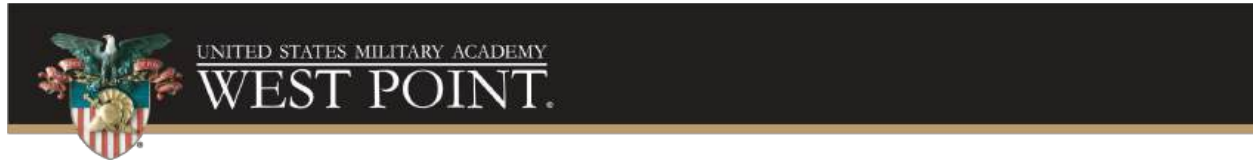
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- a. **Distribution of Labor:** It works well to have one cadet on the pump who is also reading the extensometer, one cadet reading the load meter, and another cadet recording the data. You can also have someone record the load meter with a video.
- b. **Applying Load to the Specimen:** Using a slow and steady motion, stressing rate between 70 MPa and 690 MPa /min. (ASTM A370-92, Section 11.4.3 and ASTM E8-94, Section 7.4.3.2), load the specimen using the manual pump. Maintain a constant load speed.
- c. **Recording Displacement Data:** Use the data sheet provided to record data. Record the applied load at each displacement increment. **If you miss a specific elongation point, simply record the actual elongation. If you miss several, do NOT panic, pick up at the closest increment on the data sheet and proceed with the test.** Use the following displacement increments:
 - (1) From $\delta = 0$ to $\delta = 0.2$ mm, take readings in 0.02 mm increments
 - (2) From $\delta = 0.2$ to $\delta = 1.0$ mm, take readings in 0.1 mm increments
 - (3) From $\delta = 1.0$ to $\delta = 12.0$ mm, take readings in 0.50 mm increments. Continue loading at the same loading rate until rupture.
- d. **Peak Load:** Record the largest load value by selecting the “PEAK” button on the load meter.
- e. **Extensometer:** Carefully remove the extensometer after failure. Be careful as the edges of the rupture surface may be sharp!
- f. **Specimen:** Remove the specimen and take a final picture of the broken specimen.

Outstanding Student Work

Fair-to-Good Student Work

Poor Student Work



LAB #2: TORSION LAB

CE364, 26-2, Torsion Test Report (100 points) – DUE: 11 FEB 2026 (Day 1) 12 FEB 2026 (Day 2)

In this laboratory exercise, your team conducted a torsion test on two samples with the same cross-section but made of different materials. The purpose of this report is to present the results of your experimental testing clearly and concisely. You will need to compare the measured values from the experiment to theoretical values from equations and specified material properties. Include a cover page IAW Dean's DAAW. **Each cadet must annotate next to their signature on the lab cover page the time they spent working on the lab report outside of class.** The main body of your report (Title, Introduction, Theory, Results, Conclusions) should be approximately 2 pages. Include figures, tables, and appendices after the main body. Be sure to include the following items in each of the sections.

Theory

Include a brief discussion and present relevant equations for the following items.

1. Polar moment of inertia
2. Relationship between Torque and Shear Stress
3. Relationship between Angle of Twist and Shear Strain
4. Relationship between Shear Stress and Shear Strain.
5. Relationship between the Shear Modulus and Young's Modulus
6. Shear Yield Strength based on the 0.2% offset method
7. Relationship between Axial Yield Strength and Shear Yield Strength

Hint Shear strength is 57.7% the tensile strength. i.e. $\tau_y = 0.577 \cdot \sigma_y$ (We will learn more about this relationship later in the course)

****Include assumptions needed to apply these equations****

Results

Answer the following questions in the results section:

1. Describe the materials' shear stress – shear strain graph in words only without a figure. In your description, report the key material properties: yield shear stress, yield shear strain, fracture shear stress, fracture shear strain, and Shear Modulus of Elasticity.
2. Identify the materials you tested based on your stress-strain graph and property values. Use specified material properties from the Hibbeler textbook or from other resources. Justify your selection and compare the theoretical values to experimental results.
3. Discuss any differences between your experimental and theoretical values. Explain why your answers make sense or why they are different. Report the percent difference for each value.
4. Report the average and standard deviation for the ultimate shear strength values for the material in torsion. If you were designing a structure with this material, what ultimate strength value would you use and why?
5. Compare your calculated Modulus of Rigidity value from the rear cover of your textbook to the Modulus of Rigidity measured in the Torsion Lab. Discuss any differences. Which value do you believe is more accurate and why?
6. Explain any sources of error which may have impacted the results. Identify any assumptions which may have impacted the validity of the results.

****Your response to the questions does not need to be in a specific order. Just address each of the above items.****

Figures

Include the following figures in your report

1. Figures of the experimental tests
2. Figures of the samples before and after testing
3. For both samples, create a shear stress-shear strain graph for all your data points similar to your axial stress-strain graph from the Tension Lab (You may plot both samples on the same graph to better compare the two materials). Include shear stress (in MPa) on the vertical axis and shear strain in (radians) on the horizontal axis. Label the key points on each graph to include ultimate shear stress (τ_u), fracture shear stress (τ_f), fracture shear strain (γ_f), yield shear stress (τ_y), yield shear strain (γ_y), and Shear Modulus of Elasticity (G).

CE364, 26-2, Torsion Test Report (100 points) – DUE: 11 FEB 2026 (Day 1) 12 FEB 2026 (Day 2)

4. For both samples, create a shear stress-shear strain graph focusing only on the elastic region with shear stress (in MPa) on the vertical axis and shear strain (radians) on the horizontal axis. Label the key points to include yield shear stress (τ_y), yield shear strain (γ_y), and Shear Modulus (G). (You may plot both samples on the same graph to better compare the two materials)

****Add lines and textboxes to the graph in PowerPoint or Excel, then “group” the figure and paste special as an “Enhanced Metafile Image” in Word.****

Tables

Include the following tables of results

1. Table with experimental values and theoretical material specified values for yield strength and shear modulus.
2. Table of ultimate shear strength for all groups in the section including the average and standard deviation.

Appendix

- A. Sample Calculations on Green/White Engineering Paper or Mathcad
 - a. Specimen average radius (r , mm) and polar moment of inertia (J , mm⁴).
 - b. Shear stress (τ , MPa) and shear strain (γ , radians) calculations for the fifth data point.
 - c. Shear Modulus of Elasticity (G , GPa) based on selecting two data points appropriately within the elastic region.
 - d. Axial Modulus of Elasticity (E , GPa) based on your shear modulus of elasticity. Assume a Poisson's ratio (ν) of 0.3.
- B. Process the lab data on a spreadsheet that includes a column for angle of twist (ϕ , radians), applied torque (T , N-m), shear stress (τ , MPa), and shear strain (γ , radians). Ensure your units are correct.

Torsion Test

Diameter 1 = _____ mm
 Diameter 2 = _____ mm
 Diameter 3 = _____ mm
 Average Diameter = _____ mm
 Average Radius, c = _____ mm
 $J = \pi r^4/2 =$ _____ mm⁴
 Gage Length, L = 50 mm
 Rupture Torque = _____ N-m
 Turns at Rupture = _____ #

Ultimate Torque = _____ N-m

Ult Shear Stress = _____ MPa = N/mm²

Ultimate Shear Stress from other Groups in Section

Group 1 =	
Group 2 =	
Group 3 =	
Group 4 =	
Group 5 =	
Group 6 =	

"Φ" = Angle of Twist		"T" = Torque	"Φ" = Angle of Twist		"T" = Torque
(radians)	(degrees)	(N-m)	(radians)	(degrees)	(N-m)
0.000	0		Now increments of 0.01 radians		
0.005	0.29		0.11	6.30	
0.010	0.57		0.12	6.88	
0.015	0.86		0.13	7.45	
0.020	1.15		0.14	8.02	
0.025	1.43		0.15	8.59	
0.030	1.72		0.16	9.17	
0.035	2.01		0.17	9.74	
0.040	2.29		0.18	10.31	
0.045	2.58		0.19	10.89	
0.050	2.86		0.20	11.46	
0.055	3.15		STOP - Reset Torsometer		
0.060	3.44		0.21	12.03	
0.065	3.72		0.22	12.61	
0.070	4.01		0.23	13.18	
0.075	4.30		0.24	13.75	
0.080	4.58		0.25	14.32	
0.085	4.87		0.26	14.90	
0.090	5.16		0.27	15.47	
0.095	5.44		0.28	16.04	
0.100	5.73		0.29	16.62	
STOP - Reset Torsometer			0.30	17.19	
			At Rupture:		

*Hint: Angle of Twist = # Turns * 6 °*

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Purpose: The torsion test provides information on the shear strength and ductility of a material.

Pre-Test Requirements

a. **Safety glasses:** Put them on, regardless of how cool they may or may not look.

b. **Torsion Testing Setup:**

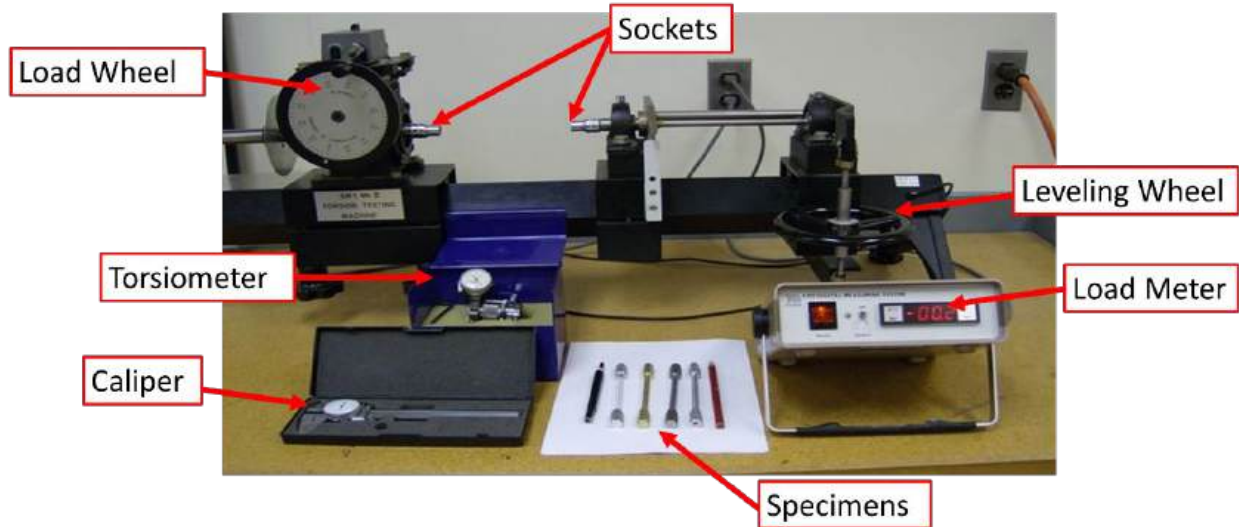


Figure 1 Torsion Testing Machine

c. **Specimens:**

(1) You are provided with two specimens for testing.



Figure 2 Torsion Specimen

(2) Score the specimen with a marker pen by making a line down the longitudinal axis.

(3) Use calipers (see Figure) to measure the diameter of the specimen at three locations. Record the diameter to an accuracy of 0.02 mm. Make sure your units are mm, not inches!

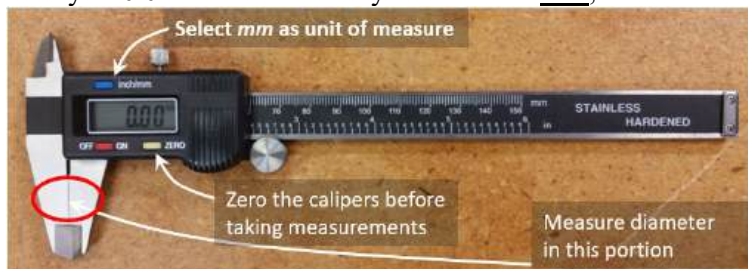
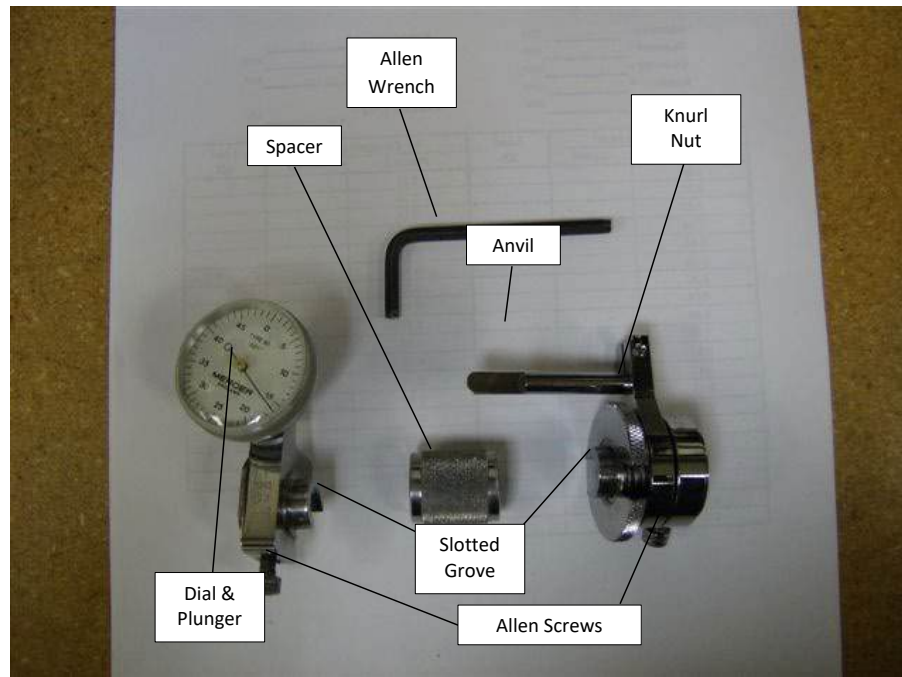


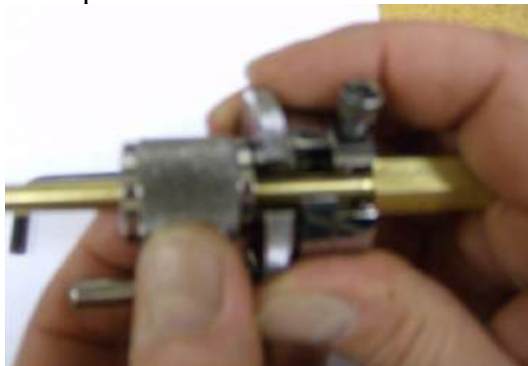
Figure 3 Digital Calipers

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Torsiometer: Mount the specimen into the Torsiometer as described below.



- (1) Back out allen screws so specimen will fit into grooves.
- (2) On Anvil & Knurl nut part loosen knurl nut and align all 3 grooved parts.
- (3) Place specimen in grooves.
- (4) Take spacer and slide over specimen until it makes contact with knurl nut alignment boss.



- (5) Slightly snug down allen screw on a Knurl nut part and ensure that Torsiometer is centered on specimen. **Stay away from filet.**
- (6) Take Dial and Plunger mount on specimen and slide so that spacer comes in contact with Dial & Plunge's alignment boss.** **NOTE** at this point there should be **NO** lateral movement between the Spacer, Knurl nut and the Dial.

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- (7) Slightly snug down allen screw on Dial.
- (8) Ensure that Torsiometer is centered on specimen, tighten both allen screws $\frac{1}{4}$ turn.
- (9) With Dial on Torsiometer facing you rotate anvil and fully depress plunger on Dial.



- (10) Tighten Knurl nut clockwise or toward you. Specimen is now ready to mount.



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TORSION TEST EXPERIMENTAL PROCEDURES
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d. Torsion Testing Apparatus:

(1) **DO NOT PLAY WITH THE LEVELING WHEEL. YOU WILL DAMAGE LOAD CELL!!**

(2) Mount the $\frac{7}{16}$ inch Whitworth hexagonal sockets on the input shaft and the deflection arm.

(3) Insert the specimen into the socket on the deflection arm, then move the input shaft forward until the socket engages the specimen. You may have to rotate the input handwheel until the socket and specimen line up. **BE CAREFUL NOT TO PUT ANY AXIAL LOAD ON THE SPECIMEN!**

e. The E101 Meter: Make sure that the E101 Meter is plugged in and on, and the units set to “SI.” The scale should be set so that only one decimal point is showing.

f. Pre-Test Final Checks:

(1) Take up the slack, rotating the input wheel *clockwise* until you notice a small amount of load (between 0.1 and 0.4 N-m). Slowly take the load back to zero; if the load does not go back to zero, manually verify that the slack is removed and zero the E101 meter again.

(2) Zero the dial gage on the Torsiometer.

(3) Zero the clicker counter.

1. Pure Torsion Test

a. Use the data sheet provided to record data. Record the applied torque at each angle of twist increment. Slowly load the specimen by rotating the input handwheel *clockwise*.

Note - each hash mark on the torsiometer is 0.001 radians.

Note – you will have to reset the torsiometer at the noted angle of twist values

(1) From $\phi = 0$ to $\phi = 0.10$ radians, take readings in 0.005 radian increments.

(2) From $\phi = 0.10$ radians to $\phi = 0.25$ radians, take readings in 0.01 radian increments.

(3) After $\phi = 0.25$ radians, remove the torsiometer. Continue loading the specimen to failure. Recorded the final failure torque and the reading on the clicker counter.

Note - Each click is equivalent to 6 degrees rotation.

b. Remove the test specimen from the torsion testing apparatus. Note the failure surfaces and any other observations.

Outstanding Student Work

Fair-to-Good Student Work

Poor Student Work



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LAB #3: BEAM LAB

CE364, 26-2, Beam Lab Report (100 points) – DUE: 4 MAR 2026 (Day 1) 5 MAR 2026 (Day 2)

In this laboratory exercise, your team conducted a flexural test on four beams with different cross-sections. Assume the beams are made of Aluminum Alloy 6061-T6. The purpose of this report is to present the results of your experimental testing clearly and concisely. You will need to compare the measured values from the experiment to theoretical values from equations and specified material properties. Follow the report format previously provided on Canvas. Include a cover page IAW Dean's DAAW. The main body of your report (Title, Introduction, Theory, Results, Conclusions) should be approximately 2 pages. Include figures, tables, and appendices after the main body. Be sure to include the following items in each of the sections.

Theory

Include a brief discussion and present relevant equations for the following items.

1. Area moment of inertia for each shape (I)– No need to show the entire calculation in the report. This will be in the Appendix.
2. Maximum first moment of outward area for each shape (Q)
3. Determining maximum shear and moment
4. Relationship between Flexural Normal Stress (σ) and Flexural Normal Strain (ϵ).
5. Flexural Normal Stress (σ) Equation as a function of height (y) from the NA of the beam
6. Flexural Shear Stress (τ) Equation as a function of first moment of outward area (Q)

Results

Answer the following questions in the results section:

- a. Discuss how close your calculated strain and measured strain are for each shape. Explain why they might be different.
- b. Discuss the difference between the performance of each of the beams. Which cross-section is the strongest with respect to normal strength? Which cross-section is the strongest with respect to shear strength?
- c. Quantify the difference in strength between the different shapes. What is the effect of the beam cross-section on normal and shear stress and the normal strain?
- d. What assumptions did you make when conducting this lab and comment on their validity.

****Your response to the questions does not need to be in a specific order. Just address each of the above items.****

Figures

Include the following figures in your report:

1. Figures of the experimental tests
2. Figures of the samples
3. Drawings of the cross-section of each shape with dimensions. Either computer aided drawings or neat hand sketches.
4. Complete shear and moment diagram for your group's loading (only draw one V&M diagram)
5. Plot of the normal stress distribution showing the variation of stress through the depth of the cross-section. Include all 4 shapes on this figure. Show normal stress (σ) on the x-axis and distance from the neutral axis (y) on the y-axis.
6. Plot of the shear stress distribution showing the variation of stress through the depth of the cross-section. Include all 4 shapes on this figure. Show shear stress (τ) on the x-axis and distance from the neutral axis (y) on the y-axis.

****Note: "up" is positive and "down" is negative. Tension normal stress is positive, and compression normal stress is negative. See example Figures 1 and 2 below. ****

Tables

Include the following tables of results

1. Table with cross-sectional properties, theoretical stress values, and experimental stress values for each cross-section. See sample Table 1 below

Appendix

- A. Sample Calculations on Green Engineering Paper or Mathcad
 - a. Shear and Moment Diagram (only draw one V&M diagram)
 - b. Cross-sectional area for each shape (A)
 - c. Area moment of inertia for each shape (I)
 - d. Maximum first moment of outward area for each shape (Q)
 - e. Maximum Flexural Normal Stress (σ) for each beam
 - f. Maximum Flexural Shear Stress (τ) for each beam
 - g. Flexural Normal Strain (ϵ) at the center of the beam (ie location of the strain gage) for each beam.
- B. Raw data from the lab shown in Table 2.
- C. Process the lab data on a spreadsheet that includes a column for distance from the NA (y), the location of the neutral axis ($\bar{y}$), thickness of the beam (t), first outward moment of area (Q), normal stress (σ), and shear stress (τ). The stress values are based on the internal moment and internal shear at the point of loading. Ensure units are correct! Plot this data on two figures (reference above for the two required figures) with stress on the x-axis and distance from the neutral axis (y) on the y-axis.

Table 1: Comparison between theoretical and experimental stress values

Shape	A	I	Q @ NA	Theoretical Normal Stress @ Top	Theoretical Shear Stress @ N.A.	Experimental Normal Stress @ Top
	(in ²)	(in ⁴)	(in ³)	(psi)	(psi)	(psi)
Rectangular						
HSS						
I-Beam						
T-Beam						

Table 2: Raw data from the lab

Beam Cross-Section	Load Position from Support (inches)	Measured Strain (micro-strain)
Rectangular		
HSS-Beam		
I-Beam		
T-Beam		

CE364, 26-2, Beam Lab (100 points)

Procedure:

1. Setup your lab group at a testing station that has four beams with supports, a strain box, and loads. You may have to share cross sections with a different group when they are complete.
2. Complete a sketch of the beam cross-sections provided and use the calipers take appropriate measurements (record these on the next sheet)
3. Position the supports for the beam for a clear span of 17 inches. Make sure the strain gauge is at mid span.
4. Position the flexible tubing with the weight hanger on the beam at the midspan of the beam.
5. Sketch the beam with supports and load and take appropriate measurements to determine internal shears and moments. Include these dimensions on your sketch.
6. Zero the strain gage box using the *Balance* button as necessary. See instructor as needed.
7. Load the weight hanger with 2ea 2-kg weights and record the strain at the center of the span.
8. Repeat for all three beam cross-sections

SETUP

As shown in Figure 3, separate two desks by about 15-inches. Place beam in triangular supports at marks on beam: unsupported span should be 17-inches. Place the weight hanger on the wire hanger which is located at mid-span of the beam. The hanger will be placed on the top strain gage – this is okay because the SG has been coated with a protective coating.

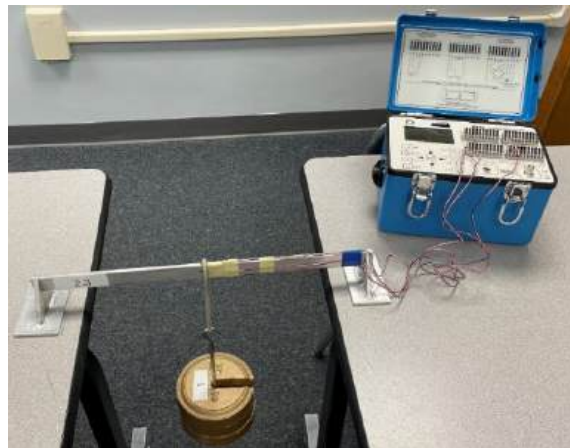


Figure 1 Setup Overview for Bending Lab

Connect the SG to any channel on the strain gage indicator box (see Figure 4). Connect **Red** to P+, **White** to S-, and **Black** to D(350). **Note: for this lab you are only recording strain at the top surface**

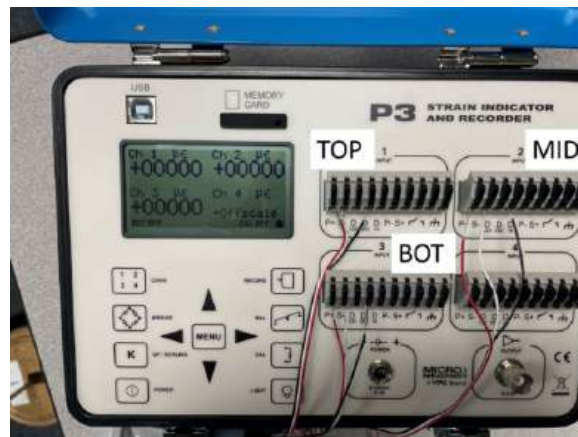


Figure 2 Connect SGs to Switch and Balance Box

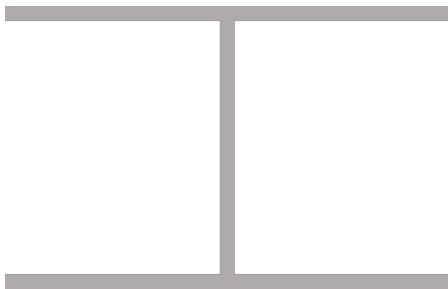
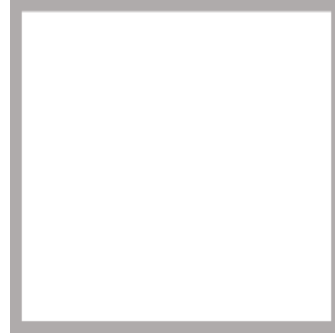
Zero the three strain gages before cadets begin the experiment. To zero the gages, press the Balance button “BAL” and follow the directions on the screen. Ensure that the strain readings are in the correct magnitude. If the strain gage readings are incorrect, adjust the scale factor till they match. “K”-Button (2.450 as starting point).

CE364, 26-2, Beam Lab (100 points)

Raw Data Record

Beam Cross-section Sketches

- Show all dimensions in inches



CE364, 26-2, Beam Lab (100 points)

Beam Span Sketch with Load

- Show all dimensions in inches
- Clearly show selected position of load with respect to support

Loaded Beam Results

Beam Cross-Section	Load Position from Support (inches)	Measured Strain (micro-strain)
Rectangular		
HSS-Beam		
I-Beam		
T-Beam		

Outstanding Student Work

Fair-to-Good Student Work

Poor Student Work



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LAB #4: THIN WALLED PRESSURE VESSELS

CE364, 26-2, Thin Walled Pressure Vessel Lab Report (100 points) – DUE: 20 APR 2026 (Day 1) & 21 APR 2026 (Day 2)

In this laboratory exercise, your team will apply a strain gage to a soda can to measure the internal pressure in the can. Prior to opening the soda can, you will estimate the strain you will measure when you open your soda can. You will then open the can and record the actual strain. Assuming that the soda can is a thin-walled pressure vessel, you can determine the original internal pressure. Assume the can is made of Aluminum Type 3104-H19 (Properties provided in Lab procedure). The purpose of this report is to present the results of your experimental testing clearly and concisely. You will need to compare the measured values from the experiment to theoretical values from equations and specified material properties. Follow the report format provided previously. Include a cover page IAW Dean's *DAAW*. **Each cadet must annotate next to their signature on the lab cover page the time they spent working on the lab report outside of class.** The main body of your report (Title, Introduction, Theory, Results, Conclusions) should be approximately 2 pages. Include figures, tables, and appendices after the main body. Be sure to include the following items in each of the sections.

Theory

Include a brief discussion and present relevant equations for the following items:

1. Plane strain transformation equations – include normal strain (ϵ_x) and shear strain (γ_{xy}), Generalized Hooke's Law for normal stress (σ_x & σ_y) and shear stress (τ_{xy})
2. Shear modulus (G)
3. Hoop stress and Longitudinal Stress (σ_{hoop} & σ_{long})

Results

- a. Which starting estimate – strain, internal pressure, or factor of safety -was most accurate compared to the measured strain reading? Justify with percent difference calculations. Provide explanation for why this method was good for estimating.
- b. Is the soda can manufactured with a reasonable factor of safety? Explain why or why not.
- c. Are soda cans efficient containers? Is there potentially a better design? Provide at least one pro and con for each improvement.
- d. What assumptions have you made in conducting this lab? Comment on each of their validity.
- e. Explain any sources of error which may have impacted the results.

****Your response to the questions does not need to be in a specific order. Just address each of the above items.****

Figures

Include the following figures in your report

1. Figures of the experimental tests
2. Figures of the sample before and after testing

Tables

Include the following tables of results

1. Table with internal pressure, hoop stress, factor of safety and strain

	Factor of Safety	Internal Pressure (psi)	Hoop Stress (ksi)	Hoop Strain ($\mu\epsilon$)
Strain Guess				
Pressure Guess				
Factor of Safety Guess				
Experimental Results				

**CE364, 26-2, Thin Walled Pressure Vessel Lab Report (100 points) – DUE: 20 APR 2026 (Day 1)
& 21 APR 2026 (Day 2)**

Appendix

A. Pre-Lab Questions:

- a. What type of pressure vessel is a soda can? What type of strain gage is your group using and what is it measuring? What type of strain gage is the instructor group data using and what is it measuring?
- b. What is your guess or estimate for the maximum strain gage reading of the TWPV? Explain how you determined your answer.
- c. What is your guess or estimate for the internal pressure of the TWPV? Explain how you determined your answer.
- d. What is your guess or estimate for the factor of safety of the TWPV? Justify your answer. You may find it useful to research common factors of safety used in other engineering applications.
- e. Did you find some of the questions in part (c-e) easier to answer than others? What was the hardest question? Why do you think the question was difficult to answer? Discuss the difference between a guessed and an estimated value.

B. Estimate Calculations for the Unopened Can. Complete neatly and to standard on green engineering paper. Only submit one set of answers per group. Provide commentary along with each step in the calculations. It is critical to provide detailed explanations of each calculation.

***HINT: The maximum allowable stress for a ductile material such as aluminum may be determined using the Tresca Maximum-Shear-Stress Theory*

*(Mechanics of Materials Section 10.7. More to follow on Lesson 30)***

$$\sigma_{max} = \sigma_1 \leq \sigma_Y$$

- a. What is your estimated thickness of the aluminum can? Measure part of the can and provide justification.
 - b. Provide hand calculations for your estimate of the maximum strain gage reading based on your estimated internal pressure from part 2c. Identify the direction and magnitude of the maximum strain gage reading. Include the factor of safety, hoop stress, and longitudinal stress.
 - c. Provide hand calculations for your estimate of the maximum strain gage reading based on your estimated factor of safety from part 2d. Identify the direction and magnitude of the maximum strain gage reading. Include internal pressure, hoop stress, and longitudinal stress.
- C. Calculations for Opened Can. Using the strain readings from the soda can, calculate the following values. Complete calculations neatly and to standard on green engineering paper. Only submit one set of answers per group. Provide commentary along with each step in the calculations. It is critical to provide detailed explanations of each calculation.**
- a. Calculate the normal strains and shear strains in the x- and y-direction based on the strain rosette readings (i.e. use strain transformation) from the instructor data provided.
 - b. Compare your measured strain from the linear gage to the instructor data provided from the rosette. Which strain does it most closely match?
 - c. Calculate the principal strains and determine the strains in the hoop direction and longitudinal direction.
 - d. Calculate the internal pressure.
 - e. Determine the hoop stress and longitudinal stress (Use Generalized Hooke's Law).
 - f. Calculate the principal stresses on the soda can.
 - g. Calculate the factor of safety of the soda can according to the Tresca Maximum-Shear-Stress Theory.

If you want to learn a lot more about the amazing soda can, we recommend watching this YouTube video.
<https://www.youtube.com/watch?v=hUhis2FBuw>

CE364, 26-2, Thin Walled Pressure Vessel Lab (100 points)

Procedure:

In this lab, you will be applying a strain gauge to a soda can to measure the pressure in the can. Prior to opening the soda can, you will be making a guess and estimate of the strain you will measure when you open your soda can. You will then open the can and record the actual strain. Assuming that the soda can is a Thin Walled Pressure vessel, you can determine the original internal pressure. Below are the estimated material properties of the aluminum can.

Material Data

Aluminum Type 3104-H19

Yield Strength = 37.7 ksi

Modulus = 10,010 ksi

Poisson's Ratio = 0.34

Example Videos:

Micro-Measurements - <https://www.youtube.com/watch?v=U0c9vzkW1to>

Michigan Tech - <https://www.youtube.com/watch?v=SCuH3AnZgFU>

PART I: Strain Gage Mounting Instructions (LSN 27)

* CAUTION: DO NOT TOUCH the surface once preparation has begun.

* KEEP YOUR WORK AREA CLEAN (close bottles tightly immediately after use)



1. *Degrease* the surface of the can using the degreaser. *Select* a location (1.5"x 3") free of dents, scrapes, or imperfections. *Spray* directly over workstation mat. *Wipe* dry with gauze after application.



CE364, 26-2, Thin Walled Pressure Vessel Lab (100 points)

2. Apply the **M-PREP CONDITIONER A** (Bottle with Red Nozzle) and *sand* the surface of the can with the sandpaper. Avoid over sanding the thin metal. *Clean* the surface with the **M-PREP CONDITIONER A** and gauze. *Wipe* surface with single strokes in one direction until surface is completely clean.



3. Apply **NEUTRALIZER** (Bottle with Blue Nozzle) and *wipe* with a single stroke of gauze until clean



4. Apply 2" long piece of masking tape to the edge of the sanded area in the can's longitudinal direction

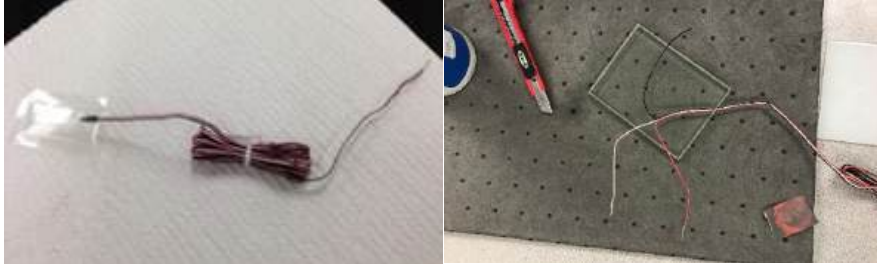


5. *Degrease* the glass plate and *wipe* dry with gauze. *Place* on the mat clean side up. Do NOT touch clean side.

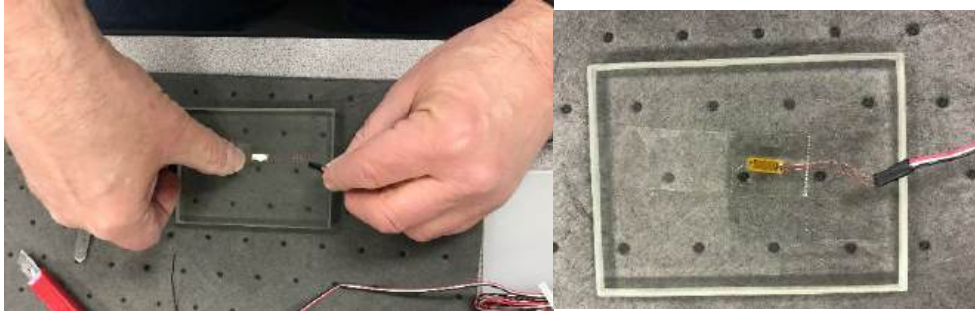


CE364, 26-2, Thin Walled Pressure Vessel Lab (100 points)

6. *Unpackage* strain gage, carefully. *LEAVE PLASTIC COVERING ON THE STRAIN GAGE*. *Unwind 6-7"* of wire from both ends. *Peel* apart the 3 terminal lead wires (Red, White, and Black) from each other. *If needed, *use* a utility knife to start separation*



7. *Remove* plastic covering from the gage and carefully *place* it on the glass with the shiny side up. *Discard* the initial 3" of cellophane tape to eliminate contaminants. *Tape* the gage to the glass using a new 3" piece of cellophane tape. *Apply* the tape so that the gage is centered. Do NOT secure the two ends of the tape so that you can remove the gage from the glass.



8. *Pull* the tape back off the glass at a 30° angle so that the strain gage comes off the glass with the tape still attached to the gage.
9. *Tape* the strain gage rosette to the can aligning one strain gage in the direction of **HOOP STRESS** and ~~one strain gage in the direction of the LONGITUDINAL STRESS~~. *Ensure* the wire end of the strain gage is just contacting the edge of the masking tape on the can. (This is to insulate the wires from the can).



10. *Peel* back the tape opposite the wired tab at a 30° angle to expose the bonding surface of the strain gage. *Secure* the tape.



CE364, 26-2, Thin Walled Pressure Vessel Lab (100 points)

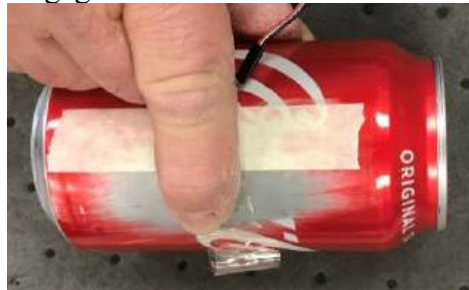
11. *Brush* the **M-BOND 200 CATALYST** (Purple Bottle) on to the gage with one swipe. Do NOT apply excess to the surface of the gage. *Allow* catalyst to Dry for at LEAST one minute.



12. *Squeeze* 2-3 drops of M-BOND 200 ADHESIVE onto the mat to remove air and contaminants from the nozzle. *Apply* 1-2 drops of M-BOND 200 ADHESIVE at the junction of the tape and the can in the area where the strain gage will be mounted.



13. Immediately, *rotate the tape to a 30 angle* (roll or press back the tape)) over the installation area. Slowly and firmly, make a single stroke *application* of the tape and gage to the can. *Apply* downward pressure with your thumb and maintain pressure for at least 2 minutes. *You want to ensure that you have complete contact with no air voids between the gage and the can.*



14. *Remove* thumb and wait one more minute. *Peel* back the tape from opposite end of the wire tabs. *Pull* directly back onto itself slowly. Do NOT pull the wires from the tabs.
15. *Secure* the wire to the can using another piece of masking tape. *Create* a relief loop as shown in the figure below. *Ensure* the wires are not touching each other or the can.



CE364, 26-2, Thin Walled Pressure Vessel Lab (100 points)

PART II: Pre-Lab Questions (Come to LSN29 with the following answers prepared)

Please cite your sources using the inline citation format.

- What type of pressure vessel is a soda can? What type of strain gage is your group using and what is it measuring? What type of strain gage is the instructor group data using and what is it measuring?
- What is your guess or estimate for the maximum strain gage reading of the TWPV? Explain how you determined your answer.
- What is your guess or estimate for the internal pressure of the TWPV? Explain how you determined your answer.
- What is your guess or estimate for the factor of safety of the TWPV? Justify your answer. You may find it useful to research common factors of safety used in other engineering applications.
- Did you find some of the questions in part (c-e) easier to answer than others? What was the hardest question? Why do you think the question was difficult to answer? Discuss the difference between a guessed and an estimated value.

PART III: Estimate Calculations for the Unopened Can (During LSN29)

*Complete neatly and to standard on green engineering paper. Provide commentary along with each step in the calculations. It is critical to provide detailed explanations of each calculation. *

HINT: The maximum allowable stress for a ductile material such as aluminum may be determined using the Tresca Maximum-Shear-Stress Theory (Mechanics of Materials Section 10.7)

$$\sigma_{max} = \sigma_1 \leq \sigma_Y$$

- What is your estimated thickness of the aluminum can? Measure part of the can and provide justification.
- Provide hand calculations for your estimate of the maximum strain gage reading based on your estimated internal pressure from part 2c. Identify the direction and magnitude of the maximum strain gage reading. Include the factor of safety, hoop stress, and longitudinal stress.
- Provide hand calculations for your estimate of the maximum strain gage reading based on your estimated factor of safety from part 2d. Identify the direction and magnitude of the maximum strain gage reading. Include internal pressure, hoop stress, and longitudinal stress.

PART IV: Recording Strain Gage Data (During LSN29)

Prior to recording the strain gage, verify your answers to the Pre-Lab Questions with your instructor.

- Connect your strain gage to a strain box setup.
 - Connect all three strain gages to channels 1-3
 - For each individual strain gage, use the Quarter Bridge setup, connect the **red** wire to terminal (P+), the **black** wire to terminal (S-), and the **white** wire to terminal (D350).
- Power up the box by pressing ϕ . Ensure channels 1-3 are reading a number to indicate the wires are connected correctly.
- Press **BAL** and follow the prompts to zero the strain reading.
- Pop the can and read the strain reading in micro-strain.

PART V: Experimental Calculations for Opened Can (Due w/ Lab Report 20/21 APR)

Using the experimental results, calculate the actual internal pressure of the soda can and actual factor of safety. Complete neatly and to standard on green engineering paper. Provide commentary along with each step in the calculations. It is critical to provide detailed explanations of each calculation.

- a. Calculate the normal strains and shear strains in the x- and y-direction based on the strain rosette readings (i.e. use strain transformation) from the instructor data provided.
- b. Compare your measured strain from the linear gage to the instructor data provided from the rosette. Which strain does it most closely match?
- c. Calculate the principal strains and determine the strains in the hoop direction and longitudinal direction.
- d. Calculate the internal pressure.
- e. Determine the hoop stress and longitudinal stress (Use Generalized Hooke's Law).
- f. Calculate the principal stresses on the soda can.
- g. Calculate the factor of safety of the soda can according to the Tresca Maximum-Shear-Stress Theory.

If you want to learn a lot more about the amazing soda can, we recommend watching this YouTube video. <https://www.youtube.com/watch?v=hUhsi2FBuw>

Outstanding Student Work

Fair-to-Good Student Work

Poor Student Work



TAB M DESIGN PROJECT



Balsa Beam Competition CE364 (100 points)



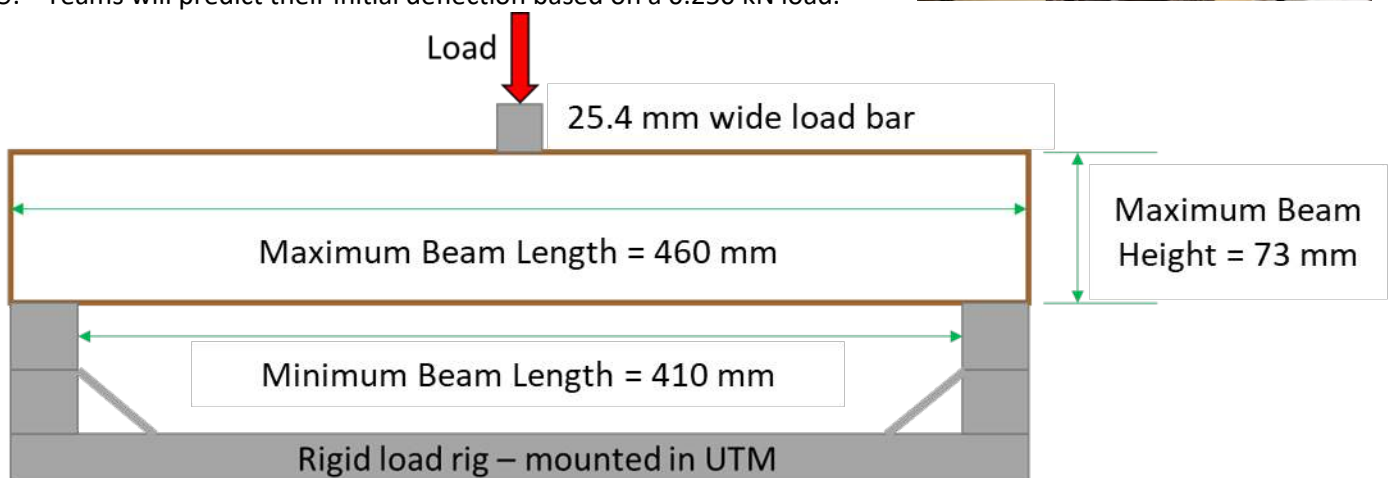
*"Members that are slender and support loads that are applied perpendicular to their longitudinal axis are called **beams**. ... Beams are considered among the most important of all structural elements. They are used to support the floor of a building, the deck of a bridge, or the wing of an aircraft." - R.C. Hibbeler, Statics and Mechanics of Materials, 6th ed. p.495.*

Introduction

Teams will **design**, **build**, and **DESTROY** a balsa beam using the Engineering Design Process. Balsa wood is a material with a high strength-to-weight ratio. Winners will be declared for the most efficient beam: the beam that carries the greatest load and weighs the least. Teams can also earn bonus points for the "best looking beam" and for making the most accurate predication of their failure load through analysis. Teams will submit a beam analysis and design prior to testing. Teams will hand-calculate the deflection, failure mechanism, and maximum load of their beams.

Design Requirements

1. Beams will be constructed from only balsa wood and hot glue. Balsa wood, cutting tools, and high temperature hot glue guns will be provided on Lesson 31.
2. Two Balsa wood sheets will be provided: one 36-inch x 3-inch x 1/4-inch and one 36-inch x 3-inch x 1/8-inch. No additional material may be used in constructing the beam (measure twice...cut once!).
3. The balsa wood may be cut to create a beam with a constant cross-section along the entire length.
4. The beam's cross-section is limited to 76 mm wide by 73 mm tall.
5. Beams must span 410 mm between supports as shown below.
6. The maximum length of the beam may not exceed 460 mm.
7. Balsa wood properties: $\sigma_{fail} = 6.65 \text{ MPa}$, $\tau_{fail} = 3 \text{ MPa}$, $E = 94 \text{ MPa}$.
8. The beams will be tested in a Universal Testing Machine (same machines used for the Axial Stress Lab). These machines will apply a load through a hydraulic pump system. The beams will be braced in the out-of-plane directions. The load will be applied through a 25.4 mm wide load bar, which you are permitted to model as a concentrated (point) load.
9. Teams will predict their initial deflection based on a 0.250 kN load.





Balsa Beam Competition CE364 (100 points)



Directions – There are two submissions for the EDP, which must be submitted in accordance with the Dean's current DAAW, including an appropriate cover page for each submission. Teams will be provided their balsa wood and storage box with utility knife and high temperature hot glue gun during LSN31. Materials will be returned during testing on LSN35.

Submission 1: Due 5 MAY (Day 1) or 6 MAY (Day 2) NLT 2359 on Canvas

- a. *Problem Definition* (10 points) Write a summary of the problem, objectives, facts, assumptions, and constraints. List and define the screening, evaluation, and weighting criteria.
- b. *Preliminary Calculations* (15 points)
 - i. Draw shear and moment diagram of their beam in terms of variables, following D/CEE presentation standards as initially introduced in MC300. Assume a load of P (kN) and a length of L (mm).
 - ii. Write an equation for the maximum shear and moment in terms of P and L .
 - iii. Write equations for the maximum bending normal stress and horizontal shear stress in terms of the variables. Your bending normal stress equation should be in terms of P , L , I , and c . Your shear stress equation should be in terms of P , L , Q , I , and t .
 - iv. Answer the following question: What section properties do you need to maximize (or minimize) to achieve the best design?
- c. *Conceptual Design* (25 points)
 - i. Produce a unique concept (i.e., three group members $\rightarrow$ three beam concepts). Provide a sketch and description of your concepts: specifically, different cross-sections and how the balsa wood will be cut. Use a straight edge!
 - ii. For each concept, calculate the area moment of inertia (I) and the maximum first moment of the outward area (Q).
 - iii. For each concept, calculate the maximum bending normal stress and horizontal shear stress under a load of 1 kN. Show your calculations with units. Indicate where on the cross section and along the beam each maximum stress will occur. (We recommend using both hand calcs & Excel).
- d. *Assessment and Decision Making* (10 pts)
 - i. Write a description of the pros and cons of each concept from part c and write a short paragraph comparing your concepts.
 - ii. Prepare a **weighted** decision matrix that evaluates the different beam concepts.
 - iii. Clearly indicate your team's selected design in a summary statement.
- e. *Detailed Design* (25 points)
 - i. Provide a detailed sketch of the 'as built' beam. Use the actual dimensions of your final design. Include a cross-section view and an elevation view.
 - ii. Calculations for the area moment of inertia (I) and the maximum first moment of the area (Q) of the 'as built' beam.
 - iii. Estimate the maximum deflection of your beam with a load of 0.250 kN.
 - iv. Estimate the maximum load your beam can sustain before failing from horizontal shear.
 - v. Estimate the maximum load your beam can sustain before failing from bending normal stress.
 - vi. What is the predicted failure load of your beam? Discuss how it will fail and how you identified the failure mode.



Balsa Beam Competition CE364 (100 points)



Submission 2: Due NLT start of class period on LSN35

- f. *Implementation* (15 points)
- Construct a beam from balsa wood and hot glue in accordance with your final design.
 - Create a PowerPoint quad-chart slide as shown. Use the EDP template on Canvas. Your team will brief from this slide in class on LSN35. Email the slide to your instructor prior to your class start time.
 - Return storage box with utility knife and high temperature hot glue gun. No credit will be given for this assignment until all materials are returned.

Team Name Cadet Names Picture of Beam	Beam Cross-Section Figure w/dimensions I, C, and Q values
Predicted Failure Load (kN), Failure Mode, and Location	Predicted Maximum Deflection and Location at 0.250 kN Load

Bonus Points!

- ❖ +3 pts (each section) for the beam that can carry the GREATEST load with the lowest weight.
- ❖ +3 pts (each section) for the “best looking” beam (be creative!)
- ❖ +3 pts (each section) for the closest failure load prediction
(Note: +2 and +1 for each award for 2nd and 3rd place also awarded)

